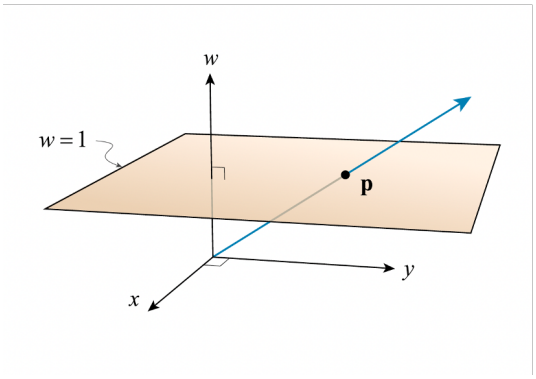
Geometric Algebra (GA)

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- *https://github.com/Daniel-G-W-Hug/ga/tree/develop*
  (The Geometric Algebra C++ Template Library including Python and Lua wrappers)
- *https://github.com/Daniel-G-W-Hug/ga/blob/develop/ga_docu/0_ga_docu.pdf*
  (This document)



Projective point. Source: „Foundations of Projective Geometric Algebra“, presentation Eric Lengyel, 05/2024

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1 GA library

1.1 Introduction to the ga library

The ga library is intended to make Geometric Algebra (GA) more accessible by providing functionality for numerical calculations required for simulation of physical systems.

It implements functionality for two- and three-dimensional spaces with Euclidean geometry. For this purpose, it provides data types and operations to handle the algebras of two- and three-dimensional Euclidean space ($EGA2D$, $EGA3D$) as well as for projective two- and three-dimensional Euclidean space ($PGA2DP$, $PGA3DP$). The latter two are modelled in a three- and four-dimensional space, respectively, since both include an additional projective dimension. On top of the projective dimension the conformal algebras ($CGA2DC$, $CGA3DC$) add a second one, the round dimension, and are thus modelled in a four- and five-dimensional space. It makes point pairs, circles and spheres directly representable as blades, next to the flat objects already known from PGA. The library also implements four-dimensional spacetime algebra ($STA4DS$).

For the ga library we use linear spaces $\mathbb{G}(p, n, z)$, where the arguments signify: p = number of basis vectors squaring to +1, n = number of basis vectors squaring to -1, and z = number of basis vectors squaring to 0. The linear space $\mathbb{G}$ is an extension of a linear vector space to a linear space of multivectors in the respective basis (for details see [Macdonald, 2010]).

- $EGA2D/3D$: $\mathbb{G}(2,0,0)/\mathbb{G}(3,0,0)$ (extends $\mathbb{R}^2/\mathbb{R}^3$)
- $PGA2DP/3DP$: $\mathbb{G}(2,0,1)/\mathbb{G}(3,0,1)$ (projective geometry)
- $CGA2DC/3DC$: $\mathbb{G}(3,1,0)/\mathbb{G}(4,1,0)$ (conformal geometry; the signature applies after diagonalization, the library stores the null basis, see subsection 1.2.14)
- $STA4DS$: $\mathbb{G}(1,3,0)$ (Spacetime algebra with signature (-1,-1,-1,+1))

The library efficiently handles fully populated multivectors, even- and odd-grade multivectors as well as their components like scalars, vectors, bivectors, trivectors, quadvectors (in the five-dimensional $CGA3DC$), and pseudoscalars of the corresponding linear spaces. It also provides operations to work with these objects. It was designed with extensibility in mind.

The core implementation language of the library is C++. The documentation is focused on the C++ implementation exclusively. However, the library provides wrappers for python (3.10+) in `ga_py` that is fully generated from the C++ implementation using nanobind, i.e. it covers almost 100% of C++ data types and functionality. There also is a wrapper for lua in `ga_lua`.

The main library is a header-only library in the `ga` sub-folder, which can be used by including either

```
#include ga/ga_ega.hpp // and/or
#include ga/ga_pga.hpp
#include ga/ga_cga.hpp
#include ga/ga_sta.hpp
```

and by making the corresponding namespaces accessible by stating

```
using namespace hd::ga;      // and
using namespace hd::ga::ega; // either/or
using namespace hd::ga::pga;
using namespace hd::ga::cga;
using namespace hd::ga::sta;
```

in your application code. For a simple usage example please have a look either at `ga_test/src/ga_ega_test.cpp` or similar.

1.1.1 Data types – user abstractions for each algebra

At the heart of the library there are user defined types that model mathematical objects of the respective geometric algebra. They are realized using type tagging in C++ to maximize reuse and to avoid code duplication. All types are defined in `ga/detail/type_t`.

To give an example for $EGA2D$, there are type aliases defined in the file `ga/detail/type_t/ga_type2d.hpp` which provide types for scalars, vectors, pseudoscalars, even-grade multivectors and full multivectors with the corresponding number of components. We have to distinguish between scalars and pseudoscalars at type level in order to allow for complement operations (defined later on) that can distinguish those types for each algebra. Following types are provided:

```
namespace hd::ga {

template <typename T> using Scalar2d
```

```
                            = Scalar_t<T,scalar2d_tag>;
template <typename T> using Vec2d = Vec2_t<T,vec2d_tag>;
template <typename T> using PScalar2d
                            = Scalar_t<T,pscalar2d_tag>;
template <typename T> using MVec2d_E = MVec2_t<T,mvec2d_e_tag>;
template <typename T> using MVec2d = MVec4_t<T,mvec2d_tag>;

} // namespace hd::ga
```

Based on these template aliases types like `Vec2d<T>` can be used directly. However, for increased user convenience additional template specializations have been defined in `ga/ga_usr_types.hpp` as

```
using scalar2d = Scalar2d<value_t>;
using vec2d = Vec2d<value_t>;
using pscalar2d = PScalar2d<value_t>;
using mvec2d_e = MVec2d_E<value_t>;
using mvec2d = MVec2d<value_t>;
```

where `value_t` is defined for the whole library in `ga/ga_value_t.hpp`. Thus, the user can simply select `value_t=double` (set as default value) or `float` for the library and the user application at one location. There is no need to use template parameters at all, if the user-defined types like `scalar2d, vec2d,pscalar2d,mvec2d_e`, and `mvec2d` are used for the application.

All algebras follow the same pattern. Just to give a second example for $PGA3DP$ following user types are defined: `scalar3dp`, `vec3dp`, `bivec3dp`, `trivec3dp`, `pscalar3dp`, `mvec3dp_e`, `mvec3dp_u`, and `mvec3dp`, where we additionally have defined types for bivectors, trivectors and uneven-grade multivectors, as required to completely implement the algebra.

The conformal algebras follow that pattern with the suffixes `2dc` and `3dc`, where the five-dimensional space of $CGA3DC$ requires the additional quadvector type:

```
// CGA2DC (16 components)
scalar2dc, vec2dc, bivec2dc, trivec2dc,
pscalar2dc, mvec2dc_e, mvec2dc_u, mvec2dc

// CGA3DC (32 components)
scalar3dc, vec3dc, bivec3dc, trivec3dc, quadvec3dc,
pscalar3dc, mvec3dc_e, mvec3dc_u, mvec3dc
```

To work with the user defined algebraic objects, the operations documented in the next subsection using these types are defined in the ga library (and some more only documented in source code). You will find operations required to work with linear (multi-)vector spaces and operations specific to geometric algebra. In the following replace `algebra` with any of {`ega2d`, `ega3d`, `pga2dp`, `pga3dp`, `cga2dc`, `cga3dc`, `sta4ds`}:

- `ga_algebra_ops_basics.hpp` contains basic operations like reversions, conjugations, complements, duals, norms, normalization and additionally for PGA bulk, weight and unitization operations. For CGA the split is four-part (round and flat bulk and weight, each with its norm), and the conformal conjugate cconj is provided next to the Clifford conjugation conj.

- `ga_algebra_ops_products.hpp` contains product operations like dot, wedge, commutator, transwedge and geometric products, and their regressive counterparts, as well as contractions, expansions, and inverses of blades.

- `ga_algebra_ops.hpp` contains general operations like angle, exp, sqrt, project_onto, reflect_from, is_congruent, and sandwich products for motion operators provided as rotate (EGA), move2dp/3dp (PGA), and transform (STA and CGA, kept apart by their namespaces); the functions get_rotor (EGA, STA), get_motor (PGA), and get_boost (STA) provide the input arguments for the motion operations. In CGA the motors come from get_translation, get_rotation and get_dilation, in $CGA3DC$ additionally from get_transversion and get_loxodromic; invert_on applies an inversion in a sphere or a circle as a flector sandwich. The exponential map and its inverse are named exp, log and sqrt w.r.t. the geometric product (EGA, STA), and rexp, rlog and rsqrt w.r.t. the regressive geometric product (PGA, CGA).

- `ga_algebra_ops_mechanics.hpp` (PGA only) contains basics for rigid body dynamics calculations like inertia, its inverse and right hand sides for the dynamics equations.

All product expressions for inner, outer, interior and geometric products as well as their regressive counterparts are generated by programs in the subfolder `ga/ga_prdxpr`. User input to configure product generation is provided in the header files of the corresponding algebra. The generated coefficient expressions are used in the source code to implement dot products, wedge products, commutator products, contractions, expansions, geometric products or their regressive counterparts, as well as sandwich products describing

rotors and motors of the corresponding algebra. If certain products are required for your application, but not yet implemented in the ga library, they can be easily generated within `ga/ga_prdxpr/src_prdxpr` as needed and added with minimal effort.

1.1.2 Comparing values – equality, closeness and congruence

Computed objects are rarely bit-identical, and three different questions can be asked of two of them: whether they hold the same value up to a few roundings (`operator==`), whether they hold the same value up to the error accumulated along the expressions that produced them (`is_close`), or whether they describe the same object of projective space and differ only by a scalar factor (`is_congruent`). Each question has its own test. All three work componentwise on the coefficients and all three are relative to the size of their operands. With a_i and b_i the components of a and b, the scale of the pair is

$$s = \max\left(|a_1|, \ldots, |a_n|, |b_1|, \ldots, |b_n|, 1\right), \tag{1}$$

the largest magnitude occurring in either operand, floored at one so the test stays absolute near zero.

Equality compares against $\varepsilon = 5\,\varepsilon_m$, where ε_m is the machine epsilon of `value_t`:

$$\texttt{a == b} \quad \Longleftrightarrow \quad |a_i - b_i| < \varepsilon\, s \quad \text{for all } i. \tag{2}$$

Let $\mathrm{ulp}(x)$ denote the distance from x to the next representable number. Since $\varepsilon_m = \mathrm{ulp}(1)$, the product $\varepsilon\, s$ lies between $\mathrm{ulp}(s)$ and $2\,\mathrm{ulp}(s)$, so the window is five to ten units in the last place of the operands at any magnitude. A window fixed in absolute terms is only equivalent to that near magnitude one: at a coordinate of order 10^6 one ulp is already $2 \cdot 10^{-10}$, and an absolute ε of order ε_m there asks for agreement closer than two neighbouring `double` values, which no computation can deliver. The scale is taken over the whole object and not per component, because a component that should vanish carries the rounding of the coordinates it was computed from rather than its own.

`is_close(a,b,rel_tol)` is the same test with a wider budget, by default $\varepsilon_c = 4.5 \cdot 10^3\, \varepsilon_m$, which for `value_t=double` evaluates to $9.99 \cdot 10^{-13}$. The factor is an error budget of some 4500 roundings, i.e. three to four significant digits given up to accumulation, and it is the right comparison for two independently computed quantities – results of wedge, dual and motor chains

– whose difference has grown over many operations instead of one.

`is_congruent(a,b,rel_tol)` asks whether $a = k\,b$ holds for some $k \neq 0$:

$$|a_i - k\,b_i| < \varepsilon_c\, s \quad \text{for all } i, \qquad k = \frac{a_j}{b_j} \;\text{ with }\; |b_j| = \max_i |b_i|. \tag{3}$$

The factor is read off the largest component of b, so the rounding of a small coefficient is never amplified onto the dominant ones. Any two non-zero scalars are congruent, as are any two non-zero pseudoscalars. For objects of a projective space congruence and equality are different questions until the object is unitized, and section 1.7 shows where the difference matters physically.

None of the three relations is transitive: three objects can agree pairwise within the window without the outer two agreeing with each other, so `a == b` and `b == c` do not give `a == c`. Equality of library types is therefore not an equivalence relation, and the types must not be sorted, passed to `std::unique`, or used as keys of an associative container. Where exact equality is meant, the coefficients are compared directly.

1.1.3 Product types – inner, interior, outer, geometric

As mentioned before, we distinguish inner, interior, outer and geometric products.

- Inner ($a \cdot b$) and interior products, like contractions and expansions ($a \vee b^\star$, $a \wedge b^\star$), are grade reducing operations. The inner product combines arguments of the same type and grade, while the interior products involve duals of one of their arguments, combining and linking arguments of different grades. Inner products are linked to the metric of the modelled space. They provide the capability of projecting objects like vectors on other objects like vectors or planes.

- The outer product ($a \wedge b$) is a grade-increasing operation adding the grades of its arguments. It extends the space that is taken by one object by the space that is taken by another object (e.g. two linear independent vectors, representing a linear subspace each, span a plane that is defined by the additive combination of those sub-spaces). The outer product (also called the wedge product) connects subspaces represented by their operands and creates new subspaces defined by the span of the subspaces of its operands. It establishes a join-operation between the involved subspaces.

The outer product also allows for direct calculation of linear independence: if vectors are linear independent their outer product is unequal zero, if they are linear dependent their outer product is zero.

- The geometric product ($a \wedge b$) is the combination of both types of products in one expression. They can result in multivectors consisting of parts with different grades as result of the operation. The main advantage of the geometric product is that it is invertible under certain circumstances.

- Regressive products ($a \circ b \equiv \underline{\overline{a} \cdot \overline{b}}$, $a \vee b \equiv \underline{\overline{a} \wedge \overline{b}}$, $\mathrm{rcmt}(a,b) \equiv \underline{\mathrm{cmt}(\overline{a},\overline{b})}$, $a \barwedge b \equiv \underline{\overline{a} \barwedge \overline{b}}$) use complement operations on their arguments before and after applying the operation[1]. The regressive wedge product finds common lower-dimensional subspaces, i.e. it effectively finds the intersection of the sub-spaces represented by its arguments, thus effectively creating a meet-operation between the involved spaces. It can be used to find the intersection of sub-spaces, like the intersection point between a line and a plane, as a very simple product operation.

- Sandwich products ($R \barwedge u \barwedge \widetilde{R}$ for EGA and STA, or $M \barwedge u \barwedge \underset{\sim}{M}$ for PGA and CGA) are used to implement orthogonal transformations of geometric objects u. They realize rotations for EGA and general motions including rotation and translation for PGA by concatenating two consecutive reflections to implement the resulting orthogonal transformations. In $2D$ the consecutive reflections are across lines, while in $3D$ the consecutive reflections are across planes. This can be achieved algebraically by multiplying the object symmetrically from the left- and right-hand side with the appropriate (typically unitized) objects, we either call rotors R in EGA or motors M in PGA. In CGA the reflections are taken across spheres and circles as well, and the motors M then realize the conformal transformations, which include the rigid motions of PGA as the subgroup that fixes the point at infinity.

Following bi-argument operations `op(a,b)` = a op b and their regressive variants `rop(a,b)` = $\overline{\overline{a}\ \mathrm{op}\ \overline{b}}$ are implemented (with `op(a,b)` or `rop(a,b)` as the name of the implemented function). Using `A` vs. `a` implies `gr(A)` $\geq$ `gr(a)`, where `gr(arg)` returns the grade of the argument `arg`:

- Inner product, delivering a scalar: $a \cdot b$, `dot(a,b)`

[1] For definition of the left and right complement operations marked with under-line and over-line, see next section.

- Outer product (wedge product) of a j- and a k-vector, delivering a $j+k$-vector in $span(a,b)$: $a \wedge b$, `wdg(a,b)`
- Geometric product: $a \owedge b$, `operator*(a,b)`
  - For vectors a and b: $a \owedge b = a \cdot b + a \wedge b$
  - For a vector a and a multivector B: $a \owedge B = B \gg a + a \wedge B$
  - For a multivector A and a vector b: $A \owedge b = b \ll A + A \wedge b$
- Commutator product (the asymmetric part of the geometric product): $\mathrm{cmt}(a,b) \equiv \frac{1}{2}(a \owedge b - b \owedge a)$, `cmt(a,b)`
- Regressive commutator product (the asymmetric part of the regressive geometric product):
  $\mathrm{rcmt}(a,b) \equiv \frac{1}{2}(\underline{a} \owedge \underline{b} - \underline{b} \owedge \underline{a}) = \frac{1}{2}(a \vee b - b \vee a)$, `rcmt`(a, b)
- Regressive inner product, delivering a pseudoscalar: $a \circ b$, `rdot(a,b)` (for comparison of the dot product and the regressive dot product for $PGA3DP$ see table 9 at the end of section 1.2)
- Regressive outer product (regressive wedge product): $a \vee b$, `rwdg(a,b)`
- Regressive geometric product: $a \vee b$, `rgpr(a,b)`, exclusively for PGA and CGA
- Left (bulk) contraction: $a \lfloor B = a \ll B = a_\star \vee B$, `operator<<(a,B)`, which implements `l_bulk_contract(a,B)`, exclusively for PGA
- Right (bulk) contraction: $B \rfloor a = B \gg a = B \vee a^\star$, `operator>>(B,a)`, which implements `r_bulk_contract(B,a)`, exclusively for PGA
- Left weight contraction: $a_\star \vee B$, `l_weight_contract(a,B)`
- Right weight contraction: $B \vee a^\star$, `r_weight_contract(B,a)`
- Left bulk expansion: $A_\star \wedge b$, `l_bulk_expand(A,b)`
- Right bulk expansion: $a \wedge B^\star$, `r_bulk_expand(a,B)`
- Left weight expansion: $A_\star \wedge b$, `l_weight_expand(A,b)`
- Right bulk expansion: $a \wedge B^\star$, `r_weight_expand(a,B)`

where the items from the left contraction downwards are all considered interior products. The product operation is performed *after* applying a dualization operation to one operand.

All products operate directly on the subspaces occupied by the objects that serve as arguments for the product operation.

The product tables for all products are generated in `ga/ga_prdxpr`. They can be derived directly from simple rules that define the products between basis vectors. The rules are generated for the geometric product, the outer (or wedge) product, the dot product, the complements and duals of the respective algebra. All resulting product expressions can be calculated from those few basic rules. Coefficient expressions for specific argument types like scalars, vector, bivectors, etc. can be generated based on the product tables using configuration options of `ga/ga_prdxpr` for each algebra.

PGA explicitly means *Projective Geometric Algebra* in the context of the ga library. Lengyel calls it in his book either *Projective Geometric Algebra* or sometimes *Rigid Geometric Algebra*, due to the transformations modelling rigid body motions. There is also an alternative approach called *Plane-based Geometric Algebra* (also abbreviated *PGA*) that is based on a space that is dual to the space modelled and used here. The alternative approach was introduced by [Gunn, 2020] and further promoted for applications in projective geometry by [Dorst and De Keninck, 2022]. That model uses dual representations of objects represented in the algebra (it uses $\mathbb{G}^*(3,0,1)$, a space dual to $\mathbb{G}(3,0,1)$). Planes are used instead of vectors to model everything, e.g. a point is modelled by three planes intersecting in a point. Both representations are mathematically equivalent, i.e. they are dual to each other. They are linked by the definition of complements and the interconnection between geometric and regressive geometric product, as defined above: $a \vee b \equiv \overline{\underline{a} \wedge \underline{b}}$. In the plane-based approach the geometric product is used to operate on complements and duals, which are defined as the primary objects used for the approach, while the approach introduced by Lengyel in [Lengyel, 2024b] favors the use of points, vectors, bivectors and planes plus the explicit use of regressive products and complements as well as duals for applying geometric operations to these objects. The latter approach unifies the use of the wedge product for join operations and of the regressive wedge product for meet operations for applications.

[Lengyel, 2024a] points out mathematical inconsistencies in applications of geometric algebra as used in the past quite often by different authors. Fur-

ther, hints on mathematical inconsistencies in typical GA approaches can be found as well in [Kritchevsky, 2024]. The goal of this work is to be as consistent and mathematically rigorous as possible implementing GA functionality, and at the same time providing an intuitively applicable framework for modelling physics and applications.

1.2 Basics of the mathematical approach

1.2.1 Basis vectors for each modelled algebra

EGA2D: The following is provided for Euclidean geometric algebra of two-dimensional space using an orthonormal basis $\boldsymbol{e}_1$, $\boldsymbol{e}_2$:

$$(\boldsymbol{e}_1)^2 = \boldsymbol{e}_1 \wedge \boldsymbol{e}_1 = \boldsymbol{e}_1 \cdot \boldsymbol{e}_1 = \boldsymbol{e}_1^2 = +1, \text{ and } \boldsymbol{e}_2^2 = +1$$

$$s_{2d} = s\,\mathbf{1} \tag{4a}$$

$$v_{2d} = v_x\boldsymbol{e}_1 + v_y\boldsymbol{e}_2 \tag{4b}$$

$$ps_{2d} = ps\,\boldsymbol{e}_{12} = ps\,\mathbb{1} \tag{4c}$$

equation (4a) is the scalar part s_{2d}, equation (4b) contains the vector part v_{2d}, and equation (4c) contains the pseudoscalar part ps_{2d}. The index $2d$ is omitted when clear from context. The four basis elements are (shown per grade):

$$\{\ \mathbf{1},\ \boldsymbol{e}_1, \boldsymbol{e}_2,\ \boldsymbol{e}_{12}\ \}$$

Using this basis a multivector M of $2d$-space is defined as

$$M = s\,\mathbf{1} + v_x\boldsymbol{e}_1 + v_y\boldsymbol{e}_2 + ps\,\boldsymbol{e}_{12} \tag{5}$$

where equation (5) contains three parts: the scalar part $s\,\mathbf{1}$ (basis element is the scalar $\mathbf{1}$; if $\mathbf{1}$ is not shown, it is implicitly assumed for scalar values), the vector part v (basis elements $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$) and the pseudoscalar part $ps\,\boldsymbol{e}_{12} = ps\,\mathbb{1}$ (basis element is $I_{2d} = \boldsymbol{e}_{12}$, which is sometimes written as $\mathbb{1}$ to show its character as pseudoscalar of this space. I_{2d} is a bivector in two-dimensional Euclidean space).

EGA3D: For Euclidean geometric algebra of three-dimensional space using an orthonormal basis $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, $\boldsymbol{e}_3$ there are:

$$(\boldsymbol{e}_1)^2 = \boldsymbol{e}_1 \wedge \boldsymbol{e}_1 = \boldsymbol{e}_1 \cdot \boldsymbol{e}_1 = \boldsymbol{e}_1^2 = +1, \boldsymbol{e}_2^2 = +1, \text{ and } \boldsymbol{e}_3^2 = +1$$

$$s_{3d} = s\,\mathbf{1} \tag{6a}$$

$$v_{3d} = v_x\boldsymbol{e}_1 + v_y\boldsymbol{e}_2 + v_z\boldsymbol{e}_3 \tag{6b}$$

$$B_{3d} = b_x\boldsymbol{e}_{23} + b_y\boldsymbol{e}_{31} + b_z\boldsymbol{e}_{12} \tag{6c}$$

$$ps_{3d} = ps\,\boldsymbol{e}_{123} = ps\,\mathbb{1} \tag{6d}$$

equation (6a) is the scalar part s_{3d}, equation (6b) is the vector part v_{3d}, equation (6c) is the bivector part B_{3d}, and equation (6d) is the pseudoscalar

part ps_{3d}. The index $3d$ is omitted when clear from context. Compared to the $2d$-case it is obvious, that all parts depend on context, specifically on the dimensionality of the modelled space, and thus need to be defined and treated accordingly.[2] The eight basis elements are (shown per grade):

$$\begin{aligned}\{\ & \mathbf{1},\\ & \boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3,\\ & \boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12},\\ & \boldsymbol{e}_{123}\ \}\end{aligned}$$

Using this basis a multivector M of $3d$-space is defined as

$$M = s\,\mathbf{1} + v_x\boldsymbol{e}_1 + v_y\boldsymbol{e}_2 + v_z\boldsymbol{e}_3 + b_x\boldsymbol{e}_{23} + b_y\boldsymbol{e}_{31} + b_z\boldsymbol{e}_{12} + ps\,\boldsymbol{e}_{123} \tag{7}$$

where equation (7) contains four parts: the scalar component $s\,\mathbf{1}$ (basis element is the scalar $\mathbf{1}$), the vector part v (basis elements $\boldsymbol{e}_1$, $\boldsymbol{e}_2$ and $\boldsymbol{e}_3$), the bivector part B (basis elements $\boldsymbol{e}_{23}$, $\boldsymbol{e}_{31}$ and $\boldsymbol{e}_{12}$) and the pseudoscalar part $ps\,\boldsymbol{e}_{123} = ps\,\mathbb{1}$ (basis element is $I_{3d} = \boldsymbol{e}_{123}$, which is sometimes written as $\mathbb{1}$ to show its character as pseudoscalar of this space. I_{3d} is a trivector in three-dimensional Euclidean space).

PGA2DP: Projective geometric algebra of $2D$ Euclidean space is modelled by embedding it in a three-dimensional space. Using an orthonormal basis $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, $\boldsymbol{e}_3$ there are following basis vectors:

$$(\boldsymbol{e}_1)^2 = \boldsymbol{e}_1 \wedge \boldsymbol{e}_1 = \boldsymbol{e}_1 \cdot \boldsymbol{e}_1 = \boldsymbol{e}_1^2 = +1, \boldsymbol{e}_2^2 = +1, \text{ and } \boldsymbol{e}_3^2 = 0$$

$$s_{2dp} = s\,\mathbf{1} \tag{8a}$$

$$v_{2dp} = v_x\boldsymbol{e}_1 + v_y\boldsymbol{e}_2 + v_z\boldsymbol{e}_3 \tag{8b}$$

$$B_{2dp} = b_x\boldsymbol{e}_{31} + b_y\boldsymbol{e}_{32} + b_z\boldsymbol{e}_{12} \tag{8c}$$

$$ps_{2dp} = ps\,\boldsymbol{e}_{321} = ps\,\mathbb{1} \tag{8d}$$

The most obvious difference is the basis vector $\boldsymbol{e}_3$ that squares to zero and the pseudoscalar of that space that is chosen to be $\boldsymbol{e}_{321}$. The trivector $\boldsymbol{e}_{321}$ squares to 0, because of its component $\boldsymbol{e}_3$ contained as a factor. Equation (8a) is the scalar part s_{2dp}, equation (8b) is the vector part v_{2dp}, equation (8c) is the bivector part B_{2dp}[3], and equation (8d) is the pseudoscalar part

[2] *Hint*: Since C++ is a statically typed language, those types need to be well-defined and distinguishable from each other. This is especially important for the different scalar and pseudoscalar types. The ga library uses type tagging extensively for that purpose.

[3] Here we deviate from Lengyel's approach chosen in [Lengyel, 2024b]. Since we want to model physics, it is important to generate unified handling for two- and three-dimensional cases. The chosen sequence of basis bivectors assures that lines, which are used to model forces and moments later on, have their directions encoded in a unified way.

ps_{2dp}. The index $2dp$ is omitted when clear from context. The eight basis elements are (shown per grade):

$$\begin{aligned}\{\ & \mathbf{1},\\ & \boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3,\\ & \boldsymbol{e}_{31}, \boldsymbol{e}_{32}, \boldsymbol{e}_{12},\\ & \boldsymbol{e}_{321}\ \}\end{aligned}$$

Using this basis a multivector M of two-dimensional projective space ($2dp$) is defined as

$$M = s\,\mathbf{1} + v_x\boldsymbol{e}_1 + v_y\boldsymbol{e}_2 + v_z\boldsymbol{e}_3 + b_x\boldsymbol{e}_{31} + b_y\boldsymbol{e}_{32} + b_z\boldsymbol{e}_{12} + ps\,\boldsymbol{e}_{321} \tag{9}$$

where equation (9) contains four parts: the scalar component $s\,\mathbf{1}$ (basis element is the scalar $\mathbf{1}$), the vector part v (basis elements $\boldsymbol{e}_1$, $\boldsymbol{e}_2$ and $\boldsymbol{e}_3$), the bivector part B (basis elements $\boldsymbol{e}_{31}$, $\boldsymbol{e}_{32}$ and $\boldsymbol{e}_{12}$) and the pseudoscalar part $ps\,\boldsymbol{e}_{321} = ps\,\mathbb{1}$ (basis element is $I_{2dp} = \boldsymbol{e}_{321}$, which is sometimes written as $\mathbb{1}$ to show its character as pseudoscalar of this space. It is a trivector in two-dimensional Euclidean projective space that squares to zero).

In $2D\ PGA$ a projective point Q is modelled as a vector

$$Q_{pga2dp} = \underbrace{q_x\boldsymbol{e}_1 + q_y\boldsymbol{e}_2}_{position} + \underbrace{q_z\boldsymbol{e}_3}_{weight} \tag{10}$$

where the position is encoded in the bulk part[4] of the point, and the weight is the projective component q_z that must be unitized (=scaled to 1.0) in order to project the point into the Euclidean space. The unitized point $Q_{unitized} = Q/q_z$ is the sum of the origin $O_{2dp} = \boldsymbol{e}_3$ and the vector $q = (q_x\boldsymbol{e}_1 + q_y\boldsymbol{e}_2)/q_z$ of weight zero from the origin to the point, $Q_{unitized} = O_{2dp} + q$ (fig. 1).

[4]Bulk and weight parts are defined in subsection 1.2.3.

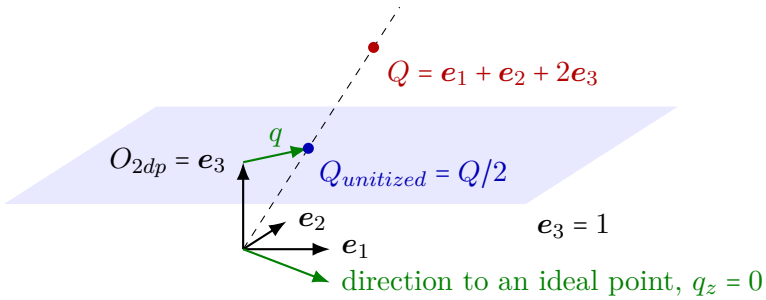


Figure 1: The point $Q = \boldsymbol{e}_1 + \boldsymbol{e}_2 + 2\boldsymbol{e}_3$ of *PGA2DP* in the three-dimensional embedding space. Its multiples (dashed) represent the same point; unitizing divides by the weight $q_z = 2$ and lands in the plane $\boldsymbol{e}_3 = 1$: $Q_{unitized} = Q/2 = O_{2dp} + q$, with the origin $O_{2dp} = \boldsymbol{e}_3$ and q of weight zero (green). A vector with $q_z = 0$ never reaches that plane; it is the direction to an ideal point.

A line L is modelled as a bivector

$$L_{pga2dp} = \underbrace{l_x\boldsymbol{e}_{31} + l_y\boldsymbol{e}_{32}}_{direction} + \underbrace{l_z\boldsymbol{e}_{12}}_{position} \tag{11}$$

where the weight part encodes the two-dimensional direction vector (l_x, l_y) of the line and the bulk part encodes the position of the line: for a unitized line, l_z is the signed distance of the origin $O_{2dp} = (0, 0, 1)$ from the line (fig. 2). When $l_z = 0$ the line passes through the origin.

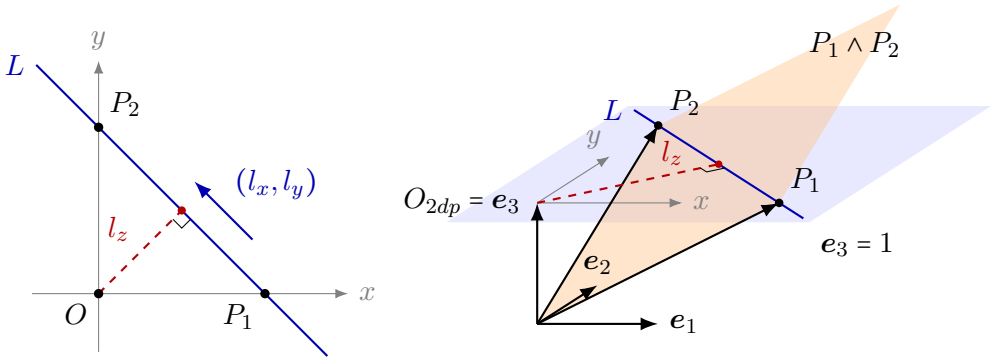


Figure 2: Left: the line $L = P_1 \wedge P_2$ of *PGA2DP* through $P_1 = (2, 0)$ and $P_2 = (0, 2)$: $L = (l_x, l_y, l_z) = (-2, 2, 4)$, unitized $(-0.707, 0.707, 1.414)$. The weight (l_x, l_y) is the direction from P_1 to P_2; the bulk $l_z = \sqrt{2}$ of the unitized line is the signed distance of the origin, positive because O lies to the left of the direction. Right: in the embedding space of fig. 1, the bivector $P_1 \wedge P_2$ spans the plane through the origin and the points P_1, P_2 (orange); L is where it meets the plane $\boldsymbol{e}_3 = 1$ (blue). That plane carries the axes x and y, parallel to $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$, and the distance l_z.

PGA3DP: Projective geometric algebra of $3D$ Euclidean space is modelled by embedding it in a four-dimensional space. Using an orthonormal basis $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, $\boldsymbol{e}_3$, $\boldsymbol{e}_4$ there are following basis vectors:

$$(\boldsymbol{e}_1)^2 = \boldsymbol{e}_1 \wedge \boldsymbol{e}_1 = \boldsymbol{e}_1 \cdot \boldsymbol{e}_1 = \boldsymbol{e}_1^2 = +1, \boldsymbol{e}_2^2 = +1, \boldsymbol{e}_3^2 = +1, \text{ and } \boldsymbol{e}_4^2 = 0$$

$$s_{3dp} = s\,\mathbf{1} \tag{12a}$$

$$v_{3dp} = v_x\boldsymbol{e}_1 + v_y\boldsymbol{e}_2 + v_z\boldsymbol{e}_3 + v_w\boldsymbol{e}_4 \tag{12b}$$

$$B_{3dp} = b_{vx}\boldsymbol{e}_{41} + b_{vy}\boldsymbol{e}_{42} + b_{vz}\boldsymbol{e}_{43} + b_{mx}\boldsymbol{e}_{23} + b_{my}\boldsymbol{e}_{31} + b_{mz}\boldsymbol{e}_{12} \tag{12c}$$

$$t_{3dp} = t_x\boldsymbol{e}_{423} + t_y\boldsymbol{e}_{431} + t_z\boldsymbol{e}_{412} + t_w\boldsymbol{e}_{321} \tag{12d}$$

$$ps_{3dp} = ps\,\boldsymbol{e}_{1234} = ps\,\mathbb{1} \tag{12e}$$

Now we have a basis vector $\boldsymbol{e}_4$ that squares to zero. The bivector has six components, and there is a trivector t. The pseudoscalar $\boldsymbol{e}_{1234}$ is a quadvector that squares to 0, because of its component $\boldsymbol{e}_4$ contained as a factor. Equation (12a) is the scalar part s_{3dp}, equation (12b) is the vector part v_{3dp}, equation (12c) is the bivector part B_{3dp}, equation (12d) is the trivector part t_{3dp}, and equation (12e) is the pseudoscalar part ps_{3dp}. The index $3dp$ is omitted when clear from context. The sixteen basis elements of *PGA3DP* are (shown per grade):

$$\begin{aligned}\{\ &\mathbf{1},\\ &\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4,\\ &\boldsymbol{e}_{41}, \boldsymbol{e}_{42}, \boldsymbol{e}_{43}, \boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12},\\ &\boldsymbol{e}_{423}, \boldsymbol{e}_{431}, \boldsymbol{e}_{412}, \boldsymbol{e}_{321},\\ &\boldsymbol{e}_{1234}\ \}\end{aligned}$$

A multivector M of three-dimensional projective space ($3dp$) is defined as

$$\begin{aligned}M = {} & s\,\mathbf{1}\\ & + v_x\boldsymbol{e}_1 + v_y\boldsymbol{e}_2 + v_z\boldsymbol{e}_3 + v_w\boldsymbol{e}_4\\ & + b_{vx}\boldsymbol{e}_{41} + b_{vy}\boldsymbol{e}_{42} + b_{vz}\boldsymbol{e}_{43} + b_{mx}\boldsymbol{e}_{23} + b_{my}\boldsymbol{e}_{31} + b_{mz}\boldsymbol{e}_{12}\\ & + t_x\boldsymbol{e}_{423} + t_y\boldsymbol{e}_{431} + t_z\boldsymbol{e}_{412} + t_w\boldsymbol{e}_{321}\\ & + ps\,\boldsymbol{e}_{1234}\end{aligned} \tag{13}$$

where equation (13) contains five parts: the scalar component $s\,\mathbf{1}$ (basis element is the scalar $\mathbf{1}$), the vector part v (basis elements $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, $\boldsymbol{e}_3$, $\boldsymbol{e}_4$), the bivector part B (basis elements $\boldsymbol{e}_{41}$, $\boldsymbol{e}_{42}$, $\boldsymbol{e}_{43}$, $\boldsymbol{e}_{23}$, $\boldsymbol{e}_{31}$, $\boldsymbol{e}_{12}$), the trivector part t (basis elements $\boldsymbol{e}_{423}$, $\boldsymbol{e}_{431}$, $\boldsymbol{e}_{412}$, $\boldsymbol{e}_{321}$) and the pseudoscalar part $ps\,\boldsymbol{e}_{1234} = ps\,\mathbb{1}$ (basis element is $I_{3dp} = \boldsymbol{e}_{1234}$, which is sometimes written as $\mathbb{1}$ to show its character as pseudoscalar of this space. It is a quadvector in

three-dimensional Euclidean projective space that squares to zero).

In $3D\ PGA$ a projective point Q is modelled as a vector

$$Q_{pga3dp} = \underbrace{q_x\boldsymbol{e}_1 + q_y\boldsymbol{e}_2 + q_z\boldsymbol{e}_3}_{position} + \underbrace{q_w\boldsymbol{e}_4}_{weight} \tag{14}$$

where the position is encoded in the bulk part of the point, and the weight is the projective component q_w that must be unitized (=scaled to 1.0) in order to project the point into the Euclidean space. The split of the unitized point is the same as in $2D$, $Q_{unitized} = Q/q_w = O_{3dp} + q$ with the origin $O_{3dp} = \boldsymbol{e}_4$.

A line L is modelled as a bivector

$$L_{pga3dp} = \underbrace{l_{vx}\boldsymbol{e}_{41} + l_{vy}\boldsymbol{e}_{42} + l_{vz}\boldsymbol{e}_{43}}_{direction} + \underbrace{l_{mx}\boldsymbol{e}_{23} + l_{my}\boldsymbol{e}_{31} + l_{mz}\boldsymbol{e}_{12}}_{moment} \tag{15}$$

where the weight part encodes the three-dimensional direction vector of the line $l_v = (l_{vx}, l_{vy}, l_{vz})$, and the bulk part encodes the moment of the line $l_m = (l_{mx}, l_{my}, l_{mz})$ when interpreted as a vector. Both are encoded in the bivector object that captures all parameters needed to model the line. For any point of the line, unitized as $P_{unitized} = O_{3dp} + p$, the bulk is $p \wedge l_v$. In $PGA2DP$ a line is the plane through the origin of the embedding space spanned by two of its points (fig. 2); in $PGA3DP$ it is likewise a two-dimensional subspace, of the four-dimensional embedding space, which cannot be drawn. Fig. 3 therefore shows the line in Euclidean space.

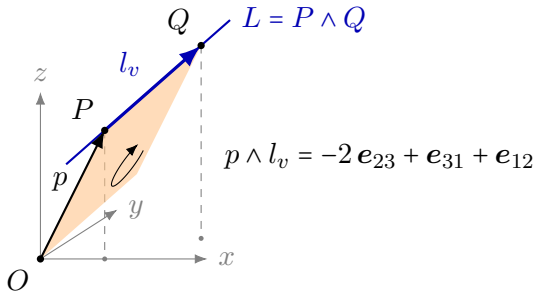


Figure 3: A line of $PGA3DP$ in general position, the wedge product $L = P \wedge Q$ of $P = (1, 0, 2)$ and $Q = P + l_v = (2, 1, 3)$: $L = \boldsymbol{e}_{41} + \boldsymbol{e}_{42} + \boldsymbol{e}_{43} - 2\,\boldsymbol{e}_{23} + \boldsymbol{e}_{31} + \boldsymbol{e}_{12}$. Its weight is $l_v = (1, 1, 1)$, its bulk the oriented area $p \wedge l_v$ (orange), the same for every point of L and as a vector the moment $l_m = (-2, 1, 1)$; bulk and weight are defined in subsection 1.2.3. The bulk's magnitude $\sqrt{6}$ is the distance $\sqrt{2}$ of L from O times $\|l_v\| = \sqrt{3}$. Dashed: drops onto the x-y plane.

Additionally, there are planes T that are modelled as trivectors

$$T_{pga3dp} = \underbrace{t_x \boldsymbol{e}_{423} + t_y \boldsymbol{e}_{431} + t_z \boldsymbol{e}_{412}}_{normal} + \underbrace{t_w \boldsymbol{e}_{321}}_{position} \tag{16}$$

where the first three components represent the normal vector of the plane $n = (t_x, t_y, t_z)$. For a unit normal the fourth component t_w is the signed distance of the origin $O_{3dp} = (0, 0, 0, 1)$ from the plane, and every point p of the plane satisfies $n \cdot p + t_w = 0$ (fig. 4).

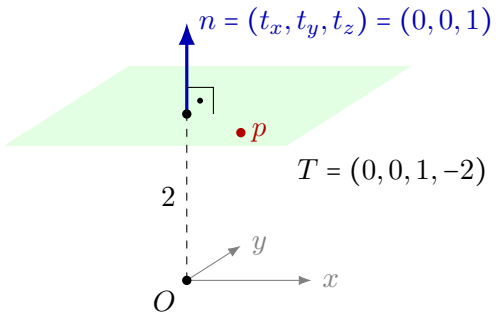


Figure 4: The plane $z = 2$ of $PGA3DP$, the join of three of its points: $T = \boldsymbol{e}_{412} - 2\,\boldsymbol{e}_{321}$, i.e. $(t_x, t_y, t_z, t_w) = (0, 0, 1, -2)$. The weight is the normal $n = (t_x, t_y, t_z)$, perpendicular to the plane. For a unit normal the bulk t_w is the signed distance of the origin from the plane, here -2 because O lies on the side opposite to n. Every point p of the plane satisfies $n \cdot p + t_w = 0$.

Many more examples of how geometric objects are modelled in projective geometric algebra and which operations can be used to manipulated them can be found in Lengyel's book [Lengyel, 2024b]. Browne explains very well how Grassmann algebra and projective geometry and geometric algebra are related in [Browne, 2012], p. 216ff. It is helpful to fully grasp and understand that relation in order to see what is possible using the algebra effectively to model physical phenomena.

STA4DS: Space-time algebra of four-dimensional Minkowski space is modelled using an orthonormal basis $\boldsymbol{g}_1$, $\boldsymbol{g}_2$, $\boldsymbol{g}_3$, $\boldsymbol{g}_4$ with the mixed (Minkowski) signature, where the three space-like directions square to -1 and the time-like

direction $\boldsymbol{g}_4$ squares to +1:

$$(\boldsymbol{g}_1)^2 = \boldsymbol{g}_1 \wedge \boldsymbol{g}_1 = \boldsymbol{g}_1 \cdot \boldsymbol{g}_1 = \boldsymbol{g}_1^2 = -1, \boldsymbol{g}_2^2 = -1, \boldsymbol{g}_3^2 = -1, \text{ and } \boldsymbol{g}_4^2 = +1$$

$$s_{4ds} = s\,\mathbf{1} \tag{17a}$$

$$v_{4ds} = v_x\boldsymbol{g}_1 + v_y\boldsymbol{g}_2 + v_z\boldsymbol{g}_3 + v_w\boldsymbol{g}_4 \tag{17b}$$

$$B_{4ds} = b_{vx}\boldsymbol{g}_{14} + b_{vy}\boldsymbol{g}_{24} + b_{vz}\boldsymbol{g}_{34} + b_{mx}\boldsymbol{g}_{23} + b_{my}\boldsymbol{g}_{31} + b_{mz}\boldsymbol{g}_{12} \tag{17c}$$

$$t_{4ds} = t_x\boldsymbol{g}_{234} + t_y\boldsymbol{g}_{314} + t_z\boldsymbol{g}_{124} + t_w\boldsymbol{g}_{123} \tag{17d}$$

$$ps_{4ds} = ps\,\boldsymbol{g}_{1234} = ps\,\mathbb{1} \tag{17e}$$

Unlike PGA, where one basis vector squares to zero (degenerate metric), $STA4DS$ has a non-degenerate metric; but in contrast to the all-positive Euclidean signature of EGA, its signature is indefinite (mixed signs). Equation (17a) is the scalar part s_{4ds}, equation (17b) is the vector part v_{4ds}, equation (17c) is the bivector part B_{4ds}, equation (17d) is the trivector part t_{4ds}, and equation (17e) is the pseudoscalar part ps_{4ds}. The index $4ds$ is omitted when clear from context. The sixteen basis elements of $STA4DS$ are (shown per grade):

$$\begin{aligned}\{\ &\mathbf{1},\\ &\boldsymbol{g}_1, \boldsymbol{g}_2, \boldsymbol{g}_3, \boldsymbol{g}_4,\\ &\boldsymbol{g}_{14}, \boldsymbol{g}_{24}, \boldsymbol{g}_{34}, \boldsymbol{g}_{23}, \boldsymbol{g}_{31}, \boldsymbol{g}_{12},\\ &\boldsymbol{g}_{234}, \boldsymbol{g}_{314}, \boldsymbol{g}_{124}, \boldsymbol{g}_{123},\\ &\boldsymbol{g}_{1234}\ \}\end{aligned}$$

The grade structure mirrors that of $PGA3DP$ (both are four-dimensional algebras with sixteen basis elements); the difference is the vector metric, where $STA4DS$ replaces the one degenerate and three positive directions of $3D\ PGA$ by three negative and one positive direction. A multivector M of space-time algebra ($4ds$) is defined as

$$\begin{aligned}M = &\ s\,\mathbf{1}\\ &+ v_x\boldsymbol{g}_1 + v_y\boldsymbol{g}_2 + v_z\boldsymbol{g}_3 + v_w\boldsymbol{g}_4\\ &+ b_{vx}\boldsymbol{g}_{14} + b_{vy}\boldsymbol{g}_{24} + b_{vz}\boldsymbol{g}_{34} + b_{mx}\boldsymbol{g}_{23} + b_{my}\boldsymbol{g}_{31} + b_{mz}\boldsymbol{g}_{12}\\ &+ t_x\boldsymbol{g}_{234} + t_y\boldsymbol{g}_{314} + t_z\boldsymbol{g}_{124} + t_w\boldsymbol{g}_{123}\\ &+ ps\,\boldsymbol{g}_{1234}\end{aligned} \tag{18}$$

where equation (18) contains five parts: the scalar component $s\,\mathbf{1}$ (basis element is the scalar $\mathbf{1}$), the vector part v (basis elements $\boldsymbol{g}_1$, $\boldsymbol{g}_2$, $\boldsymbol{g}_3$, $\boldsymbol{g}_4$), the bivector part B (basis elements $\boldsymbol{g}_{14}$, $\boldsymbol{g}_{24}$, $\boldsymbol{g}_{34}$, $\boldsymbol{g}_{23}$, $\boldsymbol{g}_{31}$, $\boldsymbol{g}_{12}$), the trivector part t (basis elements $\boldsymbol{g}_{234}$, $\boldsymbol{g}_{314}$, $\boldsymbol{g}_{124}$, $\boldsymbol{g}_{123}$) and the pseudoscalar part

$ps\,\boldsymbol{g}_{1234} = ps\,\mathbb{1}$ (basis element is $I_{4ds} = \boldsymbol{g}_{1234}$, which is sometimes written as $\mathbb{1}$ to show its character as pseudoscalar of this space. It is a quadvector in four-dimensional space-time that squares to -1 under the geometric product).

In STA a vector models a space-time event or, more generally, a four-vector

$$X_{sta4ds} = \underbrace{x_x\boldsymbol{g}_1 + x_y\boldsymbol{g}_2 + x_z\boldsymbol{g}_3}_{space} + \underbrace{x_w\boldsymbol{g}_4}_{time} \tag{19}$$

where the three space-like components are carried by $\boldsymbol{g}_1$, $\boldsymbol{g}_2$ and $\boldsymbol{g}_3$, and the time-like component by $\boldsymbol{g}_4$. Typical examples are the four-position, the four-velocity and the four-momentum. The geometric square $X^2 = X \wedge X$ is the squared Minkowski interval; it is positive for time-like, negative for space-like and zero for light-like vectors.

The bivectors carry the generators of the Lorentz transformations. The three bivectors containing the time-like direction ($\boldsymbol{g}_{14}$, $\boldsymbol{g}_{24}$, $\boldsymbol{g}_{34}$) generate boosts (relative motion between frames), while the three purely space-like bivectors ($\boldsymbol{g}_{23}$, $\boldsymbol{g}_{31}$, $\boldsymbol{g}_{12}$) generate spatial rotations. As for the rotors in EGA, the even-grade part of the algebra (scalar, bivector and pseudoscalar – the eight even basis elements) forms the group of rotors R that implement Lorentz transformations of any object X via the sandwich product $transform(X, R) = R \wedge X \wedge \widetilde{R}$ based on the geometric product, in complete analogy to the rotations in EGA.

CGA2DC: Conformal geometric algebra of $2D$ Euclidean space is modelled by embedding it in a four-dimensional space. On top of the projective dimension known from $PGA2DP$ it adds a second one, the round dimension. Both added basis vectors square to zero, but in contrast to PGA they are not orthogonal to each other:

$$(\boldsymbol{e}_1)^2 = \boldsymbol{e}_1 \wedge \boldsymbol{e}_1 = \boldsymbol{e}_1 \cdot \boldsymbol{e}_1 = \boldsymbol{e}_1^2 = +1, \boldsymbol{e}_2^2 = +1, \boldsymbol{e}_3^2 = 0, \boldsymbol{e}_4^2 = 0, \text{ and } \boldsymbol{e}_3 \cdot \boldsymbol{e}_4 = -1$$

$$s_{2dc} = s\,\mathbf{1} \tag{20a}$$

$$v_{2dc} = v_x\boldsymbol{e}_1 + v_y\boldsymbol{e}_2 + v_w\boldsymbol{e}_3 + v_u\boldsymbol{e}_4 \tag{20b}$$

$$B_{2dc} = b_{vx}\boldsymbol{e}_{31} + b_{vy}\boldsymbol{e}_{32} + b_{vz}\boldsymbol{e}_{12} + b_{mx}\boldsymbol{e}_{14} + b_{my}\boldsymbol{e}_{24} + b_{mz}\boldsymbol{e}_{34} \tag{20c}$$

$$t_{2dc} = t_x\boldsymbol{e}_{314} + t_y\boldsymbol{e}_{324} + t_z\boldsymbol{e}_{124} + t_w\boldsymbol{e}_{321} \tag{20d}$$

$$ps_{2dc} = ps\,\boldsymbol{e}_{1234} = ps\,\mathbb{1} \tag{20e}$$

The pair $\boldsymbol{e}_3$, $\boldsymbol{e}_4$ is a null pair: each of the two squares to zero, while their dot product is -1. The metric is therefore not diagonal in this basis, and it

is non-degenerate ($det(\mathbf{G}) = -1$), unlike the degenerate metric of PGA. The sixteen basis elements of $CGA2DC$ are (shown per grade):

$$\begin{aligned}\{\ & \mathbf{1},\\ & \boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4,\\ & \boldsymbol{e}_{31}, \boldsymbol{e}_{32}, \boldsymbol{e}_{12}, \boldsymbol{e}_{14}, \boldsymbol{e}_{24}, \boldsymbol{e}_{34},\\ & \boldsymbol{e}_{314}, \boldsymbol{e}_{324}, \boldsymbol{e}_{124}, \boldsymbol{e}_{321},\\ & \boldsymbol{e}_{1234}\ \}\end{aligned}$$

The round dimension is written trailing in every blade name, and the eight basis elements of $PGA2DP$ appear with identical names and orientations, so every $PGA2DP$ object embeds verbatim.[5] A multivector M of two-dimensional conformal space ($2dc$) is defined as

$$\begin{aligned}M = {} & s\,\mathbf{1}\\ & + v_x\boldsymbol{e}_1 + v_y\boldsymbol{e}_2 + v_w\boldsymbol{e}_3 + v_u\boldsymbol{e}_4\\ & + b_{vx}\boldsymbol{e}_{31} + b_{vy}\boldsymbol{e}_{32} + b_{vz}\boldsymbol{e}_{12} + b_{mx}\boldsymbol{e}_{14} + b_{my}\boldsymbol{e}_{24} + b_{mz}\boldsymbol{e}_{34}\\ & + t_x\boldsymbol{e}_{314} + t_y\boldsymbol{e}_{324} + t_z\boldsymbol{e}_{124} + t_w\boldsymbol{e}_{321}\\ & + ps\,\boldsymbol{e}_{1234}\end{aligned} \tag{21}$$

The two added basis vectors carry the two distinguished points of the conformal model: $\boldsymbol{e}_3$ is the origin e_o and $\boldsymbol{e}_4$ is the point at infinity e_∞. A Euclidean point $p = (p_x, p_y)$ is embedded as the vector

$$q_{2dc} = p_x\boldsymbol{e}_1 + p_y\boldsymbol{e}_2 + \boldsymbol{e}_3 + \frac{p^2}{2}\boldsymbol{e}_4 \qquad \text{with} \qquad q_{2dc} \wedge q_{2dc} = 0 \tag{22}$$

i.e. the embedded points are exactly the null vectors of the algebra, and they lie on a cone through the origin of the four-dimensional modelling space. Setting $p = 0$ gives $q_{2dc} = \boldsymbol{e}_3$, which identifies $\boldsymbol{e}_3$ with the origin. For $p \neq 0$ the embedded point can be scaled homogeneously by $2/p^2$, which gives

$$\frac{2}{p^2}\,q_{2dc} = \frac{2p_x}{p^2}\boldsymbol{e}_1 + \frac{2p_y}{p^2}\boldsymbol{e}_2 + \frac{2}{p^2}\boldsymbol{e}_3 + \boldsymbol{e}_4 \tag{23}$$

As $|p|$ becomes arbitrarily large, the first three components approach zero and only $\boldsymbol{e}_4$ is left. Every point infinitely far away from the origin is thus represented by $\boldsymbol{e}_4$, which is the point at infinity. It carries no direction, unlike

[5] The library types store their components positionally, so the field names of a reused template do not always match the labels used here (e.g. `vec2dc` keeps v_w and v_u in the fields named `z` and `w`). The field layout of each type is documented in `ga/detail/type_t/ga_type2dc.hpp`.

the points at infinity of PGA, which fill a whole horizon: all directions lead to this one point [Lengyel, 2024b].

The grade of a blade encodes the kind of object it represents. *Round* objects carry a radius, *flat* objects contain the point at infinity and are the objects of $PGA2DP$ lifted by $\boldsymbol{e}_4$. A vector is a round point, a bivector a dipole (a pair of points) or a flat point, and a trivector a circle or a line.

A round point A with center $p = (p_x, p_y)$ and radius r is modelled as the vector

$$A_{cga2dc} = \underbrace{p_x\boldsymbol{e}_1 + p_y\boldsymbol{e}_2}_{position} + \underbrace{\boldsymbol{e}_3}_{weight} + \underbrace{\frac{p^2 + r^2}{2}\boldsymbol{e}_4}_{round\ part} \tag{24}$$

which is the point embedding (22) for $r = 0$. Only r^2 enters, so a negative value is admissible as well and describes a round point with imaginary radius.

A circle C with center p and radius r is modelled as the trivector

$$C_{cga2dc} = \underbrace{p_y\boldsymbol{e}_{314} - p_x\boldsymbol{e}_{324} - \frac{p^2 - r^2}{2}\boldsymbol{e}_{124}}_{flat\ part} - \underbrace{\boldsymbol{e}_{321}}_{round\ weight} \tag{25}$$

where the round weight is chosen as -1, so that the center can be read off the first two components directly. Note that r^2 enters with the sign opposite to the one it has in the round point (24), which holds for every round object of codimension one. A vanishing round weight leaves the flat part alone, and the object is a line: a line is a circle through the point at infinity.

A dipole D is modelled as a bivector, which splits the same way,

$$D_{cga2dc} = \underbrace{d_{vx}\boldsymbol{e}_{31} + d_{vy}\boldsymbol{e}_{32} + d_{vz}\boldsymbol{e}_{12}}_{carrier\ line} + \underbrace{d_{mx}\boldsymbol{e}_{14} + d_{my}\boldsymbol{e}_{24} + d_{mz}\boldsymbol{e}_{34}}_{flat\ part} \tag{26}$$

with the first three components carrying the line through its two points and the last three their position on it. A vanishing carrier line leaves a flat point, which is the $PGA2DP$ point at the Euclidean position p lifted by $\boldsymbol{e}_4$

$$F_{cga2dc} = -\left(p_x\boldsymbol{e}_{14} + p_y\boldsymbol{e}_{24} + \boldsymbol{e}_{34}\right) \tag{27}$$

A dipole with imaginary radius has no real points; it arises for instance as the meet of two circles that do not intersect.

CGA3DC: Conformal geometric algebra of $3D$ Euclidean space is modelled by embedding it in a five-dimensional space, with $\boldsymbol{e}_4$ the projective dimension (the same slot as in *PGA3DP*) and $\boldsymbol{e}_5$ the round dimension:

$$(\boldsymbol{e}_1)^2 = \boldsymbol{e}_1 \wedge \boldsymbol{e}_1 = \boldsymbol{e}_1 \cdot \boldsymbol{e}_1 = \boldsymbol{e}_1^2 = +1, \boldsymbol{e}_2^2 = +1, \boldsymbol{e}_3^2 = +1, \boldsymbol{e}_4^2 = 0, \boldsymbol{e}_5^2 = 0,$$
$$\text{and } \boldsymbol{e}_4 \cdot \boldsymbol{e}_5 = -1$$

$$s_{3dc} = s\,\mathbf{1} \tag{28a}$$
$$v_{3dc} = v_x\boldsymbol{e}_1 + v_y\boldsymbol{e}_2 + v_z\boldsymbol{e}_3 + v_w\boldsymbol{e}_4 + v_u\boldsymbol{e}_5 \tag{28b}$$
$$B_{3dc} = b_{vx}\boldsymbol{e}_{41} + b_{vy}\boldsymbol{e}_{42} + b_{vz}\boldsymbol{e}_{43} + b_{mx}\boldsymbol{e}_{23} + b_{my}\boldsymbol{e}_{31} + b_{mz}\boldsymbol{e}_{12} + b_{px}\boldsymbol{e}_{15} + b_{py}\boldsymbol{e}_{25} + b_{pz}\boldsymbol{e}_{35} + b_{pw}\boldsymbol{e}_{45} \tag{28c}$$
$$t_{3dc} = t_{vx}\boldsymbol{e}_{415} + t_{vy}\boldsymbol{e}_{425} + t_{vz}\boldsymbol{e}_{435} + t_{mx}\boldsymbol{e}_{235} + t_{my}\boldsymbol{e}_{315} + t_{mz}\boldsymbol{e}_{125} + t_{px}\boldsymbol{e}_{423} + t_{py}\boldsymbol{e}_{431} + t_{pz}\boldsymbol{e}_{412} + t_{pw}\boldsymbol{e}_{321} \tag{28d}$$
$$Q_{3dc} = q_x\boldsymbol{e}_{4235} + q_y\boldsymbol{e}_{4315} + q_z\boldsymbol{e}_{4125} + q_w\boldsymbol{e}_{3215} + q_u\boldsymbol{e}_{1234} \tag{28e}$$
$$ps_{3dc} = ps\,\boldsymbol{e}_{12345} = ps\,\mathbb{1} \tag{28f}$$

Equation (28e) is the quadvector part Q_{3dc}, which the five-dimensional space requires in addition to the parts already known from the four-dimensional algebras. The thirty-two basis elements of *CGA3DC* are (shown per grade):

$$\begin{aligned}
&\{\ \mathbf{1},\\
&\ \boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4, \boldsymbol{e}_5,\\
&\ \boldsymbol{e}_{41}, \boldsymbol{e}_{42}, \boldsymbol{e}_{43}, \boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12}, \boldsymbol{e}_{15}, \boldsymbol{e}_{25}, \boldsymbol{e}_{35}, \boldsymbol{e}_{45},\\
&\ \boldsymbol{e}_{415}, \boldsymbol{e}_{425}, \boldsymbol{e}_{435}, \boldsymbol{e}_{235}, \boldsymbol{e}_{315}, \boldsymbol{e}_{125}, \boldsymbol{e}_{423}, \boldsymbol{e}_{431}, \boldsymbol{e}_{412}, \boldsymbol{e}_{321},\\
&\ \boldsymbol{e}_{4235}, \boldsymbol{e}_{4315}, \boldsymbol{e}_{4125}, \boldsymbol{e}_{3215}, \boldsymbol{e}_{1234},\\
&\ \boldsymbol{e}_{12345}\ \}
\end{aligned}$$

Per grade the blades of *PGA3DP* come first and keep their names and orientations, followed by the blades of one grade less lifted by the round dimension $\boldsymbol{e}_5$. The point embedding generalizes equation (22) with $\boldsymbol{e}_4 = e_o$ and $\boldsymbol{e}_5 = e_\infty$:

$$q_{3dc} = p_x\boldsymbol{e}_1 + p_y\boldsymbol{e}_2 + p_z\boldsymbol{e}_3 + \boldsymbol{e}_4 + \frac{p^2}{2}\boldsymbol{e}_5 \qquad \text{with} \qquad q_{3dc} \wedge q_{3dc} = 0 \tag{29}$$

Scaling by $2/p^2$ carries over unchanged,

$$\frac{2}{p^2}\,q_{3dc} = \frac{2p_x}{p^2}\boldsymbol{e}_1 + \frac{2p_y}{p^2}\boldsymbol{e}_2 + \frac{2p_z}{p^2}\boldsymbol{e}_3 + \frac{2}{p^2}\boldsymbol{e}_4 + \boldsymbol{e}_5 \tag{30}$$

so $\boldsymbol{e}_4$ is the origin and $\boldsymbol{e}_5$ the point at infinity, for the same reasons as in *CGA2DC*.

The grades carry one more object kind than in $CGA2DC$: a vector is a round point, a bivector a dipole or a flat point, a trivector a circle or a line, and a quadvector a sphere or a plane. The flats are again the objects of $PGA3DP$ lifted by $\boldsymbol{e}_5$.

A round point A with center $p = (p_x, p_y, p_z)$ and radius r is the vector

$$A_{cga3dc} = \underbrace{p_x\boldsymbol{e}_1 + p_y\boldsymbol{e}_2 + p_z\boldsymbol{e}_3}_{position} + \underbrace{\boldsymbol{e}_4}_{weight} + \underbrace{\frac{p^2 + r^2}{2}\boldsymbol{e}_5}_{round\ part} \tag{31}$$

and a sphere S with the same center and radius is the quadvector

$$S_{cga3dc} = \underbrace{p_x\boldsymbol{e}_{4235} + p_y\boldsymbol{e}_{4315} + p_z\boldsymbol{e}_{4125} - \frac{p^2 - r^2}{2}\boldsymbol{e}_{3215}}_{flat\ part} - \underbrace{\boldsymbol{e}_{1234}}_{round\ weight} \tag{32}$$

the codimension-one round object, with the same -1 weight and the same reversed sign of r^2 as the circle in $CGA2DC$. With a vanishing round weight it becomes a plane with unit normal n and signed distance d from the origin

$$P_{cga3dc} = \underbrace{n_x\boldsymbol{e}_{4235} + n_y\boldsymbol{e}_{4315} + n_z\boldsymbol{e}_{4125}}_{normal} - \underbrace{d\,\boldsymbol{e}_{3215}}_{position} \tag{33}$$

which are the components of the $PGA3DP$ plane (16), lifted.

Circles and dipoles carry ten components each, but only six degrees of freedom. Both split into a carrier part and a flat part, as the dipole does in $CGA2DC$: the circle into its carrier plane and a flat part, the dipole into its carrier line and the position of its two points on it. The flat objects of these two grades keep the $PGA3DP$ components verbatim. A line through the position p with direction v and moment $m = p \times v$ is the trivector

$$L_{cga3dc} = \underbrace{v_x\boldsymbol{e}_{415} + v_y\boldsymbol{e}_{425} + v_z\boldsymbol{e}_{435}}_{direction} + \underbrace{m_x\boldsymbol{e}_{235} + m_y\boldsymbol{e}_{315} + m_z\boldsymbol{e}_{125}}_{moment} \tag{34}$$

and a flat point at the position p is the bivector

$$F_{cga3dc} = p_x\boldsymbol{e}_{15} + p_y\boldsymbol{e}_{25} + p_z\boldsymbol{e}_{35} + \boldsymbol{e}_{45} \tag{35}$$

The chapter on conformal geometry treats these objects, their properties and the conformal transformations acting on them.

Geometric algebra seems complex and a little overwhelming at first. Especially so, when new objects beyond vectors and several specific products are introduced. Expressions tend to quickly become large for fully occupied multivectors. However, it is important to note that it is rarely required to use full multivectors in the respective algebra. For most applications it is sufficient to use a blade or k-vector exclusively, which represents only one grade of the multivector, e.g. a vector or a bivector, which can be represented with very few coefficients. This reduces the effort in terms of memory and computational requirements significantly – typically reduced to a level that is required anyway to solve the underlying geometrical question at hand.

1.2.2 The complement

The complement operation used in the library is defined in [Lengyel, 2024b]. It is uniquely determined with respect to the outer product (*not* with respect to the geometric product, as commonly used for dualization by many authors working with GA). This ensures consistent signs of expressions, operators, complements and duals. The left complement $\underline{u}$ and the right complement $\overline{u}$ are defined for a blade u as

$$\underline{u} \wedge u = I_n \qquad (\texttt{l_cmpl(u)}) \tag{36a}$$

$$u \wedge \overline{u} = I_n \qquad (\texttt{r_cmpl(u)}) \tag{36b}$$

with $I_n = \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \ldots \wedge \boldsymbol{e}_n$ as the pseudoscalar for an n-dimensional algebra, and the wedge symbol ($\wedge$) as the operator symbol of the outer product. The effect of the complement is to turn basis elements not contained in the object u into the constituting basis elements of the complement of the object $\underline{u}$ or $\overline{u}$. Connecting the object and its complement with the wedge products creates an object that fills the available space in the sense of requiring all basis elements of the space to represent it. The full space itself is represented by the pseudoscalar, since the pseudoscalar I_n is formed by the product of all n basis vectors of the space. An implication of the definition is that the complement ($\underline{u}$ or $\overline{u}$) is orthogonal to u, since we exclusively use orthogonal basis vectors to model our space.

The complement is defined as given above in equations (36) for any basis element. It is extended to full multivectors A and B by linearity as

$$\overline{sA} = s\overline{A}, \quad \overline{A+B} = \overline{A}+\overline{B} \tag{37a}$$

$$\underline{sA} = s\underline{A}, \quad \underline{A+B} = \underline{A}+\underline{B} \tag{37b}$$

where s is a scalar. The left and right complements fulfill

$$\overline{\underline{u}} = u \tag{38}$$

thus the left and right complement operations are inverses to each other. This is valid independent of the sequence of application, independent of the grade of the argument a, and independent of the dimension of the space modelled in the algebra. In spaces of odd dimension, where we only have one complement, applying the complement twice of course returns the original object as well. Fig. 5 shows both complements of a vector in $EGA2D$, built by linearity from those of the basis vectors, and the single complement of a basis vector in $EGA3D$.

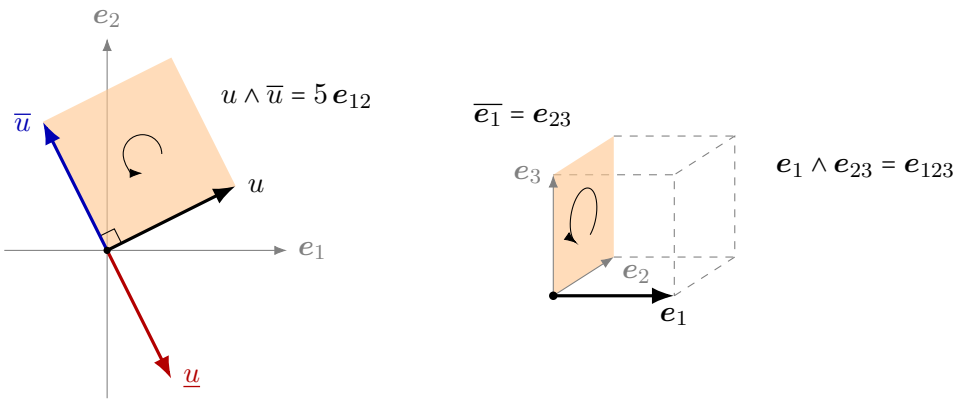


Figure 5: Left: in $EGA2D$ the right complement turns a vector by +90° and the left complement by −90°. For $u = 2\boldsymbol{e}_1 + \boldsymbol{e}_2$ they are $\overline{u} = -\boldsymbol{e}_1 + 2\boldsymbol{e}_2$ and $\underline{u} = \boldsymbol{e}_1 - 2\boldsymbol{e}_2$, and both fill the plane together with u: $u \wedge \overline{u} = \underline{u} \wedge u = 5\,\boldsymbol{e}_{12} = \|u\|^2 I_2$ (orange, for the right complement). Right: $EGA3D$ has odd dimension and a single complement. That of $\boldsymbol{e}_1$ is the plane element $\boldsymbol{e}_{23}$, orthogonal to $\boldsymbol{e}_1$, and $\boldsymbol{e}_1 \wedge \boldsymbol{e}_{23} = \boldsymbol{e}_{123}$ fills the space (the unit cube).

However, applying the same complement operation several times in spaces of even dimension does not ensure that you get back the original object. The result is dependent on the grade of the basis vector and the dimension n of the space. When applying the same complement twice we get

$$\overline{\overline{u}} = \underline{\underline{u}} = (-1)^{gr(u)(n-gr(u))} u \tag{39}$$

i.e. this involves grade-dependent sign changes. So it is favorable to use left and right complements as inverses to each other.

Based on the given complement definition it is possible to define regressive products using rules analog to de Morgan's rules in Boolean algebra as follows

(definitions can be found in [Lengyel, 2024b] and in [Browne, 2012]):

$$a \vee b \equiv \underline{\overline{a} \wedge \overline{b}} \tag{40a}$$
$$a \vee b \equiv \overline{\underline{a} \wedge \underline{b}} \tag{40b}$$

The same complement is applied to both arguments first, the resulting expressions are combined with the wedge product, and the inverse complement operation is used to transform the outcome into the final result, the regressive product. The inverted wedge symbol ($\vee$) is used for the regressive wedge product. Both definitions (40a) and (40b) are equivalent. Equation (40a) is used to derive expressions for regressive products in `ga/ga_prdxpr`.

There are some additional formulas for complements which are useful for working with GA expressions and thus are provided here for reference. Taking the right and left complements of (40a) and (40b) we end up with

$$\overline{a \vee b} = \overline{\underline{\overline{a} \wedge \overline{b}}} = \overline{a} \wedge \overline{b} \tag{41a}$$
$$\underline{a \vee b} = \underline{\overline{\underline{a} \wedge \underline{b}}} = \underline{a} \wedge \underline{b} \tag{41b}$$

Substituting $a = \underline{u}$ and $b = \underline{v}$ into (41a) and $a = \overline{u}$ and $b = \overline{v}$ into (41b) and exchanging the left- and right-hand sides of the equations we end up with

$$u \wedge v = \overline{\underline{u} \vee \underline{v}} \tag{42a}$$
$$u \wedge v = \underline{\overline{u} \vee \overline{v}} \tag{42b}$$

which shows that the outer and the regressive outer product can be expressed by each other.

The complement operation is used to define a unique dualization operation that works for cases with non-degenerate as well as for cases with degenerate metric, like in projective geometric algebra PGA. This is the main advantage compared to the usual procedure of dualization by multiplying the object with the pseudoscalar (or its inverse) using the geometric product, since the latter breaks down for spaces with degenerate metrics. For an overview on all complements in EGA see subsection 1.2.9, in PGA see subsection 1.2.10, and in STA see subsection 1.2.13.

1.2.3 The metric

The metric is defined by the dot product of the basis vectors as

$$\mathfrak{g}_{ij} \equiv \boldsymbol{e}_i \cdot \boldsymbol{e}_j \tag{43}$$

It is a square matrix that is the identity matrix I_n for EGA, since all basis vectors square to $\boldsymbol{e}_i \cdot \boldsymbol{e}_i = +1$, and $\boldsymbol{e}_i \cdot \boldsymbol{e}_j = 0$ for $i \neq j$. It is invertible in this case. However, for PGA one basis vector squares to zero, and thus there is one zero on the main diagonal of the corresponding matrix of the metric, which is therefore not invertible (its determinant is zero). It is called degenerate for that reason.

The extended metric $\mathbf{G}$ is a so-called exomorphism matrix that fulfills

$$\mathbf{G}(A \wedge B) = \mathbf{G}(A) \wedge \mathbf{G}(B) \tag{44}$$

for any multivector A and B (see [Lengyel, 2024b], pp. 61ff.). This exomorphism is defined with respect to the wedge product. The wedge product itself is defined independent of any metric. However, this property defines an extended metric, and the resulting extended metric can in turn be used to define a dot product for components of any grade of a multivector. The dot product $dot(a, b) = a^T \mathbf{G} b$ is defined as a metric product (see eq. (76) further below).

Regarding $\mathbf{G}(u) = \mathbf{G}u$ as a unary operation on u we can define the corresponding regressive operation by first taking the complement of the argument and then applying the opposed complement on the result. This defines the regressive metric $\mathbb{G}$ (named antimetric by Lengyel). It is defined as

$$\mathbb{G}u \equiv \underline{\mathbf{G}\overline{u}} = \overline{\mathbf{G}\underline{u}} \tag{45}$$

$\mathbf{G}$ and $\mathbb{G}$ are related by

$$\mathbf{G}\mathbb{G} = det(\mathfrak{g}) I_n \tag{46}$$

Thus, if all basis vectors square to ± 1, we have $\mathbf{G}\mathbb{G} = \pm I_n$, and both are invertible. However, if one basis vector squares to zero, we get $\mathbf{G}\mathbb{G} = 0$, which means that none of them is invertible, i.e. we have a degenerate metric.

For $\mathbb{G}$ there is the relation

$$\mathbb{G}(A \vee B) = \mathbb{G}(A) \vee \mathbb{G}(B) \tag{47}$$

that is complementary to equation (44). It is an exomorphism defined with respect to the regressive wedge product.

In projective geometric algebra the product $\mathbf{G}u$ defines the *bulk* of an object u. It is defined as

$$u_\bullet = \mathbf{G}u \qquad (\texttt{bulk(u)}) \tag{48}$$

The bulk only contains components that *do not* contain the projective dimension ($\boldsymbol{e}_3$ for $2D\ PGA$ or $\boldsymbol{e}_4$ for $3D\ PGA$) as a factor. The *bulk* $u_\bullet$ *contains positional information* on u relative to the origin. Zero bulk ($u_\bullet = 0$) means that u contains the origin.

The product $\mathbb{G}u$ defines the *weight* of an object u. It is defined as

$$u_\circ = \mathbb{G}u \qquad (\texttt{weight(u)}) \tag{49}$$

The weight selects components that *do* indeed contain the projective dimension ($\boldsymbol{e}_3$ for $2D\ PGA$ or $\boldsymbol{e}_4$ for $3D\ PGA$) as a factor. The *weight* $u_\circ$ *contains directional information* on u, i.e. its attitude and direction independent of its position. Zero weight ($u_\circ = 0$) means that u is contained in the horizon, i.e. it is infinitely far away.

This effective split for each object into bulk and weight in projective algebras can be written as

$$u = u_\bullet + u_\circ \tag{50}$$

This split is also relevant for the corresponding norms

$$\|u\| = \|u\|_\bullet + \|u\|_\circ = \sqrt{A \cdot A} + \sqrt{A \circ A} \tag{51}$$

where the left-hand side is called the *geometric norm* and the right-hand side consists of the sums of the *bulk* and *weight norms*. The *bulk norm* returns a scalar value of the corresponding algebra, whereas the *weight norm* provides a pseudoscalar value of the respective algebra. The dot product (symbol '$\cdot$') and its regressive counterpart (symbol '$\circ$') are defined in tables 5 and 9 in subsection 1.2.10.

Consequently, the geometric norm is an object consisting of two distinct parts, a scalar and a pseudoscalar of the respective algebra, defined here as a *dual number* in the corresponding algebra. When we apply *unitization* to geometric objects we divide the object by its weight, so that the weight becomes one. Where the weight is a single component (a point, a pseudoscalar, a dual number) it is divided out with its sign, and the unitized object has the weight +1. Where the weight has several components (a line, a plane, a general multivector) the weight norm becomes one and the orientation of the object is kept. Both, bulk and weight norms, have a positive argument under the square root due to the definitions of the inner product and the regressive inner product.

In conformal geometric algebra there are two distinguished basis vectors instead of one, the origin e_o and the point at infinity e_∞, and the split becomes a double one. A component belongs to the *flat* part when its blade contains e_∞ and to the *round* part otherwise, and it belongs to the *weight* when its blade contains e_o and to the *bulk* otherwise. This gives four parts

$$u = u_\bullet + u_\circ + u_\blacksquare + u_\square \tag{52}$$

where $u_\bullet$ is the round bulk, $u_\circ$ the round weight, $u_\blacksquare$ the flat bulk and $u_\square$ the flat weight (`round_bulk(u)`, `round_weight(u)`, `flat_bulk(u)` and `flat_weight(u)`). Each part has its own norm, the Euclidean length of its components

$$\|u\|_\bullet, \qquad \|u\|_\circ, \qquad \|u\|_\blacksquare, \qquad \|u\|_\square \tag{53}$$

The round weight carries the homogeneous weight of a round object, so *unitization* divides by the round weight here. A round weight with a single component (a round point, a sphere, a circle in the plane) is divided out with its sign, which gives the round weight +1. A round weight with several components (a dipole, a circle in space) is scaled to $\|u\|_\circ = 1$, and the orientation of the object is kept.

Unlike in *PGA*, the four parts are not the products $\mathbf{G}u$ and $\mathbb{G}u$. The conformal metric is non-degenerate with $det(\mathfrak{g}) = -1$, so both matrices are invertible, and the general relation $\mathbf{G}\mathbb{G} = det(\mathfrak{g})I_n$ together with $\mathbb{G} = -\mathbf{G}$ gives $\mathbf{G}^2 = I_n$. Neither matrix annihilates anything, and both act as a signed permutation of the basis blades (see tables 18 and 22). The four parts are selected by the two distinguished basis vectors as stated above. The norms of the round objects, which measure the distance of an object from the origin and its radius, are built from these part norms and are treated in the chapter on conformal geometry.

1.2.4 The dual

In *EGA* the dual operation (marked by a filled star *) is identical with the complement (reason: the Euclidean metric is the identity matrix; this changes for *PGA* as shown below). Therefore, in 2*D EGA* for the left and right dual operations are defined as

$$u^\star = \overline{u} \qquad \text{(right dual: } \texttt{r_dual(u)}\text{)} \tag{54}$$

$$u_\star = \underline{u} \qquad \text{(left dual: } \texttt{l_dual(u)}\text{)} \tag{55}$$

while in 3*D EGA*, an odd-dimensional algebra, where no distinction between left and right operations exists, we get

$$u^\star = \overline{u} \qquad \text{(dual: } \texttt{dual(u)}\text{)} \tag{56}$$

where the overline in $\overline{u}$ is the unique complement operation.

Now for PGA: Using the metric and the right complement, the *(right) bulk dual* $A^\star$ is defined as

$$A^\star = \overline{\mathbf{G}A} = \overline{A_\bullet} \qquad (\texttt{r_bulk_dual(u)}) \tag{57}$$

where A is an arbitrary multivector, $\mathbf{G}$ the extended metric and $\mathbf{G}A$ a matrix-vector-product between them. $A^\star$ is also called the Hodge dual. The multiplication with the metric happens before the application of the complement operation.

There is also a corresponding *(left) bulk dual* $A_\star$ operation using the metric and the left complement. It is defined as

$$A_\star = \underline{\mathbf{G}A} = \underline{A_\bullet} \qquad (\texttt{l_bulk_dual(u)}) \tag{58}$$

Where the left and right operations only differ for $3D$ PGA, while for $2D$ PGA, a three-dimensional and thus an odd-dimensional algebra, there is no difference between the left and right operations.

The Hodge star (symbolized by a filled star $\star$) is used as dualization operator in both cases. The lower or upper position symbolizes the left and right application. The difference between the Hodge dual and the complement is in both cases that multiplying the multivector with the extended metric happens *before* taking the corresponding complement. However, this does not lead to different results in spaces where the metric is the identity matrix. It only yields different results in spaces with a more involved metric, e.g. in projective spaces with a non-invertible (degenerate) metric as in PGA, or in the conformal spaces of CGA, whose metric is a signed permutation.

The effect of applying the metric before taking the complement is equivalent to applying the complement operation on the bulk part of the object u exclusively, and not on the full object u. Of course, the dualization can be applied to the weight part as well. This operation (symbolized by an unfilled star "$\star$") uses the regressive extended metric and is defined for the right weight dual $A^\star$ using the right complement as

$$A^\star = \overline{\mathbb{G}A} = \overline{A_\circ} \tag{59}$$

for an arbitrary multivector A. The left weight dual $A_\star$ is defined as

$$A_\star = \underline{\mathbb{G}A} = \underline{A_\circ} \tag{60}$$

using the left complement on the weight part of A.

In CGA the metric is non-degenerate, so there is no bulk/weight split of the dual and $A^\star = \overline{\mathbf{G}A}$ is the only metric dual, as in EGA and STA. The dualization with the regressive extended metric stays available and is called the *antidual* (`r_antidual(A)`, `l_antidual(A)`) there

$$A^* = \overline{\mathbb{G}A}, \qquad A_\star = \underline{\mathbb{G}A} \tag{61}$$

It reuses the unfilled star of the weight dual, which cannot occur in a non-degenerate algebra. Since $\mathbb{G} = -\mathbf{G}$ holds in both conformal algebras, the antidual is the negation of the dual at every grade. It is the form that appears in the defining equations of the conformal properties.

Relations to the (regressive) geometric product are given by

$$A^\star = \widetilde{A} \wedge I_n = \widetilde{A} \wedge \mathbb{1} \tag{62a}$$
$$A_\star = I_n \wedge \widetilde{A} = \mathbb{1} \wedge \widetilde{A} \tag{62b}$$
$$A^* = \underset{\sim}{A} \vee \mathbf{1} \tag{62c}$$
$$A_\star = \mathbf{1} \vee \underset{\sim}{A} \tag{62d}$$

with $I_n = \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \ldots \wedge \boldsymbol{e}_n = \mathbb{1}$ as the pseudoscalar of the n-dimensional space. $\widetilde{A}$ is the reversion operation, and $\underset{\sim}{A}$ is the regressive reversion operation applied to A (e.g. with $A = \boldsymbol{e}_1 \wedge \boldsymbol{e}_2$ the reverse is $\widetilde{A} = \boldsymbol{e}_2 \wedge \boldsymbol{e}_1$), and "$\wedge$" and "$\vee$" are the symbols for the geometric and regressive geometric products. As the complement, the reverse is defined for any basis element. It is extended to full multivectors A and B by linearity as

$$\widetilde{sA} = s\widetilde{A}, \quad \widetilde{A+B} = \widetilde{A} + \widetilde{B}, \quad \widetilde{A \wedge B} = \widetilde{B} \wedge \widetilde{A} \tag{63a}$$
$$\underset{\sim}{sA} = s\underset{\sim}{A}, \quad \underset{\sim}{A+B} = \underset{\sim}{A} + \underset{\sim}{B}, \quad \underset{\sim}{A \vee B} = \underset{\sim}{B} \vee \underset{\sim}{A} \tag{63b}$$
$$\widetilde{\widetilde{A}} = A, \quad \text{and} \quad \underset{\sim\sim}{A} = A \tag{63c}$$

where s is a scalar. The reverse and regressive reverse are their own inverses, i.e. applying the same reverse twice returns the original object.

For an overview on all bulk and weight duals in PGA see tables 7 and 11 in subsection 1.2.10. The reverses can be found in tables 8 and 12 in that subsection as well. The corresponding left and right (metric) duals and the reverse for $STA4DS$ are given in tables 16 and 17 in subsection 1.2.13; since $STA4DS$ is non-degenerate there is no bulk/weight split. The duals and antiduals for the conformal algebras, which are non-degenerate as well, are

listed in tables 20 and 24, their reverses in tables 21 and 25 in subsection 1.2.14.

Following identities link the expansions and contractions and thus the outer and inner products:

$$(A \wedge B)^{\star} = A^{\star} \vee B^{\star} \tag{64a}$$
$$A \wedge B^{\star} = (A \cdot B) \wedge \mathbb{1}, \quad \text{when } \texttt{gr(A)} = \texttt{gr(B)} \tag{64b}$$
$$A \wedge B^{*} = A \circ B, \quad \text{when } \texttt{gr(A)} = \texttt{gr(B)} \tag{64c}$$

$$(A \vee B)^{\star} = A^{\star} \wedge B^{\star} \tag{64d}$$
$$A \vee B^{\star} = A \cdot B, \quad \text{when } \texttt{gr(A)} = \texttt{gr(B)} \tag{64e}$$
$$A \vee B^{*} = (A \circ B) \vee \mathbf{1}, \quad \text{when } \texttt{gr(A)} = \texttt{gr(B)} \tag{64f}$$

The influence of the metric for the dualization operation can be seen in its effect for the different algebras which also depends on the dimension of the modelled space:

- $EGA2D$: $A_{\star}$ = `l_dual(A)`, $A^{\star}$ = `r_dual(A)`
- $EGA3D$: $A_{\star}$ = `dual(A)`, $A^{\star}$ = `dual(A)`
- $PGA2DP$: $A_{\star}$ = `bulk_dual(A)`, $A^{\star}$ = `bulk_dual(A)`
- $PGA2DP$: $A_{\star}$ = `weight_dual(A)`, A^{*} = `weight_dual(A)`
- $PGA3DP$: $A_{\star}$ = `l_bulk_dual(A)`, $A^{\star}$ = `r_bulk_dual(A)`
- $PGA3DP$: $A_{\star}$ = `l_weight_dual(A)`, A^{*} = `r_weight_dual(A)`
- $STA4DS$: $A_{\star}$ = `l_dual(A)`, $A^{\star}$ = `r_dual(A)`
- $CGA2DC$: $A_{\star}$ = `l_dual(A)`, $A^{\star}$ = `r_dual(A)`
- $CGA2DC$: $A_{\star}$ = `l_antidual(A)`, A^{*} = `r_antidual(A)`
- $CGA3DC$: $A_{\star}$ = `dual(A)`, $A^{\star}$ = `dual(A)`
- $CGA3DC$: $A_{\star}$ = `antidual(A)`, A^{*} = `antidual(A)`

In even-dimensional spaces ($EGA2D$, $PGA3DP$, $STA4DS$, $CGA2DC$) we have to distinguish left and right operations, while in odd-dimensional spaces ($EGA3D$, $PGA2DP$, $CGA3DC$) we don't, since they produce the same result. The "left" operations are typically used on the left-hand side of the

binary operation and the "right" operations on the right-hand side of the product operator. The only deviation from that pattern occurs for the definition of the regressive operations, as defined in (40a) or (40b).

In a two-dimensional algebra, like $EGA2D$, the effect of the right dual of a vector u is to turn the vector by +90°, i.e. into the orientation of $\boldsymbol{e}_{12}$, while the left dual turns the vector by −90°, i.e. into the direction against the orientation of $\boldsymbol{e}_{12}$. This statement is valid for both right- and left-handed coordinate systems.

Complement or dual: In EGA both operations return the same blade, so the result does not show which one a statement uses; its meaning does. An identification that uses the metric — a direction and the hyperplane perpendicular to it, the cross product, a gradient turned by 90°, a field vector and its induction bivector — is a dual, in EGA as well. A statement without a metric — orientation, the blade spanned by the remaining basis vectors, join, meet and incidence — is a complement. Where the choice is not obvious, the statement is formulated in an algebra in which the two differ, and the operation under which it still holds is the one to use: in PGA the bulk dual annihilates every component along the degenerate direction, in STA dual and complement differ in sign on the bivectors $\boldsymbol{g}_{k4}$.

The horizon of $PGA2DP$ is an example. With the origin $O_{2dp} = \boldsymbol{e}_3$

$$H_{2dp} = \overline{O_{2dp}} = -\boldsymbol{e}_{12}, \qquad O_{2dp} \wedge H_{2dp} = \mathbb{1} \tag{65}$$

while the two duals of the origin are

$$O_{2dp}{}^{\star} = 0, \qquad O_{2dp}{}^{\star\!\star} = -\boldsymbol{e}_{12} \tag{66}$$

The horizon is the line spanned by the two ideal directions $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$, the basis vectors the origin does not contain. That is an incidence statement without a metric, hence a complement. The bulk dual returns zero, since the origin lies entirely in the degenerate direction. The weight dual agrees with the complement only because the origin is pure weight; for a point $P = x\,\boldsymbol{e}_1 + y\,\boldsymbol{e}_2 + w\,\boldsymbol{e}_3$

$$P^{\star} = -y\,\boldsymbol{e}_{31} + x\,\boldsymbol{e}_{32}, \qquad P^{\star\!\star} = -w\,\boldsymbol{e}_{12}, \qquad \overline{P} = P^{\star} + P^{\star\!\star} \tag{67}$$

The orientation of H_{2dp} is the one the complement fixes, $O_{2dp} \wedge H_{2dp} = +\mathbb{1}$; the join $\boldsymbol{e}_1 \wedge \boldsymbol{e}_2 = \boldsymbol{e}_{12}$ spans the same line with the opposite orientation. The same holds in $PGA3DP$, with $H_{3dp} = \overline{\boldsymbol{e}_4} = \boldsymbol{e}_{321}$ and $\boldsymbol{e}_4{}^{\star} = 0$. The STA case, where only the dual reproduces the field equation, is worked through in subsection 1.9.1.

Conjugations: There are grade-selective operations which can be useful for handling grade-dependent signs. These operations are defined for each algebra, but are shown here for the example of a four-dimensional algebra.

Grade involution of an arbitrary multivector M for its grade r-component is defined as $\widehat{M_r} = (-1)^r M_r$, effectively inverting the sign of odd-grade blades (function gr_inv()):

$$\widehat{M} = \langle M\rangle_0 - \langle M\rangle_1 + \langle M\rangle_2 - \langle M\rangle_3 + \langle M\rangle_4 \tag{68}$$

Reversion of the component M_r is defined by the pattern $\widetilde{M_r} = (-1)^{(\frac{r(r-1)}{2})} M_r$ (function rev()) resulting in

$$\widetilde{M} = \langle M\rangle_0 + \langle M\rangle_1 - \langle M\rangle_2 - \langle M\rangle_3 + \langle M\rangle_4 \tag{69}$$

While the Clifford conjugation of the grade r multivector-component M_r is defined by $M_r^* = (-1)^{\frac{r(r+1)}{2}} M_r$ (function conj()). It is effectively a composition of reversion and grade involution ($M^* = \widehat{\widetilde{M}} = \widetilde{\widehat{M}}$) resulting in

$$M^* = \langle M\rangle_0 - \langle M\rangle_1 - \langle M\rangle_2 + \langle M\rangle_3 + \langle M\rangle_4 \tag{70}$$

These three operations are what the inverse is built from. For a k-blade A the inverse w.r.t. the geometric product is $A^{-1} = \widetilde{A}/\langle A \wedge \widetilde{A}\rangle_0$. For a multivector of mixed grade, [Hitzer and Sangwine, 2016] give closed forms that multiply M by its grade involution, reversion and Clifford conjugation in a fixed sequence until the product becomes a scalar. The library implements them as the function *inv*(); the regressive counterpart *rinv*() is that inverse carried through the complement.

CGA adds a conjugation that does not select by grade, the *conformal conjugate* $u^\dagger$ (function *cconj*()). It keeps the round part of an object and negates its flat part, i.e. it negates every component whose blade contains the point at infinity

$$u^\dagger = u_\bullet + u_\circ - u_\blacksquare - u_\square \tag{71}$$

using the four parts of the split defined in subsection 1.2.3. It is an involution as well. Geometrically it reflects the center of a round object through the origin and adjusts its radius accordingly, so that for a unitized object u the product $u \cdot u^\dagger$ returns the squared distance between the origin and the center of u.

Before we go on to the geometric product, let's return to eqn. (16) for a moment, which provided the definition of a plane as trivector in projective

space. We are going to use this example to show what can be done geometrically in PGA using the definitions provided above:

The trivector defining a plane can be created in multiple ways, e.g. either by wedge products between three unitized projective points P, Q, R that span the plane as $pl = P \wedge Q \wedge R$ as their common subspace, or alternatively as wedge products between a unitized projective point P and two direction vectors u and v that span the plane as a bivector such that $pl = P \wedge u \wedge v$. In the latter case, the plane pl can be interpreted as bound bivector as suggested by [Browne, 2012]. For this case it is also easy to shift the plane to contain the unitized point $P + x$ instead of P. This can be achieved easily by applying the rules for the wedge product as $pl_{shifted} = P \wedge u \wedge v + x \wedge u \wedge v = (P + x) \wedge u \wedge v$.

The attitude of the plane, i.e. its orientation in space, is given by a bivector that can be calculated using the attitude function $\mathrm{att}(pl)$. It is defined for $3D$ projective space as $\mathrm{att}(u) = u \vee \overline{\boldsymbol{e}_4}$ and as such can be interpreted as the intersection of the plane with the horizon $H_{3dp} = \overline{\boldsymbol{e}_4}$. It yields the result $\mathrm{att}(pl) = t_x \boldsymbol{e}_{23} + t_y \boldsymbol{e}_{31} + t_z \boldsymbol{e}_{12}$.

The definition of a plane as given in eqn. (16) implies that the normal direction n of the plane pl can be recovered from its trivector definition by applying the left weight dual to the plane, such that $n = pl_{\star}$. The left weight dual operation was introduced in eqn. (60). It can also be found in table 11.

The reverse is also true, i.e. if the plane normal n is known, the trivector representing the plane can be directly found by applying the right bulk dual operation to the normal n, such that $pl_O = n^{\star}$. The right bulk dual operation was defined in eqn. (57), and can be found in table 11. This operation yields a plane that contains the origin and has the required orientation with n as normal. The signed distance to the origin could now be given as coefficient t_w to the bulk component $\boldsymbol{e}_{312}$ in case the plane needs to be shifted parallel.

As demonstrated, this enables quite a number of relevant geometric operations by applying rather simple operations to the modelled objects in $3D$ PGA.

1.2.5 Geometric products

Now let's move on in more detail to the geometric products. The multiplicative identity of the geometric product and the wedge product is the scalar

basis element $\mathbf{1}$:

$$\mathbf{1} \barwedge a = a \barwedge \mathbf{1} = a \tag{72a}$$
$$\mathbf{1} \wedge a = a \wedge \mathbf{1} = a \tag{72b}$$

The multiplicative identity of the regressive geometric product and the regressive wedge product is the pseudoscalar basis element $\mathbb{1} = I_n$ of the respective algebra:

$$\mathbb{1} \barvee a = a \barvee \mathbb{1} = a \tag{73a}$$
$$\mathbb{1} \vee a = a \vee \mathbb{1} = a \tag{73b}$$

The geometric product combines the dimensions that are present in an object representation, e.g. for $PGA2DP$:

$$\begin{aligned} \boldsymbol{e}_1 \barwedge \boldsymbol{e}_1 &= 1 \\ \boldsymbol{e}_2 \barwedge \boldsymbol{e}_2 &= 1 \\ \boldsymbol{e}_3 \barwedge \boldsymbol{e}_3 &= 0 \end{aligned} \tag{74}$$

The regressive geometric product combines the dimensions that are <u>not</u> present in an object representation, e.g. for $PGA2DP$:

$$\begin{aligned} \overline{\boldsymbol{e}_1} \barvee \overline{\boldsymbol{e}_1} &= \mathbb{1} \\ \overline{\boldsymbol{e}_2} \barvee \overline{\boldsymbol{e}_2} &= \mathbb{1} \\ \overline{\boldsymbol{e}_3} \barvee \overline{\boldsymbol{e}_3} &= 0 \end{aligned} \tag{75}$$

where eqns. (75) can be shown to be equivalent to eqns. (74) by taking the left complement of both sides and applying the complement definition (40a) to it.

The inner product between multivectors A and B is defined as

$$A \cdot B = A^T \mathbf{G} B \tag{76}$$

with $\mathbf{G}$ as the extended metric and using standard matrix multiplication on the right-hand side. It satisfies the identity

$$A \cdot B = \langle A \barwedge \tilde{B} \rangle_0 = \langle B \barwedge \tilde{A} \rangle_0 \tag{77}$$

where $\barwedge$ stands for the geometric product within the grade projection operator $\langle\rangle_k$ for grade k.

Using the various notations used in literature, the contraction is defined as

$$A \rfloor B = A \ll B = A_\star \vee B \quad \text{(left contracton)} \tag{78a}$$
$$B \lfloor A = B \gg A = B \vee A^\star \quad \text{(right contracton)} \tag{78b}$$

with $\star$ denoting the Hodge dual which is formed by the respective metric and complement operation (in spaces of even dimension the left or right complement respectively, and in spaces of odd dimension the metric and complement function regardless of the side of the operand). For blades A and B, they satisfy

$$A \rfloor B = A \ll B = \langle B \barwedge \tilde{A} \rangle_{gr(B)-gr(A)} \tag{79a}$$
$$B \lfloor A = B \gg A = \langle \tilde{A} \barwedge B \rangle_{gr(B)-gr(A)} \tag{79b}$$

and fulfill the requirement that $A \rfloor B = A \lfloor B = A \cdot B$ whenever A and B have the same grade. In this case the contractions reduce to the inner product.

The geometric product between a vector v and a multivector B — a blade of any grade or a sum of them, since both sides are linear in B — is defined in terms of interior and exterior products, i.e. the right contraction "$\lfloor$", the left contraction "$\rfloor$" and wedge product "$\wedge$" as follows:

$$v \barwedge B = B \lfloor v + v \wedge B \tag{80a}$$
$$B \barwedge v = v \rfloor B + B \wedge v \tag{80b}$$

or sometimes expressed using the right shift "$\gg$" or left shift "$\ll$" operators for the contractions (as implemented in the ga library[6])

$$v \barwedge B = B \gg v + v \wedge B \tag{81a}$$
$$B \barwedge v = v \ll B + B \wedge v \tag{81b}$$

or sometimes by exclusively using the operations of the exterior algebra (wedge product "$\wedge$", bulk dual "$\star$" and regressive wedge product "$\vee$")

$$v \barwedge B = B \vee v^\star + v \wedge B \tag{82a}$$
$$B \barwedge v = v_\star \vee B + B \wedge v \tag{82b}$$

as an alternative. Each one is equivalent.

[6]Due to low C++ operator precedence, always make sure to put operator expressions using the shift operators in parentheses! Otherwise, you might get unexpected results.

Be careful, when using the equivalent formulas like the right-hand side of equations (82) when deriving regressive products. Resulting formulas may not be valid! The equivalent formulas for the regressive products must be derived directly from the regressive product table. To give one example, here is the corresponding formula for the regressive geometric product for the case of a regressive product between a vector v and a bivector B:

$$v \vee B = -B_{\star} \wedge v + \mathrm{rcmt}(v, B) \tag{83a}$$
$$B \vee v = -v \wedge B^{\star} + \mathrm{rcmt}(B, v) \tag{83b}$$

It is obviously not the result one would get from creating the regressive versions directly from the right-hand side of equations (82). The pre-conditions for the validity of the equivalence between the geometric product and the corresponding right-hand sides might change when taking the complements.

For arguments of equal grade the contractions reduce to the dot product, so that one can write specifically for two vectors u and v:

$$u \mathbin{\vartriangleleft} v = v \gg u + u \wedge v = u \cdot v + u \wedge v \tag{84}$$

Every vector v can be decomposed into parts $v_{\parallel}$ parallel to a k-blade B and $v_{\perp}$ perpendicular to B, such that $v = v_{\parallel} + v_{\perp}$. For equation (81a): $B \gg v$ is a $(k-1)$-blade. If $v_{\parallel} \neq 0$ then $B \gg v$ represents a subspace of B. $v \wedge B$ is a $(k+1)$-blade. If $v_{\perp} \neq 0$ then $v \wedge B$ represents $span(v, B)$.

So we get for two vectors u and v, and a bivector B, representing a plane in $3D\ EGA$:

$$v = v_{\parallel} + v_{\perp} \tag{85a}$$
$$(v \mathbin{\vartriangleleft} u) \mathbin{\vartriangleleft} u^{-1} = \underbrace{(v \cdot u) \mathbin{\vartriangleleft} u^{-1}}_{v_{\parallel}=\text{project_onto(v,u)}} + \underbrace{(v \wedge u) \mathbin{\vartriangleleft} u^{-1}}_{v_{\perp}=\text{reject_from(v,u)}} \tag{85b}$$
$$(v \mathbin{\vartriangleleft} B) \mathbin{\vartriangleleft} B^{-1} = \underbrace{(B \gg v) \mathbin{\vartriangleleft} B^{-1}}_{v_{\parallel}=\text{project_onto(v,B)}} + \underbrace{(v \wedge B) \mathbin{\vartriangleleft} B^{-1}}_{v_{\perp}=\text{reject_from(v,B)}} \tag{85c}$$

While for $3D\ PGA$ the projection and rejection are directly derived from the orthogonal projection as defined by [Lengyel, 2024b], p. 99f.:

$$\text{project_onto(a, b)} = \text{ortho_proj3dp(a, b)} = b \vee (a \wedge b^{\star}) \tag{86}$$

where a could be a point and b could either be a line (a bivector) or plane (a trivector), or a could be a line and b a plane. The rejection is then calculated

as reject_from(a, b) = a − project_onto(a, b). Read from the inside out, eqn. (86) is a construction in three steps; fig. 6 shows them for a point and a plane.

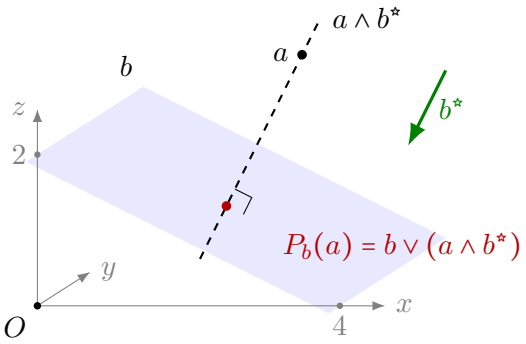


Figure 6: Eqn. (86) as three steps, for the point $a = (3, 1, 3)$ and the plane b: $x + 2z = 4$, $b = 2\,\boldsymbol{e}_{423} + 4\,\boldsymbol{e}_{412} - 8\,\boldsymbol{e}_{321}$. The weight dual $b^\star$ is the direction of the plane's normal, an ideal point (green). Its join with a, $a \wedge b^\star$, is the line through a perpendicular to b (dashed), and the meet of that line with b is the projection $P_b(a) = (2, 1, 1)$. The same steps project a point onto a line, or a line onto a plane.

1.2.6 Projections and rejections

The identity $a = a_\parallel + a_\perp$ says nothing about whether a projection is correct: wherever the rejection is defined as the difference it holds by construction. Three properties do – how the result depends on the scale of either argument, which form the expression takes for an even-grade source, and whether the difference is a blade at all.

Scale: The projective expression is quadratic in the object projected onto – the weight dual is linear in b and the meet brings it in a second time – so a plane carrying an arbitrary weight scales the projection onto it:

$$(\lambda b) \vee (a \wedge (\lambda b)^\star) = \lambda^2\, b \vee (a \wedge b^\star) \tag{87}$$

The result must depend on the subspace that b represents, not on the representative passed for it, so the library divides $\|b\|_\circ^2$ back out. What it does not divide out is the scale of a: a projection is linear in what is projected, and the weight of a projected point or the foreshortened weight of a projected line carry meaning. The division is written out in either algebra: by $\|b\|^2$ in EGA and STA, and by $\|b\|_\circ^2$ in PGA, where the metric is degenerate and a plane has no inverse that could absorb it.

Grade: Eqn. (85c) holds for a target blade of any grade, but its source is a vector: it rests on the vector-blade product eqn. (81a), whose split into contraction plus wedge does not survive a source of higher grade – at grade two the middle part of the geometric product is carried by neither term. For a source of any grade the library computes the projection from one of two expressions, and from the same two in every algebra. Where the grades differ it is eqn. (86), taken with the weight dual in PGA, whose metric is degenerate, and with the metric dual in EGA and STA; it is provided as `ortho_proj2dp`, `ortho_proj3dp`, `ortho_proj3d` and `ortho_proj4ds`, and `project_onto` calls it. Where the grades are equal the projection can only be a multiple of the target, and it is

$$P_b(a) = \frac{a \cdot b}{\|b\|^2}\, b \qquad (gr(a) = gr(b)) \tag{88}$$

Neither of the two carries a grade-dependent sign, and each is the cheaper form in its own case.

The classical formulation does, and that is the reason it is not used here. Written with the contraction and the inverse, the projection of a blade a of grade $j = gr(a)$ is

$$P_b(a) = (-1)^{\frac{j(j-1)}{2}}\, (b \gg a) \wedge b^{-1} \tag{89}$$

and the sign compensates the reversions that its two operations carry. The contractions of eqns. (79a) and (79b) take $\widetilde{a}$ and not a, the choice that makes $a \cdot a = \|a\|^2$ positive definite, and the inverse is $b^{-1} = \widetilde{b}/\|b\|^2$. On a blade the reversion is a scalar, $\widetilde{a} = (-1)^{j(j-1)/2} a$, which leaves the factor above: $+1$ up to grade one, -1 at grades two and three, $+1$ again at grade four. For a vector source it is $+1$ and eqn. (89) reduces to eqn. (85c).

A sign of that kind is easy to get wrong, because nothing obvious notices it. The negated result still lies in b, it still scales correctly in either argument, and it still sums with its rejection to a. Only idempotence separates the two, $P_b(P_b(a)) = P_b(a)$: without the sign, projecting b onto itself returns $\widetilde{b}$ instead of b.

Eqn. (86) avoids the question altogether. It is built from wedge, antiwedge and dual, none of which carries a reversion, so there is nothing to compensate at any combination of grades: point onto line, point onto plane and line onto plane are one expression, and the only correction it takes is the quadratic factor of the previous paragraph, which is the same for every grade. It also uses no inverse, which is what makes it the only one of the two available in

a degenerate metric, and where the grades differ it is the cheaper of the two, since the classical product materialises grades that are afterwards discarded. At equal grades it would serve as well, but there it computes a dual and two products where eqn. (88) needs a dot product, a division and a scaling – which is why that case keeps its own expression, sign-free for a reason of its own: a multiple of the target is all it can be.
Both expressions take the target to be a blade, and from four dimensions on that is a condition rather than a formality. A bivector in four dimensions is in general a sum of two blades and spans no plane at all, $B \wedge B \neq 0$ being the test, so there is no subspace to project onto: the two formulations then return different results, and the classical one is not even idempotent. The question cannot arise in three dimensions, where every bivector is simple, nor for a hyperplane target in any dimension – but it does arise for a line in $3D\ PGA$ and for a 2-plane in STA. The library answers it with `is_simple`, the test above, and rejects a target that fails it. The source is free: both expressions are linear in a, so a source of grade j that is not simple projects as the sum of its blade parts and carries the same sign.

Whether the difference is a blade: For a vector source the decomposition is exact and both parts are vectors, because the space splits in two, $V = b \oplus b^{\perp}$. At grade two it splits in three,

$$\Lambda^2(V) = \Lambda^2(b) \ \oplus \ (b \wedge b^{\perp}) \ \oplus \ \Lambda^2(b^{\perp}) \tag{90}$$

and the projection reaches only the first summand, so the difference $a - P_b(a)$ collects the mixed part as well. Whether that remainder is still a blade – a line or a plane rather than a sum of two of them – depends on one condition: b must be a *hyperplane*. Then $\Lambda^2(b^{\perp}) = 0$, only two summands remain, and every element $v \wedge n$ of the mixed part is simple. Otherwise the remainder is in general not simple, and in four dimensions a bivector represents a line or a 2-plane only if $B \wedge B = 0$. Accordingly the library provides a projection of a line onto a plane but none of a line onto a line: there the subtraction returns an object that is no line, and projecting onto the orthogonal complement instead loses the mixed part, so that it no longer adds back up to a.

In $3D\ PGA$ the rejection of a line from a plane can therefore be read geometrically. It is the line through the point where the original line pierces the plane, perpendicular to that plane. The piercing point comes for free: the projection lies in the plane, so it contributes nothing to the meet, and the whole of it is carried by the rejection,

$$\text{reject_from}(a, b) \vee b = a \vee b \tag{91}$$

A line lying in the plane rejects to zero, a line already perpendicular to the plane rejects to itself, and a line parallel to the plane has no piercing point – there the rejection loses its attitude and what remains is an ideal line carrying the offset.

1.2.7 STA metric, complements, and duality

In this subsection we document what was learned while implementing STA on top of the machinery that was already available from implementing EGA and PGA.

The space-time algebra $STA4DS = \mathbb{G}(1,3,0)$ uses the Minkowski signature $(\boldsymbol{g}_1^2, \boldsymbol{g}_2^2, \boldsymbol{g}_3^2, \boldsymbol{g}_4^2) = (-1,-1,-1,+1)$ (the three space-like directions $\boldsymbol{g}_1$, $\boldsymbol{g}_2$, $\boldsymbol{g}_3$ square to -1, the time-like direction $\boldsymbol{g}_4$ squares to $+1$). The non-Euclidean vector metric introduces a few specific requirements that do not show up in EGA (identity metric) or PGA (degenerate metric), and that are worth recording carefully.

Extended metric: The only metric input to an algebra is the *vector metric* $\mathfrak{g}$, i.e. the signature itself. The extended metric $\mathbf{G}$ lifts $\mathfrak{g}$ to every grade as the unique wedge exomorphism (44), seeded on the grade-1 basis vectors:

$$\mathbf{G}(\boldsymbol{g}_i) = \mathfrak{g}_{ii} \tag{92a}$$

$$\mathbf{G}(a \wedge b) = \mathbf{G}(a) \wedge \mathbf{G}(b) \implies \mathbf{G}(e_S) = \prod_{i \in S} \mathfrak{g}_{ii} \tag{92b}$$

where e_S is the basis blade spanned by the index set S. Write this product $P(e_S) = \prod_{i \in S} \mathfrak{g}_{ii}$. For example, in $STA4DS$ the bivector $\boldsymbol{g}_{14}$ has $P(\boldsymbol{g}_{14}) = \mathfrak{g}_{11}\,\mathfrak{g}_{44} = (-1)(+1) = -1$, while $\boldsymbol{g}_{23}$ has $P(\boldsymbol{g}_{23}) = \mathfrak{g}_{22}\,\mathfrak{g}_{33} = (-1)(-1) = +1$. The scalar maps to 1, the pseudoscalar to $det(\mathfrak{g})$, and there is exactly one such inner product on the whole exterior algebra derived from the requirement of an exomorphism.

Note that *the extended metric is independent of how a blade squares under the geometric product.* The extended metric is a property of the *wedge* structure seeded by the signature, not of the geometric product. This is visible in PGA: its extended-metric diagonal carries the exomorphism values $1/0$ even though its bivectors square to -1 under the geometric product. Physical *characteristics* (space-like/time-like) do not belong in the extended metric: causal character lives in the geometric square (see below), not in the metric. Finally, the extended metric *defines* the dot product and hence $nrm_sq()$

at every grade through $\boldsymbol{g}_S \cdot \boldsymbol{g}_S = \mathbf{G}(\boldsymbol{g}_S) = P$. This scalar P is the blade's *reverse-norm*, the scalar part of the blade times its *reverse* $\langle \widetilde{A} \wedge A \rangle_0$.

Metric and regressive metric: Following Lengyel, the metric carries two exomorphisms: the metric $\mathbf{G}$ is a wedge exomorphism, while the regressive metric $\mathbb{G}$ is a regressive wedge exomorphism (47) with $\mathbb{G}(e_S) = \prod_{i \notin S} \mathfrak{g}_{ii}$ (the complementary product). For *every* algebra they are related by the ordinary matrix product $\mathbf{G}\mathbb{G} = det(\mathfrak{g})\, I_n$. For $STA4DS$ we have $det(\mathfrak{g}) = (-1)(-1)(-1)(+1) = -1$, and since every basis blade squares to ± 1 we get $\mathbf{G}^2 = I_n$, so the relation collapses to $\mathbb{G} = -\mathbf{G}$: the regressive metric is the negated metric. The determinant is signature-parity invariant ($det = (-1)^q$ with q the number of negative directions), so the physically equivalent signatures $(-,-,-,+)$ and $(+,+,+,-)$ both give -1.

Geometric square B^2: The *geometric square* of a blade is $B^2 = \langle u \wedge u \rangle_0$, the blade times itself under the geometric product. It relates to the metric value P only by the reversion sign $\sigma(k) = (-1)^{k(k-1)/2}$ (with k the grade):

$$B^2(e_S) = \sigma(k)\, P(e_S) \tag{93}$$

so $B^2 = P$ at grades $k \equiv 0, 1 \pmod 4$ and $B^2 = -P$ at grades $2, 3$. Two worked examples in $STA4DS$ ($\boldsymbol{g}_1^2 = \boldsymbol{g}_2^2 = \boldsymbol{g}_3^2 = -1$, $\boldsymbol{g}_4^2 = +1$) are

$$\boldsymbol{g}_{14}: \quad P = \boldsymbol{g}_1 \cdot \boldsymbol{g}_4 = -1, \qquad B^2 = \sigma(2)\, P = +1 \;=\; \boldsymbol{g}_{14} \wedge \boldsymbol{g}_{14} \tag{94a}$$

$$\boldsymbol{g}_{23}: \quad P = \boldsymbol{g}_2 \cdot \boldsymbol{g}_3 = +1, \qquad B^2 = \sigma(2)\, P = -1 \;=\; \boldsymbol{g}_{23} \wedge \boldsymbol{g}_{23} \tag{94b}$$

The metric uses the *reverse*-norm, which removes that $\sigma(k)$: the reverse of a grade-k blade is $\widetilde{e}_S = \sigma(k)\, e_S$, so

$$nrm_sq(e_S) = \langle \widetilde{e}_S \wedge e_S \rangle_0 = \sigma(k)\, \langle e_S \wedge e_S \rangle_0 = \sigma(k)\, B^2 = \sigma(k)^2\, P = P \tag{95}$$

because $\sigma(k)^2 = 1$. A blade thus has two natural "squares" that differ exactly by the reversion sign: the geometric square $B^2 = \sigma(k)\, P$ (blade $\wedge$ blade) and the reverse-norm P (reverse $\wedge$ blade). The reverse-norm behaves like a norm; the dot product, the contraction for equal grades, and $nrm_sq()$ all use P. The geometric square B^2 is the physically meaningful quantity: a bivector is a *boost* plane when $B^2 > 0$ and a *rotation* plane when $B^2 < 0$, and the rotor exponential splits on its sign ($\cosh / \sinh$ vs. $\cos / \sin$). Causal character — `is_timelike`, `is_spacelike`, `is_lightlike` — therefore reads B^2, *not* $nrm_sq()$. Keeping B^2 and the metric value P apart avoids encoding "time-like" or "space-like" into the metric. Both squares are tabulated for every basis blade in table 14 below.

Dualization: There are two commonly used ways to dualize a blade or multivector A:

1. *complement after metric* (this library, after Lengyel): $A^\star = \overline{\mathbf{G}A}$ — multiply by the extended metric, then take the purely combinatorial complement. This is the (bulk) dual already defined in (57) and (58).

2. *pseudoscalar multiplication*: $A \wedge I_n$ and $A \wedge I_n^{-1}$.

In a non-degenerate algebra the two relate by reversion, $A^\star = \widetilde{A} \wedge I_n$ (cf. (62a)). So plain $A \wedge I_n$ equals $A^\star$ where $\sigma(k) = +1$ (grades $0, 1, 4$) and $-A^\star$ at grades $2, 3$. Pseudoscalar multiplication needs I_n to be invertible; in $STA4DS$, $I_n \wedge I_n = -1$, so $I_n^{-1} = -I_n$. In a *degenerate* algebra (PGA, and any projective algebra) a null vector sits inside I_n, so $I_n \wedge I_n = 0$ and I_n has no inverse — the scheme is structurally impossible there. The complement-after-metric definition survives because I_n never appears in it; it is the definition used throughout the library for all algebras.

Handedness (left vs. right dual): $STA4DS$ is four-dimensional (even), so the left and right complements/duals differ on the odd grades and agree on the even grades:

$$A_\star = A^\star \ \text{(grades } 0, 2, 4), \qquad A_\star = -A^\star \ \text{(grades } 1, 3) \tag{96}$$

The non-metric complements are exact mutual inverses for any signature, $\overline{(A)} = \underline{(\underline{A})} = A$, as already stated in (38). Both complements and duals are listed for every basis blade in tables 15 and 16.

Dual round trip: Composing the two metric duals yields a clean global sign, the determinant:

$$(A^\star)_\star = (A_\star)^\star = det(\mathfrak{g})\, A \tag{97}$$

For a Euclidean algebra ($det = +1$) this is the identity — the left dual undoes the right dual. For $STA4DS$ ($det = -1$) it is $-A$: the left dual is the inverse only up to the sign det. The exact inverse applies the metric on the *dual* side instead of the primal side. Since $\mathbf{G}^2 = I_n$, the inverse of $A^\star = \overline{\mathbf{G}A}$ is the *un-dual* (Lengyel: antidual), one per handedness:

$$r_undual(D) = \underline{\overline{(D)}^\star} \quad \text{inverts } A^\star\text{: } r_undual(A^\star) = A \tag{98a}$$

$$l_undual(D) = \overline{\underline{(D)}_\star} \quad \text{inverts } A_\star\text{: } l_undual(A_\star) = A \tag{98b}$$

They coincide on the even grades and differ by a sign on the odd grades. Equivalently $r_undual = det \cdot (\,\cdot\,)_\star$ and, symmetrically, $l_undual = det \cdot (\,\cdot\,)^\star$, i.e. $= -(\,\cdot\,)_\star$ and $-(\,\cdot\,)^\star$ respectively for $STA4DS$. The convention-clean statement is therefore "each un-dual undoes the dual of the same handedness": for Euclidean signatures the right un-dual *is* the left dual (and the left un-dual the right dual). The following table summarizes the regime per algebra:

algebra	$det(\mathfrak{g})$	r_undual is	l_undual is
$EGA2D/EGA3D$ (Euclidean)	+1	the left dual $(\,\cdot\,)_\star$	the right dual $(\,\cdot\,)^\star$
$STA4DS$ (Minkowski)	−1	the antidual $(-(\,\cdot\,)_\star)$	the antidual $(-(\,\cdot\,)^\star)$
$CGA2DC/CGA3DC$ (conformal)	−1	the left antidual $(\,\cdot\,)_\star$	the right antidual $(\,\cdot\,)^\star$
PGA (degenerate)	0	no inverse; bulk/weight split (see below)	

In CGA the un-dual is a function of the library rather than only a name. The round trip (97) applies there unchanged, and since $det(\mathfrak{g}) = -1$ composing the two duals again returns $-A$. The antidual of eqn. (61) differs from the dual by exactly that sign, so composing a dual with the antidual of the opposite handedness is the exact inverse:

$$(A^\star)_\star = -A, \qquad (A^\star)_\star = -(A^\star)_\star = A \tag{99}$$

and with the handedness exchanged $(A_\star)^\star = A$ in the same way. Both hold for every basis blade. In the five-dimensional $CGA3DC$ left and right coincide, and the single antidual inverts the single dual.

There is also a round trip valid for *every* algebra including PGA: the metric-free complement pair $\underline{(\overline{A})} = \overline{(\underline{A})} = A$, which carries no metric structure.

Signature duality: Because $\mathbb{G} = -\mathbf{G}$, the regressive metric negates every vector square: $\mathbf{G}$ realises the signature $(-,-,-,+)$ while $\mathbb{G}$ realises $(+,+,+,-)$. Each signature is an exomorphism of exactly *one* product — $\mathbf{G}$ of the wedge, $\mathbb{G}$ of the regressive wedge. So, relative to the library's wedge convention, the "space" is the signature whose metric respects the wedge product, $(-,-,-,+)$, and its "dual space" is $(+,+,+,-)$. The whole structure is symmetric under the exchange ($\wedge \leftrightarrow \vee$, $\mathbf{G} \leftrightarrow \mathbb{G}$, space $\leftrightarrow$ dual space): taking the regressive wedge as the primary product (working in the dual algebra) swaps the roles. This is why the two signatures are physically equivalent: each is "the space" for one of the two products, the other being its dual space. Both formulations are actively used to describe phenomena in spacetime research.

Bridge to a projective STA: The statement $\mathbf{G}\mathbb{G} = det(\mathfrak{g})\, I_n$ is universal; the determinant decides the regime. For $STA4DS$ (non-degenerate, $det = -1 \neq 0$) we get $\mathbb{G} = -\mathbf{G}$, a global rescaling with no split. For a potential *projective* STA (degenerate, e.g. $\mathbb{G}(1,3,1)$ with $det = 0$) we would get $\mathbf{G}\mathbb{G} = 0$, so $\mathbf{G}$ and $\mathbb{G}$ have disjoint support — exactly the PGA bulk/weight split, with $A^\star_\bullet = \overline{\mathbf{G}A}$ and $A^\star_\circ = \overline{\mathbb{G}A}$. Pseudoscalar multiplication becomes impossible (no I_n^{-1}), but complement-after-metric can still be used and simply splits in two, as in the case of PGA. The dual is always complement-after-metric; the only question is whether $\mathbf{G}$ and $\mathbb{G}$ are proportional (non-degenerate) or supported on complementary subspaces (degenerate).

Dependency map: The single root for STA is the vector signature $\mathfrak{g}$ (plus the basis and its canonical order). Two independent branches grow from it, and the preceding paragraphs depend on keeping them apart:

- **Combinatorial/signature-direct** (*not* via the extended metric): the wedge product "$\wedge$" (orientation only), the complement $\underline{(\cdot)}/\overline{(\cdot)}$ (metric-blind), the regressive wedge product "$\vee$", and the geometric product "$\barwedge$" (which uses the signature directly via $\boldsymbol{g}_i \barwedge \boldsymbol{g}_i = \mathfrak{g}_{ii}$).
- **Extended-metric branch** ($\mathbf{G}$ = wedge exomorphism, seeded by $\mathfrak{g}$): the extended metric and regressive metric ($\mathbf{G}\mathbb{G} = det\, I_n$), from which follow the dot product/$nrm_sq()$ (the reverse-norm), the left and right duals ($A^\star = \overline{\mathbf{G}A}$), and in turn the contractions, the expansions, and the dual round trip/un-dual.
- **Geometric-square branch** (causal/rotor physics, *not* the metric): the geometric square $B^2 = \langle u \barwedge u \rangle_0 = \sigma(k)\, P$, from which follow causal character ($\mathrm{sign}(B^2)$), the rotor exp / sqrt branch, and the transform $R \barwedge X \barwedge \widetilde{R}$.

What moves when something changes: flipping the signature $(-,-,-,+) \leftrightarrow (+,+,+,-)$ moves both metric and geometric-square branches (but the causal-character labels are signature-parity invariant); adding a null direction (projective STA, $det = 0$) changes only the extended-metric branch (the dual splits into bulk/weight) and leaves the geometric-square branch untouched; changing the complement convention moves the complements, the dual, the regressive wedge product "$\vee$", and the contractions.

1.2.8 Transwedge products

[Lengyel, 2025] has shown that the geometric product can be derived using operations from Grassmann algebra exclusively. These are the wedge prod-

uct, the complement and the duality operations. *This demonstrates that the geometric product is not a fundamental operation in its own right, but can be derived from more fundamental operations defined in Grassmann algebra.*

For that purpose Lengyel introduced the transwedge product of order k by following definition:

$$a \underset{k}{\Lsh} b = \sum_{c \in \mathcal{B}_k} (\underline{c} \vee a) \wedge (b \vee c^\star) \tag{100}$$

where $\mathcal{B}_k$ is the set of all basis elements of order k. For *EGA3D* it means for example $\mathcal{B}_0 = \{1\}$, $\mathcal{B}_1 = \{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$, $\mathcal{B}_2 = \{\boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12}\}$, and $\mathcal{B}_3 = \{\boldsymbol{e}_{123}\}$. From these expressions the geometric product can be formed by

$$a \wedge b = \sum_{k=0}^{n} (-1)^{k(k-1)/2} a \underset{k}{\Lsh} b \tag{101}$$

For $k = 0$ this yields the outer product (wedge product). For higher values of k other products are generated which sum up to the full geometric product. Thus, the transwedge products helps to decompose the geometric product. The most interesting for applications are the wedge product resulting for $k = 0$ and the transwedge product of order $k = 1$.

The regressive wedge product in equation (100) can be replaced by the wedge product using the complement-based definitions given in equations (38) to (42b). That means for example that equation (100) can be transformed to the equivalent expression

$$a \underset{k}{\Lsh} b = \sum_{c \in \mathcal{B}_k} (\underline{c \wedge \overline{a}}) \wedge (\underline{\overline{b} \wedge \overline{c^\star}}) \tag{102}$$

expressing it by using the wedge product exclusively.

Since every product can be assigned a corresponding regressive product in analogy to definition (40a) or (40b), the transwedge product (100) also has a regressive form:

$$a \underset{k}{\Rsh} b \equiv \overline{\underline{a} \underset{k}{\Lsh} \underline{b}} = \sum_{c \in \mathcal{B}_k} \overline{(\underline{c} \vee \underline{a}) \wedge (\underline{b} \vee c^\star)} \tag{103}$$

which in turn can be expressed in terms of the wedge product exclusively as:

$$a \underset{k}{\Rsh} b = \sum_{c \in \mathcal{B}_k} \overline{(\underline{c \wedge a}) \wedge (\underline{b \wedge \overline{c^\star}})} \tag{104}$$

Equations (102) and (104) are used for calculation of the product tables of the (regressive) transwedge products of order k and the resulting (regressive)

geometric products in `ga/ga_prdxpr`.

The transwedge product is very useful to show that the geometric product can be derived from primitives of Grassmann algebra. When (regressive) transwedge product expressions are calculated, it can be shown that each transwedge product and the resulting coefficient expressions are equivalent to specific combinations of wedge, commutator or dot products, and their regressive counterparts. This was done for the (regressive) transwedge product of order $k = 1$ in the `ga/ga_<algebra>_ops_products.hpp` header of every algebra, i.e. for *ega2d*, *ega3d*, *pga2dp*, *pga3dp*, *sta4ds*, *cga2dc* and *cga3dc*.

Equation (101) holds for every metric. The metric comes into the equation via the dual in equation (100). The product tables computed from (102) and (104) reproduce the (regressive) geometric product of each of the seven algebras cell for cell, for the Euclidean signature of EGA, the degenerate signature of PGA, the mixed signature of STA and the non-orthogonal null-pair metric of CGA alike. The null-pair case is the one that shows what the decomposition does: there $\boldsymbol{e}_4 \wedge \boldsymbol{e}_5 = -1 + \boldsymbol{e}_{45}$ is multi-vector-valued, while every summand of (102) is a single blade. The orders k separate the grades, and the multi-term geometric product is their sum.

1.2.9 Complements, duals, and reverses in *EGA*

Left and right complements defined as $\underline{u} \wedge u = u \wedge \overline{u} = \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge ... \wedge \boldsymbol{e}_n = I_n$. Complements are independent of the metric. They just depend on presence or absence of a basis vector in a blade. The complement provides the missing basis vectors required to fill the complete space with the wedge product, i.e. to fill the pseudoscalar I_n of the respective algebra.

Left and right (metric) duals are defined in a similar way, but the complement operation is done *after* the basis vector was multiplied with the metric $\mathbf{G}$ of the space in regular matrix multiplication. The left dual is defined as $(\underline{\mathbf{G}u}) \wedge u = u_\star \wedge u = I_n$ and the right dual is defined as $u \wedge (\overline{\mathbf{G}u}) = u \wedge u^\star = I_n$.

Complements and duals are indistinguishable in *EGA* since the metric is the identity matrix. However, in *PGA* they differ since the metric selects certain basis vectors that are no longer present after the matrix multiplication with the metric (see subsection 1.2.10).

In spaces of odd dimension left and right complements deliver the same results and thus are replaced by the complement or dual operation regardless whether we consider multiplying from the left or right side with the wedge product. Therefore, no left complement and left dual are given in the table for 3*D EGA* below.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_{12} = \mathbb{1}$
$\overline{\boldsymbol{u}}$, $\boldsymbol{u}^\star$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_2$	$-\boldsymbol{e}_1$	$\mathbf{1}$
$\underline{\boldsymbol{u}}$, $\boldsymbol{u}_\star$	$\boldsymbol{e}_{12}$	$-\boldsymbol{e}_2$	$\boldsymbol{e}_1$	$\mathbf{1}$

Table 1: Right and left complements, as well as right and left duals in 2*D EGA* for every basis element.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_{12} = \mathbb{1}$
$\widetilde{\boldsymbol{u}}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$-\boldsymbol{e}_{12}$

Table 2: Reverse in 2*D EGA* for every basis element.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_{23}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{123} = \mathbb{1}$
$\overline{\boldsymbol{u}}$, $\boldsymbol{u}^\star$	$\boldsymbol{e}_{123}$	$\boldsymbol{e}_{23}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\mathbf{1}$

Table 3: Complements and duals in 3*D EGA* for every basis element. Complements and duals are the same in *EGA* since the metric is the identity matrix.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_{23}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{123} = \mathbb{1}$
$\widetilde{\boldsymbol{u}}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$-\boldsymbol{e}_{23}$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{12}$	$-\boldsymbol{e}_{123}$

Table 4: Reverse in 3*D EGA* for every basis element.

1.2.10 Dot products, complements, duals, and reverses in *PGA*

Note: *All* regressive dot products (symbol "$\circ$") and *weight duals of scalar* **1** (symbols "*", "$_\star$") get a pseudoscalar zero for statical type correctness in C++ (also written here as 0, but with actual type `pscalar2dp` ($\mathbb{1} = \boldsymbol{e}_{321}$) or `pscalar3dp` ($\mathbb{1} = \boldsymbol{e}_{1234}$)).

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{32}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{321} = \mathbb{1}$
$\boldsymbol{u} \cdot \boldsymbol{u}$	$\mathbf{1}$	$\mathbf{1}$	$\mathbf{1}$	0	0	0	$\mathbf{1}$	0
$\boldsymbol{u} \circ \boldsymbol{u}$	0	0	0	$\mathbb{1}$	$\mathbb{1}$	$\mathbb{1}$	0	$\mathbb{1}$

Table 5: Dot and regressive dot products in $2D$ *PGA* for every basis element.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{32}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{321} = \mathbb{1}$
$\overline{\boldsymbol{u}}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{32}$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{12}$	$-\boldsymbol{e}_2$	$\boldsymbol{e}_1$	$-\boldsymbol{e}_3$	$\mathbf{1}$

Table 6: Complements in $2D$ *PGA* for every basis element.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{32}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{321} = \mathbb{1}$
$\boldsymbol{u}^\star$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{32}$	$-\boldsymbol{e}_{31}$	0	0	0	$-\boldsymbol{e}_3$	0
$\boldsymbol{u}^*$	0	0	0	$-\boldsymbol{e}_{12}$	$-\boldsymbol{e}_2$	$\boldsymbol{e}_1$	0	$\mathbf{1}$

Table 7: Bulk duals and weight duals for every basis element (=complement applied to bulk and weight).

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{32}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{321} = \mathbb{1}$
$\widetilde{\boldsymbol{u}}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{32}$	$-\boldsymbol{e}_{12}$	$-\boldsymbol{e}_{321}$
$\underset{\sim}{\boldsymbol{u}}$	$\mathbf{-1}$	$-\boldsymbol{e}_1$	$-\boldsymbol{e}_2$	$-\boldsymbol{e}_3$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{32}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{321}$

Table 8: Reverse and regressive reverse in $2D$ *PGA* for every basis element.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_{41}$	$\boldsymbol{e}_{42}$	$\boldsymbol{e}_{43}$	$\boldsymbol{e}_{23}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{423}$	$\boldsymbol{e}_{431}$	$\boldsymbol{e}_{412}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{1234} = \mathbb{1}$
$\boldsymbol{u} \cdot \boldsymbol{u}$	$\mathbf{1}$	$\mathbf{1}$	$\mathbf{1}$	$\mathbf{1}$	0	0	0	0	$\mathbf{1}$	$\mathbf{1}$	$\mathbf{1}$	0	0	0	$\mathbf{1}$	0
$\boldsymbol{u} \circ \boldsymbol{u}$	0	0	0	0	$\mathbb{1}$	$\mathbb{1}$	$\mathbb{1}$	$\mathbb{1}$	0	0	0	$\mathbb{1}$	$\mathbb{1}$	$\mathbb{1}$	0	$\mathbb{1}$

Table 9: Dot and regressive dot products in $3D$ *PGA* for every basis element.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_{41}$	$\boldsymbol{e}_{42}$	$\boldsymbol{e}_{43}$	$\boldsymbol{e}_{23}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{423}$	$\boldsymbol{e}_{431}$	$\boldsymbol{e}_{412}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{1234} = \mathbb{1}$
$\overline{\boldsymbol{u}}$	$\boldsymbol{e}_{1234}$	$\boldsymbol{e}_{423}$	$\boldsymbol{e}_{431}$	$\boldsymbol{e}_{412}$	$\boldsymbol{e}_{321}$	$-\boldsymbol{e}_{23}$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{12}$	$-\boldsymbol{e}_{41}$	$-\boldsymbol{e}_{42}$	$-\boldsymbol{e}_{43}$	$-\boldsymbol{e}_1$	$-\boldsymbol{e}_2$	$-\boldsymbol{e}_3$	$-\boldsymbol{e}_4$	$\mathbf{1}$
$\underline{\boldsymbol{u}}$	$\boldsymbol{e}_{1234}$	$-\boldsymbol{e}_{423}$	$-\boldsymbol{e}_{431}$	$-\boldsymbol{e}_{412}$	$-\boldsymbol{e}_{321}$	$-\boldsymbol{e}_{23}$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{12}$	$-\boldsymbol{e}_{41}$	$-\boldsymbol{e}_{42}$	$-\boldsymbol{e}_{43}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\mathbf{1}$

Table 10: Right and left complements in $3D$ *PGA* for every basis element.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_{41}$	$\boldsymbol{e}_{42}$	$\boldsymbol{e}_{43}$	$\boldsymbol{e}_{23}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{423}$	$\boldsymbol{e}_{431}$	$\boldsymbol{e}_{412}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{1234} = \mathbb{1}$
$\boldsymbol{u}^\star$	$\boldsymbol{e}_{1234}$	$\boldsymbol{e}_{423}$	$\boldsymbol{e}_{431}$	$\boldsymbol{e}_{412}$	0	0	0	0	$-\boldsymbol{e}_{41}$	$-\boldsymbol{e}_{42}$	$-\boldsymbol{e}_{43}$	0	0	0	$-\boldsymbol{e}_4$	0
$\boldsymbol{u}_\star$	$\boldsymbol{e}_{1234}$	$-\boldsymbol{e}_{423}$	$-\boldsymbol{e}_{431}$	$-\boldsymbol{e}_{412}$	0	0	0	0	$-\boldsymbol{e}_{41}$	$-\boldsymbol{e}_{42}$	$-\boldsymbol{e}_{43}$	0	0	0	$\boldsymbol{e}_4$	0
$\boldsymbol{u}^*$	0	0	0	0	$\boldsymbol{e}_{321}$	$-\boldsymbol{e}_{23}$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{12}$	0	0	0	$-\boldsymbol{e}_1$	$-\boldsymbol{e}_2$	$-\boldsymbol{e}_3$	0	$\mathbf{1}$
$\boldsymbol{u}_*$	0	0	0	0	$-\boldsymbol{e}_{321}$	$-\boldsymbol{e}_{23}$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{12}$	0	0	0	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	0	$\mathbf{1}$

Table 11: Right/left bulk duals and right/left weight duals in $3D$ *PGA* for every basis element (=complement applied to bulk and weight).

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_{41}$	$\boldsymbol{e}_{42}$	$\boldsymbol{e}_{43}$	$\boldsymbol{e}_{23}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{423}$	$\boldsymbol{e}_{431}$	$\boldsymbol{e}_{412}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{1234} = \mathbb{1}$
$\widetilde{\boldsymbol{u}}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$-\boldsymbol{e}_{41}$	$-\boldsymbol{e}_{42}$	$-\boldsymbol{e}_{43}$	$-\boldsymbol{e}_{23}$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{12}$	$-\boldsymbol{e}_{423}$	$-\boldsymbol{e}_{431}$	$-\boldsymbol{e}_{412}$	$-\boldsymbol{e}_{321}$	$\boldsymbol{e}_{1234}$
$\underset{\sim}{\boldsymbol{u}}$	$\mathbf{1}$	$-\boldsymbol{e}_1$	$-\boldsymbol{e}_2$	$-\boldsymbol{e}_3$	$-\boldsymbol{e}_4$	$-\boldsymbol{e}_{41}$	$-\boldsymbol{e}_{42}$	$-\boldsymbol{e}_{43}$	$-\boldsymbol{e}_{23}$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{12}$	$\boldsymbol{e}_{423}$	$\boldsymbol{e}_{431}$	$\boldsymbol{e}_{412}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{1234}$

Table 12: Reverse and regressive reverse in $3D$ *PGA* for every basis element.

1.2.11 Solving basic geometric tasks in *PGA*

To demonstrate the benefit of using the *PGA* approach vs. the classical vector algebra approach let us look into very basic recurring geometric tasks that would partially involve solving system of equations in the classical approach. They come as simple product expressions in *PGA*.

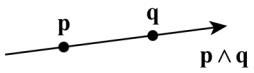
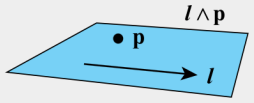
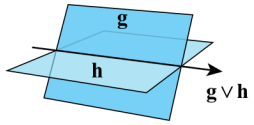
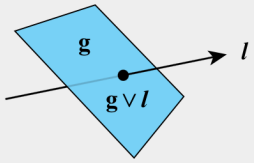
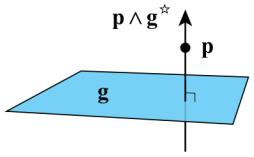
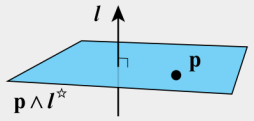
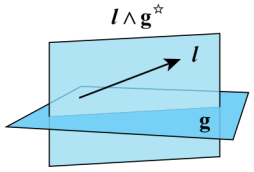
Operation	Illustration
$\mathbf{p} \wedge \mathbf{q}$ Line containing points $\mathbf{p}$ and $\mathbf{q}$.	
$\boldsymbol{l} \wedge \mathbf{p}$ Plane containing line $\boldsymbol{l}$ and point $\mathbf{p}$.	
$\mathbf{g} \vee \mathbf{h}$ Line where planes $\mathbf{g}$ and $\mathbf{h}$ intersect.	
$\mathbf{g} \vee \boldsymbol{l}$ Point where plane $\mathbf{g}$ and line $\boldsymbol{l}$ intersect.	
$\mathbf{p} \wedge \mathbf{g}^{\star}$ Line containing point $\mathbf{p}$ and orthogonal to plane $\mathbf{g}$.	
$\mathbf{p} \wedge \boldsymbol{l}^{\star}$ Plane containing point $\mathbf{p}$ and orthogonal to line $\boldsymbol{l}$.	
$\boldsymbol{l} \wedge \mathbf{g}^{\star}$ Plane containing line $\boldsymbol{l}$ and orthogonal to plane $\mathbf{g}$.	

Figure 7: Basic tasks addressed with *PGA* by using product expressions without need to solve equations. (Source: https://terathon.com/blog/transwedge-product.html, Eric Lengyel, 2025.)

1.2.12 Comparison of approaches in different algebras

Application		**Description**
Create line $\boldsymbol{l}$ from points P, Q	VA	$P = (p_x, p_y, p_z)$ $\boldsymbol{l}: \vec{x}(t) = (Q - P) \cdot t + P$ (explicit), $ax + by = c$ (implict)
	EGA	$P = (p_x, p_y, p_z)$, a vector represents a $1D$ subspace $\boldsymbol{l}: x(t) = P \cdot t$ (explicit, line through origin and P) $\boldsymbol{l}: x(t) = (Q - P) \cdot t + P$ (explicit, line in direction $Q - P$ through P) $\boldsymbol{l}: (x - P) \wedge (Q - P) = 0$ (implicit, line in direction $Q - P$ through P)
	PGA	$P = (p_x, p_y, p_z, 1)$ $\boldsymbol{l} = P \wedge Q$ (line through points P and Q, bivector)
Create plane $\boldsymbol{T}$ from points P, Q, R	VA	$\vec{u} = Q - P$, $\vec{v} = R - P$ $\boldsymbol{T}: x(s,t) = \vec{u} \cdot s + \vec{v} \cdot t + P$
	EGA	$u = Q - P$, $v = R - P$, $B = u \wedge v$ $\boldsymbol{T} = P \wedge Q$ (plane through O, P, Q; bivector) $\boldsymbol{T}: (x - P) \wedge B = 0$ (plane through P, Q, R; bivector)
	PGA	$\boldsymbol{T} = P \wedge Q \wedge R$ (plane through P, Q, R, trivector)
Intersect two lines $\boldsymbol{l_1}$, $\boldsymbol{l_2}$	VA	$\boldsymbol{l_1}: \vec{x}(s) = \vec{u} \cdot s + P$, $\boldsymbol{l_2}: \vec{y}(t) = \vec{v} \cdot t + Q$, solve system in $2D$ for s_i and t_i: $\vec{u} \cdot s + P = \vec{v} \cdot t + Q$ intersection point $I = \vec{x}(s_i) = \vec{y}(t_i)$
	EGA	see VA, # of solutions: 1, none or infinitely many
	PGA	Intersect arbitrary lines l_1, l_2 in a common plane intersection point I: $I = l_1 \vee l_2$ (projective point)
Intersect two planes $\boldsymbol{T_1}$, $\boldsymbol{T_2}$	VA	$x(s,t) = \vec{u} \cdot s + \vec{v} \cdot t + P$, $y(m,n) = \vec{m} \cdot p + \vec{n} \cdot q + Q$ solve system in $3D$: $\vec{u} \cdot s + \vec{v} \cdot t + P = \vec{m} \cdot p + \vec{n} \cdot q + Q$
	EGA	see VA
	PGA	Intersect arbitrary planes T_1, T_2 intersection line I: $I = T_1 \vee T_2$ (projective line, bivector)

Table 13: Comparison of common geometric construction tasks across three approaches. **VA** = Vector Algebra (classical approach using vectors and parametric equations); **EGA** = Euclidean Geometric Algebra; **PGA** = Projective (Euclidean) Geometric Algebra.

PGA can define geometric primitives like lines or planes directly from points. It does not require solving systems of equations for simple geometrical tasks, like the intersection of geometric objects. They can be solved easily by using a suitable product expression. Another key advantage is that the intersections, e.g. of two lines or a line and a plane always have a solution in projective geometric algebra. There is no need to distinguish different cases, as required in *VA* or *EGA*, where one might get either none, one or infinitely many solutions depending on the specific geometric case. The intersection point might be at infinity, but in projective algebra it is always well-defined.

1.2.13 Dot products, complements, duals, and reverses in *STA*

The tables below collect the per-basis-element values for the space-time algebra $STA4DS = \mathbb{G}(1,3,0)$. The Minkowski signature is $(\boldsymbol{g}_1^2, \boldsymbol{g}_2^2, \boldsymbol{g}_3^2, \boldsymbol{g}_4^2) = (-1,-1,-1,+1)$. They are the space-time analogue of the EGA and PGA tables above; the underlying concepts (extended metric as wedge exomorphism, the reverse-norm P versus the geometric square B^2, left/right complements and duals) were developed in subsection 1.2.7. $STA4DS$ is four-dimensional (even), so left and right complements/duals are distinguished; it is non-degenerate ($det(\mathfrak{g}) = -1$), so — unlike PGA — there is *no* bulk/weight split, and every basis blade squares to ±1.

Table 14 lists two distinct "squares" side by side: the dot product $\boldsymbol{u} \cdot \boldsymbol{u} = \mathbf{G}(\boldsymbol{u}) = P$ (the extended-metric reverse-norm, equal to $nrm_sq(\boldsymbol{u})$), and the geometric square $\boldsymbol{u} \wedge \boldsymbol{u} = B^2 = \sigma(k)\, P$. They agree at grades $0, 1, 4$ and differ by sign at grades $2, 3$ — the distinction that drives causal character and the rotor exponential (see eqn. (93)).

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{g}_1$	$\boldsymbol{g}_2$	$\boldsymbol{g}_3$	$\boldsymbol{g}_4$	$\boldsymbol{g}_{14}$	$\boldsymbol{g}_{24}$	$\boldsymbol{g}_{34}$	$\boldsymbol{g}_{23}$	$\boldsymbol{g}_{31}$	$\boldsymbol{g}_{12}$	$\boldsymbol{g}_{234}$	$\boldsymbol{g}_{314}$	$\boldsymbol{g}_{124}$	$\boldsymbol{g}_{123}$	$\boldsymbol{g}_{1234} = \mathbb{1}$
$\boldsymbol{u} \cdot \boldsymbol{u}$	1	−1	−1	−1	1	−1	−1	−1	1	1	1	1	1	1	−1	−1
$\boldsymbol{u} \wedge \boldsymbol{u}$	1	−1	−1	−1	1	1	1	1	−1	−1	−1	−1	−1	−1	1	−1

Table 14: Reverse-norm $\boldsymbol{u} \cdot \boldsymbol{u} = P = nrm_sq(\boldsymbol{u})$ (extended metric) and geometric square $\boldsymbol{u} \wedge \boldsymbol{u} = B^2 = \sigma(k)\, P$ in $STA4DS$ for every basis element. They differ by sign at grades 2 and 3.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{g}_1$	$\boldsymbol{g}_2$	$\boldsymbol{g}_3$	$\boldsymbol{g}_4$	$\boldsymbol{g}_{14}$	$\boldsymbol{g}_{24}$	$\boldsymbol{g}_{34}$	$\boldsymbol{g}_{23}$	$\boldsymbol{g}_{31}$	$\boldsymbol{g}_{12}$	$\boldsymbol{g}_{234}$	$\boldsymbol{g}_{314}$	$\boldsymbol{g}_{124}$	$\boldsymbol{g}_{123}$	$\boldsymbol{g}_{1234} = \mathbb{1}$
$\overline{\boldsymbol{u}}$	$\boldsymbol{g}_{1234}$	$\boldsymbol{g}_{234}$	$\boldsymbol{g}_{314}$	$\boldsymbol{g}_{124}$	$-\boldsymbol{g}_{123}$	$\boldsymbol{g}_{23}$	$\boldsymbol{g}_{31}$	$\boldsymbol{g}_{12}$	$\boldsymbol{g}_{14}$	$\boldsymbol{g}_{24}$	$\boldsymbol{g}_{34}$	$-\boldsymbol{g}_1$	$-\boldsymbol{g}_2$	$-\boldsymbol{g}_3$	$\boldsymbol{g}_4$	1
$\underline{\boldsymbol{u}}$	$\boldsymbol{g}_{1234}$	$-\boldsymbol{g}_{234}$	$-\boldsymbol{g}_{314}$	$-\boldsymbol{g}_{124}$	$\boldsymbol{g}_{123}$	$\boldsymbol{g}_{23}$	$\boldsymbol{g}_{31}$	$\boldsymbol{g}_{12}$	$\boldsymbol{g}_{14}$	$\boldsymbol{g}_{24}$	$\boldsymbol{g}_{34}$	$\boldsymbol{g}_1$	$\boldsymbol{g}_2$	$\boldsymbol{g}_3$	$-\boldsymbol{g}_4$	1

Table 15: Right and left complements in $STA4DS$ for every basis element. Complements are independent of the metric.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{g}_1$	$\boldsymbol{g}_2$	$\boldsymbol{g}_3$	$\boldsymbol{g}_4$	$\boldsymbol{g}_{14}$	$\boldsymbol{g}_{24}$	$\boldsymbol{g}_{34}$	$\boldsymbol{g}_{23}$	$\boldsymbol{g}_{31}$	$\boldsymbol{g}_{12}$	$\boldsymbol{g}_{234}$	$\boldsymbol{g}_{314}$	$\boldsymbol{g}_{124}$	$\boldsymbol{g}_{123}$	$\boldsymbol{g}_{1234} = \mathbb{1}$
$\boldsymbol{u}^\star$	$\boldsymbol{g}_{1234}$	$-\boldsymbol{g}_{234}$	$-\boldsymbol{g}_{314}$	$-\boldsymbol{g}_{124}$	$-\boldsymbol{g}_{123}$	$-\boldsymbol{g}_{23}$	$-\boldsymbol{g}_{31}$	$-\boldsymbol{g}_{12}$	$\boldsymbol{g}_{14}$	$\boldsymbol{g}_{24}$	$\boldsymbol{g}_{34}$	$-\boldsymbol{g}_1$	$-\boldsymbol{g}_2$	$-\boldsymbol{g}_3$	$-\boldsymbol{g}_4$	−1
$\boldsymbol{u}_\star$	$\boldsymbol{g}_{1234}$	$\boldsymbol{g}_{234}$	$\boldsymbol{g}_{314}$	$\boldsymbol{g}_{124}$	$\boldsymbol{g}_{123}$	$-\boldsymbol{g}_{23}$	$-\boldsymbol{g}_{31}$	$-\boldsymbol{g}_{12}$	$\boldsymbol{g}_{14}$	$\boldsymbol{g}_{24}$	$\boldsymbol{g}_{34}$	$\boldsymbol{g}_1$	$\boldsymbol{g}_2$	$\boldsymbol{g}_3$	$\boldsymbol{g}_4$	−1

Table 16: Right and left (metric) duals in $STA4DS$ for every basis element ($\boldsymbol{u}^\star = \overline{\mathbf{G}\boldsymbol{u}}$, $\boldsymbol{u}_\star = \underline{\mathbf{G}\boldsymbol{u}}$). Unlike in EGA they differ from the complements, since the Minkowski metric flips signs.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{g}_1$	$\boldsymbol{g}_2$	$\boldsymbol{g}_3$	$\boldsymbol{g}_4$	$\boldsymbol{g}_{14}$	$\boldsymbol{g}_{24}$	$\boldsymbol{g}_{34}$	$\boldsymbol{g}_{23}$	$\boldsymbol{g}_{31}$	$\boldsymbol{g}_{12}$	$\boldsymbol{g}_{234}$	$\boldsymbol{g}_{314}$	$\boldsymbol{g}_{124}$	$\boldsymbol{g}_{123}$	$\boldsymbol{g}_{1234} = \mathbb{1}$
$\widetilde{\boldsymbol{u}}$	1	$\boldsymbol{g}_1$	$\boldsymbol{g}_2$	$\boldsymbol{g}_3$	$\boldsymbol{g}_4$	$-\boldsymbol{g}_{14}$	$-\boldsymbol{g}_{24}$	$-\boldsymbol{g}_{34}$	$-\boldsymbol{g}_{23}$	$-\boldsymbol{g}_{31}$	$-\boldsymbol{g}_{12}$	$-\boldsymbol{g}_{234}$	$-\boldsymbol{g}_{314}$	$-\boldsymbol{g}_{124}$	$-\boldsymbol{g}_{123}$	$\boldsymbol{g}_{1234}$

Table 17: Reverse in $STA4DS$ for every basis element. The reverse sign is the grade-only factor $\sigma(k) = (-1)^{k(k-1)/2}$, independent of the metric.

1.2.14 Dot products, complements, duals, and reverses in *CGA*

The tables below collect the per-basis-element values for the conformal algebras $CGA2DC = \mathbb{G}(3,1,0)$ and $CGA3DC = \mathbb{G}(4,1,0)$. Both are stored in the null basis of subsection 1.2.1, where the two added basis vectors square to zero and are not orthogonal to each other. The signature refers to the diagonal basis $\boldsymbol{e}_+$, $\boldsymbol{e}_-$ with $\boldsymbol{e}_+^2 = +1$ and $\boldsymbol{e}_-^2 = -1$, which is linked to the stored one by

$$\boldsymbol{e}_4 = \frac{1}{2}(\boldsymbol{e}_- - \boldsymbol{e}_+), \qquad \boldsymbol{e}_5 = \boldsymbol{e}_- + \boldsymbol{e}_+ \tag{105a}$$

$$\boldsymbol{e}_- = \boldsymbol{e}_4 + \frac{1}{2}\boldsymbol{e}_5, \qquad \boldsymbol{e}_+ = -\boldsymbol{e}_4 + \frac{1}{2}\boldsymbol{e}_5 \tag{105b}$$

written for $CGA3DC$; in $CGA2DC$ read $\boldsymbol{e}_3$ and $\boldsymbol{e}_4$ in place of $\boldsymbol{e}_4$ and $\boldsymbol{e}_5$. All coefficients are dyadic, so the change of basis is exact in binary floating point. The determinant of the metric is -1. Both algebras are therefore non-degenerate, and the bulk and weight duals of PGA collapse into the single dual already used in EGA and STA.[7]

The extended metric is the wedge exomorphism (44) seeded on the grade-1 metric, as in every other algebra. Here the grade-1 metric is a signed permutation, and so is the extended metric: every basis blade is mapped to exactly one signed partner blade, given in the first row of tables 18 and 22. A blade whose partner is a different blade therefore has $\boldsymbol{u} \cdot \boldsymbol{u} = 0$, and its metric content sits off the diagonal, e.g. $\boldsymbol{e}_3 \cdot \boldsymbol{e}_4 = -1$ in $CGA2DC$. Where PGA annihilates a blade containing the projective basis vector, CGA pairs it with the blade containing the round one, which is why the norms generalize instead of splitting.

Complements do not depend on the metric. $CGA2DC$ is four-dimensional, so left and right complements differ; $CGA3DC$ is five-dimensional, where the two coincide and the single complement is an involution. This is the same distinction by even and odd dimension that separates $EGA2D$, $PGA3DP$ and $STA4DS$ from $EGA3D$ and $PGA2DP$. The dual is the complement taken after the metric, and the antidual the same construction with the regressive extended metric $\mathbb{G}$. In both conformal algebras $\mathbb{G} = -\mathbf{G}$ holds, so

$$\boldsymbol{u}^\star = \overline{\mathbb{G}\boldsymbol{u}} = -\overline{\mathbf{G}\boldsymbol{u}} = -\boldsymbol{u}^\star \tag{106}$$

at every grade. The unfilled star denotes the antidual here; the weight duals it marks in PGA do not exist in a non-degenerate algebra. The antidual

[7] The four-part bulk/weight split of the conformal objects is a different decomposition and is treated in the chapter on conformal geometry.

is the form that appears in the defining equations of the conformal properties.

The reverse sign is the grade-only factor $\sigma(k) = (-1)^{k(k-1)/2}$, the regressive reverse the same factor taken on the complementary grade, $\sigma(n-k)$. In the five-dimensional $CGA3DC$ the two coincide at every grade, so $\widetilde{\boldsymbol{u}} = \underset{\sim}{\boldsymbol{u}}$ there.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{32}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{14}$	$\boldsymbol{e}_{24}$	$\boldsymbol{e}_{34}$	$\boldsymbol{e}_{314}$	$\boldsymbol{e}_{324}$	$\boldsymbol{e}_{124}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{1234} = \mathbb{1}$
$\mathbf{G}\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$-\boldsymbol{e}_4$	$-\boldsymbol{e}_3$	$\boldsymbol{e}_{14}$	$\boldsymbol{e}_{24}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{32}$	$-\boldsymbol{e}_{34}$	$-\boldsymbol{e}_{314}$	$-\boldsymbol{e}_{324}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{124}$	$-\boldsymbol{e}_{1234}$
$\boldsymbol{u}\cdot\boldsymbol{u}$	$\mathbf{1}$	$\mathbf{1}$	$\mathbf{1}$	0	0	0	0	$\mathbf{1}$	0	0	$\mathbf{-1}$	$\mathbf{-1}$	$\mathbf{-1}$	0	0	$\mathbf{-1}$
$\boldsymbol{u}\circ\boldsymbol{u}$	$-\mathbb{1}$	$-\mathbb{1}$	$-\mathbb{1}$	0	0	0	0	$-\mathbb{1}$	0	0	$\mathbb{1}$	$\mathbb{1}$	$\mathbb{1}$	0	0	$\mathbb{1}$

Table 18: Extended metric partner $\mathbf{G}\boldsymbol{u}$, dot and regressive dot products in $2D$ CGA for every basis element. A blade whose partner is a different blade has $\boldsymbol{u}\cdot\boldsymbol{u} = 0$.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{32}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{14}$	$\boldsymbol{e}_{24}$	$\boldsymbol{e}_{34}$	$\boldsymbol{e}_{314}$	$\boldsymbol{e}_{324}$	$\boldsymbol{e}_{124}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{1234} = \mathbb{1}$
$\overline{\boldsymbol{u}}$	$\boldsymbol{e}_{1234}$	$-\boldsymbol{e}_{324}$	$\boldsymbol{e}_{314}$	$\boldsymbol{e}_{124}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{24}$	$-\boldsymbol{e}_{14}$	$\boldsymbol{e}_{34}$	$-\boldsymbol{e}_{32}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$-\boldsymbol{e}_{2}$	$\boldsymbol{e}_{1}$	$-\boldsymbol{e}_{3}$	$-\boldsymbol{e}_{4}$	$\mathbf{1}$
$\underline{\boldsymbol{u}}$	$\boldsymbol{e}_{1234}$	$\boldsymbol{e}_{324}$	$-\boldsymbol{e}_{314}$	$-\boldsymbol{e}_{124}$	$-\boldsymbol{e}_{321}$	$\boldsymbol{e}_{24}$	$-\boldsymbol{e}_{14}$	$\boldsymbol{e}_{34}$	$-\boldsymbol{e}_{32}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{2}$	$-\boldsymbol{e}_{1}$	$\boldsymbol{e}_{3}$	$\boldsymbol{e}_{4}$	$\mathbf{1}$

Table 19: Right and left complements in $2D$ CGA for every basis element. Complements are independent of the metric.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{32}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{14}$	$\boldsymbol{e}_{24}$	$\boldsymbol{e}_{34}$	$\boldsymbol{e}_{314}$	$\boldsymbol{e}_{324}$	$\boldsymbol{e}_{124}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{1234} = \mathbb{1}$
$\boldsymbol{u}^\star$	$\boldsymbol{e}_{1234}$	$-\boldsymbol{e}_{324}$	$\boldsymbol{e}_{314}$	$-\boldsymbol{e}_{321}$	$-\boldsymbol{e}_{124}$	$-\boldsymbol{e}_{32}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{34}$	$\boldsymbol{e}_{24}$	$-\boldsymbol{e}_{14}$	$-\boldsymbol{e}_{12}$	$\boldsymbol{e}_{2}$	$-\boldsymbol{e}_{1}$	$-\boldsymbol{e}_{4}$	$-\boldsymbol{e}_{3}$	$\mathbf{-1}$
$\boldsymbol{u}_\star$	$\boldsymbol{e}_{1234}$	$\boldsymbol{e}_{324}$	$-\boldsymbol{e}_{314}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{124}$	$-\boldsymbol{e}_{32}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{34}$	$\boldsymbol{e}_{24}$	$-\boldsymbol{e}_{14}$	$-\boldsymbol{e}_{12}$	$-\boldsymbol{e}_{2}$	$\boldsymbol{e}_{1}$	$\boldsymbol{e}_{4}$	$\boldsymbol{e}_{3}$	$\mathbf{-1}$
$\boldsymbol{u}^\star$	$-\boldsymbol{e}_{1234}$	$\boldsymbol{e}_{324}$	$-\boldsymbol{e}_{314}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{124}$	$\boldsymbol{e}_{32}$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{34}$	$-\boldsymbol{e}_{24}$	$\boldsymbol{e}_{14}$	$\boldsymbol{e}_{12}$	$-\boldsymbol{e}_{2}$	$\boldsymbol{e}_{1}$	$\boldsymbol{e}_{4}$	$\boldsymbol{e}_{3}$	$\mathbf{1}$
$\boldsymbol{u}_\star$	$-\boldsymbol{e}_{1234}$	$-\boldsymbol{e}_{324}$	$\boldsymbol{e}_{314}$	$-\boldsymbol{e}_{321}$	$-\boldsymbol{e}_{124}$	$\boldsymbol{e}_{32}$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{34}$	$-\boldsymbol{e}_{24}$	$\boldsymbol{e}_{14}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{2}$	$-\boldsymbol{e}_{1}$	$-\boldsymbol{e}_{4}$	$-\boldsymbol{e}_{3}$	$\mathbf{1}$

Table 20: Right and left duals ($\boldsymbol{u}^\star = \overline{\mathbf{G}\boldsymbol{u}}$) and antiduals ($\boldsymbol{u}^\star = \overline{\mathbb{G}\boldsymbol{u}}$) in $2D$ CGA for every basis element. There is no bulk/weight split, and the antidual is the negation of the dual.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{32}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{14}$	$\boldsymbol{e}_{24}$	$\boldsymbol{e}_{34}$	$\boldsymbol{e}_{314}$	$\boldsymbol{e}_{324}$	$\boldsymbol{e}_{124}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{1234} = \mathbb{1}$
$\widetilde{\boldsymbol{u}}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{32}$	$-\boldsymbol{e}_{12}$	$-\boldsymbol{e}_{14}$	$-\boldsymbol{e}_{24}$	$-\boldsymbol{e}_{34}$	$-\boldsymbol{e}_{314}$	$-\boldsymbol{e}_{324}$	$-\boldsymbol{e}_{124}$	$-\boldsymbol{e}_{321}$	$\boldsymbol{e}_{1234}$
$\underset{\sim}{\boldsymbol{u}}$	$\mathbf{1}$	$-\boldsymbol{e}_1$	$-\boldsymbol{e}_2$	$-\boldsymbol{e}_3$	$-\boldsymbol{e}_4$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{32}$	$-\boldsymbol{e}_{12}$	$-\boldsymbol{e}_{14}$	$-\boldsymbol{e}_{24}$	$-\boldsymbol{e}_{34}$	$\boldsymbol{e}_{314}$	$\boldsymbol{e}_{324}$	$\boldsymbol{e}_{124}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{1234}$

Table 21: Reverse and regressive reverse in $2D$ CGA for every basis element. They differ at the grades 1 and 3.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_5$	$\boldsymbol{e}_{41}$	$\boldsymbol{e}_{42}$	$\boldsymbol{e}_{43}$	$\boldsymbol{e}_{23}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{15}$	$\boldsymbol{e}_{25}$	$\boldsymbol{e}_{35}$	$\boldsymbol{e}_{45}$
$\mathbf{G}\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$-\boldsymbol{e}_5$	$-\boldsymbol{e}_4$	$\boldsymbol{e}_{15}$	$\boldsymbol{e}_{25}$	$\boldsymbol{e}_{35}$	$\boldsymbol{e}_{23}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{41}$	$\boldsymbol{e}_{42}$	$\boldsymbol{e}_{43}$	$-\boldsymbol{e}_{45}$
$\boldsymbol{u}\cdot\boldsymbol{u}$	$\mathbf{1}$	$\mathbf{1}$	$\mathbf{1}$	$\mathbf{1}$	0	0	0	0	0	$\mathbf{1}$	$\mathbf{1}$	$\mathbf{1}$	0	0	0	$\mathbf{-1}$
$\boldsymbol{u}\circ\boldsymbol{u}$	$-\mathbb{1}$	$-\mathbb{1}$	$-\mathbb{1}$	$-\mathbb{1}$	0	0	0	0	0	$-\mathbb{1}$	$-\mathbb{1}$	$-\mathbb{1}$	0	0	0	$\mathbb{1}$

$\boldsymbol{u}$	$\boldsymbol{e}_{415}$	$\boldsymbol{e}_{425}$	$\boldsymbol{e}_{435}$	$\boldsymbol{e}_{235}$	$\boldsymbol{e}_{315}$	$\boldsymbol{e}_{125}$	$\boldsymbol{e}_{423}$	$\boldsymbol{e}_{431}$	$\boldsymbol{e}_{412}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{4235}$	$\boldsymbol{e}_{4315}$	$\boldsymbol{e}_{4125}$	$\boldsymbol{e}_{3215}$	$\boldsymbol{e}_{1234}$	$\boldsymbol{e}_{12345}$
$\mathbf{G}\boldsymbol{u}$	$-\boldsymbol{e}_{415}$	$-\boldsymbol{e}_{425}$	$-\boldsymbol{e}_{435}$	$-\boldsymbol{e}_{423}$	$-\boldsymbol{e}_{431}$	$-\boldsymbol{e}_{412}$	$-\boldsymbol{e}_{235}$	$-\boldsymbol{e}_{315}$	$-\boldsymbol{e}_{125}$	$\boldsymbol{e}_{321}$	$-\boldsymbol{e}_{4235}$	$-\boldsymbol{e}_{4315}$	$-\boldsymbol{e}_{4125}$	$\boldsymbol{e}_{1234}$	$\boldsymbol{e}_{3215}$	$-\mathbb{1}$
$\boldsymbol{u}\cdot\boldsymbol{u}$	$\mathbf{-1}$	$\mathbf{-1}$	$\mathbf{-1}$	0	0	0	0	0	0	$\mathbf{1}$	$\mathbf{-1}$	$\mathbf{-1}$	$\mathbf{-1}$	0	0	$\mathbf{-1}$
$\boldsymbol{u}\circ\boldsymbol{u}$	$\mathbb{1}$	$\mathbb{1}$	$\mathbb{1}$	0	0	0	0	0	0	$-\mathbb{1}$	$\mathbb{1}$	$\mathbb{1}$	$\mathbb{1}$	0	0	$\mathbb{1}$

Table 22: Extended metric partner $\mathbf{G}\boldsymbol{u}$, dot and regressive dot products in $3D$ CGA for every basis element, in two halves by grade ($\boldsymbol{e}_{12345} = \mathbb{1}$ is the pseudoscalar).

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_5$	$\boldsymbol{e}_{41}$	$\boldsymbol{e}_{42}$	$\boldsymbol{e}_{43}$	$\boldsymbol{e}_{23}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{15}$	$\boldsymbol{e}_{25}$	$\boldsymbol{e}_{35}$	$\boldsymbol{e}_{45}$
$\overline{\boldsymbol{u}}$	$\mathbb{1}$	$\boldsymbol{e}_{4235}$	$\boldsymbol{e}_{4315}$	$\boldsymbol{e}_{4125}$	$\boldsymbol{e}_{3215}$	$\boldsymbol{e}_{1234}$	$-\boldsymbol{e}_{235}$	$-\boldsymbol{e}_{315}$	$-\boldsymbol{e}_{125}$	$-\boldsymbol{e}_{415}$	$-\boldsymbol{e}_{425}$	$-\boldsymbol{e}_{435}$	$-\boldsymbol{e}_{423}$	$-\boldsymbol{e}_{431}$	$-\boldsymbol{e}_{412}$	$-\boldsymbol{e}_{321}$

$\boldsymbol{u}$	$\boldsymbol{e}_{415}$	$\boldsymbol{e}_{425}$	$\boldsymbol{e}_{435}$	$\boldsymbol{e}_{235}$	$\boldsymbol{e}_{315}$	$\boldsymbol{e}_{125}$	$\boldsymbol{e}_{423}$	$\boldsymbol{e}_{431}$	$\boldsymbol{e}_{412}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{4235}$	$\boldsymbol{e}_{4315}$	$\boldsymbol{e}_{4125}$	$\boldsymbol{e}_{3215}$	$\boldsymbol{e}_{1234}$	$\boldsymbol{e}_{12345}$
$\overline{\boldsymbol{u}}$	$-\boldsymbol{e}_{23}$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{12}$	$-\boldsymbol{e}_{41}$	$-\boldsymbol{e}_{42}$	$-\boldsymbol{e}_{43}$	$-\boldsymbol{e}_{15}$	$-\boldsymbol{e}_{25}$	$-\boldsymbol{e}_{35}$	$-\boldsymbol{e}_{45}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_5$	$\mathbf{1}$

Table 23: Complement in $3D$ *CGA* for every basis element, in two halves by grade ($\boldsymbol{e}_{12345} = \mathbb{1}$). The algebra is five-dimensional, so left and right complement coincide.

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_5$	$\boldsymbol{e}_{41}$	$\boldsymbol{e}_{42}$	$\boldsymbol{e}_{43}$	$\boldsymbol{e}_{23}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{15}$	$\boldsymbol{e}_{25}$	$\boldsymbol{e}_{35}$	$\boldsymbol{e}_{45}$
$\boldsymbol{u}^\star$	$\mathbb{1}$	$\boldsymbol{e}_{4235}$	$\boldsymbol{e}_{4315}$	$\boldsymbol{e}_{4125}$	$-\boldsymbol{e}_{1234}$	$-\boldsymbol{e}_{3215}$	$-\boldsymbol{e}_{423}$	$-\boldsymbol{e}_{431}$	$-\boldsymbol{e}_{412}$	$-\boldsymbol{e}_{415}$	$-\boldsymbol{e}_{425}$	$-\boldsymbol{e}_{435}$	$-\boldsymbol{e}_{235}$	$-\boldsymbol{e}_{315}$	$-\boldsymbol{e}_{125}$	$\boldsymbol{e}_{321}$
$\boldsymbol{u}^\star$	$-\mathbb{1}$	$-\boldsymbol{e}_{4235}$	$-\boldsymbol{e}_{4315}$	$-\boldsymbol{e}_{4125}$	$\boldsymbol{e}_{1234}$	$\boldsymbol{e}_{3215}$	$\boldsymbol{e}_{423}$	$\boldsymbol{e}_{431}$	$\boldsymbol{e}_{412}$	$\boldsymbol{e}_{415}$	$\boldsymbol{e}_{425}$	$\boldsymbol{e}_{435}$	$\boldsymbol{e}_{235}$	$\boldsymbol{e}_{315}$	$\boldsymbol{e}_{125}$	$-\boldsymbol{e}_{321}$

$\boldsymbol{u}$	$\boldsymbol{e}_{415}$	$\boldsymbol{e}_{425}$	$\boldsymbol{e}_{435}$	$\boldsymbol{e}_{235}$	$\boldsymbol{e}_{315}$	$\boldsymbol{e}_{125}$	$\boldsymbol{e}_{423}$	$\boldsymbol{e}_{431}$	$\boldsymbol{e}_{412}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{4235}$	$\boldsymbol{e}_{4315}$	$\boldsymbol{e}_{4125}$	$\boldsymbol{e}_{3215}$	$\boldsymbol{e}_{1234}$	$\boldsymbol{e}_{12345}$
$\boldsymbol{u}^\star$	$\boldsymbol{e}_{23}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{15}$	$\boldsymbol{e}_{25}$	$\boldsymbol{e}_{35}$	$\boldsymbol{e}_{41}$	$\boldsymbol{e}_{42}$	$\boldsymbol{e}_{43}$	$-\boldsymbol{e}_{45}$	$-\boldsymbol{e}_1$	$-\boldsymbol{e}_2$	$-\boldsymbol{e}_3$	$\boldsymbol{e}_5$	$\boldsymbol{e}_4$	$-\mathbf{1}$
$\boldsymbol{u}^\star$	$-\boldsymbol{e}_{23}$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{12}$	$-\boldsymbol{e}_{15}$	$-\boldsymbol{e}_{25}$	$-\boldsymbol{e}_{35}$	$-\boldsymbol{e}_{41}$	$-\boldsymbol{e}_{42}$	$-\boldsymbol{e}_{43}$	$\boldsymbol{e}_{45}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$-\boldsymbol{e}_5$	$-\boldsymbol{e}_4$	$\mathbf{1}$

Table 24: Dual ($\boldsymbol{u}^\star = \overline{\mathbf{G}\boldsymbol{u}}$) and antidual ($\boldsymbol{u}^\star = \overline{\mathbb{G}\boldsymbol{u}}$) in $3D$ *CGA* for every basis element, in two halves by grade ($\boldsymbol{e}_{12345} = \mathbb{1}$).

$\boldsymbol{u}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_5$	$\boldsymbol{e}_{41}$	$\boldsymbol{e}_{42}$	$\boldsymbol{e}_{43}$	$\boldsymbol{e}_{23}$	$\boldsymbol{e}_{31}$	$\boldsymbol{e}_{12}$	$\boldsymbol{e}_{15}$	$\boldsymbol{e}_{25}$	$\boldsymbol{e}_{35}$	$\boldsymbol{e}_{45}$
$\widetilde{\boldsymbol{u}} = \underset{\sim}{\boldsymbol{u}}$	$\mathbf{1}$	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_5$	$-\boldsymbol{e}_{41}$	$-\boldsymbol{e}_{42}$	$-\boldsymbol{e}_{43}$	$-\boldsymbol{e}_{23}$	$-\boldsymbol{e}_{31}$	$-\boldsymbol{e}_{12}$	$-\boldsymbol{e}_{15}$	$-\boldsymbol{e}_{25}$	$-\boldsymbol{e}_{35}$	$-\boldsymbol{e}_{45}$

$\boldsymbol{u}$	$\boldsymbol{e}_{415}$	$\boldsymbol{e}_{425}$	$\boldsymbol{e}_{435}$	$\boldsymbol{e}_{235}$	$\boldsymbol{e}_{315}$	$\boldsymbol{e}_{125}$	$\boldsymbol{e}_{423}$	$\boldsymbol{e}_{431}$	$\boldsymbol{e}_{412}$	$\boldsymbol{e}_{321}$	$\boldsymbol{e}_{4235}$	$\boldsymbol{e}_{4315}$	$\boldsymbol{e}_{4125}$	$\boldsymbol{e}_{3215}$	$\boldsymbol{e}_{1234}$	$\boldsymbol{e}_{12345}$
$\widetilde{\boldsymbol{u}} = \underset{\sim}{\boldsymbol{u}}$	$-\boldsymbol{e}_{415}$	$-\boldsymbol{e}_{425}$	$-\boldsymbol{e}_{435}$	$-\boldsymbol{e}_{235}$	$-\boldsymbol{e}_{315}$	$-\boldsymbol{e}_{125}$	$-\boldsymbol{e}_{423}$	$-\boldsymbol{e}_{431}$	$-\boldsymbol{e}_{412}$	$-\boldsymbol{e}_{321}$	$\boldsymbol{e}_{4235}$	$\boldsymbol{e}_{4315}$	$\boldsymbol{e}_{4125}$	$\boldsymbol{e}_{3215}$	$\boldsymbol{e}_{1234}$	$\mathbb{1}$

Table 25: Reverse in $3D$ *CGA* for every basis element, in two halves by grade ($\boldsymbol{e}_{12345} = \mathbb{1}$). In five dimensions the regressive reverse is the same operation.

1.3 Differential calculus

The algebra as used so far is pointwise: products of multivectors at a single location. A *field* assigns a multivector to every point of space, and differentiating one asks for a derivative that is itself an element of the algebra. That operator is the *vector derivative*

$$\nabla = \sum_i \boldsymbol{e}_i \, \partial_i \,, \tag{107}$$

with $\partial_i = \partial/\partial x_i$ and the coordinates numbered in the order of the basis vectors, $(x_1, x_2, x_3) = (x, y, z)$ for $(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3)$. It is a vector whose components are the partial derivatives, acting on a field through the geometric product. Because it multiplies rather than scales, applying it to a k-vector field produces parts of grade $k-1$ and $k+1$ at once.

The juxtaposition in eqn. (107) scales each basis vector by a scalar differential operator, so no product between multivectors has occurred yet. It does occur as soon as ∇ is applied to a field, and it is the *geometric* product,

$$\nabla \overset{\wedge}{} A = \Big(\sum_i \boldsymbol{e}_i \, \partial_i\Big) \overset{\wedge}{} A = \sum_i \boldsymbol{e}_i \overset{\wedge}{} (\partial_i A) \;, \tag{108}$$

The subscript names the product the operator applies, as $\exp_{\wedge}$ and $\exp_{\vee}$ do elsewhere in this text; the symbol itself then appears only on the right, where products are actually formed. The definition eqn. (107) carries no subscript, since the same ∇ is applied in $\nabla \cdot X$ and $\nabla \wedge X$ as well: the product belongs to the application, not to the operator. That choice of product is what returns both grades at once — the inner and the outer product each return one of them, and eqn. (114) separates them again. Where the field is a scalar the product reduces to scaling and $\nabla\varphi$ is written plain.

Resolving the field in the same way, $A = \sum_j A_j \boldsymbol{e}_j$ with scalar coefficients A_j, and substituting it into eqn. (108) carries the derivative onto the coefficients and leaves the basis vectors to be multiplied out:

$$\nabla \overset{\wedge}{} A = \sum_i \boldsymbol{e}_i \overset{\wedge}{} \partial_i\Big(\sum_j A_j \boldsymbol{e}_j\Big) = \sum_{i,j} (\boldsymbol{e}_i \overset{\wedge}{} \boldsymbol{e}_j) \, \partial_i A_j \,. \tag{109}$$

In three dimensions that is a sum of nine terms, one per pair (i, j), and the algebra sorts them by itself. The three terms with $i = j$ have $\boldsymbol{e}_i \overset{\wedge}{} \boldsymbol{e}_i = 1$ and contribute a scalar. The six with $i \neq j$ have $\boldsymbol{e}_i \overset{\wedge}{} \boldsymbol{e}_j = -\boldsymbol{e}_j \overset{\wedge}{} \boldsymbol{e}_i$ and pair into

bivectors, leaving

$$\nabla_{\wedge} A = \underbrace{(\partial_1 A_1 + \partial_2 A_2 + \partial_3 A_3)}_{\text{grade 0}} + \underbrace{(\partial_2 A_3 - \partial_3 A_2)\, \boldsymbol{e}_{23} + (\partial_3 A_1 - \partial_1 A_3)\, \boldsymbol{e}_{31} + (\partial_1 A_2 - \partial_2 A_1)\, \boldsymbol{e}_{12}}_{\text{grade 2}} . \quad (110)$$

One product returns both. The grade-0 coefficient is the divergence; the grade-2 coefficients are those of the curl, the same three numbers attached to the planes $\boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12}$ instead of to the axes $\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3$.

Taking $A = x\,(\boldsymbol{e}_1 + \boldsymbol{e}_2)$, a field varying along x alone, the only non-zero partial derivative is $\partial_1 A = \partial A / \partial x = \boldsymbol{e}_1 + \boldsymbol{e}_2$, and the same two substitutions run to the end in one line:

$$\begin{aligned} \nabla_{\wedge} A = \sum_i \boldsymbol{e}_i \wedge (\partial_i A) = \boldsymbol{e}_1 \wedge (\partial_1 A) = \boldsymbol{e}_1 \wedge (\boldsymbol{e}_1 + \boldsymbol{e}_2) \\ = \boldsymbol{e}_1 \wedge \boldsymbol{e}_1 + \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 = \underbrace{1}_{\text{grade 0}} + \underbrace{\boldsymbol{e}_{12}}_{\text{grade 2}} . \end{aligned} \quad (111)$$

The field spreads at unit rate and circulates at unit rate in the $\boldsymbol{e}_{12}$ plane. Classically the same two numbers are two computations of different type, $\operatorname{div} A = 1$ and $\operatorname{curl} A = \boldsymbol{e}_3$, and the axis $\boldsymbol{e}_3$ names the plane $\boldsymbol{e}_{12}$ only because three dimensions permit it.

Eqn. (107) is a special case. In general the sum runs over the *reciprocal* basis $\{\boldsymbol{e}^i\}$, the frame defined by $\boldsymbol{e}^i \cdot \boldsymbol{e}_j = \delta^i_j$,

$$\nabla = \sum_i \boldsymbol{e}^i\, \partial_i , \quad (112)$$

which coincides with $\{\boldsymbol{e}_i\}$ when that basis is orthonormal and the metric is the identity. The two frames separate as soon as either condition fails: under a non-Euclidean signature they differ by a sign per direction, which is eqn. (115) below, and under a non-orthogonal or position-dependent frame they differ outright.

A curvilinear frame fails the second condition, and in one further respect: its basis vectors are functions of position, so differentiating a field $A = \sum_j A^j \boldsymbol{e}_j$ differentiates them as well,

$$\partial_i A = \sum_j \left(\partial_i A^j\right) \boldsymbol{e}_j + \sum_j A^j\, (\partial_i \boldsymbol{e}_j) . \quad (113)$$

Only the first sum survives for a constant frame, which is why the coefficient form eqn. (109) could be read off directly above. The second sum carries what a tensor treatment expresses through connection coefficients.

1.3.1 Inner and outer derivative

Splitting eqn. (107) by grade gives the two halves of the vector derivative,

$$\nabla_{\wedge} X = \nabla \cdot X + \nabla \wedge X \,, \tag{114}$$

the *inner derivative* lowering the grade of a k-vector field to $k-1$ and the *outer derivative* raising it to $k+1$. The two braced groups of eqn. (110) are these parts for a vector field: the scalar is $\nabla \cdot A$ and the bivector $\nabla \wedge A$, which in the example eqn. (111) are the 1 and the $\boldsymbol{e}_{12}$. Both are defined in every algebra and mean the same thing in each.

The classical operators of vector analysis are single grades of this pair rather than operators in their own right. In three dimensions:

field	$\nabla \cdot X$	$\nabla \wedge X$
scalar	0	vector = grad
vector	scalar = div	bivector = $(\text{curl})_\star$
bivector	vector	trivector
trivector	bivector	0

So grad is the outer derivative of a scalar field and div the inner derivative of a vector field, while curl is not a graded part of ∇ at all: $\nabla \wedge A$ is the *dual* of the curl, the same coordinates carrying a plane rather than an axis. Recovering the classical vector takes the dual, $(\nabla \wedge A)_\star$[8], and the dual carries the metric where the complement does not. In EGA the distinction is invisible, the identity metric making the two maps agree at every grade; they part on precisely those blades whose extended-metric entry is -1, of which $STA4DS$ has eight of sixteen. Multiplication by $\mathbb{1}$ is a third map again, agreeing with the dual only up to grade one, since $\mathbb{1} \wedge X = (\widetilde{X})_\star$ carries a reversion.

[8]The *left* dual throughout. In odd-dimensional algebras the left and right duals coincide, and only one of them is needed — $EGA3D$ here. In even dimensions they differ by a sign on the odd grades and agree on the even ones: in $STA4DS$ the duals of a vector and of a trivector flip sign between the two conventions while the dual of a bivector does not, and in $EGA2D$ the dual of a vector flips while those of the scalar and the pseudoscalar do not. The side therefore has to be named for signs to compose consistently across an expression, which is why both are carried rather than one.

Reading eqn. (114) as "gradient equals divergence plus curl" misstates it three times over: the identity holds for a *vector* field, its left-hand side is the vector derivative rather than the gradient, and its second term is the dual of the curl. For a scalar field the inner part vanishes identically, and only there is $\nabla\varphi$ the gradient.[9]

Of the three classical operators only curl is tied to three dimensions. The outer derivative of a vector field is a pseudoscalar in $2D$ — a scalar curl — a bivector in $3D$, and a trivector in $STA4DS$, whereas $\nabla\cdot$ and $\nabla\wedge$ carry over unchanged.

1.3.2 The spacetime derivative

In $STA4DS$ time is a coordinate rather than a parameter, so eqn. (107) runs over all four directions and there is no separate time derivative. The metric is not the identity, so the reciprocal basis of eqn. (112) no longer coincides with the basis itself and the distinction becomes visible,

$$\nabla = \sum_\mu \boldsymbol{g}^\mu \partial_\mu \,, \qquad \boldsymbol{g}^\mu = \sum_\nu \eta^{\mu\nu} \boldsymbol{g}_\nu \,, \tag{115}$$

with $\eta^{\mu\nu}$ the inverse metric, here diagonal with the entries of the signature, so that each reciprocal vector is its own basis vector up to a sign, $\boldsymbol{g}^\mu = \pm\boldsymbol{g}_\mu$, positive for the time direction and negative for the three spatial ones. Writing the contraction with a second index rather than a repeated one keeps the statement a definition of $\boldsymbol{g}^\mu$ per direction; the sums here are written out, as everywhere in this text. The form places the signature in one location instead of distributing its signs over the individual terms. The inhomogeneous and homogeneous Maxwell equations become the graded parts of a single statement about the electromagnetic bivector F,

$$\nabla \cdot F = \mu_0 J \,, \qquad \nabla \wedge F = 0 \,, \qquad \nabla \wedge F = \mu_0 J \,. \tag{116}$$

One caution follows from the signature rather than from the algebra: a Minkowski norm cannot decide whether a result vanishes. Contributions

[9] This is a statement about grades, not about the metric. The two halves of eqn. (114) are the grade projections $\langle \nabla \wedge X\rangle_{k-1}$ and $\langle \nabla \wedge X\rangle_{k+1}$ of one geometric product, so for a scalar field, $k = 0$, the first selects a grade that does not exist and is therefore empty. The extended metric is meanwhile defined on every grade and is what makes that geometric product — and with it $\langle a \wedge b\rangle_0 = a \cdot b$ for vectors — well defined. What vanishes is the projection, not the product. The other projection meanwhile takes all of it: a wedge with a scalar is ordinary scalar multiplication, so $\nabla \wedge \varphi$ carries a grade-one part and nothing else, and $\langle \nabla \wedge \varphi\rangle_1$ is the entire product.

of opposite causal character cancel in the norm, so a field that does not solve eqn. (116) can still return $\|\nabla F\| = 0$ while carrying non-zero components. Tests for zero are componentwise.

Applying the vector derivative twice gives the second vector derivative. As ∇ is itself a vector its square splits like any other, $\nabla_{\wedge}\nabla = \nabla\cdot\nabla + \nabla\wedge\nabla$, and the outer part vanishes for a sufficiently regular field because the partial derivatives commute. What remains is a scalar operator, the d'Alembertian

$$\Box = \nabla\cdot\nabla = \nabla^2\,, \tag{117}$$

grade-preserving because a scalar operator cannot be anything else. It is the same construction as the Laplacian of subsection 1.3.4 and differs from it only through the metric that ∇ carries, which enters the square along with it: written out under the signature $(-,-,-,+)$,

$$\Box = \partial_4^2 - \left(\partial_1^2 + \partial_2^2 + \partial_3^2\right)\,. \tag{118}$$

The difference is not one of labelling. An indefinite signature makes the operator hyperbolic where the Euclidean one is elliptic — a wave operator rather than a potential operator, propagating at finite speed along the light cone — which is why it is named separately.

It carries the potential form of eqn. (116). Writing the field as the outer derivative of a potential, $F = \nabla\wedge A$, and fixing the gauge by the Lorenz condition $\nabla\cdot A = 0$ — which makes $\nabla\wedge A = \nabla_{\wedge}A$, so that $\nabla_{\wedge}F = \nabla_{\wedge}\nabla_{\wedge}A = \Box A$ — turns eqn. (116) into

$$\Box A = \mu_0 j\,. \tag{119}$$

The same operator arises a second way, without a Minkowski signature. Keep time a parameter and work with the spatial vector derivative alone, whose square $\nabla^2 = \Delta$ is a scalar. The wave operator then factors into two first-order ones,

$$\left(\frac{1}{u}\partial_t + \nabla\right)\left(\frac{1}{u}\partial_t - \nabla\right) = \frac{1}{u^2}\partial_t^2 - \Delta\,, \tag{120}$$

for a propagation speed u and in either order, the cross terms cancelling because ∂_t and ∇ commute [Jancewicz, 1988]. What the factorisation needs is that ∇ be a single algebraic object whose square is a scalar. Classical vector analysis has no square root of the Laplacian to offer, div and grad being separate operators between different spaces, and Δ only their composition.

The two routes end at the same operator. In three dimensions with time as a parameter it is assembled by hand from eqn. (120); in *STA4DS* the signature

performs the same factorisation, and $\Box = \nabla^2$ needs no assembly. A first-order field equation such as eqn. (116) is then the more primitive statement, with the second-order eqn. (119) obtained by applying ∇ once more — rather than the wave equation being primary and its first-order form a factorisation of it.

1.3.3 Evaluating the derivative numerically

For a field given as a function, the partial derivatives in eqn. (107) are approximated by finite differences, with the stencil weights generated for arbitrary node sets and arbitrary derivative order. Three properties of that layer matter to a reader.

Explicit and compact schemes. An explicit scheme forms a weighted sum of field values and is evaluable at a single point. A compact (Padé) scheme couples the derivative values at neighbouring nodes and reaches a higher order from the same stencil width, at the cost of being implicit: it is defined on a grid and cannot be evaluated pointwise.

Uniform and non-uniform grids. Node coordinates enter the weight generation directly, so a stretched grid needs no separate treatment; the uniform case is the special one, not the only one. The equidistant schemes hold their nodes as multiples of the step, which makes those weights independent of h; applying them rescales by h^{-d} for a d-th derivative.

Where the derivative is taken. The development point need not be one of the sample nodes. Sampling on the half-offsets $\pm\frac{1}{2}h$, $\pm\frac{3}{2}h$ and developing at their centre gives the derivative midway between the samples, the arrangement of a staggered lattice. Values of the other derivative order may be supplied alongside the field values, which shortens the stencil by one node: $f''(x_0)$ from $\{x_0, x_0 + h\}$ with $f'(x_0)$ prescribed reaches the order of the three-node one-sided form. A boundary condition enters that way rather than as an equation of its own.

The criterion for a correct derivative is the observed *order*, not a fixed tolerance. At any given step size the residual of a correct expression is the truncation error of the scheme, so a threshold chosen at one step size accepts an expression that is merely close. Halving h must reduce the residual by 2^p for a scheme of order p; that ratio distinguishes an exact identity from an approximate one, and a bare threshold does not.

1.3.4 Identities

The second derivatives vanish for any sufficiently regular field,

$$\nabla \wedge (\nabla \wedge X) = 0\,, \qquad \nabla \cdot (\nabla \cdot X) = 0\,, \tag{121}$$

which contains curl grad = 0 and div curl = 0 as the three-dimensional special cases. The Laplacian $\Delta = \nabla^2$ preserves grade, and $\Delta\, r^{-1} = 0$ away from the origin.

Products differentiate by rules that depend on the grades involved; the rules below and the standard-field derivatives that follow them are those of [Jancewicz, 1988]. With A, B vector fields and $\hat{B}$ a bivector field,

$$\nabla \cdot (A \cdot \hat{B}) = (\nabla \wedge A) \cdot \hat{B} - A \cdot (\nabla \cdot \hat{B})\,, \tag{122}$$
$$\nabla \wedge (A \cdot B) = (A \cdot \nabla)B + (B \cdot \nabla)A - B \cdot (\nabla \wedge A) - A \cdot (\nabla \wedge B)\,, \tag{123}$$
$$\nabla \cdot (A \wedge B) = B(\nabla \cdot A) - A(\nabla \cdot B) + (A \cdot \nabla)B - (B \cdot \nabla)A\,. \tag{124}$$

The corresponding rule for $\nabla \cdot (A \wedge \hat{B})$ carries a term $\hat{B} \times (\nabla \wedge A)$ in the commutator product, since both factors there are bivectors and neither the inner nor the outer product returns a bivector.

For the position field r with a a constant vector, $\hat{B}$ a constant bivector and $r = \|r\|$:

$$\nabla(r \cdot a) = a\,, \qquad \nabla_{\wedge} r = 3\,, \qquad \nabla\, r^k = k\, r^{k-2}\, r\,, \tag{125}$$
$$\nabla \cdot (a \wedge r) = -2a\,, \qquad \nabla \wedge (a \wedge r) = 0\,, \qquad \nabla \cdot (\hat{B} \wedge r) = \hat{B}\,. \tag{126}$$

The divergence in eqn. (126) is not zero although a is constant; the position factor carries it. The two derivatives $\nabla_{\wedge}(a \wedge r) = -a$ and $\nabla_{\wedge}(r \wedge a) = 3a$ differ because a and r do not commute.

1.4 Modelling motion

Every rigid motion is a composition of reflections, which is where the sandwich products and the half angles of this section come from. Reflecting a vector x in the hyperplane through the origin orthogonal to a unit vector n — a line in $2D$, a plane in $3D$ — is the sandwich

$$x' = -n \wedge x \wedge n \tag{127}$$

(the library's `reflect_on`). The minus sign is required: the sandwich alone flips the component lying *in* the mirror and leaves the normal component unchanged, returning $-x'$. Reflecting once more, in a mirror with unit normal m, and collecting the factors gives

$$x'' = -m \wedge (-n \wedge x \wedge n) \wedge m = (m \wedge n) \wedge x \wedge \widetilde{(m \wedge n)} \tag{128}$$

since $\widetilde{(m \wedge n)} = n \wedge m$. The two reflections have become a single sandwich with one even-grade object $R = m \wedge n$, whose scalar part is the cosine of the angle θ between the two mirrors and whose bivector part lies in the plane they span. It turns by 2θ — *twice* the angle between the mirrors — and that factor of two is where the half angles of this section come from.

In PGA the same construction is carried out with planes (lines in $2D$) and the regressive product. A reflection in a unitized mirror p is

$$X' = p \vee X \vee \underset{\sim}{p} \tag{129}$$

without a minus sign, because a projective object is defined only up to a non-zero scalar factor, so its overall sign is not observable — whereas the EGA vector x above is a value and its sign is. Two such reflections compose into the motor

$$M = p_2 \vee p_1 \tag{130}$$

which yields both Euclidean isometries at once: mirrors that *intersect* give a rotation about their common line by twice the dihedral angle between them, while *parallel* mirrors give a translation perpendicular to them by twice their distance. A translation is in this sense not an operation of its own kind but a rotation whose axis has moved to infinity, and that is what lets PGA treat both with one sandwich product. Fig. 8 shows both cases in the plane.

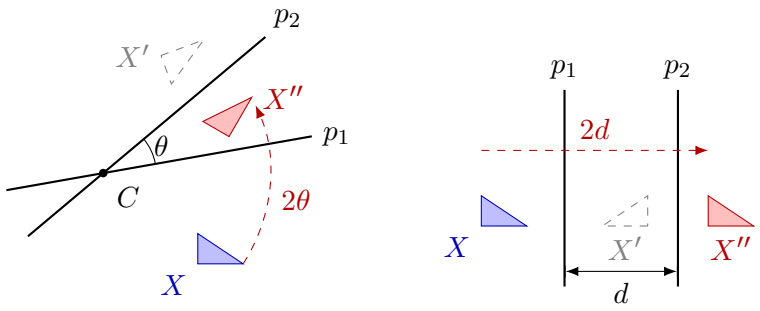


Figure 8: Two reflections compose into the motor $M = p_2 \vee p_1$, drawn in the plane, where the mirrors are lines. Left: mirrors meeting at the angle $\theta = 30°$ turn X about their intersection C by $2\theta = 60°$, from p_1 towards p_2. Right: parallel mirrors at the distance d move X by $2d$, perpendicular to them. The first reflection gives the mirror-inverted X'; the second restores the orientation in $X'' = M \vee X \vee \underset{\sim}{M}$.

For translational motion in physical contexts we typically use the following naming convention for scalars or vectors:

$$\begin{aligned} s(t) &= s \\ \dot{s}(t) &= v(t) = \dot{s} = v \\ \ddot{s}(t) &= \dot{v}(t) = a(t) = \ddot{s} = \dot{v} = a \end{aligned}$$

while for rotational motion we typically use:

$$\begin{aligned} \phi(t) &= \phi \\ \dot{\phi}(t) &= \omega(t) = \dot{\phi} = \omega \\ \ddot{\phi}(t) &= \dot{\omega}(t) = \ddot{\phi} = \dot{\omega} \end{aligned}$$

Especially in *PGA*, we will be able to combine the translational and rotational aspects in one physical variable. Thus, we will use a convention of using capital letters for physical quantities to distinguish them from the classical convention, especially so for bivectors and even- or odd-grade multivectors that do not occur in classical descriptions. We will use in *PGA*:

$B(t) = B$	(bivector)
$\dot{B}(t) = \Omega(t) = \dot{B} = \Omega$	(rate of change of bivector, "bivector speed")
$\ddot{B}(t) = \dot{\Omega}(t) = \ddot{B} = \dot{\Omega}$	(rate of change of bivector speed)

In order to model motion in *PGA* we need to work with the exponential function applying it in the context of the geometric product (symbol "$\wedge$")

and the regressive geometric product (symbol "∨").

$$\exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} \qquad = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \frac{x^7}{7!} + \cdots \tag{131}$$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} \quad = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \frac{x^{10}}{10!} + \frac{x^{12}}{12!} - \frac{x^{14}}{14!} + \cdots \tag{132}$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \quad = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!} + \frac{x^{13}}{13!} - \frac{x^{15}}{15!} + \cdots \tag{133}$$

If we substitute $x = i\phi$ with i as the imaginary unit ($i^2 = -1$), we get Euler's formula:

$$\begin{aligned} \exp(i\phi) &= \sum_{n=0}^{\infty} \frac{(i\phi)^n}{n!} \\ &= 1 + i\phi + \frac{(i\phi)^2}{2!} + \frac{(i\phi)^3}{3!} + \frac{(i\phi)^4}{4!} + \frac{(i\phi)^5}{5!} + \frac{(i\phi)^6}{6!} + \frac{(i\phi)^7}{7!} + \cdots \qquad (134) \\ &= 1 + i\phi - \frac{\phi^2}{2!} - i\frac{\phi^3}{3!} + \frac{\phi^4}{4!} + i\frac{\phi^5}{5!} - \frac{\phi^6}{6!} - i\frac{\phi^7}{7!} + \cdots \\ &= 1 - \frac{\phi^2}{2!} + \frac{\phi^4}{4!} - \frac{\phi^6}{6!} + \cdots + i(\phi - \frac{\phi^3}{3!} + \frac{\phi^5}{5!} - \frac{\phi^7}{7!} + \cdots) \qquad (135) \end{aligned}$$

$$\exp(i\phi) = \cos(\phi) + i\sin(\phi) \tag{136}$$

Applying this for geometric algebra we have to distinguish between calculating the powers in the Taylor series using the geometric product (in EGA) and using the regressive geometric product (in PGA). For the rotation case we assume a unitized bivector B and a scalar ϕ. Based on the geometric product we know that $B \wedge B = -\mathbf{1}$, and for its regressive counterpart we know that $B \vee B = -\mathbb{1}$. Thus, B plays a role equivalent to i in both cases. In addition, taking any value to the power of 0 results in the neutral element of the product, which is $\mathbf{1}$ for the geometric product, and $\mathbb{1}$ for the regressive

geometric product. Taking the powers then works like this:

$$
\begin{aligned}
&\dots \\
\frac{(B\phi)^2_{\wedge}}{2!} &= \frac{1}{2!}(B\phi \wedge B\phi) = \frac{\phi^2}{2!}(B \wedge B) = \frac{\phi^2}{2!}(-\mathbf{1}) = -\frac{\phi^2}{2!} && \text{(137a)}\\
\frac{(B\phi)^3_{\wedge}}{3!} &= \frac{\phi^3}{3!}(B \wedge B \wedge B) = \frac{\phi^3}{3!}(-\mathbf{1} \wedge B) = -\frac{\phi^3}{3!}B && \text{(137b)}\\
\frac{(B\phi)^4_{\wedge}}{4!} &= \frac{\phi^4}{4!}(B \wedge B \wedge B \wedge B) = \frac{\phi^4}{4!}(-\mathbf{1} \wedge -\mathbf{1}) = \frac{\phi^4}{4!} && \text{(137c)}\\
&\dots \\
\frac{(B\phi)^2_{\vee}}{2!} &= \frac{1}{2!}(B\phi \vee B\phi) = \frac{\phi^2}{2!}(B \vee B) = \frac{\phi^2}{2!}(-\mathbb{1}) = -\frac{\phi^2}{2!}\mathbb{1} && \text{(137d)}\\
\frac{(B\phi)^3_{\vee}}{3!} &= \frac{\phi^3}{3!}B \vee B \vee B = \frac{\phi^3}{3!}(-\mathbb{1} \vee B) = -\frac{\phi^3}{3!}B && \text{(137e)}\\
\frac{(B\phi)^4_{\vee}}{4!} &= \frac{\phi^4}{4!}B \vee B \vee B \vee B = \frac{\phi^4}{4!}(-\mathbb{1} \vee -\mathbb{1}) = \frac{\phi^4}{4!}\mathbb{1} && \text{(137f)}\\
&\dots
\end{aligned}
$$

Thus, the result of exponentiating a bivector B with respect to the geometric product is (for EGA):

$$
\begin{aligned}
\exp_{\wedge}(B\phi) &= \sum_{n=0}^{\infty} \frac{(B\phi)^n_{\wedge}}{n!} \\
&= 1 + B\phi + \frac{(B\phi)^2_{\wedge}}{2!} + \frac{(B\phi)^3_{\wedge}}{3!} + \frac{(B\phi)^4_{\wedge}}{4!} + \frac{(B\phi)^5_{\wedge}}{5!} + \frac{(B\phi)^6_{\wedge}}{6!} + \cdots && \text{(138)}\\
&= 1 + B\phi - \frac{\phi^2}{2!} - B\frac{\phi^3}{3!} + \frac{\phi^4}{4!} + B\frac{\phi^5}{5!} - \frac{\phi^6}{6!} - B\frac{\phi^7}{7!} + \cdots \\
&= 1 - \frac{\phi^2}{2!} + \frac{\phi^4}{4!} - \frac{\phi^6}{6!} + \cdots + B(\phi - \frac{\phi^3}{3!} + \frac{\phi^5}{5!} - \frac{\phi^7}{7!} + \cdots)
\end{aligned}
$$

$$
\exp_{\wedge}(B\phi) = \cos(\phi) + \sin(\phi)B \qquad \text{(139)}
$$

The last step of moving B after the brackets is allowed, since scalar factors commute with any object using the regular product. The result structurally is an even-grade multivector with a scalar and a bivector component. We get the correct sign and angle of rotation if we replace ϕ by $-\frac{1}{2}\phi$, which was not considered above to keep the formulas as simple as possible. The substitution yields the final form of a rotor R in EGA with B as unitized bivector and ϕ

the rotation angle:

$$R = \exp_{\wedge}(-\frac{\phi}{2}B) = \cos(-\frac{\phi}{2}) + \sin(-\frac{\phi}{2})B$$
$$R = \cos(\frac{\phi}{2}) - \sin(\frac{\phi}{2})B \tag{140}$$

For the case using the regressive geometric product it becomes (for PGA):

$$\begin{aligned}\exp_{\vee}(B\phi) &= \sum_{n=0}^{\infty} \frac{(B\phi)_{\vee}^{n}}{n!} \\ &= \mathbb{1} + B\phi + \frac{(B\phi)_{\vee}^{2}}{2!} + \frac{(B\phi)_{\vee}^{3}}{3!} + \frac{(B\phi)_{\vee}^{4}}{4!} + \frac{(B\phi)_{\vee}^{5}}{5!} + \frac{(B\phi)_{\vee}^{6}}{6!} + \cdots \\ &= \mathbb{1} + B\phi - \mathbb{1}\frac{\phi^2}{2!} - B\frac{\phi^3}{3!} + \mathbb{1}\frac{\phi^4}{4!} + B\frac{\phi^5}{5!} - \mathbb{1}\frac{\phi^6}{6!} - B\frac{\phi^7}{7!} + \cdots \\ &= \mathbb{1}(1 - \frac{\phi^2}{2!} + \frac{\phi^4}{4!} - \frac{\phi^6}{6!} + \cdots) + B(\phi - \frac{\phi^3}{3!} + \frac{\phi^5}{5!} - \frac{\phi^7}{7!} + \cdots)\end{aligned} \tag{141}$$

$$\exp_{\vee}(B\phi) = \cos(\phi)\mathbb{1} + \sin(\phi)B \tag{142}$$

The last step of moving $\mathbb{1}$ and B after the brackets is allowed, since scalar factors, like the results of the trigonometric functions, commute with any object using the regular product. The resulting motor structurally is an even-grade multivector with a bivector and a pseudoscalar of grade 4 for $3D$ PGA.

The two exponentials are separate functions in the library, and the names say which product they are taken with respect to. The exponential w.r.t. the geometric product $\exp_{\wedge}$ is `exp`, with `log` and `sqrt` as its inverse and its square root (EGA, STA). The exponential w.r.t. the regressive geometric product $\exp_{\vee}$ is `rexp`, with `rlog` and `rsqrt` (PGA, CGA). The unprefixed names stay reserved for the geometric product in every algebra.

To create a rotor that generates a rotation of angle ϕ we use

$$M = \exp_{\vee}\left(\frac{\phi}{2}B\right) \tag{143}$$

where the factor $\frac{1}{2}$ is needed to create a resulting angle ϕ through the double-sided sandwich product. Typically, in applications the angle ϕ is directly contained as a scalar factor in the bivector B while the factor $\frac{1}{2}$ is kept explicit.

In $2D$ PGA we have the same expression, but the argument of the exponential to create a motor for rotation is not a bivector B, but a unitized point

P. Here $P \vee P = -\mathbb{1}$ is valid, and equation (142) becomes $\exp_\vee(P\phi) = \cos(\phi)\mathbb{1} + \sin(\phi)P$ instead, which leads to a motor M that structurally is an odd-grade multivector with a vector and a pseudoscalar of grade 3. The resulting rotation direction of the motor depends on the chosen pseudoscalar. The pseudoscalar for $PGA2DP$ was chosen as $\boldsymbol{e}_{321}$ in order to make signs of rotation angles (and thus turning directions) consistent between $2D$ and $3D$ PGA. For the actual motor we get

$$M = \exp_\vee\left(\frac{\phi}{2}P\right) \tag{144}$$

in this case.

To model a translation $\delta = (\delta_x, \delta_y, \delta_z, 0)$ in $3D$ PGA we use a special bivector $B = B_{tra}$ that does not square to $-\mathbb{1}$, but instead squares to zero. That makes sure that the expansion of the Taylor series of the exponential function $\exp_\vee(B_{tra})$ only contains the first two terms. The translation is resulting from two consecutive reflections across parallel planes perpendicular to δ with the distance $\frac{1}{2}\|\delta\|_\bullet$ relative to each other. They intersect in a line at infinity that is used as "rotation axis". This however yields a translation, since the axis is infinitely far away. The derivation of the formula for the bivector will be done later in this chapter (see eqn. (188)). We use $B_{tra} = \delta^\star \vee \overline{\boldsymbol{e}_4} = \delta^\star \vee H_{3dp} = att(\delta^\star)$. This yields $B_{tra} = (\delta_x\boldsymbol{e}_{23} + \delta_y\boldsymbol{e}_{31} + \delta_z\boldsymbol{e}_{12})$. The resulting motor is $M = \exp_\vee(\frac{1}{2}B_{tra}) = \mathbb{1} + \frac{1}{2}B_{tra} = \mathbb{1} + \frac{1}{2}(0, 0, 0, \delta_x, \delta_y, \delta_z)$. The motor M is an even-grade multivector containing a pseudoscalar component, which is a quadvector in $3D$ PGA, and a bivector component.

To model a translation $\delta = (\delta_x, \delta_y, 0)$ in $2D$ PGA we need a special vector as argument of the exponential. It represents a direction to a point at infinity. The "rotation" around that infinitely far away point generates a translation (actually it is resulting from two consecutive reflections across parallel lines perpendicular to δ with the distance $\frac{1}{2}\|\delta\|_\bullet$ relative to each other). The derivation of the special vector will be done later in this section (see eqn. (195)). The argument of the exponential $\exp_\vee$ for a translation is the direction vector $P = \delta^\star \vee \overline{\boldsymbol{e}_3} = \delta^\star \vee H_{2dp} = att(\delta^\star) = (-\delta_y\boldsymbol{e}_1 + \delta_x\boldsymbol{e}_2)$. P points in a direction perpendicular to the translation δ towards infinity. The main difference of this special point P compared to the one used for the rotation above is that $P_z = 0$. This results in $P \vee P = 0$ which causes the expansion of the exponential to only contain the first two terms: $M = \exp_\vee(\frac{1}{2}P) = \mathbb{1} + \frac{1}{2}P = \mathbb{1} + \frac{1}{2}(-\delta_y, \delta_x, 0)$. The motor M is an odd-grade multivector containing a pseudoscalar, which is a trivector in $2D$ PGA, and a vector component, i.e. it is structurally the same as for the rotation case in $2D$ PGA above.

Eqn. (139) allows us to model rotations in EGA, where the bivector B is the complement of the axis of rotation and $\theta = -2\phi$ is the angle of the rotation around that axis. If we directly want to specify the angle of rotation θ we need to substitute ϕ by $-\frac{\phi}{2}$ in eqn. (139). The rotated object X can then be obtained from the original object X_0 by:

$$X = R \wedge X_0 \wedge \widetilde{R} = \exp_{\wedge}(-\frac{\phi}{2}B) \wedge X_0 \wedge \exp_{\wedge}(\frac{\phi}{2}B) \quad (\text{for } EGA) \tag{145}$$

If we want to model combined translations and rotations in PGA, we need to look into another extension of the exponential function. We follow the approach defined by [Lengyel, 2024b] in chapter 3.2 and 3.6.2, where it is shown that a general motor in PGA can be expressed as a screw motion, i.e. a rotation around a unitized line $\boldsymbol{l}$ by an angle ϕ and a translation along $\boldsymbol{l}$ by a displacement δ as

$$\begin{aligned} M = \exp_{\vee}\left[(\frac{\delta}{2}\mathbf{1} + \frac{\phi}{2}\mathbb{1}) \vee \boldsymbol{l}\right] &= \cos_{\vee}(\frac{\delta}{2}\mathbf{1} + \frac{\phi}{2}\mathbb{1}) + \sin_{\vee}(\frac{\delta}{2}\mathbf{1} + \frac{\phi}{2}\mathbb{1}) \vee \boldsymbol{l} \\ &= -\frac{\delta}{2}\mathbf{1}\sin(\frac{\phi}{2}) + \mathbb{1}\cos(\frac{\phi}{2}) + \left(\frac{\delta}{2}\mathbf{1}\cos(\frac{\phi}{2}) + \mathbb{1}\sin(\frac{\phi}{2})\right) \vee \boldsymbol{l} \\ M &= -\frac{\delta}{2}\mathbf{1}\sin(\frac{\phi}{2}) - \frac{\delta}{2}\boldsymbol{l}^{\star}\cos(\frac{\phi}{2}) + \boldsymbol{l}\sin(\frac{\phi}{2}) + \mathbb{1}\cos(\frac{\phi}{2}) \end{aligned} \tag{146}$$

where M structurally is an even-grade multivector with scalar, bivector and pseudoscalar parts in $3D$ PGA.

For pure rotational motion with $\delta = 0$ and $\phi \neq 0$ we get

$$M_{rot} = \boldsymbol{l}\sin(\frac{\phi}{2}) + \mathbb{1}\cos(\frac{\phi}{2}) \tag{147}$$

while for pure translational motion with $\delta \neq 0$ and $\phi = 0$ we get

$$M_{tra} = -\frac{\delta}{2}\boldsymbol{l}^{\star} + \mathbb{1} \tag{148}$$

The argument in the exponential function in eqn. (146) for all cases results in a bivector. This can be derived from inspecting the product table for the regressive geometric product. However, this time we only have a common factor of $\frac{1}{2}$, and not $-\frac{1}{2}$ as in EGA as required argument of the exponential function. This will impact a sign for the transformation in the regressive case for PGA compared to the corresponding transformations in EGA. The resulting transformation is then obtained from

$$X = M \vee X_0 \vee \underset{\sim}{M} \quad (\text{for } PGA) \tag{149}$$

We can think of the screw motion used to model motion in $3D\ PGA$ as a combination of a rotation by an angle ϕ around the unitized line $\boldsymbol{l}$ and by a translation from a displacement δ along that same line $\boldsymbol{l}$. The latter part results from a rotation around a line at infinity given by $\boldsymbol{l}^*$. Thus, the bivector representing the line $\boldsymbol{l}$ can always be split into the parts creating the rotation (movement around that line) and the parts creating the translation (movement along that line).

1.4.1 Motion in *EGA*

In order to dive deeper into modelling of motions we first start with the *EGA* case. It is identical to the derivation shown in [Dorst and De Keninck, 2023] for their plane-based approach. This will help to better recognize the differences to the *PGA* approach used here.

Before going into more detail, we need to prepare some formulas we later need for our derivations:

For *EGA* the rotor will now be written as M instead of R to make it easy to compare to our *PGA*-derivations using motors M later on. M as a rotor in *EGA* is an orthonormal transformation and thus satisfies

$$M \wedge \widetilde{M} = \widetilde{M} \wedge M = 1 \tag{150}$$

(remember: $M^{-1} = \widetilde{M}$), and therefore

$$\frac{d}{dt}(M \wedge \widetilde{M}) = \dot{M} \wedge \widetilde{M} + M \wedge \dot{\widetilde{M}} = 0 \tag{151}$$

It follows that $M \wedge \dot{\widetilde{M}} = -\dot{M} \wedge \widetilde{M}$, which results in

$$\dot{\widetilde{M}} = -\widetilde{M} \wedge \dot{M} \wedge \widetilde{M} \tag{152}$$

Starting from $\widetilde{M} \wedge M = 1$ instead yields the same result for $\dot{\widetilde{M}}$.

In order to model rigid body mechanics we consider two coordinate frames:

- A *world frame* (index w) that is normally chosen to be an inertial frame.
- A *body frame* (index b) that is attached to a rigid body. Typically, at its center of gravity, but this is not strictly required. Rigid body's internal forces are in equilibrium and thus need not be considered explicitly. It is sufficient to describe the translational motion of the body's center of gravity and its rotational motion with respect to the center of gravity. The resulting motion of the body frame relative to the world frame then fully covers rigid body motion (kinematics and dynamics), which depends on distribution of the mass of the body as well as on external forces and moments acting on the body.
- To describe the motion we use a time dependent motor $M = M(t)$. At time $t = 0$ we call it $M_0 = M(t = 0)$. M_0 brings an element X of the body to its position X_0 at time $t = 0$.

World frame: From eqn. (140) we know the form of a general rotor in EGA and use the ansatz for a time dependent rotor $M(t)$ after the object X was brought into its initial position by M_0 as $X_0 = M_0 \wedge X \wedge \widetilde{M_0}$. If there is no difference between world and body frames at $t = 0$ then $M_0 = 1$:

$$M(t) = \exp_{\wedge}(-\frac{1}{2}B_w(t)) \wedge M_0 \tag{153}$$

with $B_w(t)$ as a time-dependent bivector that models the rotational plane and the turning angle. The derivative of M with respect to time t is

$$\dot{M}(t) = -\frac{1}{2}\dot{B}_w(t) \wedge \exp_{\wedge}(-\frac{1}{2}B_w(t)) \wedge M_0 = -\frac{1}{2}\dot{B}_w(t) \wedge M(t) \tag{154}$$

where the rate $\dot{B}_w$ was moved in front of the exponential. That step requires B_w and $\dot{B}_w$ to commute, and they do so exactly when the plane of rotation is held fixed: writing $B_w = \theta\hat{B}$ with $\hat{B}$ of unit magnitude gives $[B_w, \dot{B}_w] = \theta^2[\hat{B}, \dot{\hat{B}}]$, and $\hat{B} \cdot \dot{\hat{B}} = 0$ makes any non-vanishing $\dot{\hat{B}}$ a *different* plane, which in $3D$ does not commute with $\hat{B}$. So the step is exact for a rotation about a fixed axis, where $\dot{B}_w \parallel B_w$, and only there.

The rate in the world frame is therefore *defined* by the relation

$$\dot{M} = -\frac{1}{2}\Omega_w \wedge M \tag{155}$$

which needs no assumption: M is a unit rotor, so $\dot{M} \wedge \widetilde{M}$ is a bivector and (155) can be solved for Ω_w at every instant. For a fixed plane of rotation $\Omega_w = \dot{B}_w$; when the plane turns, the two differ by the differential of the exponential, whose leading term is

$$\Omega_w = \dot{B}_w - \frac{1}{2}\mathrm{cmt}(B_w, \dot{B}_w) + \dots \tag{156}$$

This is the correction a Lie-group integrator carries, and the reason the generator of a finite displacement is not the accumulated velocity.
By multiplying this from the right with "$\wedge\, \widetilde{M}$" this can be transformed to

$$\Omega_w = -2\dot{M} \wedge \widetilde{M} \tag{157}$$

Body frame: Here we first move the object in the *body frame* and then transform it into the world frame by the transformation M_0. In this case eqn. (153) becomes

$$M(t) = M_0 \wedge \exp_{\wedge}(-\frac{1}{2}B_b(t)) \tag{158}$$

which in turn results in

$$\dot{M}(t) = -\frac{1}{2} M_0 \wedge \exp_{\wedge}(-\frac{1}{2} B_b(t)) \wedge \dot{B}_b(t) = -\frac{1}{2} M(t) \wedge \dot{B}_b(t) \tag{159}$$

where the rate $\dot{B}_b$ was moved behind the exponential, which again requires B_b and $\dot{B}_b$ to commute. The rate in the *body frame* is therefore *defined* by

$$\dot{M} = -\frac{1}{2} M \wedge \Omega_b \tag{160}$$

which needs no assumption either. Ω_b is the same angular velocity as Ω_w, expressed in the body frame instead of the world frame (eqn. 163a below). For a fixed plane of rotation $\Omega_b = \dot{B}_b$; otherwise

$$\Omega_b = \dot{B}_b + \frac{1}{2} \mathrm{cmt}(B_b, \dot{B}_b) + \dots \tag{161}$$

with the sign opposite to eqn. (156), because the exponential now stands to the right of M_0. The test case "ega3d: the rate of a rotor and the rate of its generator" in `ga_test/src/ga_ega_contract_test.hpp` takes both rates from the rotor itself: for a fixed plane they equal $\dot{B}$, for a turning plane the uncorrected $\dot{B}$ is off at first order and eqns. (156) and (161) hold at second order. Multiplying eqn. (160) from the left by "$\widetilde{M} \wedge$" gives

$$\Omega_b = -2\widetilde{M} \wedge \dot{M} \tag{162}$$

From equations (155) and (160) we can see how world and body frame rates are connected to each other: $\Omega_w \wedge M = M \wedge \Omega_b$, which can be transformed into

$$\Omega_w = M \wedge \Omega_b \wedge \widetilde{M} \tag{163a}$$
$$\Omega_b = \widetilde{M} \wedge \Omega_w \wedge M \tag{163b}$$

Now we derive the rate of change of a point $X(t)$, i.e. its speed $\dot{X}(t)$, in the *world frame*, first assuming that $X(t=0) = X_0 = const.$, i.e. as a constant element that is transformed to become X by a time dependent motor $M(t)$. During the derivation we will use the associativity of the geometric product and the neutral element 1 of the geometric product. We can write

$$\begin{aligned}
\dot{X}(t) &= \frac{d}{dt} X(t) = \frac{d}{dt}(M \wedge X_0 \wedge \widetilde{M}) \\
&= \dot{M} \wedge X_0 \wedge \widetilde{M} + M \wedge X_0 \wedge \dot{\widetilde{M}} \\
&\overset{(152)}{=} \dot{M} \wedge 1 \wedge X_0 \wedge \widetilde{M} - M \wedge X_0 \wedge (\widetilde{M} \wedge \dot{M} \wedge \widetilde{M}) \\
&\overset{(150)}{=} \dot{M} \wedge (\widetilde{M} \wedge M) \wedge X_0 \wedge \widetilde{M} - (M \wedge X_0 \wedge \widetilde{M}) \wedge \dot{M} \wedge \widetilde{M} \\
&= (\dot{M} \wedge \widetilde{M}) \wedge (M \wedge X_0 \wedge \widetilde{M}) - (M \wedge X_0 \wedge \widetilde{M}) \wedge (\dot{M} \wedge \widetilde{M})
\end{aligned}$$

From eqn. (157) we can conclude that $-\frac{1}{2}\Omega_w(t) = \dot{M} \wedge \widetilde{M}$, resulting in

$$\begin{aligned} \dot{X}(t) &= (-\frac{1}{2}\Omega_w(t)) \wedge X(t) - X(t) \wedge (-\frac{1}{2}\Omega_w(t)) \\ &= -\frac{1}{2}(\Omega_w \wedge X - X \wedge \Omega_w) \\ &= -\mathrm{cmt}(\Omega_w, X) \\ \dot{X}_w &= \mathrm{cmt}(X_w, \Omega_w) \end{aligned} \tag{164}$$

with $\mathrm{cmt}(X_w, \Omega_w)$ as the commutator product between X_w and Ω_w.

So, if the bivector rate of change Ω_w is known, the rate of change of X_w, which is $\dot{X}_w$, can be calculated from eqn. (164) using the commutator product.

It can be shown that $\dot{M} \wedge \widetilde{M}$ is always a bivector: From equations (157) and (151), and the linearity of the involved operations, we get

$$\Omega_w = -2\dot{M} \wedge \widetilde{M} = 2M \wedge \dot{\widetilde{M}} = 2(\widetilde{\dot{M} \wedge \widetilde{M}}) = -\widetilde{\Omega}_w \tag{165}$$

Considering the grade-dependent sign-reversal of the reverse in $3D$ EGA, Ω_w must either be a bivector or a trivector. Since $\dot{M}$ and $\widetilde{M}$ are even-grade multivectors, their geometric product results in an even-grade multivector as well, which does not have a trivector part. Thus, Ω_w and therefore $\dot{M} \wedge \widetilde{M}$ must indeed be a bivector.

This also assures, that the grade of X and $\dot{X}$ are identical. If X is a vector then $\dot{X}$ is a vector as well. This statement is true for any grade of X as long as Ω_w is a bivector.

If X_0 is not a constant element, but a function of time, which means that X_0 is a time dependent element in the *body frame* $X_0 = X_b(t)$, we have to add another term to the derivative in eqn. (164) as follows:

$$\begin{aligned} \dot{X}(t) &= \frac{d}{dt}(M \wedge X_b \wedge \widetilde{M}) \\ &= \dot{M} \wedge X_b \wedge \widetilde{M} + M \wedge X_b \wedge \dot{\widetilde{M}} + M \wedge \dot{X}_b \wedge \widetilde{M} \\ &= \mathrm{cmt}(M \wedge X_b \wedge \widetilde{M}, \Omega_w) + M \wedge \dot{X}_b \wedge \widetilde{M} \\ \dot{X}_w &= \mathrm{cmt}(X_w, \Omega_w) + (\dot{X}_b)_w \end{aligned} \tag{166}$$

Eqn. (166) is to be considered in the world frame. The same $\dot{X}$ derivation can be carried out entirely in the *body frame*, starting from the body-frame ansatz eqn. (158) and the corresponding rate eqns. (160) and (162) — note the swapped order of M and Ω_b relative to the world frame — in place of their world-frame counterparts. The algebra is identical and yields exactly the same results as eqns. (164) and (166), with Ω_w replaced by Ω_b, so we do not reproduce it here — in the same spirit as the body-frame derivation we skip for *PGA* below.

In *EGA* rotation as a Euclidean isometry can be described with sandwich products based on the geometric product. The corresponding sandwich product is available in the library either by using the geometric product and the grade operator *gr*() to extract the grade of the results directly, or by using the function `rotate(X,R)` which takes the object X and the rotor R as arguments and returns the rotated object.

1.4.2 Motion in *PGA*

Now we are looking into *PGA*. Lengyel has shown in [Lengyel, 2024b] that Euclidean isometries (covering translations as well as rotations) can be described with sandwich products based on the regressive geometric product in *PGA*. If we do that, we can stay in the space $\mathbb{G}(3,0,1)$ and do not need to switch to its dual space in order to implement these orthonormal transformations as already described above.

Here we repeat the derivation of the time-derivative $\dot{X}$ of a projective point $X(t)$ for the approach of *PGA* used here: We will use the regressive geometric product, the regressive reversion, and $\mathbb{1}$ as its neutral element. The derivation is then completely analogous to the *EGA* case with the given substitutions.

M as a motor in *PGA* is an orthonormal transformation and satisfies

$$M \vee \underset{\sim}{M} = \underset{\sim}{M} \vee M = \mathbb{1} \tag{167}$$

as the defining equation for a motor in *PGA* (remember: $M^{-1} = \underset{\sim}{M}$), leads to

$$\frac{d}{dt}(M \vee \underset{\sim}{M}) = \dot{M} \vee \underset{\sim}{M} + M \vee \dot{\underset{\sim}{M}} = 0 \tag{168}$$

It follows that $M \vee \dot{\underset{\sim}{M}} = -\dot{M} \vee \underset{\sim}{M}$, which results in

$$\dot{\underset{\sim}{M}} = -\underset{\sim}{M} \vee \dot{M} \vee \underset{\sim}{M} \tag{169}$$

Starting from $\underset{\sim}{M} \vee M = \mathbb{1}$ instead yields the same result for $\dot{\underset{\sim}{M}}$.

However, we still have to take additional aspects into account:

First, Lengyel has shown that a motor in $3D$ *PGA* can be created from the exponential in eqn. (146) as $M = \exp_{\vee}\left[(\frac{\delta}{2}\mathbf{1} + \frac{\phi}{2}\mathbb{1}) \vee \boldsymbol{l}\right]$ where it is easy to show that the regressive geometric product of the line $\boldsymbol{l}$, a bivector, with the scalar part yields a bivector and the same is true for the regressive geometric product between the pseudoscalar and the line. Thus, it is sufficient to consider the exponential function of a bivector argument for the derivation for $3D$ *PGA*. The resulting motor M will be an even-grade multivector in this case.

Second, for $2D$ *PGA* the motor is derived from $M = \exp_{\vee}\left[\frac{\phi}{2}P\right]$ with the unitized point P, a vector, as an argument. The resulting motor M will be an odd-grade multivector. We need to distinguish these two cases later

on. However, since the equations are structurally similar (an exponential function with either a vector or a bivector as argument), we only do the derivation for the $3D$ case once and then will interpret the results for the $2D$ case separately afterwards.

World frame: From eqn. (146) we know the form of a general motor in PGA and use the ansatz for a time dependent motor $M(t)$ after the object X was brought into its initial position by M_0 as $X_0 = M \vee X \vee \underset{\sim}{M}$. If there is no difference between world and body frames at $t = 0$ then $M_0 = \mathbb{1}$:

$$M(t) = \exp_{\vee}(\frac{1}{2}B_w(t)) \vee M_0 \tag{170}$$

with $B_w(t)$ as a time-dependent bivector that models the rotational plane and the turning angle. The derivative of M with respect to time t is

$$\dot{M}(t) = \frac{1}{2}\dot{B}_w(t) \vee \exp_{\vee}(\frac{1}{2}B_w(t)) \vee M_0 = \frac{1}{2}\dot{B}_w(t) \vee M(t) \tag{171}$$

where the rate $\dot{B}_w$ was moved in front of the exponential, which requires B_w and $\dot{B}_w$ to commute. The condition is the one derived for the rotor (156) and it is no weaker here: the generator of a motor is a screw, and two screws commute when they share their axis — pure translations commute with one another whatever their directions, since they form an abelian subalgebra — so the step is exact for a motion about a *fixed* screw axis and for a purely translational one, and approximate otherwise. A rotation about a parallel but offset axis already fails it. The plane $PGA2DP$ is no exception: its generators form the non-commutative $\mathfrak{se}(2)$, in which a rotation and a translation never commute (199).

The rate in the world frame is therefore *defined* by

$$\dot{M} = \frac{1}{2}\Omega_w \vee M \tag{172}$$

which holds unconditionally, M being a unit motor. For a fixed screw axis $\Omega_w = \dot{B}_w$; otherwise the two differ by the differential of the exponential, here

$$\Omega_w = \dot{B}_w + \frac{1}{2}\mathrm{rcmt}(B_w, \dot{B}_w) + \dots \tag{173}$$

the sign following the $+\frac{1}{2}B_w$ in the exponent, against the $-\frac{1}{2}B_w$ of the rotor in (156).
By multiplying this from the right with "$\vee \underset{\sim}{M}$" this can be transformed to

$$\Omega_w = 2\dot{M} \vee \underset{\sim}{M} \tag{174}$$

Body frame: Here we first move the object in the *body frame* and then transform it into the world frame by the transformation M_0. In this case eqn. (170) becomes

$$M(t) = M_0 \vee \exp_\vee(\frac{1}{2}B_b(t)) \tag{175}$$

which in turn results in

$$\dot{M}(t) = \frac{1}{2}M_0 \vee \exp_\vee(\frac{1}{2}B_b(t)) \vee \dot{B}_b(t) = \frac{1}{2}M(t) \vee \dot{B}_b(t) \tag{176}$$

where the rate $\dot{B}_b$ was moved behind the exponential, which again requires B_b and $\dot{B}_b$ to commute. The rate in the *body frame* is therefore *defined* by

$$\dot{M} = \frac{1}{2}M \vee \Omega_b \tag{177}$$

which holds unconditionally. Ω_b is the same velocity as Ω_w, expressed in the body frame instead of the world frame (eqn. 180a below). For a fixed screw axis $\Omega_b = \dot{B}_b$; otherwise

$$\Omega_b = \dot{B}_b - \frac{1}{2}\mathrm{rcmt}(B_b, \dot{B}_b) + \dots \tag{178}$$

with the sign opposite to eqn. (173), because the exponential now stands to the right of M_0. The test case "pga2dp / pga3dp: the rate of a motor and the rate of its generator" in `ga_test/src/ga_pga_contract_test.hpp` takes both rates from the motor itself. For motions whose generators commute — one screw axis with any pitches, translations in any directions, in $2D$ rotations about one point — they equal $\dot{B}$. For rotations about two parallel axes, and in $2D$ for a rotation together with a translation, $\dot{B}$ is off at first order, and eqns. (173) and (178) hold at second order. Multiplying eqn. (177) from the left by "$\underset{\sim}{M} \vee$" gives

$$\Omega_b = 2\underset{\sim}{M} \vee \dot{M} \tag{179}$$

From equations (172) and (177) we can see how world and body frame rates are connected to each other: $\Omega_w \vee M = M \vee \Omega_b$, which can be transformed into

$$\Omega_w = M \vee \Omega_b \vee \underset{\sim}{M} \tag{180a}$$
$$\Omega_b = \underset{\sim}{M} \vee \Omega_w \vee M \tag{180b}$$

Now we derive the rate of change of a point $X(t)$, i.e. its speed $\dot{X}(t)$, in the *world frame*, first assuming that $X(t = 0) = X_0 = const.$, i.e. as a constant

element that is transformed by a time dependent motor $M(t)$. During the derivation we will use the associativity of the regressive geometric product and the neutral element $\mathbb{1}$ of the regressive geometric product. We can write

$$\begin{aligned}
\dot{X}(t) &= \frac{d}{dt}X(t) = \frac{d}{dt}(M \vee X_0 \vee \underset{\sim}{M}) \\
&= \dot{M} \vee X_0 \vee \underset{\sim}{M} + M \vee X_0 \vee \underset{\sim}{\dot{M}} \\
&\overset{(169)}{=} \dot{M} \vee \mathbb{1} \vee X_0 \vee \underset{\sim}{M} - M \vee X_0 \vee (\underset{\sim}{M} \vee \dot{M} \vee \underset{\sim}{M}) \\
&\overset{(167)}{=} \dot{M} \vee (\underset{\sim}{M} \vee M) \vee X_0 \vee \underset{\sim}{M} - (M \vee X_0 \vee \underset{\sim}{M}) \vee \dot{M} \vee \underset{\sim}{M} \\
&= (\dot{M} \vee \underset{\sim}{M}) \vee (M \vee X_0 \vee \underset{\sim}{M}) - (M \vee X_0 \vee \underset{\sim}{M}) \vee (\dot{M} \vee \underset{\sim}{M})
\end{aligned}$$

From eqn. (174) we can conclude that $\frac{1}{2}\Omega_w(t) = \dot{M} \vee \underset{\sim}{M}$, resulting in

$$\begin{aligned}
\dot{X}(t) &= (\frac{1}{2}\Omega_w(t)) \vee X_w(t) - X_w(t) \vee (\frac{1}{2}\Omega_w(t)) \\
&= \frac{1}{2}(\Omega_w \vee X_w - X_w \vee \Omega_w) \\
\\
\dot{X}_w &= \text{rcmt}(\Omega_w, X_w) = -\text{rcmt}(X_w, \Omega_w) \qquad (181)
\end{aligned}$$

with $\text{rcmt}(\Omega_w, X_w)$ as the regressive commutator product between Ω_w and X_w. The resulting minus sign in eqn. (181) compared to eqn. (164) is the only difference between PGA and EGA. This is caused by the different ansatz for the motors comparing equations (140) and (146).

It can be shown that $\dot{M} \vee \underset{\sim}{M}$ is either always vector or a bivector: From equations (174) and (168), and the linearity of the involved operations, we get

$$\Omega_w = 2\dot{M} \vee \underset{\sim}{M} = -2M \vee \underset{\sim}{\dot{M}} = -2(\underset{\sim}{\dot{M} \vee \underset{\sim}{M}}) = -\underset{\sim}{\Omega}_w \qquad (182)$$

Considering the grade-dependent sign-reversal of the regressive reverse in $3D$ PGA, Ω_w must either be a vector or a bivector (see table 12 at the end of section 1.2). Since $\dot{M}$ and $\underset{\sim}{M}$ are even-grade multivectors, their regressive geometric product results in an even-grade multivector as well, which does not have a vector part. Thus, Ω_w and $\dot{M} \vee \underset{\sim}{M}$ must indeed be a bivector in $3D$ PGA.

However, in $2D$ PGA due to the grade-dependent sign-reversal of the regressive reverse, Ω_w must either be a scalar or a vector (see table 8 at the end of section 1.2). Since $\dot{M}$ and $\underset{\sim}{M}$ are odd-grade multivectors, their regressive geometric product results in an odd-grade multivector as well, which does not

have a scalar part. Thus, Ω_w and $\dot{M} \vee \underset{\sim}{M}$ must indeed be a vector in $2D$ PGA.

If X_0 is not a constant element, but a function of time, which means that X_0 is a time dependent element in the *body frame* $X_0 = X_b(t)$, we have to add another term to the derivative in eqn. (181) as follows:

$$\begin{aligned} \dot{X}(t) &= \frac{d}{dt}(M \vee X_b \vee \underset{\sim}{M}) \\ &= \dot{M} \vee X_b \vee \underset{\sim}{M} + M \vee X_b \vee \dot{\underset{\sim}{M}} + M \vee \dot{X}_b \vee \underset{\sim}{M} \\ &= -\mathrm{rcmt}(M \vee X_b \vee \underset{\sim}{M}, \Omega_w) + M \vee \dot{X}_b \vee \underset{\sim}{M} \\ \dot{X}_w &= -\mathrm{rcmt}(X_w, \Omega_w) + (\dot{X}_b)_w = \mathrm{rcmt}(\Omega_w, X_w) + (\dot{X}_b)_w \end{aligned} \tag{183}$$

Eqn. (183) is to be considered in the world frame. However, we can also do our derivations in a completely equivalent way using the corresponding bivector rates and rotors of the *body frame*.

The derivation in the *body frame* does not bring new insights, so we don't reproduce that here for PGA again.

1.4.3 Motor generators in *PGA*

The final step before we move on to modelling mechanics is to show how points are moved in *PGA* using the approach described above.

In $3D$ *PGA* every bivector $B = B(s, \phi)$ or every bivector rate $\dot{B} = \Omega = \Omega(v, \omega)$ has its basis components

$$\Omega_{3D} = \underbrace{\Omega_{vx}\boldsymbol{e}_{41} + \Omega_{vy}\boldsymbol{e}_{42} + \Omega_{vz}\boldsymbol{e}_{43}}_{\text{weight}(\Omega)=\Omega_\circ} + \underbrace{\Omega_{mx}\boldsymbol{e}_{23} + \Omega_{my}\boldsymbol{e}_{31} + \Omega_{mz}\boldsymbol{e}_{12}}_{\text{bulk}(\Omega)=\Omega_\bullet} \tag{184}$$

where the *weight* $\Omega_\circ$ *carries rotational information* and the *bulk* $\Omega_\bullet$ *carries translational information.*

In general, the weight only contains components that *do* contain a factor of $\boldsymbol{e}_4$. It has *directional* information on Ω. If there is zero weight, the object is contained in the horizon (H_{3dp}). The bulk only has components that *do not* contain a factor of $\boldsymbol{e}_4$. It has *positional* information on Ω. If there is zero bulk, the object contains the origin (O_{3dp}).

We can either model a pure translation, which will only be visible in the bulk part of Ω, with the weight part being zero in that case, or we can model a pure rotation around a fixed (non-ideal) line $\boldsymbol{l}$. In that case the positional information of the fixed line of rotation becomes encoded in the bulk part of Ω and the directional information on the rotation speed vector ω is encoded in the weight part of Ω.

A mixture of a translation and a rotation would either be equivalent to a rotation around another fixed line shifted by the translational part or, in case of translation along the axis of rotation, a screw motion around and along that axis. It is sufficient to look at pure translation and pure rotation around a fixed line as separate cases. This separation is also important for defining the exponential function with respect to the regressive geometric product that converts the bivector B to the respective motor M: $M(t) = \exp_{\vee}(\frac{1}{2}B(t)) = \exp_{\vee}(\frac{1}{2}\Omega t)$.

The total bivector rate Ω can be split into

$$\Omega = \Omega_\circ + \Omega_\bullet = \Omega_{rot} + \Omega_{tra} \tag{185}$$

where eqn. (185) is called the "Euclidean split" of the bivector rate Ω because it separates the rotational and translational aspects of a moving object.

A pure rotation around an axis $\vec{\omega} = \omega$ can be modelled by a fixed (unitized) line that is scaled by the angular speed of rotation, i.e. by the bulk norm $\|\omega\|_\bullet$. If we have an arbitrary rotation axis we can either wedge a unitized point $Q = O_{3dp} + q = \boldsymbol{e}_4 + q$ that is contained in the rotational axis with the rotational speed ω, or create a unitized line $\hat{\omega}$ in the direction of the angular velocity vector ω that is later scaled with the rotational speed such that the bivector Ω_{rot} becomes

$$\begin{aligned}
\Omega_{rot} &= Q \wedge \omega = (O_{3dp} + q) \wedge \omega \\
&= \boldsymbol{e}_4 \wedge \omega + q \wedge \omega \\
&= \|\omega\|_\bullet (\boldsymbol{e}_4 \wedge \hat{\omega} + q \wedge \hat{\omega}) \\
\Omega_{rot} &= \|\omega\|_\bullet \begin{pmatrix} \hat{\omega}_x \\ \hat{\omega}_y \\ \hat{\omega}_z \\ q_y\hat{\omega}_z - q_z\hat{\omega}_y \\ q_z\hat{\omega}_x - q_x\hat{\omega}_z \\ q_x\hat{\omega}_y - q_y\hat{\omega}_x \end{pmatrix}
\end{aligned} \tag{186}$$

where $\hat{\omega}$ is the bulk normalized angular velocity vector $\hat{\omega} = \omega/\|\omega\|_\bullet$. Thus, Ω_{rot} directly represents the normalized (non-ideal) axis of rotation. It is a line of fixed points not changing due to the rotation, scaled with the angular speed of rotation $\|\omega\|_\bullet$.

Of course, Ω_{rot} could also be expressed directly by the plane of rotation pl (a trivector) when the magnitude of the plane encodes the angular speed $\|\omega\|_\bullet$. Considering that the left weight dual of the plane $pl_\star$ provides the normal vector of the plane including its magnitude, eqn. (186) can be written in the equivalent form directly in terms of the plane of rotation pl

$$\Omega_{rot} = Q \wedge pl_\star \tag{187}$$

Which of the two weight duals is taken matters: $pl^\star$ returns the *opposite* normal, so it generates the reverse rotation, and the left one is the choice that makes eqn. (187) agree with eqn. (186). The same normal in Euclidean form is the plane's weight part read off directly, `to_vec3d(pl)`, from which the ideal vector above is the lift `vec3dp(to_vec3d(pl),0.0)`.

For modelling translational speed $\vec{v} = v$ in $3D$ we need a bivector rate Ω_{tra} that does not contain a factor of $\boldsymbol{e}_4$. This can be achieved by encoding the speed of translation in the normal vector n_v of a plane and then intersecting this plane with the horizon $\overline{\boldsymbol{e}_4} = \overline{O_{3dp}} = H_{3dp}$. The plane with the corresponding attitude can be encoded by using the right bulk dual $v^\star$ to model the

plane (a trivector) with the bulk norm $\|v\|_\bullet$ as magnitude. It contains the origin O_{3dp} and results in a bivector rate Ω_{tra} of the translation as

$$\Omega_{tra} = v^\star \vee \overline{\boldsymbol{e}_4} = v^\star \vee H_{3dp} = att(v^\star) = (0,0,0,v_x,v_y,v_z) \tag{188}$$

For the case of screw motion we have the special case that the translational motion is directed in the same direction as the rotation axis, i.e. we have a special Ω_{tra}

$$\Omega_{tra,screw} = \frac{\delta}{\|\omega\|_\bullet} att(\omega^\star) = \delta\,(0,0,0,\hat{\omega}_x,\hat{\omega}_y,\hat{\omega}_z) \tag{189}$$

where $\hat{\omega}$ is the bulk normalized vector ω, which has magnitude one. This vector is scaled with the translation distance δ resulting in Ω_{screw} of

$$\Omega_{screw} = \Omega_{rot} + \Omega_{tra,screw} = \|\omega\|_\bullet \begin{pmatrix} \hat{\omega}_x \\ \hat{\omega}_y \\ \hat{\omega}_z \\ q_y\hat{\omega}_z - q_z\hat{\omega}_y \\ q_z\hat{\omega}_x - q_x\hat{\omega}_z \\ q_x\hat{\omega}_y - q_y\hat{\omega}_x \end{pmatrix} + \delta \begin{pmatrix} 0 \\ 0 \\ 0 \\ \hat{\omega}_x \\ \hat{\omega}_y \\ \hat{\omega}_z \end{pmatrix} \tag{190}$$

The components of Ω_{rot}, Ω_{tra} and Ω_{screw} are relevant for the exponential function that transforms our bivector, representing the intended motion, into a motor that transforms points accordingly.

Using for example eqn. (181) and the Euclidean split as defined in eqn. (185) we can write

$$\begin{aligned} \dot{X}_w &= \mathrm{rcmt}(\Omega_w, X_w) = \dot{x}_w \\ &= \mathrm{rcmt}(\Omega_{rot} + \Omega_{tra}, X_w) \\ \dot{x}_w &= \mathrm{rcmt}(\Omega_{rot}, X_w) + \mathrm{rcmt}(\Omega_{tra}, X_w) && \text{(191a)} \\ \dot{x}_w &= v_{rot} + v_{tra} && \text{(191b)} \end{aligned}$$

to calculate the combined speed $\dot{x}_w$ from rotational and translational effects at a position X_w.

In $2D$ PGA the equations are very similar, but the rate of change is not a rate of change bivector, but is it a rate of change vector, since the rotation is not around a fixed axis, as in $3D$, but it is around a fixed point. The translation is characterized by a $2D$ vector, whereas the rotation rate can be characterized by a single scalar value and a fixed point Q around with the rotation occurs. To show that we still get comparable equations to the $3D$ case, we will also use $\Omega = \Omega(v, \omega)$ for the rate of change. In $2D$ PGA has its basis components

$$\Omega_{2D} = \underbrace{\Omega_x \boldsymbol{e}_1 + \Omega_y \boldsymbol{e}_2}_{\text{bulk}(\Omega)=\Omega_\bullet} + \underbrace{\Omega_z \boldsymbol{e}_3}_{\text{weight}(\Omega)=\Omega_\circ} \tag{192}$$

where the *weight* $\Omega_\circ$ *carries rotational information* and the *bulk* $\Omega_\bullet$ *carries translational information*. In general, the weight only contains components that *do* contain a factor of $\boldsymbol{e}_3$ (which here takes a role equivalent to $\boldsymbol{e}_4$ in the $3D$ case). The bulk only has components that *do not* contain a factor of $\boldsymbol{e}_3$.

The total vector rate of change Ω can be split into

$$\Omega = \Omega_\bullet + \Omega_\circ = \Omega_{tra} + \Omega_{rot} \tag{193}$$

where eqn. (193) is called the "Euclidean split" of the vector rate of change Ω separating the translational and rotational aspects of a moving object.

A simple rotation in $2D$ with an angular speed ω (a simple scalar in this case) around the unitized point $Q = O_{2dp} + q = \boldsymbol{e}_3 + q$ can be modelled by multiplying the point Q with ω using the wedge product to generate the vector Ω_{rot}.

$$\Omega_{rot} = Q \wedge \omega = \boldsymbol{e}_3 \wedge \omega + q \wedge \omega = (0, 0, 1)\,\omega + (q_x, q_y, 0)\,\omega = (q_x, q_y, 1)\,\omega \tag{194}$$

For modelling translational speed $\vec{v} = v$ in $2D$ we need a vector rate of change Ω_{tra} that does not contain a factor of $\boldsymbol{e}_3$. To show this we use exactly the same equations as in the $3D$ case by encoding the speed of translation in the normal vector n_v of a line and then intersecting this line with the horizon $\overline{\boldsymbol{e}_3} = \overline{O_{2dp}} = H_{2dp}$. The double reflection on parallel lines that effectively creates the translation, necessitates the need for encoding the translation speed in a vector pointing normal to the translation direction. The line with the corresponding attitude can be encoded by using the right bulk dual $v^\star$ to model the line (a bivector) with the bulk norm $\|v\|_\bullet$ as magnitude. The vector rate of change Ω_{tra} of a pure translation is then given as

$$\Omega_{tra} = v^\star \vee \overline{\boldsymbol{e}_3} = v^\star \vee H_{3dp} = att(v^\star) = (-v_y, v_x, 0) \tag{195}$$

where the direction of the movement is encoded as the normal pointing towards the point at infinity which is the "rotation center at infinity" of the modelled translation.

We can either model a pure translation, which will only be visible in the bulk part of Ω, with the weight part being zero in that case, or we can model a pure rotation around a fixed (non-ideal) point Q. In that case the positional information of the fixed point becomes encoded in the bulk part of Ω and the rotational speed ω is encoded in the weight part of Ω. A mixture of a translation and a rotation would be equivalent to a rotation around another fixed point shifted by the translational part. So it is sufficient to look at pure translation and pure rotation around a fixed point as separate cases.

$$\Omega_{2D} = \Omega_{rot} + \Omega_{tra} = \omega \begin{pmatrix} q_x \\ q_y \\ 1 \end{pmatrix} + \begin{pmatrix} -v_y \\ v_x \\ 0 \end{pmatrix} \tag{196}$$

This separation is also important for defining the exponential function with respect to the regressive geometric product that converts the bivector B to the respective motor M: $M(t) = \exp_{\vee}(\frac{1}{2}B(t)) = \exp_{\vee}(\frac{1}{2}\Omega t)$ that enables us to actually transform points according to the vector rates of change as defined above.

Caution: The additive split $\Omega = \Omega_{rot} + \Omega_{tra}$ is a statement about *generators* — the velocity level, where rates add and the components are independent. Finite motions do not add. One must not read this split as a recipe for building a combined finite motion by adding generators (equivalently, by assuming the two motors commute); that is valid only in one special case. The next section 1.4.4 makes this precise.

1.4.4 Independent generators, coupled finite displacements

The motion generator Ω_b lives in the *Lie algebra* $\mathfrak{se}(2)$ $(2D)$ or $\mathfrak{se}(3)$ $(3D)$. The motor M itself lives in the corresponding *Lie group* $SE(2)$ or $SE(3)$. These two levels have fundamentally different structure. The vocabulary is that of Lie theory — algebra and group, the exponential and logarithmic maps between them, the adjoint and the bracket — and [Solà et al., 2018] states it compactly in the form the robotics literature uses, with the groups written as matrices; the correspondence to the motors used here is taken up in subsubsection 1.7.13. The Lie algebra $\mathfrak{se}(2)$ is three-dimensional with basis generators:

Generator	`vec2dp` encoding	Effect
Translation in x	$\{0,\ 1,\ 0\}$	body drifts in $+x$
Translation in y	$\{-1,\ 0,\ 0\}$	body drifts in $+y$
Rotation about origin	$\{0,\ 0,\ 1\}$	body spins about world origin

Table 26: Basis generators of $\mathfrak{se}(2)$ in $PGA2DP$ encoding.

A general instantaneous velocity $\Omega_b = \{-v_y,\ v_x,\ \omega\}$ means simultaneously: translate at (v_x, v_y) and rotate at ω – with no coupling between the three components. Integrating the velocity over an infinitesimal step dt gives

$$B_b \leftarrow B_b + \Omega_b \cdot dt, \tag{197}$$

so all three components of B_b accumulate independently.

When computing a *finite* displacement via $M = \exp_\vee(\frac{1}{2}B_b)$, the structure of the $2D$ plane imposes a hard constraint:

> **Chasles' theorem ($2D$):** Any rigid body displacement in $2D$ is either a pure rotation about some center, or a pure translation. There is no screw motion.

Consequently, for any B_b with $B_b.z \neq 0$, the motor $\exp_\vee(\frac{1}{2}B_b)$ is always a pure rotation – the translational components merely determine the rotation center:

$$\text{pivot} = \left(\frac{B_b.x}{B_b.z},\ \frac{B_b.y}{B_b.z}\right). \tag{198}$$

The "translation" encoded in $B_b.x$ and $B_b.y$ does not produce an independent translational displacement; it shifts the rotation center.

At the velocity level (Lie algebra) the components are independent. At the displacement level (Lie group) they are not.

In $3D$ the Lie algebra $\mathfrak{se}(3)$ is six-dimensional: three components for the rotation axis direction, three for the translation velocity. Chasles' theorem for $3D$ states:

> **Chasles' theorem ($3D$):** Any rigid body displacement in $3D$ is a *screw motion*: a rotation about an axis combined with a translation *along* that same axis (helical motion).

Exponentiating a combined $3D$ generator therefore yields a screw – rotation and translation are simultaneously present in the finite displacement. The rotation axis direction and the translational pitch are independently adjustable because there exists a direction (along the axis) that is geometrically orthogonal to the plane of rotation. In $2D$ no such direction exists: the axis is perpendicular to the entire plane, so simultaneous rotation and translation always collapse to rotation about a different center.

Lie algebra level	*PGA2DP*	*PGA3DP*
Dimension of $\mathfrak{se}$	3	6
Velocity generator Ω_b	$(-v_y,\ v_x,\ \omega)$	$(v_x,\ v_y,\ v_z,\ \omega_x,\ \omega_y,\ \omega_z)$
Components independent?	yes	yes

Table 27: Velocity generators at the Lie algebra level.

Lie group level	*PGA2DP*	*PGA3DP*
Result of $\exp_\vee(\frac{1}{2}B)$	pure rotation	screw motion
Finite pivot / screw axis	point $\left(\frac{B.x}{B.z},\ \frac{B.y}{B.z}\right)$	line (Plücker coordinates)
Role of "translation" in B	shifts rotation center	adds pitch along axis

Table 28: Finite displacements at the Lie group level.

The accumulated generator B_b (and B_w) is a Lie algebra element – its components are additive and independent *as generators*. The pivot decoded from B_b as $(B_b.x/B_b.z,\ B_b.y/B_b.z)$ is meaningful as the instantaneous rotation center of the *finite* displacement $\exp_\vee(\frac{1}{2}B_b)$, not of the instantaneous velocity Ω_b. When $B_b = B_{b,0} + \Omega_b \cdot t$ (uniform motion from initial condition $B_{b,0}$):

- If $B_{b,0} = \mathbf{0}$: the decoded pivot equals the body-frame pivot of Ω_b for all t, i.e. $(B_b.x/B_b.z,\ B_b.y/B_b.z) = (\Omega_b.x/\Omega_b.z,\ \Omega_b.y/\Omega_b.z) = Q_b$ – constant and geometrically meaningful.
- If $B_{b,0} \neq \mathbf{0}$ with $B_{b,0}.z = 0$ (a translational initial offset): the decoded pivot starts at infinity (pure translation at $t = 0$) and converges asymptotically to Q_b as $t \to \infty$. This makes the decoded pivot hard to interpret and should be avoided in visualizations intended to explain the encoding.

Design rule: use $B_{b,0} = \mathbf{0}$ (body starts at the M_0 reference pose) so that the accumulated generator retains a clean geometric meaning.

Composing motions — multiply motors, do not add generators: The same caveat governs how separate motions *compose*. A finite motion is applied by its motor $M = \exp_{\vee}(\frac{1}{2}B)$, and two successive motions compose by the regressive geometric product (motor multiplication), which is order-dependent:

$$M_2 \vee M_1 \neq M_1 \vee M_2 \tag{199a}$$

$$\text{and} \quad \exp_{\vee}\big(\tfrac{1}{2}(A+B)\big) \neq \exp_{\vee}\big(\tfrac{1}{2}A\big) \vee \exp_{\vee}\big(\tfrac{1}{2}B\big) \tag{199b}$$

$$\text{unless} \quad \operatorname{rcmt}(A,B) = 0 \tag{199c}$$

Adding generators equals multiplying motors only when the generators commute under the regressive product. For a pure translation and a pure rotation this happens *exclusively* in the $3D$ screw case (translation *along* the rotation axis); in $2D$ a nonzero rotation and a translation never commute (Chasles: the result is always a pure rotation about a shifted center). Consequently, writing $B = B_{rot} + B_{tra}$ and exponentiating yields the *single* screw — or, in $2D$, the single rotation about a shifted pivot — generated by that combined twist; it is *not* the composition "translate, then rotate". To realize that composition, multiply the two motors in the intended order, $M = M_{rot} \vee M_{tra}$. Fig. 9 shows the three results for a translation and a quarter turn.

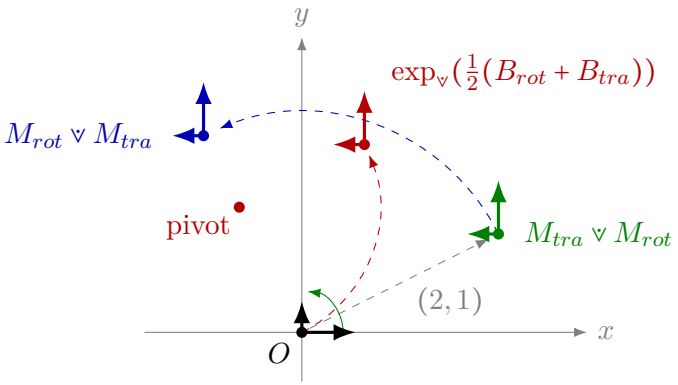


Figure 9: A translation by $(2,1)$ and a turn by 90° about O, combined in three ways; the body frame starts at O (black). Translating, then rotating (blue) ends at $(-1,2)$; rotating in place, then translating (green) ends at $(2,1)$; both have the same orientation. Adding the generators (red) gives a single rotation about the pivot $(-2/\pi, 4/\pi)$ of eqn. (198), which ends at $(2/\pi, 6/\pi)$. The library uses the green order throughout: a frame's motor is built from its pose as $M = M_{tra} \vee M_{rot}$, the rotation about the parent's origin first (`motor_from_pose2dp`, `motor_from_pose3dp`), and along a kinematic chain the parent's motor multiplies from the left, $M_{world} = M_{parent} \vee M$ (subsection 1.7.7).

1.4.5 Reading an angle off a motor

Composition and the sandwich act on the motor itself and hold for any number of turns. Turning a motor back into numbers — an angle, a heading, the error of a controller — goes through the logarithm, which is one-to-one only on part of the group. Three of its properties account for the errors met in applications. They are shown on a motor turning by ψ about a fixed unit line $\boldsymbol{l}$, $Y(\psi) = \exp_{\vee}(\frac{1}{2}\psi\boldsymbol{l})$, whose angle is read back as the weight of $2\,\mathrm{rlog}(Y)$. The numbers read off a motor are the components of its *generator*, $B = 2\,\mathrm{rlog}(M)$ with $M = \exp_{\vee}(\frac{1}{2}B)$, the screw of the whole motion. As for the rate Ω in eqn. (186), the weight of B carries the rotation: the direction of its axis line scaled by the angle. Classical texts call these three numbers the rotation vector, axis times angle; here they are the weight of the generator.

- *Range and double cover.* `rlog` inverts `rexp` on the motors, on both sheets: the angle read lies in $(-2\pi, 2\pi]$ (outside a narrow band at a full turn, see the conditioning below), so 270° reads 270° and 361° reads −359°. M and $-M$ move every object identically, since the sandwich is quadratic in M, and their logarithms differ by a full turn. The motor of a quarter turn, $Y(90°)$, reads 90°; its negative $-Y(90°)$ — the same motion, which is also $Y(450°)$, a quarter turn and a full turn — reads −270°. The number of completed turns is not contained in a motor.

- *Non-additivity.* By eqn. (199) the logarithm of a product is not the sum of the logarithms, so the components of an *absolute* logarithm do not name the motions they appear to. The paragraph "A small roll under a heading" below works this out.

- *Conditioning.* The logarithm passes the rounding of the motor's components on magnified by $(\varphi/\sin\varphi)$, φ the half angle, which grows without bound towards a full turn. Within a narrow band there `rlog` reads $-M$, the same motion the short way round; the paragraph "Conditioning near a full turn" below derives this for PGA and CGA.

The relative motor: The remedy for all three is to take the logarithm of a motion near the identity: the motion between two poses rather than a pose itself. The *relative motor* is the motor that carries a reference pose M_0 to the pose M. Like the motion in subsubsection 1.4.2, it is stated either in the body frame or in the world frame (eqns. 175 and 170):

$$M = M_0 \vee M_b = M_w \vee M_0, \qquad M_b = \exp_{\vee}(\tfrac{1}{2}B_b), \quad M_w = \exp_{\vee}(\tfrac{1}{2}B_w), \quad (200)$$

with M_b the relative motor in the body frame and M_w the relative motor in the world frame, B_b and B_w their generators. Which frame a relative motor is stated in follows from the side of M_0 it stands on. In a product the right factor acts first: with $\underset{\sim}{A \vee B} = \underset{\sim}{B} \vee \underset{\sim}{A}$ (eqn. 63b) the sandwich of a product is the sandwich of each factor in turn,

$$(A \vee B) \vee X \vee \underset{\sim}{(A \vee B)} = A \vee \big(B \vee X \vee \underset{\sim}{B}\big) \vee \underset{\sim}{A}. \tag{201}$$

A pose motor maps an object X, given in the coordinates of the body frame, into the world frame: $X \mapsto M \vee X \vee \underset{\sim}{M}$. With $M = M_0 \vee M_b$, eqn. (201) applies M_b first, while X is still expressed in the body frame, and M_0 then maps the moved object into the world frame: M_b is a motion stated in the body frame. With $M = M_w \vee M_0$, M_0 first maps X into the world frame, and M_w then moves it there: M_w is a motion stated in the world frame. Each equation of (200) is solved for the relative motor by multiplying with $\underset{\sim}{M_0}$ on the side of M_0, since $\underset{\sim}{M_0} \vee M_0 = M_0 \vee \underset{\sim}{M_0} = \mathbb{1}$ is the neutral element of $\vee$ (eqn. 167):

$$\begin{aligned} M = M_0 \vee M_b \quad &\Rightarrow \quad \underset{\sim}{M_0} \vee M = \big(\underset{\sim}{M_0} \vee M_0\big) \vee M_b = M_b, \\ M = M_w \vee M_0 \quad &\Rightarrow \quad M \vee \underset{\sim}{M_0} = M_w \vee \big(M_0 \vee \underset{\sim}{M_0}\big) = M_w. \end{aligned} \tag{202}$$

The relative motors are therefore (fig. 10 shows both for a translation)

$$M_b = \underset{\sim}{M_0} \vee M, \qquad M_w = M \vee \underset{\sim}{M_0}. \tag{203}$$

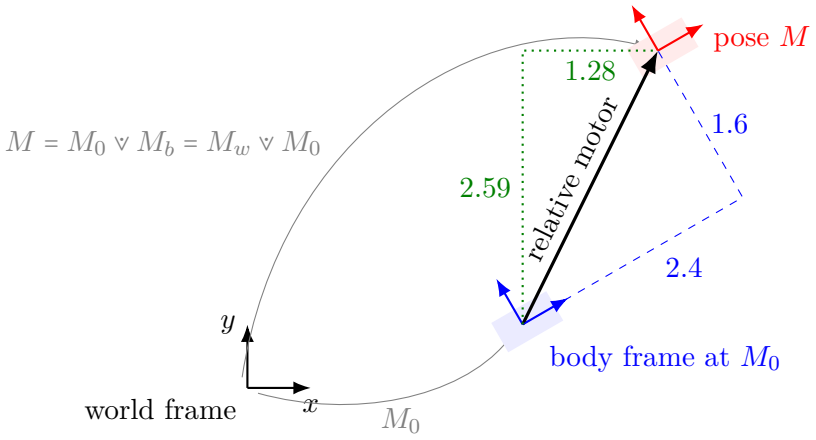


Figure 10: The relative motor of a translation. The body frame at the reference pose M_0 (blue) is turned by 30° against the world frame; the relative motor (black) carries it to the pose M (red). Stated in the body frame, $M_b = \underset{\sim}{M_0} \vee M$ is the translation $(2.4,\ 1.6)$ (blue); stated in the world frame, $M_w = M \vee \underset{\sim}{M_0} = M_0 \vee M_b \vee \underset{\sim}{M_0}$ is the translation $(1.28,\ 2.59)$ (green). One arrow, read in two frames.

Both are the identity $\mathbb{1}$ when $M = M_0$. Equating the two products for M in eqn. (200), $M_w \vee M_0 = M_0 \vee M_b$, and multiplying from the right with $\underset{\sim}{M}_0$ relates them:

$$M_w = M_0 \vee M_b \vee \underset{\sim}{M}_0, \qquad B_w = M_0 \vee B_b \vee \underset{\sim}{M}_0. \tag{204}$$

The second equation holds because the sandwich with a motor G commutes with the exponential: it maps every power in the series term by term, $G \vee B^k \vee \underset{\sim}{G} = (G \vee B \vee \underset{\sim}{G})^k$ with $\underset{\sim}{G} \vee G = \mathbb{1}$, so $G \vee \exp_\vee(B) \vee \underset{\sim}{G} = \exp_\vee(G \vee B \vee \underset{\sim}{G})$; and the sandwich leaves the pseudoscalar component, which selects the sheet, unchanged. So the generators $B_b = 2\,\mathrm{rlog}(M_b)$ and $B_w = 2\,\mathrm{rlog}(M_w)$ are the same screw — the same angle and pitch — about axis lines carried from one frame into the other, as the rates are in eqn. (180a).

A small roll under a heading: Let $\boldsymbol{l}_x$, $\boldsymbol{l}_y$ and $\boldsymbol{l}_z$ be the unit lines through the origin along the axes, $Y(\psi) = \boldsymbol{l}_z \sin a + \mathbb{1} \cos a$ a heading and $R(\varepsilon) = \boldsymbol{l}_x \sin b + \mathbb{1} \cos b$ a roll about the body's x axis, with the half angles $a = \psi/2$ and $b = \varepsilon/2$ (eqn. 147). Their product is

$$\begin{aligned} M = Y(\psi) \vee R(\varepsilon) &= w\,\mathbb{1} + v_x \boldsymbol{l}_x + v_y \boldsymbol{l}_y + v_z \boldsymbol{l}_z, \\ w &= \cos a \cos b, \\ (v_x,\ v_y,\ v_z) &= (\cos a \sin b,\ \sin a \sin b,\ \sin a \cos b), \end{aligned} \tag{205}$$

a rotation about the line through the origin with the direction v, by the angle $2\,\mathrm{atan2}(|v|, w)$. The horizontal part of $2\,\mathrm{rlog}(M)$ — the angle times the unit direction — is therefore

$$h = \frac{2\,\mathrm{atan2}(|v|, w)}{|v|} \sin b\,(\cos a,\ \sin a), \qquad |v|^2 = \sin^2 a + \cos^2 a \sin^2 b. \tag{206}$$

It points at $\psi/2$ in the world frame, whatever ε: halfway between the x axis of the world frame and that of the body frame, which lies at ψ. Its length has two limits. For $\varepsilon \to 0$, $|v| \to \sin a$ and the angle tends to ψ, so

$$\frac{|h|}{\varepsilon} \to \frac{\psi/2}{\sin(\psi/2)}, \tag{207}$$

which is the first order of $\mathrm{rlog}(Y \vee R) = \mathrm{rlog}(Y) + J^{-1}\,\mathrm{rlog}(R) + O(\varepsilon^2)$, with J^{-1} the inverse Jacobian of the exponential at $\mathrm{rlog}(Y)$ [Solà et al., 2018]: it turns a small rotation by $\psi/2$ and stretches it by this factor. Near a full turn, where $\delta = 2\pi - \psi$ is of the size of ε, the first order fails. There $w \to -1$: the motor is -1 times a small rotation by $\Theta = \sqrt{\delta^2 + \varepsilon^2}$, whose axis is tipped

from the vertical by $\arctan(\varepsilon/\delta)$, since the roll and the heading left to a full turn are of the same size; `rlog` reads it on the far sheet as a turn by $2\pi - \Theta$ about that tipped axis, and

$$\frac{|h|}{\varepsilon} \to \frac{2\pi - \Theta}{\Theta}. \tag{208}$$

Table 29 lists what `rlog` reads for $\varepsilon = 0.01$ against the two limits; eqn. (206) reproduces it to rounding at every heading. The relative motor of eqn. (203) with the heading as the reference, $M_b = \underset{\sim}{Y}(\psi) \vee M = R(\varepsilon)$, reads the roll as 0.0100000000 at every heading, the last column of the table (test case "pga3dp: reading a motion off a motor - why an absolute log misreads a roll" in `ga_test/src/ga_appl3dp_mechanics_test.hpp`).

	$\lvert h\rvert/\varepsilon$, h the horizontal part of $2\,\mathrm{rlog}(M)$				relative motor
heading ψ	read	eqn. (207)	eqn. (208)	direction of h	roll read/ε
0°	1.000	1.000	—	0.0°	1.000
45°	1.026	1.026	—	22.5°	1.000
90°	1.111	1.111	—	45.0°	1.000
180°	1.571	1.571	—	90.0°	1.000
270°	3.332	3.332	—	135.0°	1.000
350°	34.99	35.04	34.94	175.0°	1.000
355°	70.56	71.02	70.53	177.5°	1.000
359°	311.4	359.0	311.4	179.5°	1.000
360°	627.3	—	627.3	180.0°	1.000

Table 29: A roll $\varepsilon = 0.01$ under a heading ψ, $M = Y(\psi) \vee R(\varepsilon)$. Left: read from the horizontal part h of the absolute logarithm, its direction in the world frame. Right: read from the relative motor $\underset{\sim}{Y}(\psi) \vee M$, which is $R(\varepsilon)$ itself; it reads the roll about the body's x axis, to 10^{-15}, at every heading.

Conditioning near a full turn: In *PGA* a unit motor that rotates by the full angle $\theta = 2\varphi$ about the unit line $\boldsymbol{l}$ is $M = \boldsymbol{l}\sin\varphi + \mathbb{1}\cos\varphi$ (eqn. 147); a screw adds the pitch terms of eqn. (146). The logarithm takes the half angle from the size of the line part and the pseudoscalar part, $\varphi = \mathrm{atan2}(\sin\varphi, \cos\varphi)$, divides the line part by $\sin\varphi$ to obtain the unit axis, and multiplies the axis by φ. For a rotation

$$\mathrm{rlog}(M) = \frac{\varphi}{\sin\varphi}\langle M\rangle_2, \tag{209}$$

and a screw's pitch is read in the same way, as $-\langle M\rangle_0/\sin\varphi$. Each component of the motor carries a rounding error of about 10^{-16}, since it is a number of size at most one, and eqn. (209) passes that error on multiplied by $(\varphi/\sin\varphi)$.

Near the identity, $\varphi \to 0$, the factor tends to 1. Near a full turn, $\varphi \to \pi$, the line part shrinks to $\sin\varphi$ while its rounding does not, and the factor grows as $\pi/(\pi-\varphi)$: for a screw whose full angle is 10^{-6} short of a full turn it is $6.3{\cdot}10^{6}$, and the motion read back is off by $8{\cdot}10^{-11}$; 10^{-9} short of a full turn it is $6.3\cdot10^{9}$, and the motion is off by $7\cdot10^{-9}$. At a full turn itself $\sin\varphi = 0$ and the axis is undefined. (Test case "pga3dp: reading a motor near a full turn - far sheet, band, shortest" in `ga_test/src/ga_pga_contract_test.hpp`, for these numbers and those below.)

The same motion is also $-M = \boldsymbol{l}\sin(\varphi-\pi)+\mathbb{1}\cos(\varphi-\pi)$, the rotation by $\theta-2\pi$, the short way round. Its half angle $\varphi-\pi$ is near zero, where the factor is 1. `rlog` therefore reads $-M$ once the motor's pseudoscalar component $\cos\varphi$ lies within $64\,\texttt{eps} = 7.1\cdot10^{-14}$ of -1 ($\texttt{eps} = 1.1\cdot10^{-15}$, five machine epsilons, is the library's equality tolerance). Since $1+\cos\varphi \approx \frac{1}{2}(\pi-\varphi)^2$, this is a full angle within $7.5{\cdot}10^{-7}$ of a full turn. Inside that band the reading jumps by a full turn: `rlog` returns the short remainder, an angle near 0 instead of near 2π, and $\mathrm{rexp}(\mathrm{rlog}(M)) = -M$, the same motion; the screw 10^{-9} short of a full turn is read to $3\cdot10^{-17}$. Outside the band the far-sheet reading is kept, so that `rlog` inverts `rexp` on both sheets, and its error stays below about $2\cdot10^{-10}$ ($1.8\cdot10^{-10}$ just outside the band). The exact far root, $\cos\varphi = -1$, is a pure translation on the far sheet (two half turns about parallel axes give one); its line part is the translation negated, and $-M$ gives the translation itself. Before any of this, `rlog` unitizes the motor, so that a motor at another scale, a product of reflections in planes that are not unit planes, reads as the motion it performs.

The loss of digits is avoided by reading the motion rather than the motor. Part of this the library does by itself, the rest is the choice of the function that reads:

- *In the library.* `rlog` unitizes the motor and reads $-M$ inside the band, so that its error stays below about $2\cdot10^{-10}$ at every angle. `rlog_shortest`, `motion_error_body` and `motion_error_world` always read the shortest representative, whose half angle is at most $\pi/2$; the factor $(\varphi/\sin\varphi)$ is then at most $\pi/2 \approx 1.57$, at every angle. The screw 10^{-6} short of a full turn, which `rlog` reads back to $8\cdot10^{-11}$, is read by `rlog_shortest` to $6\cdot10^{-17}$; a half turn, the shortest representative's worst case, to $2\cdot10^{-16}$.
- *By the user.* A number that stands for a motion — an error, the interpolation between two poses, a direction of travel — is taken with `rlog_shortest` or from the relative motor of eqn. (203) through `motion_`

`error_body` or `motion_error_world`, and not with `rlog` of a pose that can come near a full turn. Two absolute logarithms are never subtracted; the relative motor is formed first. Where the angle itself must run past a half turn — a commanded spin, a count of turns — it is kept as a scalar state, the coordinate of a revolute joint or a value continued from read to read with `rlog_continued`, rather than taken as the full angle of $\operatorname{rlog}(M)$.

In CGA a motor is $M = \cosh(w)\,\mathbb{1} + V$ with $V = \frac{\sinh w}{w}\,t$ for a generator t with $t \vee t = \alpha\,\mathbb{1}$ and $w = \sqrt{\alpha}$ (eqn. 215), and the logarithm divides V by $\sinh(w)/w$. For a rotation by 2φ, $w = i\varphi$, so the factor $w/\sinh w = \varphi/\sin\varphi$ is the one of eqn. (209); for a translation $w = 0$ and the factor is 1; for a dilation w is real. Whether the factor costs digits depends on how w is read. Read from the central part alone, as its inverse hyperbolic cosine, the half angle near a full turn comes from $\cos\varphi$ near -1, which loses digits as $1/\sqrt{1+\cos\varphi}$, and $\sinh(w)/w$, near zero there, carries that loss divided by its own size: a rotation 10^{-3} short of a full turn is read $1.4\cdot10^{-9}$ off, one $8\cdot10^{-7}$ short $1.8\cdot10^{-3}$ off. `rlog` reads w from both parts, as the PGA logarithm reads φ by atan2: $V \vee V = \sinh^2(w)\,\mathbb{1}$ gives $\sinh w$, and $w = \log(\cosh w + \sinh w)$. The same rotations are read to $2 \cdot 10^{-15}$ (test case "cga2dc / cga3dc: reading a motor near a full turn - far sheet, band" in `ga_test/src/ga_cga_contract_test.hpp`). Of the two roots $\pm\sinh w$ it takes the one with $|\cosh w + \sinh w| \geq 1$, so that the sum does not cancel; the generator $Vw/\sinh w$ is the same for both. `rlog` reads $-M$ in the band of the PGA logarithm, where the central part lies within 64 `eps` of -1, a full angle within $7.5\cdot10^{-7}$ of a full turn, and, in $CGA3DC$, wherever the central part lies below -1 (the far sheet of a dilation, where w is complex).

The logarithm assumes a unit motor, $M \vee \underset{\sim}{M} = \mathbb{1}$. A motor built as the product of two reflections carries a scale s that depends on its factors, $M \vee \underset{\sim}{M} = s^2\,\mathbb{1}$: it performs the same transformation, but its parts are $s\cosh w$ and sV, and $\log(s\cosh w + s\sinh w) = w + \log s$. Unlike the ratio that atan2 takes in PGA, the logarithm does not divide the scale out. The translation by $(0.4, -1.1, 0.3)$, built from its two parallel planes at half that distance, has $s = 0.79$; its central part is 0.79 where a translation has 1, and read as its inverse hyperbolic cosine it is a rotation by 1.31. `rlog` therefore divides by s, the square root of $M \vee \underset{\sim}{M}$, first, and `get_translation` returns the unit motor (test case "cga3dc: a motor at another scale - read as it stands, a translation turns" in `ga_test/src/ga_cga_contract_test.hpp`).

Reading an error: Three properties make the relative motor of eqn. (203) the place to take the logarithm. Near the reference pose M_0 it is near the

identity, where the chart is well conditioned ($\varphi/\sin\varphi \approx 1$) and additive to first order, $\operatorname{rlog}(\exp_\vee(A) \vee \exp_\vee(B)) = A + B + O(|A|\,|B|)$, so the components of B_b are separate small rotations about and translations along the axes of the body frame. Its two sheets $\pm M_b$ are one motion, and the one whose pseudoscalar component — the identity's — is not negative is the shortest, with an angle in $[-\pi, \pi]$. And it is equivariant: turning the whole scene by a motor G, $M \to G \vee M$ and $M_0 \to G \vee M_0$, leaves $M_b = \underset{\sim}{M}_0 \vee \underset{\sim}{G} \vee G \vee M$ unchanged and takes M_w to $G \vee M_w \vee \underset{\sim}{G}$. The absolute logarithm has none of the three; by eqn. (206) it lies halfway between the body frame and the world frame, turned by $\psi/2$ from each. An error is therefore read as the generator of a relative motor,

$$B_b = 2\operatorname{rlog}\big(\pm \underset{\sim}{M}_0 \vee M\big), \qquad B_w = 2\operatorname{rlog}\big(\pm M \vee \underset{\sim}{M}_0\big), \tag{210}$$

with the sign that makes the pseudoscalar component non-negative. Which of the two a law needs follows from the side on which it applies its correction. $M \leftarrow M \vee \exp_\vee(\frac{1}{2}\Delta t\,\Omega_b)$, a step of eqn. (177), takes a rate Ω_b in the body frame and the error B_b: the form of a controller working with the body Jacobian. $M \leftarrow \exp_\vee(\frac{1}{2}\Delta t\,\Omega_w) \vee M$, a step of eqn. (172), takes a rate Ω_w in the world frame and the error B_w: the form of one working with the space Jacobian. A count of turns is state: it is integrated as a scalar, as a revolute joint carries its coordinate q, and is never read back from the motor.

The library provides the rule in *PGA2DP* and *PGA3DP*. `rlog_shortest` is the logarithm of the shortest representative, a full angle in $[-\pi, \pi]$; `motion_error_body`(M_0, M) and `motion_error_world`(M_0, M) are B_b and B_w of eqn. (210); `rlog_continued`(M, L_{prev}) adds to $\operatorname{rlog}(M)$ the whole turns about its own axis that bring it nearest a previous logarithm. Each is shown below on the task it serves, against the reading of `rlog` it replaces. Both readings run in the test cases "pga3dp: reading a motion off a motor - ..." of `ga_test/src/ga_appl3dp_mechanics_test.hpp`.

Interpolating between two poses, `rlog_shortest`: A motion from M_0 to M_1 follows the logarithm of the relative motor, $M(t) = M_0 \vee \exp_\vee(t\,L)$ with $L = \operatorname{rlog}(\underset{\sim}{M}_0 \vee M_1)$, and every reading of L arrives at M_1 at $t = 1$. The test turns a pose by 90° about a vertical line off the origin and builds the target twice: as that quarter turn, and as a quarter turn and a full turn, which is its negative — one target pose. Read by `rlog`, the motion is decided by how the motor was built: halfway it has turned +45° in the first case and −135° in the second, two different motions between the same two poses. `rlog_shortest` reads the shortest of $\pm\underset{\sim}{M}_0 \vee M_1$, and both turn +45° (test case "...- rlog_shortest: interpolating two poses").

A levelling servo in the body frame, `motion_error_body`: The heading of the body is free and its tilt is corrected: the horizontal part h of a rotation read off the motor drives a rotation about the horizontal axes of the body frame, $M \leftarrow M \vee \exp_\vee(-\frac{1}{2}\Delta t\, k\, h)$. Started tilted by 0.112 rad under a heading ψ, with $k = 5\,\mathrm{s}^{-1}$ for 2 s, the tilt should fall as e^{-kt}, to $4.95{\cdot}10^{-6}$. Read from the absolute logarithm, h is turned by $\psi/2$ from the axes of the body frame and stretched by $s = (\psi/2)/\sin(\psi/2)$ (eqns. 206, 207). To first order the loop is $\dot{h} = -k\, s\, R(\psi/2)\, h$, with $R(\psi/2)$ the rotation of the horizontal plane by $\psi/2$, and its eigenvalues have the real part $-k\, s \cos(\psi/2)$: it levels more slowly at 90°, only circles at 180°, and past 180° the correction drives the tilt up. The body tips over, to 1.58 at 270° and 2.97 at 350°, before the large-angle motion brings it round (table 30). Read from the relative motor with the heading as the reference pose, $M_b = \underset{\sim}{Y}(\psi) \vee M$, the tilt falls as e^{-kt} at every heading. The same holds for a full pose servo, $\Omega_b = -k\, B_b$: started 150° and 0.55 m off, it leaves an error of $7{\cdot}10^{-15}$ after 10 s, with the reference pose M_0 at headings from 0° to 359° (test case "...- motion_error_body: a levelling servo").

heading ψ	correction in the body frame			correction in the world frame	
	absolute `rlog`		B_b	B_b	B_w
	final	peak	final	final / peak	final
0°	$4.95 \cdot 10^{-6}$	0.112	$4.95 \cdot 10^{-6}$	$4.95 \cdot 10^{-6}$ / 0.112	$4.95 \cdot 10^{-6}$
90°	$4.30 \cdot 10^{-5}$	0.112	$4.95 \cdot 10^{-6}$	$9.85 \cdot 10^{-2}$ / 0.112	$4.95 \cdot 10^{-6}$
180°	$8.39 \cdot 10^{-2}$	0.112	$4.95 \cdot 10^{-6}$	3.13 / 3.14	$4.95 \cdot 10^{-6}$
270°	$3.39 \cdot 10^{-3}$	1.58	$4.95 \cdot 10^{-6}$	$9.85 \cdot 10^{-2}$ / 0.112	$4.95 \cdot 10^{-6}$
350°	$2.86 \cdot 10^{-4}$	2.97	$4.95 \cdot 10^{-6}$	$5.77 \cdot 10^{-6}$ / 0.112	$4.95 \cdot 10^{-6}$

Table 30: The levelling servo: tilt after 2 s and its peak, started at 0.112, $k = 5\,\mathrm{s}^{-1}$. The absolute logarithm applied in the world frame gives the values of its body-frame column.

The servo in the world frame, `motion_error_world`: The same correction applied on the world side, $M \leftarrow \exp_\vee(-\frac{1}{2}\Delta t\, k\, h) \vee M$, needs the error in the world frame. The absolute logarithm fails there exactly as in the body frame and with the same numbers, since it lies $\psi/2$ from both. The right error in the wrong frame fails as well: the components of B_b lie along the axes of the body frame, and applied along those of the world frame they are turned by the whole heading. The real part is then $-k\cos\psi$: the loop is neutral at 90° and 270° and tips the body over at 180°. The error $B_w = 2\,\mathrm{rlog}(M \vee \underset{\sim}{Y})$ levels as e^{-kt} at every heading, and with $\Omega_w = -k\, B_w$ the full pose servo leaves

$7 \cdot 10^{-15}$. The test also checks eqn. (204), $B_w = M_0 \vee B_b \vee \underset{\sim}{M}_0$ (test case "...- motion_error_world: the servo in the world frame").

Counting turns, `rlog_continued`: A turntable, a wheel or a heading over a long run carries more turns than a motor holds. `rlog_continued`(M, L_{prev}) returns $\mathrm{rlog}(M) + k\pi\,\boldsymbol{l}$, with $\boldsymbol{l}$ the unit axis line of M and k the integer that brings the rotation nearest to that of L_{prev}. Fed its own previous value it follows any number of turns, provided the motion changes by less than half a turn between reads. Read every 90° for 41 reads, a turntable about a vertical line off the origin reports 10.25 revolutions, every read within $9 \cdot 10^{-16}$; `rlog` alone reports the last quarter turn, 0.25. At an exact whole number of turns the motor is $\pm\mathbb{1}$ and has no axis of its own; the turns then continue along the axis of L_{prev} (test case "...- rlog_continued: counting turns"). Summing the angles of the relative motors between consecutive reads counts the same turns; `rlog_continued` keeps the generator, its axis and translation included.

Writing a motion, and other charts: The same chart is met in the other direction, and in charts other than the logarithm.

- *A perturbation is a motor.* A small rotation n — simulated sensor noise, a correction — is applied as $M \vee \exp_{\vee}(\frac{1}{2}n)$ in the body frame or $\exp_{\vee}(\frac{1}{2}n) \vee M$ in the world frame. Added to the absolute logarithm instead, it moves the body by the exponential's Jacobian at $\mathrm{rlog}(M)$, the inverse of the factor in eqn. (207): across the axis it is turned and shrunk to $\sin(\psi/2)/(\psi/2)$ of itself, 0.64 at $\psi = 180°$ and 0.03 at 350°, so noise injected that way fades under a heading.

- *A turn count and a lean do not share one generator.* At a whole number of turns the heading's motor is -1, so a small lean R under it is the orientation $-R$, a rotation about the lean's horizontal axis; the weight of every generator of it is horizontal, and none carries the turns. A path whose heading accumulates turns while it leans keeps the heading ψ as a scalar coordinate and the lean as a second one, and composes $R \vee Y(\psi)$ when it is evaluated. Interpolated as one generator, a lap that closes at a full turn under a lean of 0.007 tilted to 1.4 halfway into its last segment.

- *A winding is state.* A torsion spring on a joint that can turn freely is a count of turns, continued from the previous step with `rlog_continued`. Read off the motor alone, its potential $\frac{1}{2}k|\theta|^2$ has the period 4π and its

maximum at a full turn: wound past it, the joint unwinds towards two full turns while the energy is conserved, so an energy check does not see the missing turns. Wound 1.5 turns, the motor-only spring carries a body through 4.3 turns where the continued one turns it back at 1.5 (test case "pga3dp: a spring on a motor joint - wound past a full turn").

- *Euler angles are a chart too.* Roll, pitch and yaw read off a motor fail on their own singular set: at a pitch of ±90° roll and yaw turn about one axis and only their difference (sum) exists, so read apart they are rounding noise; a frame written that way at a pitch of 90° came back turned by 0.06. The pitch is read by atan2 against the horizontal length of the moved x axis, and on the singular set the roll is set to 0 and the yaw read from the moved y axis.

- *A readout of yaw and tilt* is taken on the images of the basis vectors, which the sandwich gives exactly for any number of turns: the yaw as the heading of the moved x axis, continued from step to step, the tilt as the angle of the moved z axis off the vertical. The components of the generator's weight are neither: the x component is no tilt about y, and after a full turn a lean of 0.6° reads as a generator of angle near 2π about the lean's axis.

`screw_axis` reads a motor through `rlog_shortest`: M and $-M$ give one screw, with an angle of at most π.

A test of such a reading belongs at more than one heading. Turning the whole scene by G — every motor M becomes $G \vee M$ — leaves an error in the body frame unchanged and turns an error in the world frame with G; checked at 0°, 90°, 180°, 270°, a full turn and a little and nearly two, this exposes each of the three properties above, which a test at heading zero cannot see. The test helpers for this are in `ga_test/src/ga_test_equivariance.hpp`.

1.4.6 Motion in STA

Motion in $STA4DS$ is the Lorentz transformation of a space-time object X, applied by the same geometric-product rotor sandwich as in EGA — not by the regressive motors of PGA:

$$X' = R \wedge X \wedge \widetilde{R}, \qquad R \wedge \widetilde{R} = 1. \tag{211}$$

The even-grade part of the algebra forms the group of rotors R, each built as the exponential of a bivector generator. What the Minkowski signature

adds to the *EGA* picture is a second kind of generator.

A boost is the space-time counterpart of a spatial rotation: a rotation mixes two space directions, a boost mixes one space direction with time. It relates the measurements of two observers in uniform relative motion — the passage from a rest frame to one moving at constant velocity. The character of a rotor follows from the geometric square of its generating plane: a space-like plane ($B^2 < 0$) exponentiates with cos and sin (a rotation, periodic in its angle), a time-like plane ($B^2 > 0$) with cosh and sinh (a boost, unbounded in its *rapidity* φ). A rotation by θ in $\boldsymbol{g}_{12}$ and a boost by φ in $\boldsymbol{g}_{14}$ therefore read

$$R_{rot} = \cos\tfrac{\theta}{2} + \sin\tfrac{\theta}{2}\,\boldsymbol{g}_{12} = \exp_{\wedge}(+\tfrac{1}{2}\,\theta\,\boldsymbol{g}_{12}), \tag{212}$$

$$R_{boost} = \cosh\tfrac{\varphi}{2} + \sinh\tfrac{\varphi}{2}\,\boldsymbol{g}_{14} = \exp_{\wedge}(+\tfrac{1}{2}\,\varphi\,\boldsymbol{g}_{14}) \tag{213}$$

These are the rotors built by `get_rotor` and `get_boost`. Both carry the half-argument with a + sign, so a positive parameter produces a positive-sense transform: the rotation turns $\boldsymbol{g}_1$ toward $\boldsymbol{g}_2$, the boost carries $\boldsymbol{g}_4$ toward $+\boldsymbol{g}_1$ with $\gamma = \cosh\varphi$. The sign deserves a remark. The corresponding rotor in *EGA* carries a minus, $\exp_{\wedge}(-\frac{1}{2}\theta\,\boldsymbol{e}_{12})$, because there the spatial basis vectors square to +1; here they square to −1, so the sandwich already supplies that flip. Transferring the *EGA* sign literally reverses the rotation. The relative speed of the boost is $\beta = \tanh\varphi$ (with $c = 1$), so no boost attains $\beta = 1$, and collinear rapidities add as $\varphi_1 + \varphi_2$, just as rotation angles do. The mutual tilt of the time and space axes that a boost produces is the origin of time dilation and length contraction.

A general Lorentz transformation is one boost and one rotation combined, carried by the two commuting simple bivectors of the logarithm $\log(R) = b_{boost} + b_{rot}$ (one time-like, one space-like). This invariant decomposition is what `exp`, `log` and `sqrt` use for a non-simple rotor: exponentiate each plane on its own branch and multiply, $R = \exp_{\wedge}(b_{boost}) \wedge \exp_{\wedge}(b_{rot})$; invert that to recover the two generators; halve both to take the square root.

1.4.7 Motion in CGA

The conformal algebras use the same sandwich as PGA,

$$X = M \vee X_0 \vee \underset{\sim}{M} \quad (\text{for } CGA) \tag{214}$$

applied by the function `transform`. What changes is the set of transformations the versors M can carry. The rigid motions of PGA are those that leave the point at infinity where it is; the conformal group adds the transformations that move it, which are the dilations, the inversions in a sphere and the transversions. All of them preserve angles, none of them preserves distances, so they are motions in a wider sense than the rigid ones.

The generators follow the parity of the algebra, as in PGA. In $CGA2DC$ the versors are even-grade and the generators are bivectors; in the five-dimensional $CGA3DC$ they are odd-grade and the generators are trivectors, the same way the motors of $PGA2DP$ are odd-grade. For a *simple* generator t — a blade, which every line, every circle and every dual flat point is — the regressive square is central,

$$t \vee t = \alpha\, \mathbb{1} \tag{215}$$

and the exponential series sums in closed form

$$\operatorname{rexp}(t) = \cosh(w)\, \mathbb{1} + \frac{\sinh(w)}{w}\, t, \qquad w = \sqrt{\alpha} \tag{216}$$

Both $\cosh(w)$ and $\sinh(w)/w$ are even in w, so one formula covers the three cases that the sign of α distinguishes: $\alpha < 0$ is elliptic and gives the trigonometric form of a rotation, $\alpha > 0$ is hyperbolic and gives the hyperbolic form of a dilation, and $\alpha = 0$ is parabolic, where the series terminates after two terms as $\operatorname{rexp}(t) = \mathbb{1} + t$ and produces a translation. In $CGA3DC$ a generator that is not simple carries a non-vanishing $t \vee t$; `rexp` rejects it, and such motors are composed as products of simple exponentials instead. `rlog` inverts the simple case on both sheets, after normalizing the versor (the two-plane translation of `get_translation` came out with a weight of 0.79 before it was normalized, and was read as a rotation), `rsqrt` halves the generator.

The library provides the following builders, each returning a unit versor. The names follow the transformation, not the algebra:

- `get_translation` moves by a displacement δ. Its generator is the parabolic one, so the motor is $M = \mathbb{1} + \frac{1}{2} B_{tra}$ exactly, as for the PGA translation above.

- `get_rotation` rotates by an angle θ. The generator is the unit flat point at the center in $CGA2DC$, and the unit line as rotation axis in $CGA3DC$, taken with the half angle $\frac{\theta}{2}$ of the double-sided sandwich.

- `get_dilation` scales by a factor $\sigma > 0$ about a center m. The generator is the dual of the flat point at m (the antidual in $CGA3DC$), which is the hyperbolic case, and the scale enters as the rapidity

$$\delta = -\frac{1}{2}\ln\sigma \tag{217}$$

  so that repeated application multiplies the scale factors. A dilation about m is the composition of two inversions in concentric spheres around m.

- `get_transversion` ($CGA3DC$) applies a special conformal transformation with the parameter t. It is a translation conjugated by the inversion in the unit sphere, and acts on a point x as

$$x' = \frac{x + x^2 t}{1 + 2\,t \cdot x + t^2 x^2} \tag{218}$$

- `get_loxodromic` ($CGA3DC$) fixes the two points of a dipole d and combines a rotation by ϕ about its carrier line $\boldsymbol{l}$ with a hyperbolic motion of rapidity δ flowing from one of the two points to the other. Its generator $\frac{\phi}{2}\boldsymbol{l} + \delta h$ is not simple, but its two parts commute, so the motor is the product of two simple exponentials

$$M = \operatorname{rexp}\left(\frac{\phi}{2}\boldsymbol{l}\right) \vee \operatorname{rexp}(\delta\, h) \tag{219}$$

  with h the normalized antidual of d. Every flow line spirals from one fixed point to the other.

[Lengyel, 2024b] reaches the two dimensions from opposite ends, and both routes end at the builders above. In three dimensions the rigid transformations are carried over from PGA unchanged: a rigid operator of $CGA3DC$ is the corresponding $PGA3DP$ operator multiplied by e_∞, which is where the odd grade of the motors comes from. The lift is componentwise, $\boldsymbol{e}_{41} \to \boldsymbol{e}_{415}$, $\boldsymbol{e}_{23} \to \boldsymbol{e}_{235}$, the scalar part onto $\boldsymbol{e}_5$ and the pseudoscalar onto $\boldsymbol{e}_{12345}$, so a rotation motor keeps its PGA components one by one, and so does the dual-number pitch of a screw motion:

$$M = \operatorname{rexp}\left[(\delta\,\boldsymbol{e}_5 + \phi\,\mathbb{1}) \vee \boldsymbol{l}\right] \tag{220}$$

The transformations that are not rigid — dilation, inversion, transversion — are then added one at a time, and the general motion with two fixed points is the loxodromic one above. What the lift costs is computation on the round objects: flat objects transform with the PGA formulas unchanged, while a round point, and much more so a dipole or a circle, needs its e_∞ component recomputed, because position and radius are mixed there.

In two dimensions the general conformal motor is written directly from a dipole instead. With d radius-normalized so that $d \vee d = -\mathbb{1}$, its dual $d^\star$ is an imaginary dipole with $d^\star \vee d^\star = \mathbb{1}$. The two commute, so a dual-number magnitude splits the exponential into a hyperbolic and an elliptic factor[10]

$$M = \operatorname{rexp}\left[(\delta\,\mathbf{1} + \phi\,\mathbb{1}) \vee d\right] = \operatorname{rexp}(\delta\, d^\star) \vee \operatorname{rexp}(\phi\, d) \tag{221}$$

The quotient δ/ϕ is the *loxodromic parameter*. For $\delta = 0$ the elliptic factor remains and the motion rotates about the two points of d; for $\phi = 0$ the hyperbolic factor remains and the motion flows from one of them to the other; any other ratio spirals between the two. The screw motion of PGA is the same expression evaluated in the degenerate metric, with d replaced by the line $\boldsymbol{l}$ and the hyperbolic factor collapsing to $\cosh\delta = 1$, $\sinh\delta = \delta$. The four-dimensional conformal algebra of the plane and the four-dimensional rigid algebra of space share their exterior structure and differ only in the metric, which is what makes the two expressions look alike.

The inversion in a sphere or a circle is not a versor of this kind. It is applied by `invert_on` as the flector sandwich $-s \vee u \vee \underset{\sim}{s}$ with the sphere s itself as the operator, and it reverses orientation, so two inversions are needed to return a motion. It maps spheres to spheres, but a sphere through the center of s becomes a plane — the round weight of the image vanishes.

Because the sandwich is quadratic in M, versors double-cover the transformations here as well: M and $-M$, and every uniform rescaling of M, act identically. Whether two versors describe the same conformal map is therefore decided by their action rather than by their components, which is what `is_same_transform` does. The objects these transformations act on — the round and flat objects, their properties and the products that construct them — are the subject of the chapter on conformal geometry.

[10] In the basis used here the hyperbolic factor carries $+\delta$, because $d \vee d^\star = -\mathbf{1}$; the book writes $-\delta$ for the same motion, its 2D basis giving the opposite sign (see eqns. (5.30)–(5.34) there).

1.5 Modelling conformal geometry in CGA

The objects of the conformal algebras were introduced with the basis in subsection 1.2.1, and the transformations acting on them in subsection 1.4.7. What remains is the geometry between the two: how the objects are built from points and from each other, what their norms measure, and which properties can be read off them. All of it is written with the products the algebra already has — the wedge product as join, the regressive wedge product as meet — so the constructions carry over from PGA with the round objects added.

1.5.1 Constructing the objects

A round object is the join of the points that lie on it. Each point enters as a round point of radius zero, the null embedding (22)/(29), and the grade grows with every point joined. Two points spanning a dipole d, three a circle c, and — in $CGA3DC$ — four a sphere s:

$$d = a_1 \wedge a_2 \,, \qquad c = a_1 \wedge a_2 \wedge a_3 \,, \qquad s = a_1 \wedge a_2 \wedge a_3 \wedge a_4 \tag{222}$$

In $CGA2DC$ the chain ends one step earlier, with three points spanning a circle, which is the top round object of the plane.

A flat object is the same join with the point at infinity appended,

$$F = a_1 \wedge e_\infty \,, \qquad \boldsymbol{l} = a_1 \wedge a_2 \wedge e_\infty \,, \qquad \boldsymbol{p} = a_1 \wedge a_2 \wedge a_3 \wedge e_\infty \tag{223}$$

which is the precise sense in which a flat is a round through the point at infinity: a line $\boldsymbol{l}$ is the circle through two points and e_∞, a plane $\boldsymbol{p}$ the sphere through three points and e_∞, and a flat point F the dipole whose second point is e_∞. The components that result are the PGA components of the same object, lifted (subsection 1.4.7).

The grade therefore does not determine the object. In $CGA3DC$ a bivector is a dipole or a flat point, a trivector a circle or a line, a quadvector a sphere or a plane; in $CGA2DC$ the pairs are dipole/flat point and circle/line. This is unlike PGA, where grade and object correspond one to one, and it is a property of the model rather than of the representation: the two kinds are separated by e_∞, which a flat carries in every one of its blades and a round does not. In the bulk/weight split of subsection 1.4.7 that is the statement that a flat object has no round part,

$$u \text{ flat} \quad \Longleftrightarrow \quad u \wedge e_\infty = 0 \quad \Longleftrightarrow \quad \text{round_bulk}(u) = \text{round_weight}(u) = 0 \tag{224}$$

the middle form because the wedge with e_∞ annihilates exactly the blades that already contain it, and the right one because those blades are the flat part. The library tests the right-hand form (`is_flat`, `is_round`), which needs no product and is therefore available below the product layer; it compares the round part against the object's own magnitude, so the answer does not depend on the object's weight. The carrier gives the same answer one layer up — the carrier car() of a flat vanishes (subsection 1.5.3).

Constructing an object from data requires a convention for its directional argument, and one rule covers both algebras: an object is given its own *direction* — its attitude — and never a normal, except where the object is a hyperplane, whose defining datum is a normal and nothing else. A dipole and a line are one-dimensional and take a direction, in the plane as in space; a plane takes its normal, and a circle in space takes the normal of the plane carrying it. In the plane a line is at once one-dimensional and of codimension one, and the direction wins, which is the choice already made for the $PGA2DP$ bivectors, whose components run along a line rather than normal to it. The rule is one invariant,

$$\mathrm{att}(\mathrm{unitize}(u)) = \text{the constructor's directional argument} \tag{225}$$

so that the two points of a dipole of radius r about p lie at $p \pm r\,\boldsymbol{d}$, and a line through p with direction $\boldsymbol{d}$ contains $p + t\,\boldsymbol{d}$. Each constructor's orientation is fixed to satisfy it.

Meeting objects works the same way as in PGA, with the regressive wedge product. In $CGA3DC$ a sphere met with a plane through its center gives a circle, and met with a line through its center a dipole,

$$c = s \vee \boldsymbol{p}\,, \qquad d = s \vee \boldsymbol{l} \tag{226}$$

and in $CGA2DC$ the meet of two circles is the dipole of their intersection points. Where PGA produces a point at infinity when two lines are parallel, CGA produces an object with an imaginary radius when two circles do not intersect: the meet always exists, and the radius decides whether its points are real.

One configuration escapes that test. Two *concentric* circles, or two concentric spheres, meet in an object that keeps a round bulk — so it is not flat — but has no round weight at all, and an object without round weight has no centre and therefore no radius to take. Its squared radius is not negative, it is undefined, and dividing by the vanishing weight would return an infinity

that the sign test reads as an ordinary miss. The condition is intrinsic to the meet and needs no comparison of the two centres,

$$u \text{ degenerate} \quad \Longleftrightarrow \quad \text{round_weight}(u) = 0 \tag{227}$$

which the library tests as `is_degenerate` — relative to the object's own magnitude, since a meet computed through a chain lands near zero rather than at it. Any intersection then reads as three cases in order: degenerate, imaginary, real. Only the first is new; a merely near-concentric pair keeps a small round weight and a large negative squared radius, which the second case already reports correctly as no intersection.

The incidence test is carried out with the usual wedge product, as everywhere else in this document: the wedge of two objects vanishes if and only if one of them is contained in the other. A round point q lies on a sphere s exactly when

$$q \wedge s = 0 \tag{228}$$

and the same test with a circle, a dipole, a plane or a line answers the corresponding question. For two points a_1, a_2 of radius zero the dot product measures their distance directly,

$$a_1 \cdot a_2 = -\tfrac{1}{2}\,\delta^2 \qquad \Longrightarrow \qquad \delta = \sqrt{-2\,(a_1 \cdot a_2)} \tag{229}$$

with δ the Euclidean distance between them, unitized points assumed. It is the radius-zero case of the dot product of two round points, which in general returns $a_1 \cdot a_2 = -\frac{1}{2}(\delta^2 + r_1^2 + r_2^2)$.

Objects can also be written down directly, without joining points, and the library provides a constructor per kind: `round_point`, `flat_point`, `dipole`, `circle` and, in $CGA3DC$, `sphere`, `line` and `plane`. Two conventions of these forms are worth restating, because both are easy to get wrong and both are invisible to any test that compares objects up to a scalar factor. The round object of codimension one — the circle in $CGA2DC$, the sphere in $CGA3DC$ — carries the radius with the sign *opposite* to that of a round point, and its round weight is chosen as -1: compare the round point (31) with the sphere (32). The first convention follows from the object being the antidual of a vector that carries $(p^2 - r^2)/2$ in its infinity component, the second lets the center be read off the leading components directly. Both correspond to the derivation in [Lengyel, 2024b], and the extracted radius is positive for a real object in either case.

1.5.2 Center and radius

A round object carries a radius, and the algebra returns it from the object itself. The regressive dot product of the object with itself is the *radius norm* (`radius_nrm_sq`),

$$u \circ u = \|u\|_{\oslash}^2 \tag{230}$$

which for a unitized object is the squared radius directly. In general it is divided by the round weight, which is the quantity unitization scales to one,

$$r^2 = \frac{u \circ u}{\|u\|_{\circ}^2} \qquad (\texttt{radius_sq}) \tag{231}$$

The denominator of (231) is a sum of squares. It is therefore positive for every round object, and zero exactly for the flat ones, which is why the sign of the radius comes from the antidot square alone,

$$\operatorname{signum}(r^2) = \operatorname{signum}(u \circ u) \tag{232}$$

with signum the classical three-valued signum, $\operatorname{signum}(0) = 0$, so that the degenerate case below is included. It is provided as `signum()` in `ga_usr_utilities.hpp`, next to `sign()`, which is deliberately a different function: `sign()` returns +1 for a zero argument, so that a sign factor never vanishes. The object itself carries r^2 and never r: the constructor forms of subsection 1.2.1 contain the radius squared only, so r and $-r$ produce the same object and a sign of the radius is not there to be recovered. What the sign of r^2 does decide is the character of the object:

- $r^2 > 0$: a real object with radius $r = \sqrt{r^2}$, taken positive.
- $r^2 = 0$: the degenerate case. A round point of radius zero is an ordinary Euclidean point, and a dipole or circle of radius zero is the tangential case, its two points or its whole circle contracted to one point.
- $r^2 < 0$: an *imaginary* radius $r = i\sqrt{-r^2}$. Such an object exists in the algebra and has a center and a carrier, but no real points.

This is why (231) stops at r^2 and takes no square root: the sign is the result, and the caller tests it before extracting points. Two unit circles show all three cases — met at center distance 1.5 they give $r^2 = +0.4375$, at distance 2 they are tangent and give $r^2 = 0$, and at distance 5 they give $r^2 = -5.25$. The meet is constructed the same way in all three, and nothing has to be decided in advance.

The dot product of the object with itself returns the same quantity with the opposite sign,

$$u \cdot u = -r^2 \tag{233}$$

for a unitized round object — the round point and the codimension-one round included, whose components carry r^2 with opposite signs. Which of the two products is used therefore decides the sign, not the object. A flat object has no radius at all: its round weight vanishes, so (231) has no value there.

The distance from the origin to the center is the *center norm* (`center_nrm`). It is built from two of the four part norms of subsection 1.2.3,

$$\|u\|_\odot^2 = \|u\|_\bullet^2 + \|u\|_\square^2 \tag{234}$$

a right-triangle relation: the round bulk measures the weighted distance from the origin to the carrier, the flat weight the distance from the carrier's support to the center, and the two legs are perpendicular. The same quantity follows from the conformal conjugate of subsection 1.2.4,

$$\|u\|_\odot^2 = u \cdot u^\dagger \tag{235}$$

since conjugating reflects the center through the origin. For the unitized sphere with center $(3, 4, 0)$ and radius 2 both give $\|u\|_\odot = 5$, with $\|u\|_\bullet = 0$ and $\|u\|_\square = 5$, while $u \circ u = 4$ and $u \cdot u = -4$.
Both quantities are conformal: the squared radius is signed and the center is a round point, not a position. Reading the Euclidean numbers back out is the last step of a construction, and it differs by kind. For a round object the position is the center with the round weight divided out; a flat point also has a position but carries it in the flat part, over the flat weight instead, so the two cases are separated by (224). A line and a plane have no center at all. The library provides the two readouts as `radius` and `position`, the first the square root of r^2 and hence undefined for a flat object and for an imaginary one — both of which occur by construction, so the sign of the squared radius is tested before the root is taken, as in (241) below.

1.5.3 Geometric properties

Six properties are read off an object, all of them coordinate-free and all built from the products already in place. Written with the antidual in the star

role and e_o, e_∞ for origin and point at infinity, they are

$$\mathrm{car}(u) = u \wedge e_\infty \tag{236a}$$
$$\mathrm{ccr}(u) = u_\star \wedge e_\infty \tag{236b}$$
$$\mathrm{cen}(u) = \mathrm{ccr}(u) \vee u \tag{236c}$$
$$\mathrm{con}(u) = u \wedge \mathrm{car}(u)_\star \tag{236d}$$
$$\mathrm{par}(u) = (-1)^{gr(u)+1}\,\mathrm{con}(u_\star) \vee \mathrm{car}(u) \tag{236e}$$
$$\mathrm{att}(u) = u \vee \overline{e_o} \tag{236f}$$

The same equations serve both algebras. The star is the *left* antidual, hence the subscript, in the four-dimensional $CGA2DC$, where the two handednesses differ; in the five-dimensional $CGA3DC$ left and right coincide and the marker carries no information. The complement in (236f) is the right one, likewise a single operation in $CGA3DC$.

The *carrier* $\mathrm{car}(u)$ is the lowest-dimensional flat object containing u: the plane of a circle, the line of a dipole, the flat point of a round point. It raises the grade by one and annihilates a flat object, since a flat already contains e_∞ and the wedge with it vanishes — car of a plane or a line is zero, which is the test for flatness. The *cocarrier* $\mathrm{ccr}(u)$ is the carrier of the antidual and completes the pair used by the remaining properties.

The *center* $\mathrm{cen}(u)$ is returned as a round point, and it carries the radius of u as well as its position. A round point is therefore its own center, and the sphere, the circle and the dipole around a given center all return that same round point. Because the center carries the radius, it is incident with the object,

$$u \wedge \mathrm{cen}(u) = 0 \tag{237}$$

which is a statement about the round point $\mathrm{cen}(u)$, not about the Euclidean center, and does not say that the center lies on the circle.

The *container* $\mathrm{con}(u)$ is the smallest codimension-one round containing u — the sphere through a circle in $CGA3DC$, the circle through a dipole in $CGA2DC$. Carrier and container factor the object between them,

$$u = \mathrm{car}(u) \vee \mathrm{con}(u) \tag{238}$$

so a circle is recovered as the meet of its plane with its sphere, and a dipole as the meet of its line with its circle. The *partner* $\mathrm{par}(u)$ keeps center and

carrier and negates r^2, turning a real object into an imaginary one and back. It is orthogonal to the object it belongs to,

$$u \cdot \mathrm{par}(u) = 0 \tag{239}$$

The *attitude* $\mathrm{att}(u)$ removes the origin content and lowers the grade by one, leaving the direction data: the weight of a round point, the axis direction of a dipole, the normal of a circle. It is the same operation as the attitude in PGA (subsection 1.2.4), and like it (236f) meets the object with the complement of the algebra's own projective basis vector, which is a different vector in each of the four algebras:

$$\mathrm{att}(u) = u \vee \overline{\boldsymbol{e}_3} \;\; (CGA2DC)\,, \qquad \mathrm{att}(u) = u \vee \overline{\boldsymbol{e}_4} \;\; (CGA3DC) \tag{240}$$

against $u \vee \overline{\boldsymbol{e}_3}$ in $2D$ PGA and $u \vee \overline{\boldsymbol{e}_4}$ in $3D$ PGA. The round dimension takes no part in it.

Together the six answer the questions one asks of a round object without decomposing it into coordinates: where it sits (cen), how big it is (231), which flat it lies in (car), how it is oriented (att), and what contains it (con).

1.5.4 Worked constructions

The pieces assembled so far answer a classical question in one chain. Two objects are met, the sign of the squared radius of the result decides whether real points exist, and if they do, the two points of the resulting dipole d are

$$p_\pm = \mathrm{cen}(d) \pm \sqrt{r^2}\,\mathrm{att}(d) \tag{241}$$

with d unitized. Center and attitude come from (236c) and (236f), the radius from (231).

For two circles in $CGA2DC$ the meet $d = c_1 \vee c_2$ runs through this chain without a single case distinction:

configuration	r^2 of $c_1 \vee c_2$	points $p_\pm$
$r = 2$ at distance 3 (secant)	+1.75	$(1.5, \pm 1.3229)$
$r = 1$ at distance 2 (tangent)	0	$(1, 0)$ twice
$r = 1$ at distance 5 (disjoint)	−5.25	none

which are the classical values: the secant case has its points at $x = 1.5$ and $y = \pm\sqrt{4 - 1.5^2}$, the tangent case has them coincide, and the disjoint case returns an object with an imaginary radius. No discriminant is formed and no

branch is taken before the meet — the meet is the same product in all three, and the sign of r^2 carries the answer that a case analysis would otherwise have to produce first.

Nor does the second object have to be a round one, or pass through the center. A line is the join of two of its points with e_∞ (223), and meeting a circle of radius 2 about the origin with the horizontal lines at $y = 1$, 2 and 3 runs the identical chain:

line	r^2 of $c \vee \boldsymbol{l}$	points $p_\pm$
$y = 1$ (half the radius, secant)	+3	$(\pm\sqrt{3}, 1)$
$y = 2$ (tangent)	0	$(0, 2)$ twice
$y = 3$ (missing)	−5	none

so the chord ends come out as $x = \pm\sqrt{r^2 - d^2}$ with d the distance of the line from the center, without that formula appearing anywhere.

CGA3DC behaves the same way with a sphere in place of the circle. The sphere of radius 2 about the origin met with the lines along $\boldsymbol{e}_1$ at $y = 1$, 2 and 3 gives exactly the three rows above, with the points at $(\pm\sqrt{3}, 1, 0)$ in the secant case. Meeting the same sphere with the *planes* $z = 1$, 2 and 3 gives a circle instead of a dipole, of squared radius 3, 0 and −5 and centered on the cutting plane at $(0, 0, z)$ — the off-center cut $r^2 - d^2$ again, now one grade higher. A larger example in the same pattern: the sphere of radius 5 met with the line $y = 3$ gives a dipole with $r^2 = 16$ at $(0, 3, 0)$ and the points $(\pm 4, 3, 0)$, the solutions of $x^2 + 3^2 = 5^2$.

What the algebra contributes here is not a shorter formula for a known case, but the absence of the case distinction: the meet of two rounds is defined for every configuration, the result is an object of the algebra either way, and the questions "do they meet" and "where" are answered by the radius and the two properties of that one object.

The chain is also planar two-link inverse kinematics. The joint between a link of length L_1 from a base point b and a link of length L_2 to a tip point t lies on both circles, so the two points of $c(b, L_1) \vee c(t, L_2)$ are the two configurations (elbow up and down, knee forward and back), and an imaginary radius says the tip is out of reach. In *CGA3DC* the joint of an arm whose links lie in a plane π through the base is the dipole $s(b, L_1) \vee s(t, L_2) \vee \pi$, the sphere meet being the circle of all joint positions and the plane picking two of them (library: `dipole_points`, `two_link_ik2dc`, `two_link_ik3dc`). The *PGA*

solution of the same triangle, and a comparison of the two routes, are in subsubsection 1.8.4: both give the two configurations as a ± pair with an equivalent reachability condition, and the meet saves the law of cosines rather than a case distinction.

1.6 Modelling geodetic coordinates in *PGA*

Positions on the earth are stated as geodetic coordinates — latitude, longitude and height — while directions, lines and distances are computed in Cartesian frames. This subsection models both sides in $3D$ *PGA*: the reference ellipsoid with the two latitudes that come with it, the earth-fixed and the local frame, the motor that carries one frame into the other, and the lines and paths that connect two places; then the maps — the projections onto the tangent plane, the transverse Mercator, the equal-area map, the equidistant map — and the transformation between two geodetic datums. It also shows where the constructions of the algebra reach their limit: the shortest path on an ellipsoid is not a plane section, so the geodesic comes from an integration and not from a product expression.

The earth is not a sphere. Modelled as an ellipsoid of revolution it is described by two radii, and every quantity derived from it is a function of the latitude alone — the surface is generated by rotating one ellipse about the polar axis, so nothing in it can depend on the longitude. That single symmetry decides most of what follows: it is what makes the three-dimensional case a two-dimensional one plus a rotation, and it is what supplies the conserved quantity of the geodesic problem.

1.6.1 The reference ellipsoid

With the equatorial radius a and the polar radius b, the flattening and the first eccentricity are

$$f = \frac{a-b}{a} \qquad \text{and} \qquad e^2 = \frac{a^2-b^2}{a^2} \tag{242}$$

and the distance of the surface from the geocenter at geodetic latitude φ is

$$R(\varphi) = \sqrt{\frac{(a^2\cos\varphi)^2 + (b^2\sin\varphi)^2}{(a\cos\varphi)^2 + (b\sin\varphi)^2}} \tag{243}$$

running from $R(0) = a$ to $R(\pi/2) = b$. For $WGS84$, $a = 6378137\,m$ and $b \approx 6356752.314\,m$.
Two different angles carry the name latitude, and the distinction is the one thing that must not be blurred (figure 11). The *geodetic* latitude φ is the angle of the surface *normal* against the equatorial plane; the *geocentric* latitude ψ is the angle of the ray from the geocenter. They are related by

$$\tan\psi = (1-e^2)\tan\varphi \tag{244}$$

and differ by up to $11.5'$ at mid-latitudes. Maps, encyclopedias and *GNSS* receivers quote the geodetic one, and a height is measured along that same normal, not along the ray. The normal does not pass through the geocenter; extended inwards it meets the polar axis below the center.

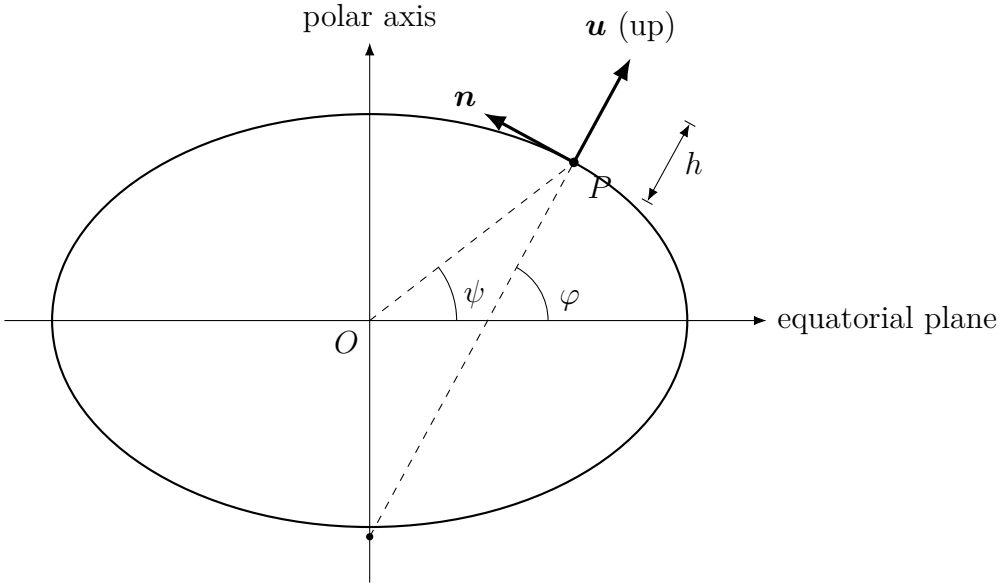


Figure 11: Meridian section of the reference ellipsoid, with the flattening greatly exaggerated. The geodetic latitude φ is the angle of the surface normal against the equatorial plane, the geocentric latitude ψ that of the ray from the geocenter O. The normal meets the polar axis below O, not at it. The height h is measured along the normal, and the local frame is (east into the page, $\boldsymbol{n}$, $\boldsymbol{u}$).

1.6.2 Position in the earth-fixed frame

The Earth-Centered, Earth-Fixed frame $ECEF$ has its origin at the geocenter, $\boldsymbol{e}_1$ in the equatorial plane through the prime meridian, $\boldsymbol{e}_2$ in the equatorial plane $90°$ east of it, and $\boldsymbol{e}_3$ along the mean rotational axis. Writing $N(\varphi)$ for the radius of curvature in the prime vertical,

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}} \tag{245}$$

a position at geodetic latitude φ and height h sits in its own meridian plane at

$$r = \big(N + h\big) \cos \varphi \,, \qquad z = \Big(N\big(1 - e^2\big) + h\Big) \sin \varphi \tag{246}$$

where r is the distance from the polar axis and z the distance along it. The factor $(1 - e^2)$ on z is what makes φ geodetic; the height enters both components alike because it runs along the normal.
Equation (246) is the whole geometry. The three-dimensional position follows by turning the meridian plane to the longitude λ,

$$X = \begin{pmatrix} r\cos\lambda \\ r\sin\lambda \\ z \end{pmatrix} \tag{247}$$

so the three-dimensional case is the two-dimensional one plus a rotation about the polar axis. The inverse of eqn. (246) is implicit in φ and h and is solved in closed form by Bowring's construction [Bowring, 1976], whose auxiliary latitude $\theta = \arctan(z\,a/r\,b)$ makes the relation explicit. It is a single pass, not an iteration, so its accuracy degrades with height (see [Karney, 2013] for the modern treatment of both problems): the round trip is exact to double precision on the surface, holds to below a millimetre up to about $100\,km$, and to about $0.26\,m$ at geostationary altitude.

1.6.3 The local frame

The local frame at a station is East-North-Up, ENU, and it is fixed by the meridian plane through the position. With $\boldsymbol{r} = (\cos\lambda, \sin\lambda, 0)$ the horizontal direction towards the position, that plane is the bivector

$$\mathbf{N} = \boldsymbol{e}_3 \wedge \boldsymbol{r} \tag{248}$$

and with $\boldsymbol{u}$ the unit surface normal the remaining two axes follow from it by contraction and by the dual,

$$\boldsymbol{n} = \boldsymbol{u} \lrcorner \mathbf{N}\,, \qquad \boldsymbol{e} = \mathbf{N}^\star \tag{249}$$

north lying in the meridian plane perpendicular to up, east being the plane's normal. No angle is taken apart and no rotation matrix is written out; the triple is orthonormal and right-handed in the order $(\boldsymbol{e}, \boldsymbol{n}, \boldsymbol{u})$. Both expressions hold on either hemisphere without a case distinction: $\boldsymbol{u}$ lies in the meridian plane, and contracting it into that plane turns it by a quarter turn within it, in the sense the orientation of $\mathbf{N}$ fixes — which is north for every latitude. Only the poles are excluded, where $\boldsymbol{r}$, and with it the meridian plane, is undefined.
Building $\mathbf{N}$ from the longitude rather than from the position is what keeps equation (248) non-degenerate. The alternative $\boldsymbol{u} \wedge \boldsymbol{e}_3$ equals $\cos\varphi \cdot (\boldsymbol{r} \wedge \boldsymbol{e}_3)$, so the two agree in direction everywhere but only the first survives at the

poles, where $\boldsymbol{u}$ becomes parallel to $\boldsymbol{e}_3$. At a pole the frame is genuinely undetermined up to a rotation about $\boldsymbol{u}$: every horizontal direction there is south, and the frame one obtains depends on the meridian along which the pole is approached. A position carries that meridian, a bare point does not.

1.6.4 The motor between the frames

The motion carrying local coordinates into $ECEF$ is a motor, composed of the translation to the station and the two rotations that orient the frame,

$$M = T(X) \vee R_{\boldsymbol{e}_3}\left(\lambda + \tfrac{\pi}{2}\right) \vee R_{\boldsymbol{e}_1}\left(\tfrac{\pi}{2} - \varphi\right) \tag{250}$$

read right to left: the second rotation tilts the equatorial frame up to the station's latitude, the first swings it onto the local meridian and puts $\boldsymbol{e}_1$ onto east, the translation moves it out to the station. The inverse map, $ECEF$ into local coordinates, is $\underset{\sim}{M}$. The latitude enters uncorrected because it already is the angle of the normal, which is what up is.
It is tempting to derive M from the frame of eqn. (249) instead, by constructing the rotation that carries the reference axes onto $(\boldsymbol{e}, \boldsymbol{n}, \boldsymbol{u})$. That construction fails. Every such alignment — $\sqrt{l_2 \vee \underset{\sim}{l_1}}$ from two lines, or the closed form $1 + \sum \boldsymbol{b}_i \boldsymbol{a}_i$ — is singular when the required rotation is by π, where its axis is undetermined, and along the meridian $\lambda = 90°$ that is exactly what is asked of it. Equation (250) has no such inversion: longitude and co-latitude are the position's own coordinates.

1.6.5 Lines between places

A line is an element of the algebra, so the line from a station to an observed object is the join of the two points,

$$L = X_A \wedge X_B \tag{251}$$

and $\mathrm{att}(L)$ is its direction. Because L is an object rather than a list of coordinates, the line a user constructs in their own frame — the only frame in which they can measure — is the same element as the one built in $ECEF$,

$$M \vee \left(O \wedge X_B^{loc}\right) \vee \underset{\sim}{M} \;\simeq\; X_A \wedge X_B \tag{252}$$

up to the weight that the join does not normalize. Elevation and azimuth are then read off the direction in the local frame rather than recomputed per frame.

1.6.6 Paths on the ellipsoid

On a sphere the shortest path between two points is a plane section: the plane through the two points and the center cuts a great circle, and the problem is one of incidence. On an oblate ellipsoid that construction fails — geodesics are not plane curves — and the two candidates separate (see figure 12).
The plane section survives as a curve, if not as the shortest one. The plane through the geocenter and the two points is the join

$$\Pi = X_A \wedge X_B \wedge O \tag{253}$$

and the curve in it is swept by a rotor built from that plane's own bivector $\hat{\mathbf{B}}$, carrying the start direction round to the end direction,

$$\boldsymbol{d}(t) = \exp\!\left(-\tfrac{1}{2}\hat{\mathbf{B}}\,t\right)\boldsymbol{d}_A \exp\!\left(\tfrac{1}{2}\hat{\mathbf{B}}\,t\right), \qquad 0 \le t \le \theta \tag{254}$$

each swept direction being pushed out to the surface along its own ray. The result is the great ellipse.
The geodesic is obtained by integration. Writing the ellipsoid as the quadratic form $\mathbf{G} = \mathrm{diag}(1/a^2, 1/a^2, 1/b^2)$, so that the surface is $X \cdot \mathbf{G}X = 1$ and its gradient is $\mathbf{G}X$, the geodesic condition is that the tangent turns only towards the surface normal and never within the surface,

$$\frac{dX}{ds} = T\,, \qquad \frac{dT}{ds} = \kappa\,\boldsymbol{u}\,, \qquad \boldsymbol{u} = \frac{\mathbf{G}X}{\|\mathbf{G}X\|}\,, \qquad \kappa = -\frac{T \cdot \mathbf{G}T}{\|\mathbf{G}X\|} \tag{255}$$

No auxiliary sphere, no meridian or prime-vertical radii and no series expansion enter: the ellipsoid appears only through $\mathbf{G}$, and $\boldsymbol{u}$ is the same gradient that supplies up in eqn. (249). For $a = b = R$ the curvature reduces to $\kappa = -1/R$, the great-circle result. The boundary value problem between two given places is solved by shooting on the initial azimuth.
The rotational symmetry of the ellipsoid leaves a conserved quantity along any geodesic,

$$r \sin\alpha = \text{const} \tag{256}$$

with α the azimuth — Clairaut's relation, stated as such in [Karney, 2013], and all that survives of the sphere's constant plane. It is the sharpest available check on an integrated path.
Measured against Vincenty's solution [Vincenty, 1975] for $WGS84$, the two curves differ by little at continental range and measurably at intercontinental range:

route	geodesic	plane section	difference
Berlin – Madrid	1 872 384.211 m	1 872 384.247 m	0.036 m
Berlin – Sydney	16 088 403.760 m	16 088 428.043 m	24.284 m

The plane section gets increasingly inaccurate with growing range. Over about 1800 km (e.g. Berlin–Madrid) it reproduces the geodesic to a few centimetres, over about 16 000 km (e.g. Berlin–Sydney) it is about 24 m too long.

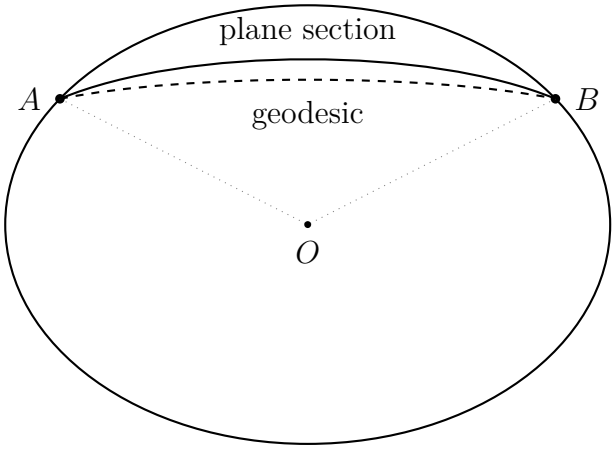


Figure 12: Schematic, with the separation greatly exaggerated. The plane section lies in the plane spanned by A, B and the geocenter O; the geodesic does not lie in any plane through O and is the shorter of the two. The real separation is 0.036 m over the 1872 km from Berlin to Madrid and 24.284 m over the 16 088 km from Berlin to Sydney.

The integration is the check, not the tool: a map needs a path to every one of its points. For that the library solves the two classical problems — the *inverse* problem, distance s_{12} and azimuths between two given places, and the *direct* problem, the end of a path of given start, azimuth and length — on the auxiliary sphere of [Karney, 2013]. With the reduced latitude $\tan\beta = (1-f)\tan\varphi$ the geodesic becomes a great circle of arc length σ and equatorial azimuth α_0, $\sin\alpha_0 = \sin\alpha\cos\beta$ being Clairaut's constant of eqn. (256), and distance and longitude follow from two integrals,

$$\begin{aligned} s &= b\int \sqrt{1+k^2\sin^2\sigma}\,d\sigma\,, \qquad k^2 = e'^2\cos^2\alpha_0 \\ \lambda &= \omega - f\sin\alpha_0 \int \frac{2-f}{1+(1-f)\sqrt{1+k^2\sin^2\sigma}}\,d\sigma \end{aligned} \tag{257}$$

ω the longitude on the auxiliary sphere. The integrands are smooth and nearly constant ($k^2 < 0.007$), so Gauss–Legendre quadrature with twelve nodes per quarter circle reaches machine precision; the series of the reference are not needed. With the two ends arranged canonically — the one farther from the equator first and in the south, the other east of it — the longitude reached increases monotonically with the departure azimuth from 0 to π, so the inverse problem is a bracketed root search that converges for every pair, nearly antipodal ones included. Both problems agree with GeographicLib to $2 \cdot 10^{-8}\,m$ over 2000 random pairs and the degenerate ones (both ends on the equator, an end at a pole):

```
auto const g = geodesic_inverse(a, b); // s12, azi1, azi2
auto const c = geodesic_direct(a, azi, s).pos;
auto const route = geodesic_path(a, b, 100.0e3);
```

1.6.7 Projections onto the tangent plane

A map of a region is a projection of the ellipsoid onto the plane that touches it at one point, the station S. In the station's local frame (eqn. (250)) that plane is $U = 0$, the plane $\Pi_0 = \boldsymbol{e}_{412}$, and every projection of this family is one construction: the line from a projection centre C through the point P, met with the plane,

$$X = (C \wedge P) \vee \Pi_0 \tag{258}$$

Unitized, the first two coordinates of X are the map coordinates in metres, x towards east and y towards north at the station. The projections differ only in where C sits (figure 13), and that choice decides what the map keeps:

projection	centre C	keeps	used for
orthographic	$\boldsymbol{e}_3$, at infinity	the globe's look	globe views
gnomonic	the geocenter	great circles straight	routes
stereographic	far end of the normal	angles (sphere)	polar maps
perspective	H above S	the view from H	satellite images

The orthographic map compresses radially towards its edge and keeps neither angles nor areas. The gnomonic map draws every great circle as a straight line, so a straight line on it is a shortest route; its scale grows fast, and less than a hemisphere fits. The stereographic map is conformal on a sphere — small shapes keep their form — and its scale grows slowly; national grids are built on it. The vertical perspective shows what a camera at height H sees and approaches the orthographic map as H grows. On a sphere all four reduce to the textbook forms of [Snyder, 1987]. On the ellipsoid the orthographic

and the perspective map are the *EPSG* methods 9840 and 9838, which define them by this construction [IOGP, 2025]; the stereographic one differs from *EPSG* 9809, which goes through a conformal sphere, by up to $3.2 \cdot 10^{-4}$ of the distance from the station.

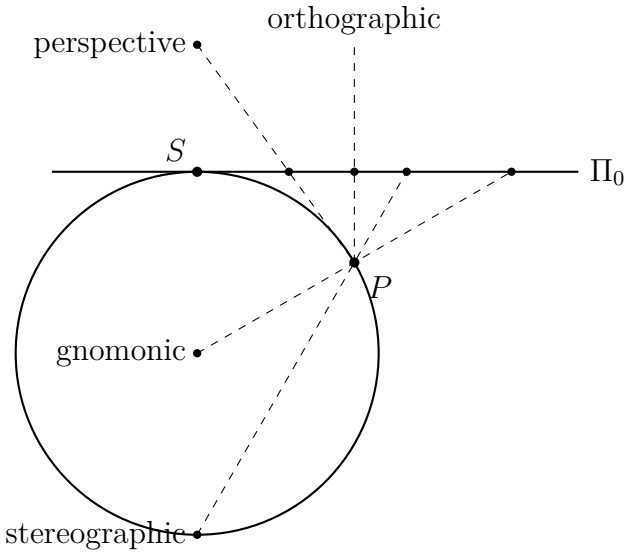


Figure 13: The four projections in a meridian section through the station S: the image of P is where the line from the centre through P meets the tangent plane Π_0. From left to right on the plane: perspective (centre at height H), orthographic (centre at infinity), stereographic (centre at the far end of the normal), gnomonic (centre at the geocenter).

What can be drawn is bounded by a plane as well. For the orthographic and the perspective map it is the polar plane of $C = (\boldsymbol{c}, w)$ with respect to the ellipsoid $X \cdot \mathbf{G}X = 1$,

$$\boldsymbol{x} \cdot \mathbf{G}\boldsymbol{c} = w\,, \qquad P \text{ visible} \iff P \cdot \mathbf{G}\boldsymbol{c} > w \tag{259}$$

which meets the ellipsoid in the horizon seen from C; for the ideal centre of the orthographic map ($w = 0$) it is the diametral plane conjugate to up. The gnomonic and the stereographic map have no horizon of their own and are cut at a chosen angular radius. An outline that leaves the region is cut where a segment crosses the bounding plane Π,

$$Q = (A \wedge B) \vee \Pi \tag{260}$$

and the inverse map — from map coordinates back to the ellipsoid — is the line from C through the image point met with the ellipsoid, a quadratic in

the line parameter.

Over a region small against the earth the four maps differ in a simple way. A centre at depth D below the plane, on the normal, maps a point at distance s from the station outward by the factor $1+s^2/(2\rho D)$, ρ the radius of curvature, so against the orthographic map the area grows by

$$\frac{\Delta A}{A} \approx \frac{2\,s^2}{\rho\,D} \tag{261}$$

— for gnomonic ($D \approx \rho$), stereographic ($D \approx 2\rho$) and perspective ($D = -H$) in the ratio $2 : 1 : -2\rho/H$. The gnomonic centre is the exception on the ellipsoid: the geocenter lies off the station's normal by $Ne^2 \sin\varphi \cos\varphi$ ($21\,km$ at $49°$), which adds a north-south tilt that cancels only over a region centred on the station.

Outlines in geodata are usually polygons with edges straight in latitude and longitude; projected, such an edge is a curve, and it is densified before it is projected. The library provides the construction as `azimuthal_map`:

```
auto const m =
    make_azimuthal_map(station, azimuthal::stereographic);
// ECEF -> the local frame, then onto the plane
auto const P = to_local(m, geo_to_ecef(p));
if (is_drawable(m, P)) {
    vec2d const x = project(m, P);
    geo_pos const q = unproject(m, x); // the way back
}
```

together with `clip_at` (eqn. (260)), `limit_outline` (the edge of the region) and `densify`.

1.6.8 Transverse Mercator and UTM

Official geodata is often delivered in grid coordinates — easting and northing in metres — of a transverse Mercator map: the conformal projection onto a cylinder that touches the ellipsoid along a central meridian λ_0, the basis of UTM and of most national grids. Reading such data needs the inverse map. It is classical numerics with no product expression, and the library implements it as such: Krüger's series in the third flattening $n = f/(2-f)$, in the form of [Karney, 2011] (figure 14).

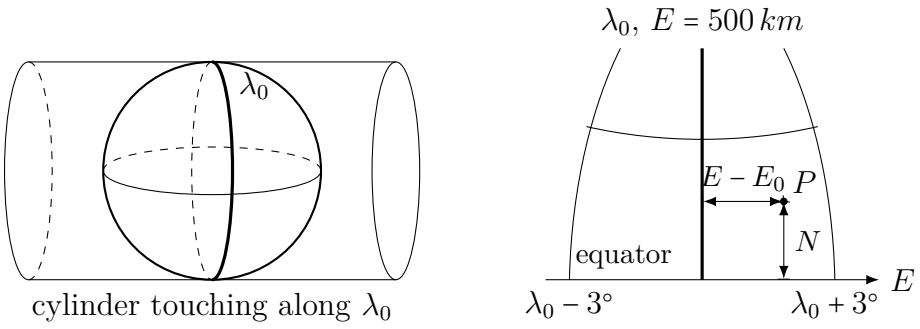


Figure 14: Transverse Mercator. Left: the cylinder touches the ellipsoid along the central meridian λ_0, its axis lying in the equatorial plane. Right: one *UTM* zone unrolled — the central meridian is the straight line $E = E_0 = 500\,km$, where the scale is $k_0 = 0.9996$; the zone's edges $\lambda_0 \pm 3°$ and the parallels are curves; N is counted from the equator.

With $\tau = \tan\varphi$, the conformal latitude $\tau' = \sinh\big(\sinh^{-1}\tau - e\tanh^{-1}(e\sin\varphi)\big)$ and λ counted from the central meridian, the spherical coordinates

$$\xi' = \arctan\frac{\tau'}{\cos\lambda}\,, \qquad \eta' = \sinh^{-1}\frac{\sin\lambda}{\sqrt{\tau'^2 + \cos^2\lambda}} \tag{262}$$

are carried to the ellipsoid by

$$\xi = \xi' + \sum_{j=1}^{6}\alpha_j \sin 2j\xi' \cosh 2j\eta'\,, \qquad \eta = \eta' + \sum_{j=1}^{6}\alpha_j \cos 2j\xi' \sinh 2j\eta' \tag{263}$$

with $\alpha_1 = n/2 - 2n^2/3 + \ldots$ (all six to order n^6 in [Karney, 2011]), and the grid coordinates are

$$E = E_0 + k_0 A\,\eta\,, \qquad N = N_0 + k_0 A\,(\xi - \xi_0) \tag{264}$$

$A = a\,(1 + n^2/4 + n^4/64 + \ldots)/(1+n)$ the rectifying radius, E_0, N_0 the false easting and northing, k_0 the scale on the central meridian and ξ_0 the value of ξ at the latitude of origin. The inverse runs the same steps backwards with a second set of coefficients β_j and recovers φ from τ' by Newton's method. A *UTM* zone z has $\lambda_0 = 6z - 183°$, $k_0 = 0.9996$, $E_0 = 500\,km$ and $N_0 = 0$ (northern) or $10\,000\,km$ (southern hemisphere). Within a zone the implementation agrees with *PROJ* to $10^{-6}\,m$ and its round trip to $3\cdot10^{-9}\,m$:

```
auto const utm = utm_zone(32); // ETRS89 / UTM 32N
// geodetic -> grid (easting, northing), and back
vec2d const en = tm_forward(p, utm, grs80);
geo_pos const q = tm_inverse(vec2d{E, N}, utm, grs80);
```

1.6.9 Lambert azimuthal equal-area

A map that compares quantities per area — population density, land use, statistics — has to keep area. The azimuthal map that does is not a central projection: no centre C in eqn. (258) produces it. On a sphere of radius R it keeps the azimuth from the centre T and places a point P at the length of the chord TP (figure 15),

$$\rho = |P - T| = 2R \sin \tfrac{c}{2} \tag{265}$$

c the angle between T and P at the centre of the sphere. The cap within c has the area $2\pi R^2(1-\cos c) = \pi\rho^2$, that of the disc of radius ρ, so area is kept ring by ring, and with the azimuth kept, everywhere. The price is shape: along the radius the map is compressed by $\cos \frac{c}{2}$ and across it stretched by $1/\cos \frac{c}{2}$.

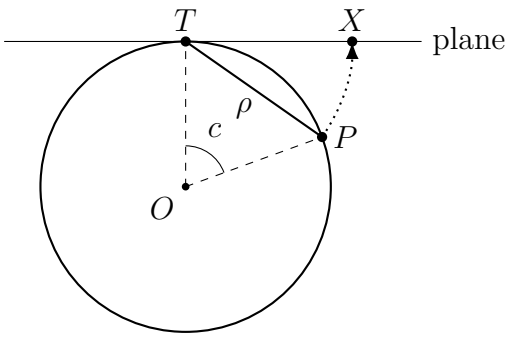


Figure 15: Lambert azimuthal equal-area on a sphere: the point P at the angle c from the centre T is drawn at the length ρ of the chord TP, swung down into the plane at T, in the direction of its azimuth.

On the ellipsoid the sphere is the authalic sphere, the sphere of the same surface area, and each geodetic latitude φ moves to the authalic latitude β, which divides the ellipsoid's area between equator and parallel in the same ratio. With the area $\pi a^2 q(\varphi)$ between the equator and φ [Snyder, 1987],

$$\begin{aligned} q(\varphi) &= (1-e^2)\left(\frac{\sin\varphi}{1-e^2\sin^2\varphi} + \frac{\tanh^{-1}(e\sin\varphi)}{e}\right) \\ \sin\beta &= \frac{q(\varphi)}{q_p}, \qquad R_q = a\sqrt{q_p/2} \end{aligned} \tag{266}$$

with $q_p = q(90°)$, and R_q is the radius of the authalic sphere. Centred at (φ_0, λ_0), the map is the chord construction on that sphere (method 9820 of

[IOGP, 2025]),

$$\begin{aligned} B &= R_q\sqrt{\frac{2}{1+\sin\beta_0\sin\beta+\cos\beta_0\cos\beta\cos\Delta\lambda}} \\ E &= E_0 + BD\cos\beta\sin\Delta\lambda \\ N &= N_0 + \frac{B}{D}\big(\cos\beta_0\sin\beta-\sin\beta_0\cos\beta\cos\Delta\lambda\big) \end{aligned} \tag{267}$$

with $\Delta\lambda = \lambda - \lambda_0$ and $D = a\cos\varphi_0/\big(\sqrt{1-e^2\sin^2\varphi_0}\,R_q\cos\beta_0\big)$, which restores the true scale at the centre ($D = 1$ at a pole). The antipode of the centre has no image; the rest of the ellipsoid fills a disc of radius $2R_q$. The inverse recovers φ from $q = q_p\sin\beta$ by Newton's method. The *EU*'s statistical grid, *ETRS89-LAEA* (*EPSG*:3035), is this map centred at $52°N$, $10°E$ with $E_0 = 4321\,km$ and $N_0 = 3210\,km$ on *GRS80*; *EPSG* lists its axes in the order (N, E). The implementation agrees with *PROJ* to $5\cdot10^{-7}\,m$, and a projected $2°\times2°$ cell has the area of the cell on the ellipsoid to $6\cdot10^{-9}$:

```
auto const eu = etrs89_laea(); // EPSG:3035
vec2d const en = laea_forward(p, eu, grs80);
geo_pos const q = laea_inverse(en, eu, grs80);
auto const m = laea_at(p0);    // centred at p0
```

1.6.10 Azimuthal equidistant

The map of everything seen from one place keeps distance and direction from its centre C: each point is drawn at its geodesic distance s_{12} from C, in the direction of the geodesic's azimuth α_1 there,

$$x = s_{12}\sin\alpha_1\,, \qquad y = s_{12}\cos\alpha_1 \tag{268}$$

So the shortest route from C to any place is a straight line of its true length, and the circles about C are the places at one distance (figure 16). That makes it the flight map of an airport and the map of radio and seismic ranges; the emblem of the United Nations is its polar aspect. Distances and directions between two other places are not kept — a route between them is a curve — and shape and area worsen with the distance. The whole earth fills a disc of radius about $20\,000\,km$, whose rim is the antipode of C. Like the equal-area map it is no central projection; it is the inverse geodesic problem in polar coordinates, here in its rigorous ellipsoidal form (*PROJ*'s `aeqd`), not the series of *EPSG* methods 1125 and 9832, which hold near the centre only. The implementation agrees with *PROJ* to $10^{-6}\,m$; a route from the centre stays

on its straight line to $4 \cdot 10^{-8}\,m$.

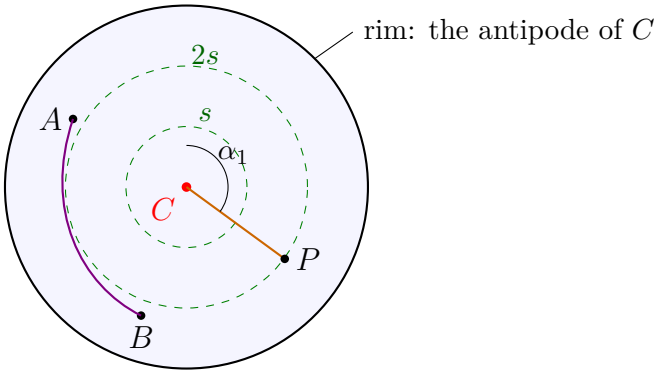


Figure 16: Azimuthal equidistant. From the centre C the route to P is the straight line of length s_{12} in the direction α_1, and the circles about C are the places at the distances s, $2s$. The route between two other places A and B is a curve. The rim is the antipode of C.

```
auto const m = aeqd_at(hub);         // centred at the hub
vec2d const x = aeqd_forward(p, m); // true s12, azi1
geo_pos const q = aeqd_inverse(x, m);
```

1.6.11 Datum transformation

Two geodetic datums — an older national one on the Bessel ellipsoid and $ETRS89$, say — place their $ECEF$ axes differently: shifted by up to some hundred metres, turned by a few arc seconds, and scaled by some parts per million. The 7-parameter Helmert transformation carries $ECEF$ coordinates from one datum to the other,

$$\boldsymbol{x}' = \boldsymbol{t} + (1 + d_s)\, R(\boldsymbol{r})\, \boldsymbol{x} \tag{269}$$

with the translation $\boldsymbol{t}$, a rotation by the small angles $\boldsymbol{r} = (r_x, r_y, r_z)$ and the scale correction d_s. In PGA the rigid part is one motor, and the scale about the geocenter a change of the point's weight,

$$X' = M \vee \left(x\, \boldsymbol{e}_1 + y\, \boldsymbol{e}_2 + z\, \boldsymbol{e}_3 + \tfrac{w}{1+d_s}\, \boldsymbol{e}_4\right) \vee \underset{\sim}{M} \tag{270}$$

with

$$M = T(\boldsymbol{t}) \vee \exp_{\vee}\left(\tfrac{1}{2}(r_x \boldsymbol{e}_{41} + r_y \boldsymbol{e}_{42} + r_z \boldsymbol{e}_{43})\right) \tag{271}$$

The rotation is generated by the line through the geocenter with the direction $\boldsymbol{r}$, and the line's weight is the angle. Published parameter sets use

two conventions, which differ in the sign of the rotation [IOGP, 2025]: *position vector* ($EPSG$ 1033, also $PROJ$'s `towgs84`) turns the point by $+\boldsymbol{r}$, *coordinate frame* ($EPSG$ 1032) turns the axes by $+\boldsymbol{r}$, i.e. the point by $-\boldsymbol{r}$. Mixing them up costs about $2|\boldsymbol{r}||\boldsymbol{x}|$, $110\,m$ for the German $DHDN$ set, so the library names one function after each: `helmert_position_vector` and `helmert_coordinate_frame`. $EPSG$ writes R in its linearized form $1 + [\boldsymbol{r}]_\times$; the motor is the exact rotation, and the two differ by at most $\frac{1}{2}|\boldsymbol{r}|^2|\boldsymbol{x}|$ ($0.33\,mm$ for $DHDN$), below the accuracy of any published parameter set. Negating all seven parameters inverts the transformation to first order only: for $DHDN$ the round trip misses by $9\,mm$.

```
// DHDN -> ETRS89: m, arc seconds, ppm as published
auto const h = make_helmert7(598.1, 73.7, 418.2,
                             0.202, 0.045, -2.455, 6.7);
ellipsoid const bessel{6377397.155, 6356078.963};
auto const P = geo_to_ecef(p, bessel);
auto const X = helmert_position_vector(P, h);
geo_pos const q = ecef_to_geo(X, grs80);
```

1.7 Modelling mechanics in PGA

The key idea is to represent physical entities using objects represented in the underlying algebra. They have a geometric interpretation and can be manipulated by applying algebraic operations. This is possible since the algebraic objects we use in geometric algebra, especially the blades resulting from outer products between vectors, represent subspaces of the overall space and thus allow for direct algebraic manipulation. At the same time, they represent or can be attributed to objects of geometric or physical meaning. This enables implementing geometrically and physically meaningful manipulations by applying algebraic operations on those subspaces or between them.

Euclidean geometric algebra (EGA) is limited to objects that contain the origin. Thus, it can be used to directly model vectors (i.e. directions), lines, and planes through the origin. In addition, it can model complex numbers, quaternions and thus rotations in $2D$ and $3D$ well. To realize them, it uses sandwich products based on the geometric product for calculating the transformation resulting from two consecutive reflections either across lines ($2D$ case) or across planes ($3D$ case). The resulting transformations are rotations. The advantage of using geometric algebra to model those rotations is the intuitive modelling, due to geometric meaning of the objects and the fact that the sandwich products can be used to transform different objects, like vectors, bivectors or rotation operators using the same sandwich product formulas in a unified fashion regardless of the type of object to be transformed. Despite those limitations EGA is already a very powerful tool to express physical ideas as shown by David Hestenes in [Hestenes, 1986]. In this work he reformulates classical mechanics using geometric algebra mainly with EGA as a tool. But he is also showing that even more is possible, e.g. for formulating the motion of rigid bodies, clearly pointing to PGA in his chapter 7 on rigid body mechanics, even if he does not call that PGA yet. This is already looking extremely similar to what is shown later in this section. His book is a rich source of applications of geometric algebra for many application areas of mechanics.

Projective geometric algebra (PGA) is indeed capable to model objects that do not necessarily contain the origin. Thus, it can be used to directly model points, directions, lines, and planes that may or may not contain the origin. In contrast to EGA, translations can additionally be implemented in PGA directly as orthogonal transformation using a sandwich product. The operation is realized by two consecutive reflections across parallel lines ($2D$) or planes ($3D$). It is based on the regressive form of the geometric product.

The reason for this deviation from EGA, which uses the geometric product for that purpose, is that the Euclidean isometries (rotation, translation, and screw motion) can only be performed in a space that is dual to the space we use for modelling our projective objects (for a more in depth explanation see [Lengyel, 2024b]). The use of the regressive geometric product on these objects is equivalent to first transforming the object into dual space by applying the complement operation, then using the geometric product in that space for the intended transformation and finally transforming the resulting transformed object back into the original space, by applying the inverse complement operation. Thus, the sandwich product transformation formulas in PGA only differ from the formulas used in EGA by using the regressive geometric product instead of the geometric product, but are otherwise the same.

With both algebras in view, which one to reach for is decided by whether position is part of the physics. EGA carries the *value* of a quantity at a point, the point being a parameter of the evaluation; PGA's gain appears where the location of an object is part of what the object *is* — a force on its line of action, a pose, a momentum line — which is what the mechanics below models throughout. That choice of the most suitable algebra is set out in more detail in subsubsection 1.10.7.

The additional degree of freedom in PGA is possible due to embedding the Euclidean space of dimension n into a projective space of dimension $n+1$, e.g. two-dimensional Euclidean space is modelled by embedding it in a three-dimensional projective space, and three-dimensional Euclidean space is modelled in a four-dimensional projective space. In the $n+1$-dimensional projective space all algebraic objects contain the origin (exactly as in EGA), but in the modelled n-dimensional Euclidean space the objects do not need to contain the origin any longer. They can be located anywhere in the modelled space including positions at infinity. This can be realized in projective geometry, since objects of dimension $n+1$ are projected into the modelled space resulting in objects of dimension n after the projection. A projective point (which models a scalar or $0D$-object) is a vector representing a ray through the origin in the embedding space (a $1D$-object), and a $1D$-object, like a line in modelled space, is created by a bivector in the embedding projective space (a $2D$-object), and so on for higher-grade objects.

Due to its larger expressiveness PGA is suitable to model physics of mass points and rigid bodies including all isometries. For example, it can model position, velocity, acceleration, force, torque, momentum, angular momentum, etc., and thus it can be used to describe kinematics as well as dynam-

ics of mass points and rigid bodies. The approach used here is inspired by [Dorst and De Keninck, 2022] and [Dorst and De Keninck, 2023], but it does not use the plane-based ansatz with planes as vectors, which are based on dual representations of the objects to be modelled. It rather uses the approach as described by [Lengyel, 2024b] using projective geometric algebra based on direct representations of points, lines and planes. Lengyel also gives reasons for choosing the latter approach and explains the background in more detail compared to the short indications given here.

Objects modelled in a projective space can be multiplied by a scalar value without changing the *geometric* meaning of the object in the modelled Euclidean space, i.e. its interpretation after the projection. Equality of modelled geometric objects is thus often only defined up to a scalar factor. This kind of equality up to a scalar factor is defined as congruence (see e.g. [Browne, 2012]). Therefore, it is important for modelling of physical facts that unitization is used consciously to avoid false interpretations. For example, a heavy point can be modelled by multiplying a *unitized* point with the scalar value of its mass, but of course the multiplication with m must happen after unitization in order to keep the scalar factor m in the calculation. Other geometric objects need to be unitized in order to focus on the intended geometric operation and avoid additional scaling, i.e. the rotation operators. They must be unitized before use as well. Thus, it is advisable to think through the geometric operations and their physical interpretation well to avoid surprises and to be consistent.

1.7.1 Modelling force and torque

[Browne, 2012] (see sections 4.1-4.5) is a very good source showing how to use Grassmann algebra and projective geometry in order to model forces and moments. Classical approaches based on vector algebra only model forces by providing vectors of adequate magnitude and direction to represent the force. However, the physically very meaningful "line of action", which determines which torque is created by the force, is not modelled mathematically in the classical approach at all. It rather needs to be derived from the description of the context in order to know where the force is actually acting. Thus, force and torque are not directly interlinked in the classical description, but must be modelled as separate physical effects.

This changes in *PGA*. Any force can be directly modelled by its line of action, which encodes the force *and* the torque generated by the force at the origin within one algebraic object. A force f is modelled by a line F. If f is the force vector of corresponding magnitude and direction, and Q is any unitized projective point on its line of action then F is an algebraic object representing the force f as well as the resulting line of action directly:

$$F = Q \wedge f \tag{272}$$

The resulting bivector F represents a line, a geometric object which contains all physically relevant information of the force f and the torque generated by f. For that reason [Dorst and De Keninck, 2023] call this object "forque". It can directly be used to recover the force vector and to calculate the resulting free torque caused by the force either at the origin or at any point in space.

But first, let's have a closer look how this is accomplished. We can separate F into its components. Each unitized projective point Q is the sum of the origin O and a vector q parallel to the projective plane (in $2D$) or projective space (in $3D$) pointing from O to Q, such that $Q = O + q$ ("projective point split"). Thus, F can be split into two bivector components

$$F = Q \wedge f = (O + q) \wedge f = \underbrace{O \wedge f}_{=\text{force}} + \underbrace{q \wedge f}_{=torque} = F_\circ + F_\bullet \tag{273}$$

where the first term encodes the force within the weight components of the bivector, while the second term encodes the resulting free torque relative to the origin caused by the action of f at Q – or alternatively anywhere along the line of action – within the bulk components of the bivector.

To see this even more clearly, let's look at the component equations of both terms written as bivectors for the case of $2D$ PGA (the corresponding basis bivectors are shown on the right-hand side in square brackets for reference):

$$O \wedge f = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \wedge \begin{pmatrix} f_x \\ f_y \\ 0 \end{pmatrix} = \begin{pmatrix} f_x \\ f_y \\ 0 \end{pmatrix} \qquad \begin{bmatrix} \boldsymbol{e}_{31} \\ \boldsymbol{e}_{32} \\ \boldsymbol{e}_{12} \end{bmatrix} \tag{274a}$$

$$q \wedge f = \begin{pmatrix} q_x \\ q_y \\ 0 \end{pmatrix} \wedge \begin{pmatrix} f_x \\ f_y \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ q_x f_y - q_y f_x \end{pmatrix} \qquad \begin{bmatrix} \boldsymbol{e}_{31} \\ \boldsymbol{e}_{32} \\ \boldsymbol{e}_{12} \end{bmatrix} \tag{274b}$$

where equation (274a) shows how the force is encoded in the first two components of the bivector and equation (274b) shows how the torque is encoded in the third component of the bivector F.

For $3D$ PGA the encoding is similar, but now requires six bivector components instead of three (the corresponding basis bivectors are shown on the right-hand side in square brackets for reference again):

$$O \wedge f = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \wedge \begin{pmatrix} f_x \\ f_y \\ f_z \\ 0 \end{pmatrix} = \begin{pmatrix} f_x \\ f_y \\ f_z \\ 0 \\ 0 \\ 0 \end{pmatrix} \qquad \begin{bmatrix} \boldsymbol{e}_{41} \\ \boldsymbol{e}_{42} \\ \boldsymbol{e}_{43} \\ \boldsymbol{e}_{23} \\ \boldsymbol{e}_{31} \\ \boldsymbol{e}_{12} \end{bmatrix} \tag{275a}$$

$$q \wedge f = \begin{pmatrix} q_x \\ q_y \\ q_z \\ 0 \end{pmatrix} \wedge \begin{pmatrix} f_x \\ f_y \\ f_z \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ q_y f_z - q_z f_y \\ q_z f_x - q_x f_z \\ q_x f_y - q_y f_x \end{pmatrix} \qquad \begin{bmatrix} \boldsymbol{e}_{41} \\ \boldsymbol{e}_{42} \\ \boldsymbol{e}_{43} \\ \boldsymbol{e}_{23} \\ \boldsymbol{e}_{31} \\ \boldsymbol{e}_{12} \end{bmatrix} \tag{275b}$$

The first three bivector components model the force while the last three components model the torque. The force vector f itself can be retrieved from F by using the attitude function att():

$$f = att(F) = F \vee \overline{\boldsymbol{e}_n} = att(F_\circ) = F_\circ \vee \overline{\boldsymbol{e}_n} \tag{276}$$

with $\boldsymbol{e}_n$ denoting the basis vector of the projective dimension that squares to zero in the respective algebra ($\boldsymbol{e}_3$ for $2D$ and $\boldsymbol{e}_4$ for $3D$). The torque generated relative to the origin O is encoded in the bulk of the bivector:

$$M_O = bulk(F) = F_\bullet \tag{277}$$

while the resulting torque from the application of f at any given point R in space can be retrieved from F, either using any known point Q on F[11], or alternatively by projecting R onto F and using the perpendicular distance vector $d_\perp$ from R towards the line F in order to calculate the resulting free torque $M_F(R)$ at the position R as

$$M_F(R) = [(Q - R) \wedge f]_\bullet = [(Q - R) \wedge att(F)]_\bullet \tag{278a}$$
$$M_F(R) = [d_\perp \wedge f]_\bullet = [(F \vee (R \wedge F^*) - R) \wedge att(F)]_\bullet \tag{278b}$$

where equation (278a) is the formula with the least computational effort using a known point Q on F, while equation (278b) uses the orthogonal projection of R onto F to calculate the perpendicular distance vector $d_\perp$.

In the plane no point on F needs to be known: the moment about R is the incidence of R with the force line,

$$R \wedge F = R \wedge (Q \wedge f) = [(Q - R) \wedge f] \wedge \boldsymbol{e}_3 = M_F(R)\, \boldsymbol{e}_{123} = -M_F(R)\, \boldsymbol{e}_{321} \tag{2D}$$
$$\tag{279}$$

a pseudoscalar, since the $\boldsymbol{e}_{12}$ part of $R \wedge Q$ has no $\boldsymbol{e}_3$ to complete a trivector with f and drops out. Its coefficient in the algebra's pseudoscalar $\boldsymbol{e}_{321}$ is $-M_F(R)$ — the orientation convention, not the physics; the library's `wdg(R, F)` returns that coefficient. Static equilibrium of a body pivoted at R reads $R \wedge \sum_i F_i = 0$ — one wedge with the sum of the force lines, where the sign does not matter. The identity is planar: in $3D$ the wedge of a point with a line is a plane, not the moment bivector, and eqn. (278a) is the form to use.

Eqn. (278a) needs a point on F; the linearity of the wedge removes that need. With $F_\bullet = q \wedge f$ the moment about the origin, and r the direction from O to R,

$$M_F(R) = F_\bullet - r \wedge att(F) \tag{280}$$

holds in $2D$ and $3D$ alike and is what the library's `moment_about(R, F)` computes, in both mechanics headers; in the plane its $\boldsymbol{e}_{12}$ component is $M_F(R)$ of eqn. (279), with the physical sign.

The *bulk* of the second term is not written because it acts trivially: $att(F)$ is an ideal vector and r a direction, so neither carries $\boldsymbol{e}_4$ ($\boldsymbol{e}_3$ in $2D$) and their wedge has no weight component to discard. It is the *ideal* r that licenses this. With the projective point R in its place the weight component is f and

[11] Which is always available from the function $sup(F)$. It provides the support point on F with the shortest distance to the origin O by orthogonally projecting the origin onto F.

the *bulk* is required,

$$M_F(R) = F_\bullet - [R \wedge att(F)]_\bullet \tag{281}$$

where both terms now carry the same weight f, so their difference is ideal and the *bulk* may be dropped from the expression as a whole:

$$M_F(R) = F - R \wedge att(F) \qquad (R \text{ unitized}) \tag{282}$$

Two force lines carrying the same force differ by an ideal bivector: the force cancels in the weight and a pure moment remains, i.e. a couple, whose moment is the same about every point. $R \wedge att(F)$ is the force f on the line through R parallel to F, so the couple by which F differs from it is the moment of F about R — and being a couple, it needs no reference point of its own.

Eqns. (280) to (282) hold for any bivector, not only for a force line: the *bulk* of a bivector is its moment about the origin and *att* its resultant force whether or not the bivector is simple. A pure couple has $att(F) = 0$, and its moment is accordingly the same about every point. The sum of two skew force lines is a screw, for which eqn. (278a) cannot be evaluated at all — a screw has no line of action, so there is no point Q on it — while the forms above return the sum of the two lines' moments.

The resulting total force and torque acting on an object are given by:

$$F_{total} = \sum_i F_i \tag{283}$$

This resulting sum will be used later for connecting kinematics and dynamics.

Things to remember:

- Adding forces represented as projective lines creates resulting force lines located at the right position to get equivalent moments automatically. When two parallel forces are added the resulting force line will be located in between such that it creates a torque equivalent to the torque initially generated by both forces. It creates an equivalent force system reducing multiple forces and moments acting on a rigid body to a single resultant force and a resultant moment about a specific point while maintaining the same external effect on the rigid body.
- Thus, the sum of forces and their induced moments can be directly created by the sum of their respective lines of action, i.e. by summing

the bivectors representing those lines. The resulting lines of action contain all relevant information incl. the resulting vector sum of the forces and the resulting moment values from the summation. There is no need to calculate these separately. The result will always be a resulting bivector that potentially contains a bound vector and a resulting moment which creates a torque. (see [Browne, 2012], p. 176).

- Neither a vector nor a bivector are located anywhere. They can be moved around freely. A line however represents a bound vector that is bound by its line. Thus, the vector can only be moved along its line of action without changing the geometric and physical meaning (see [Browne, 2012], p. 169).

- A free torque is modelled by a bivector with force components being zero, i.e. the resulting force vector (e.g. from summing several forces acting on an object) has no direction and zero length, but the bivector still might contain a resulting torque value.

- The torque calculated from either equation (278a) or (278b) is necessarily a free torque as can be seen from the formulas. The origin does not occur in the result anymore due to the difference of the two projective points involved.

1.7.2 Modelling linear and angular momentum

Momentum in the classical approach is split into linear momentum $\vec{p} = m\vec{v}$ and angular momentum $\vec{L} = \vec{x} \times \vec{p} = \vec{x} \times m\vec{v}$. Both are represented as vectors, but they represent entities with different physical meaning. So they cannot be combined into one object in the classical approach, e.g. by adding them.

The *PGA* approach to momentum is to define both, the linear and the angular aspects, within one algebraic entity. This is in complete analogy to what we have seen in the last subsection for force and moment resulting in its line of action modelled by a bivector.

Single point mass: The momentum P of a point mass m at a projective (unitized) position X that moves with velocity $\dot{X}$ is defined as a bivector P as

$$P = X \wedge m\dot{X} \tag{284}$$

It defines the overall momentum at position X including linear and angular momentum in the bivector $P(X)$, which can be shown by splitting the projective point X into its components $X = O + x$ ("projective point split"). In the $3D$-case with the origin $O = O_{3dp} = (0, 0, 0, 1)$ and the direction vector x, where $x = (X_x, X_y, X_z, 0)$, i.e. specifically with $X_w = 0$, which represents a direction towards a point at infinity. This results in

$$\begin{aligned} P(X) &= X \wedge m\dot{X} = (O + x) \wedge m\dot{X} \\ &= O \wedge m\dot{X} + x \wedge m\dot{X} \\ &= O \wedge m\dot{(O + x)} + x \wedge m\dot{(O + x)} \\ &= O \wedge m\dot{x} + x \wedge m\dot{x} && \text{(285a)} \\ &= \underbrace{P_\circ}_{\text{linear}} + \underbrace{P_\bullet}_{\text{angular}} && \text{(285b)} \end{aligned}$$

P defines an instantaneous momentum line where the weight part $P_\circ$ of the bivector P encodes the linear momentum and the bulk part $P_\bullet$ encodes the angular momentum with respect to the origin.[12]

[12] The "projective point split" works in this way, since the origin is a distinguishable point. This is in contrast to *EGA* or the classical approach where the origin is a null-vector, which has no lasting effect if it is added to another point. This is one reason why *PGA* can combine linear and angular aspects as separate physical effects in a single algebraic object. The other is that the required 6 degrees of freedom can be encoded in a bivector.

Recovering linear and angular momentum: The physical aspects encoded in the momentum line P can be retrieved in complete analogy to the case of force and torque in subsection 1.7.1:

The linear momentum vector p itself can be retrieved from P by using the attitude function att():

$$p = mv = m\dot{x} = att(P) = P \vee \overline{\boldsymbol{e}_n} = att(P_\circ) = P_\circ \vee \overline{\boldsymbol{e}_n} \tag{286}$$

with $\boldsymbol{e}_n$ denoting the basis vector of the projective dimension that squares to zero in the respective algebra ($\boldsymbol{e}_3$ for $2D$ and $\boldsymbol{e}_4$ for $3D$).

The angular momentum L generated relative to the origin O is encoded in the bulk of the bivector:

$$L_O = x \wedge m\dot{x} = bulk(P) = P_\bullet \tag{287}$$

while the resulting angular momentum relative to any given point R in space can be retrieved from the momentum line P by

$$L(R) = \left[(X - R) \wedge m\dot{X}\right]_\bullet = \left[(X - R) \wedge att(P)\right]_\bullet \tag{288}$$

using X as a known point on P. This provides the resulting angular momentum $L(R)$ with respect to R. The momentum line encodes the resulting momentum including its linear and angular aspects relative to any arbitrarily chosen point.

The linearity of the wedge removes the point X here as it does in the case of force and torque: with R a unitized projective point,

$$L(R) = P - R \wedge att(P) \tag{289}$$

which is eqn. (282) with the momentum line in place of the force line — its derivation, the role of the ideal direction and its validity for a bivector that is not simple are given there, and the library's `moment_about(R, P)` computes both. The physical fact behind it is the same one: two momentum lines carrying the same linear momentum differ by an ideal bivector, the linear momentum cancelling in the weight and leaving a pure angular momentum, which is therefore the same about every point. $R \wedge att(P)$ carries p on the line through R parallel to P, so what P differs from it by is the angular momentum about R. Two equal masses moving with opposite momenta are that case outright: their centre of mass is at rest, the momentum line is ideal, and the angular momentum is independent of the point it is taken about. The

generalisation carries more weight here: a single point mass gives a simple P, while the total momentum of a mechanism is a sum of momentum lines and in general a screw, so eqn. (288) has no point X to offer, whereas eqn. (289) returns the sum of the parts' angular momenta. Taken about the centre of mass it is $\sum_i r'_i \wedge p'_i$ with r'_i measured from there.

The equations (286) and (287) given above permit to separate the linear and angular effects as in the classical approach. However, these aspects need not be separated at all and can be calculated together in PGA.

Using eqn. (284) and the definition $X = O + x$ with X as unitized projective point, O as the chosen origin and x as the direction vector pointing from O to X we can write

$$P_O(X) = X \wedge m\dot{X} = (O + x) \wedge m(\dot{O + x}) \tag{290}$$

Compared to eqn. (284) an index O was added to P, since the formula (290) seems to confirm that the momentum depends on the choice of the origin. Now it is important to demonstrate that this is *not* the case. We are going to demonstrate that the described physics does not depend on the chosen origin and not on the resulting coordinates used for its description.

We use a transformation of the standpoint from which we describe the physical situation without actually moving the coordinate system (we still want to use the chosen coordinate system to describe the vectors and bivectors used): We go to a virtual origin $O' = Y$, but describe the situation in the same coordinate system. This results in describing the origin as $O = Y - y$ as seen from Y. Inserting that definition into eqn. (290) and using the geometric vector identity

$$X = Y + (X - Y) = Y + (O + x - (O + y)) = Y + (x - y) \tag{291}$$

we express $P(X)$ as viewed from Y by reformulating the equation for P relative to the origin O

$$\begin{aligned} P_O(X) &= X \wedge m\dot{X} = (O + x) \wedge m(\dot{O + x}) \\ &= (Y - y + x) \wedge m(\dot{Y - y + x}) \\ &= (Y + (x - y)) \wedge m(\dot{Y + (x - y)}) = P_Y(X) \end{aligned} \tag{292}$$

where the last equation obviously is the same as eqn. (290), but here Y takes the role of O and $(x - y)$ takes the role of the vector x in the old description.

The dot used for the time-derivative refers to the whole expression in parentheses. So the same physical situation was described from another viewpoint. However, the algebraic object P is unchanged, i.e. $P_O(X) = P_Y(X)$. This clearly shows that P does not depend on the chosen origin and that eqn. (290) is indeed a coordinate free definition.

Mass distributions: The total momentum of a set of mass points $m_i X_i$ is defined as the sum of all momentum blades.

$$P_{total} = \sum_i P_i \tag{293}$$

P_{total} will always be a momentum blade in $2D$ PGA, while in $3D$ PGA (a four-dimensional algebra) it could also result in a general bivector that might not be factorizable into a blade (see [Browne, 2012]). That would be the case, if $P \wedge P \neq 0\ \mathbb{1}$, i.e. when the result is a pseudoscalar that is not equal to a zero multiple of the pseudoscalar $\boldsymbol{e}_{1234}$. This test can always be used to determine whether a bivector is a blade in $3D$ PGA.

Using discrete points $X_{w,i}$ of the world frame indexed by i that can be moved by a motor $M(t) = exp_\vee[\frac{1}{2}B_w(t)] \vee M_0$ in the world frame as by eqn. (149) $X_{w,i} = M \vee X_{0,i} \vee \underset{\sim}{M}$ results in a local speed of $\dot{X}_{w,i} = \text{rcmt}(\Omega_w, X_{w,i})$ as shown in eqn. (181). Therefore, we can write the total momentum P_w in the world frame as

$$\begin{aligned} P_w &= \sum_i P_i \\ &= \sum_i X_i \wedge m_i \dot{X}_i \\ &= \sum_i X_i \wedge m_i\ \text{rcmt}(\Omega_w, X_i) \\ P_w &\equiv I_w[\Omega_w] \end{aligned} \tag{294}$$

where eqn. (294) is a linear function of Ω_w that maps the bivector rate in the world frame Ω_w to the total momentum P_w. It is a function that takes a bivector as an argument and returns a bivector (the "inertia map" I_w). It describes the influence of the mass distribution of the points on the movement via the combined Newton/Euler law

$$\dot{P}_w = F_w \tag{295}$$

(for the total force in the *world frame* $F_{total} = F_w$ see eqn. (283)). If we go from discrete mass points to a continuous mass distribution, the sum has

to be replaced by an integral over $dm = \rho dV$ (see following subsections on inertia maps for more details).

The equation $\dot{P}_w = X_w \wedge m\ddot{X}_w = F_w$ cannot be integrated directly since the second derivative $\ddot{X}_w$ cannot be isolated on the left-hand side of the equation as required for direct integration. To circumvent this we introduce another variable that can be isolated, enabling us to solve the ODE. For this purpose we use the rate of change Ω (a vector in $2D$ and a bivector in $3D$). The integration of that variable will then allow for computation of position, speed, etc. as a separate step.

Rate of change: Calculating the derivative $\dot{P}_w$ in the world frame in general yields two terms

$$\dot{P}_w = \dot{I}_w[\Omega_w] + I_w[\dot{\Omega}_w] \tag{296}$$

since in the world frame the inertia $I_w[\Omega_w]$ map will change over time due to moving points as well as due to the rate of change Ω_w. Therefore, it is reasonable to calculate the inertia map in the body frame, since it is constant in that frame, if the body frame is chosen to be fixed to the modelled rigid object. This avoids the effort to additionally calculate the time derivative of the inertia map (the first term on the right-hand side above).

Body frame: If we want to express the physical problem in the body frame, we have to transform the components of eqn. (294) accordingly. Using eqn. (180b), i.e. $I_b = \underset{\sim}{M} \vee I_w \vee M$ and eqn. (180a) $\Omega_w = M \vee \Omega_b \vee \underset{\sim}{M}$ then results in

$$\begin{aligned} P_w &= I_w[\Omega_w] \\ \underbrace{\underset{\sim}{M} \vee P_w \vee M}_{P_b} &= \underbrace{\underset{\sim}{M} \vee I_w[\underset{\sim}{M} \vee \Omega_w \vee M] \vee M}_{I_b[\Omega_b]} \end{aligned}$$

which finally results in an expression for the inertia map

$$P_b = I_b[\Omega_b] \equiv \underset{\sim}{M} \vee I_w[M \vee \Omega_b \vee \underset{\sim}{M}] \vee M \tag{297}$$

The same transformation must be applied to F_w for the right-hand side of the differential equation $\dot{P}_w = F_w$. Therefore, we get in the body frame $F_b = \underset{\sim}{M} \vee F_w \vee M$, resulting in $\dot{P}_b = F_b$.

The time derivative in the body frame is simpler, as long as we restrict ourselves to rigid bodies, resulting in only one term, since the rate of change

Ω_b is the only time dependent function in the body frame

$$\dot{P}_b = I_b[\dot{\Omega}_b] \tag{298}$$

otherwise $\dot{I}_b[\Omega_b]$ would have to be covered, as in eqn. (296) in the world frame.

The mapping to the classical inertia can be done when splitting Ω_b relative to the centroid of the body with a projective point split on Ω_b and on $I_b[\Omega_b]$ as shown by [Dorst and De Keninck, 2023] on page 25f and adjusted to our approach using regressive geometric products.

Reference points: Now we introduce some typically used reference points to describe kinematics via the bivector rate of change Ω and dynamics via the total momentum P:

- *Fixed body reference point* O_b used as origin of the body frame. When the body is moved by the motor M it will move in the world frame to $O_{b,w} = M \vee O_b \vee \underset{\sim}{M}$.
- *Relative locations* r_i *versus* O_b are positions in the body frame. Their projective counterpart in the body frame is $R_i = O_b + r_i$. When transformed to the world frame we get X_i in the world frame by the usual transformation $X_i = M \vee R_i \vee \underset{\sim}{M} = M \vee (O_b + r_i) \vee \underset{\sim}{M}$.
- *Center of mass* in the body frame is chosen to be located at O_b, such that $\sum_i m_i r_i = \vec{0}$. Thus, the origin of the body frame is located at the center of mass. The center of mass in the world frame C_w then is $C_w \equiv \sum_i m_i X_i / \sum_i m_i = \sum_i m_i (M \vee (O_b + r_i) \vee \underset{\sim}{M}) / \sum_i m_i = M \vee O_b \vee \underset{\sim}{M}$.

The last missing piece for dynamics is the time derivative of P_w expressed as a function of the time rate of change Ω_b of the inertia in the body system. Applying eqn. (181) in the body frame, which is valid for any rigid object moved by a motor M, we get

$$\frac{d}{dt} P_b = \frac{d}{dt} I_b[\Omega_b] = \text{rcmt}(\Omega_b, I_b[\Omega_b]) + I_b[\dot{\Omega}_b] \tag{299}$$

where the last term arises since Ω_b is also time-dependent. Solving eqn. (295), but in the body frame, we get

$$\dot{P}_b = \text{rcmt}(\Omega_b, I_b[\Omega_b]) + I_b[\dot{\Omega}_b] = F_b \tag{300}$$

resolving this for $\dot{\Omega}_b$ we get

$$I_b[\dot{\Omega}_b] = F_b - \text{rcmt}(\Omega_b, I_b[\Omega_b]) \tag{301}$$

$$\dot{\Omega}_b = I_b^{-1}\left[F_b - \text{rcmt}(\Omega_b, I_b[\Omega_b])\right] \tag{302}$$

Together with the ansatz in the body system eqn. (175) that can be used to calculate M(t) when we know Ω_b at a given time t from

$$M(t) = M_0 \vee \exp_\vee(\frac{1}{2}B_b(t)) = M_0 \vee \exp_\vee(\frac{1}{2}\Omega_b t)$$

and the rate of change in the body frame according to eqn. (177) we get

$$\dot{\Omega}_b = I_b^{-1}\left[F_b - \text{rcmt}(\Omega_b, I_b[\Omega_b])\right] \tag{303}$$

as an ODE that can be solved as initial value problem. Due to $\dot{\Omega}_b = \ddot{B}_b$ it actually is a 2nd order ODE (like the classical Newton/Euler equations that can be integrated directly from the instantaneous acceleration and the angular acceleration, respectively, in order to calculate speed and position). Here we rather integrate from $\ddot{B}_b$ in order to calculate $\dot{B}_b = \Omega_b$ and B_b. Thus, we require initial values for Ω_b and B_b. The actual positions of the points of the rigid body can then be computed from eqn. (149) $X_w(t) = M(t) \vee X_0 \vee \underset{\sim}{M}(t)$. The speed of the points can be calculated either from eqn. (181) $\dot{X}_w = \text{rcmt}(\Omega_w, X_w)$ or eqn. (183) $\dot{X}_w = \text{rcmt}(\Omega_w, X_w) + (\dot{X}_b)_w$ – which one is applicable depends on whether the rigid body is fixed in the body frame or moves in the body frame. The expressions in the world frame can of course be expressed in the body frame as well using eqn. (180a) $\Omega_w = M \vee \Omega_b \vee \underset{\sim}{M}$ and $X_w = M \vee X_b \vee \underset{\sim}{M}$.

Inertia map ($2D$): For that purpose we need the inertia map $I_b[\Omega_b]$, and it's inverse to set up the right-hand side of the ODE system given in equations (303). We start with the $2D$ case where the rate of change Ω_b is a vector, as shown at the end of subsection 1.4. We assume discrete points X_i and use the wedge product $X \wedge Y$ and the regressive commutator product term $m\,\text{rcmt}(\Omega_b, X)$ to express the total momentum P written in component form:

$$X \wedge Y = \begin{pmatrix} X_x \\ X_y \\ X_z \end{pmatrix} \wedge \begin{pmatrix} Y_x \\ Y_y \\ Y_z \end{pmatrix} = \begin{pmatrix} X_z Y_x - X_x Y_z \\ X_z Y_y - X_y Y_z \\ X_x Y_y - X_y Y_x \end{pmatrix} \tag{304a}$$

$$m\,\text{rcmt}(\Omega_b, X) = m \begin{pmatrix} \Omega_y X_z - \Omega_z X_y \\ \Omega_z X_x - \Omega_x X_z \\ 0 \end{pmatrix} \tag{304b}$$

Now the inertia map for a point X with mass m and rate of change Ω can

be expressed in the $2D$ case as

$$\begin{aligned} I_{2D}[\Omega] &= X \wedge m \operatorname{rcmt}(\Omega, X) \\ &= m \begin{pmatrix} X_z(\Omega_y X_z - \Omega_z X_y) \\ X_z(\Omega_z X_x - \Omega_x X_z) \\ X_x(\Omega_z X_x - \Omega_x X_z) - X_y(\Omega_y X_z - \Omega_z X_y) \end{pmatrix} \\ I_{2D}[\Omega] &= m \begin{bmatrix} 0 & X_z^2 & -X_y X_z \\ -X_z^2 & 0 & X_x X_z \\ -X_x X_z & -X_y X_z & X_x^2 + X_y^2 \end{bmatrix} \begin{bmatrix} \Omega_x \\ \Omega_y \\ \Omega_z \end{bmatrix} \end{aligned} \tag{305}$$

where eqn. (305) is linear in Ω as expected. Of course, the total inertia map needs to be summed for the contributions from each point, i.e. $I_{total} = \sum_i I_i$. Since all points of the rigid body see the same rate of change the total inertia map can be summed for all mass point contributions independent of the rate of change and the resulting total inertia map can be inverted either symbolically or numerically to get its inverse I_{2D}^{-1} as needed to set up the right-hand side of the system in eqn. (303). The inertia map in $2D$ maps the rate of change vector to the momentum bivector. The inverse inertia map than maps from a momentum bivector to the rate of change vector.

Inertia map ($3D$): Now let's do the same for the $3D$ case, where the rate of change is a bivector. The component expressions are generated from the corresponding product tables in `ga_prdxpr` and integrated into the products in the ga library.

$$X \wedge Y = \begin{pmatrix} X_x \\ X_y \\ X_z \\ X_w \end{pmatrix} \wedge \begin{pmatrix} Y_x \\ Y_y \\ Y_z \\ Y_w \end{pmatrix} = \begin{pmatrix} X_w Y_x - X_x Y_w \\ X_w Y_y - X_y Y_w \\ X_w Y_z - X_z Y_w \\ X_y Y_z - X_z Y_y \\ X_z Y_x - X_x Y_z \\ X_x Y_y - X_y Y_x \end{pmatrix} \tag{306a}$$

$$m \operatorname{rcmt}(\Omega_b, X) = m \begin{pmatrix} \Omega_{vy} X_z - \Omega_{vz} X_y + \Omega_{mx} X_w \\ \Omega_{vz} X_x - \Omega_{vx} X_z + \Omega_{my} X_w \\ \Omega_{vx} X_y - \Omega_{vy} X_x + \Omega_{mz} X_w \\ 0 \end{pmatrix} \tag{306b}$$

Now the inertia map for a point X with mass m and rate of change Ω can

be expressed in the $3D$ case as

$$I_{3D}[\Omega] = X \wedge m \operatorname{rcmt}(\Omega, X) =$$

$$m \begin{pmatrix} X_w(\Omega_{vy}X_z - \Omega_{vz}X_y + \Omega_{mx}X_w) \\ X_w(\Omega_{vz}X_x - \Omega_{vx}X_z + \Omega_{my}X_w) \\ X_w(\Omega_{vx}X_y - \Omega_{vy}X_x + \Omega_{mz}X_w) \\ X_y(\Omega_{vx}X_y - \Omega_{vy}X_x + \Omega_{mz}X_w) - X_z(\Omega_{vz}X_x - \Omega_{vx}X_z + \Omega_{my}X_w) \\ X_z(\Omega_{vy}X_z - \Omega_{vz}X_y + \Omega_{mx}X_w) - X_x(\Omega_{vx}X_y - \Omega_{vy}X_x + \Omega_{mz}X_w) \\ X_x(\Omega_{vz}X_x - \Omega_{vx}X_z + \Omega_{my}X_w) - X_y(\Omega_{vy}X_z - \Omega_{vz}X_y + \Omega_{mx}X_w) \end{pmatrix} =$$

$$m \begin{bmatrix} 0 & X_zX_w & -X_yX_w & X_w^2 & 0 & 0 \\ -X_zX_w & 0 & X_xX_w & 0 & X_w^2 & 0 \\ X_yX_w & -X_xX_w & 0 & 0 & 0 & X_w^2 \\ X_y^2 + X_z^2 & -X_xX_y & -X_xX_z & 0 & -X_zX_w & X_yX_w \\ -X_xX_y & X_x^2 + X_z^2 & -X_yX_z & X_zX_w & 0 & -X_xX_w \\ -X_xX_z & -X_yX_z & X_x^2 + X_y^2 & -X_yX_w & X_xX_w & 0 \end{bmatrix} \begin{bmatrix} \Omega_{vx} \\ \Omega_{vy} \\ \Omega_{vz} \\ \Omega_{mx} \\ \Omega_{my} \\ \Omega_{mz} \end{bmatrix} \quad (307)$$

where eqn. (307) is linear in Ω as expected. Again, the total inertia map needs to be summed for the contributions from each point, i.e. $I_{total} = \sum_i I_i$. The resulting total inertia map can be inverted numerically to get its inverse I_{3D}^{-1} as needed to set up the right-hand side of the system in eqn. (303). The inertia map in $3D$ maps the rate of change bivector to the momentum bivector. The inverse inertia map than maps from a momentum bivector to the corresponding rate of change bivector.

As shown in [Dorst and De Keninck, 2023] the inertia map is additive. That means that inertia maps of parts of a body consisting of several parts that move with the same bivector rate can be added to get the total inertia.

1.7.3 Solving the Newton-Euler equations numerically

In classical mechanics the Newton equation states that change of linear momentum $\vec{p} = m\vec{v}$ of a mass point of mass m moving with speed $\vec{v}$ is driven by an external force $\vec{f}$ acting on the mass. This results in $\dot{\vec{p}} = m\dot{\vec{v}} = m\ddot{\vec{x}}_{cm} = \vec{f}$ for translational motion in a system with constant mass m. The center of mass at position $\vec{x}_{cm}$ moves with the velocity $\vec{v}$ as a result.

The analogous equation for rotational motion, attributed to Euler, states that the temporal change of angular momentum $\vec{L} = \vec{x} \times \vec{p} = \vec{x} \times m\vec{v}$ is driven by an external angular momentum $\vec{M}$. $\vec{M}$ is given as $\vec{M} = \vec{r} \times \vec{f}$ with $\vec{r}$ as the vector pointing from the rotational axis to the point of action of the force $\vec{f}$. The equation for rotational motion $\dot{\vec{L}} = J\dot{\vec{\omega}} = \vec{M}$ for a rigid body with moment of inertia $J = \int_m r_\perp^2 dm = \int_V r_\perp^2 \rho dV$ with $r_\perp$ as the perpendicular distance of the mass point within the rigid body to the momentary rotational axis. The equation determines the temporal development of the orientation of the rigid body which freely spins with the angular speed $\|\vec{\omega}\|$ around the momentary rotation axis defined by $\vec{\omega}$ as a result.

Due to the same structure the equations of a freely moving body can be written as one system of equations that can be solved as uncoupled equations (e.g. covering cases with either translation or rotation exclusively within the identical structure). In the $2D$ case the translational part requires two degrees of freedom (DoF) and the rotational part needs one DoF resulting in a total of three DoF. It can be written compactly as a system introducing the inertia matrix I as a linear function that maps the acceleration degrees of freedom to the corresponding external force and torque components[13]:

$$\underbrace{\begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & J \end{bmatrix}}_{\text{inertia } I} \begin{pmatrix} \ddot{x}_x \\ \ddot{x}_y \\ \dot{\phi} \end{pmatrix} = \begin{pmatrix} f_x \\ f_y \\ M \end{pmatrix} \tag{308}$$

Solving such a system requires to isolate the accelerations by multiplying the

[13] Be aware, that this does not actually couple the equations! We just solve two independent equations conveniently in parallel to show the principle here. If the movement is actually coupled, coupling terms must be added to model physics correctly. In the *PGA* formulation this coupling is already captured by the rcmt term in the body-frame equation of motion (303).

system from the left with the inverse of the inertia map I^{-1} resulting in

$$\begin{pmatrix} \ddot{x}_x \\ \ddot{x}_y \\ \ddot{\phi} \end{pmatrix} = \underbrace{\begin{bmatrix} \frac{1}{m} & 0 & 0 \\ 0 & \frac{1}{m} & 0 \\ 0 & 0 & \frac{1}{J} \end{bmatrix}}_{inverse\ inertia\ I^{-1}} \begin{pmatrix} f_x \\ f_y \\ M \end{pmatrix} \tag{309}$$

Both equations result in a system of ordinary differential equation (ODE) of second order that requires two initial conditions, one for the initial position $\vec{x}(t=0) = \vec{x}_0$ and the initial speed $\vec{v}(t=0) = \vec{v}_0$ (Newton equation) and one for the initial orientation $\phi(t=0) = \phi_0$ and the initial angular speed $\omega(t=0) = \omega_0$ (Euler equation) to be integrated. In the $2D$ case ϕ and ω are scalar values, while in the $3D$ case they also become vectors each representing three degrees of freedom.

To integrate the equations the system of 2nd order must be transferred into a system of 1st order. We have a 2nd order ODE for $y(t)$ with $\ddot{y} = \frac{d^2}{dt^2}(y)$ of the general form

$$\ddot{\vec{y}} = f(\dot{\vec{y}}, \vec{y}, t) \tag{310}$$

The vector y has components that we index with n. We use an additional index to distinguish between the position (index 0) and the speed (index 1) of the respective DoF. With the substitution

$$y_{n,0} = y_n \qquad \rightarrow position \tag{311a}$$

$$y_{n,1} = \dot{y}_n \qquad \rightarrow speed \tag{311b}$$

this results in the following system of 1st order for each component n of the vector $\vec{y}$:

$$\dot{y}_{n,0} = y_{n,1} \tag{312a}$$

$$\dot{y}_{n,1} = f(y_{n,1}, y_{n,0}, t) \tag{312b}$$

If we would have to integrate for more than one point, one could easily add an index at the beginning to extend the system accordingly. However, for now we only focus on integrating the system at one point (the center of mass) at which we want to know position and speed as well as angular orientation and angular speed as function of time. The substitution above transforms

the system in eqn. (309) into a system of 1st order as follows:

$$\dot{x}_x = v_x \tag{313a}$$
$$\dot{v}_x = \frac{1}{m} f_x \tag{313b}$$
$$\dot{x}_y = v_y \tag{313c}$$
$$\dot{v}_y = \frac{1}{m} f_y \tag{313d}$$
$$\dot{\phi} = \omega \tag{313e}$$
$$\dot{\omega} = \frac{1}{J} M \tag{313f}$$

This 1st order ODE system $\dot{\vec{u}} = rhs(\vec{u}, t)$ can now be solved by direct integration, using e.g. a Runge-Kutta-Scheme of 4th order accuracy for precise time integration, and using the initial conditions defined above. As a result, we directly get the positions and velocities as function of time.

Classical body frame vs. *PGA* — what unifies: It is instructive to compare the classical body-frame formulation with the single *PGA* equation of motion (303), $\dot{\Omega}_b = I_b^{-1}[F_b - \mathrm{rcmt}(\Omega_b, I_b[\Omega_b])]$. Classically one carries two separate vector equations – Newton's for $\dot{\vec{v}}$ and Euler's for $\dot{\vec{\omega}}$ (the latter with the nonlinear gyroscopic term $\vec{\omega} \times \mathbf{I}_b\vec{\omega}$) – plus a separate orientation update via a rotation matrix or quaternion. The *PGA* formulation collapses all of this into one bivector ODE plus a motor update, as summarised in Table 31.

Classical body frame	*PGA* equivalent
Two separate equations: Newton + Euler	One unified equation for $\dot{\Omega}_b$
Coupling term $\vec{\omega} \times \vec{v}$	Encoded in $\mathrm{rcmt}(\Omega_b, I_b[\Omega_b])$
Gyroscopic term $\vec{\omega} \times \mathbf{I}_b\vec{\omega}$	Also encoded in $\mathrm{rcmt}(\Omega_b, I_b[\Omega_b])$
Inertia tensor $\mathbf{I}_b$ (3×3 or 6×6 matrix)	Inertia map $I_b[\cdot]$ (linear operator on bivectors)
Quaternion evolution $\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \otimes \vec{\omega}$	Motor $M = \exp_\vee[\frac{1}{2}B_b]$ – orientation implicit in B_b
State: $(\vec{r}, \mathbf{q}, \vec{v}, \vec{\omega})$	State: (B_b, Ω_b) – bivector pair
Inertial-frame position from $\dot{\vec{r}} = \mathbf{R}\vec{v}$	Inertial-frame points updated via sandwich product $X(t) = M \vee X_b(t) \vee \underset{\sim}{M}$

Table 31: Classical body-frame vs. *PGA* formulation: explicit correspondence.

Where the classical formulation needs one equation for $\dot{\vec{v}}$ (3 DoF), one for $\dot{\vec{\omega}}$ (3 DoF), and a separate orientation integration, the *PGA* ODE handles the first two in a single equation over the bivector $\dot{\Omega}_b$ (6 DoF) and recovers position and orientation from a motor derived directly from Ω_b. The single term $\mathrm{rcmt}(\Omega_b, I_b[\Omega_b])$ plays the combined role of both classical coupling terms ($\vec{\omega} \times \vec{v}$ and $\vec{\omega} \times \mathbf{I}_b\vec{\omega}$). As described above, the *PGA* system is first solved for $B(t)$, $\dot{B} = \Omega(t)$ and $\ddot{B} = \dot{\Omega}(t)$; the actual points and their velocities then follow in a second step from the motor and the sandwich transformation.

Basic examples for translational and rotational motion are implemented as tests in `ga_test/src/ga_appl2dp_mechanics_test.hpp` and `ga_test/src/ga_appl3dp_mechanics_test.hpp` based on the Newton-Euler equations in both approaches, the classical approach and the approach directly formulated in *PGA*, for very simple reference cases.

Constrained motion — rotation about a fixed pivot: The body-frame *PGA* formulation extends naturally to constrained motion. A common case in $2D$ is a rigid body whose rotation is constrained so that a specific body-frame point Q_b (the pivot) remains fixed in the world – for example a plate pendulum fixed at one of its corners. Two questions are to be answered independently:

1. *In which frame are the equations expressed?* (body frame vs. world frame)
2. *About which axis does the body actually rotate?* (center of mass vs. pivot)

The Euler equation $\dot{\Omega}_b = I_{cm}^{-1}[F_b - \mathrm{rcmt}(\Omega_b, I_{cm}[\Omega_b])]$ describes a *free* rigid body rotating about its own center of mass. When the pivot is fixed, the center of mass follows a circular arc of radius r around the pivot, with a displacement of the rotation point relative to the center of mass, effectively increasing the inertia relative to that pivot point. The constraint reaction also creates centripetal acceleration to keep the center of mass moving along the arc. This is felt in the body system at the center of mass as additional centrifugal acceleration. The Steiner (parallel axis) theorem gives the corrected moment of inertia in $2D$:

$$I_{\mathrm{pivot}}[2,2] \;=\; I_{cm}[2,2] + mr^2 \tag{314}$$

so the angular acceleration about the fixed pivot is $\alpha = \tau_{\mathrm{eff}}/I_{\mathrm{pivot}}[2,2]$, *not* $\tau_{\mathrm{eff}}/I_{cm}[2,2]$. This is a correction for the physical rotation axis, not for the

choice of frame – it is required regardless of whether the equations are expressed in the body frame or the world frame.

In the *PGA* implementation the scalar Steiner correction is built into the inertia constructor via an optional pivot parameter:

- $2D$ – *pivot is a point* Q_b (grade 1, `Vec2dp`):
  `get_plate_inertia(m, w, h, Q_b)` updates only $I[2,2]$ by adding $m\,r^2$ where $r^2 = Q_{b,x}^2 + Q_{b,y}^2$ is the squared distance from the body origin to Q_b. All other entries (including off-diagonal coupling terms) remain equal to I_{body}. The default $Q_b = (0,0,1)$ (body origin) gives I_{body} with no correction.

- $3D$ – *pivot is a line* L_{pivot} (grade 2, `BiVec3dp`):
  `get_cuboid_inertia(m, w, h, d, L_pivot)` computes the foot of the perpendicular $P_{\text{foot}} = (P_x, P_y, P_z)$ from the body origin to L_{pivot} and updates only the diagonal entries of the lower-left block: $I[3,0]$ += $m(P_y^2 + P_z^2)$, $I[4,1]$ += $m(P_x^2 + P_z^2)$, $I[5,2]$ += $m(P_x^2 + P_y^2)$. Off-diagonal products of inertia are not added. The default $L_{\text{pivot}} = 0$ (pivot through origin) gives no correction. The grade of the pivot entity – point in $2D$, line in $3D$ – matches the grade of the velocity generator Ω (see Tables 32 and 34).

After computing $\dot{\Omega}_b = I_{\text{pivot}}^{-1}[F_b - \text{rcmt}(\Omega_b, I_{\text{pivot}}[\Omega_b])]$ a constraint projection is still required to prevent numerical drift of off-constraint components during RK4 integration. In $2D$ the body-frame state is encoded as $B_b = \varphi \cdot Q_b$ and $\Omega_b = \omega \cdot Q_b$; since $Q_b.z = 1$ the scalar angular acceleration is

$$\alpha = (\dot{\Omega}_b)_z, \qquad \dot{\Omega}_b = \alpha \cdot Q_b \tag{315}$$

In $3D$ the analogous projection onto the normalized pivot line gives $\alpha = \dot{\Omega}_b \cdot \hat{L}_{\text{pivot}}$. The body-frame encoding $B_b = \varphi \cdot Q_b$ carries a key geometric property: the world image of Q_b is stationary for all φ. From the body-frame motor formula (175), $M(t) = M_0 \vee R(\varphi)$ with the pure pivot rotation $R(\varphi) = \exp_\vee(\frac{1}{2}B_b(t)) = \exp_\vee(\frac{1}{2}\varphi Q_b)$ and the constant base motor $M_0 = M(t{=}0)$. A finite point is invariant under a rotation about itself, so $R(\varphi) \vee Q_b \vee \underbrace{R(\varphi)} = Q_b$, and the world image of the pivot reduces to

$$M(t) \vee Q_b \vee \underbrace{M(t)} \;=\; M_0 \vee Q_b \vee \underbrace{M_0} \;=\; Q_{b,\text{world}} = \text{const} \tag{316}$$

No explicit constraint enforcement is required for the *position* of the pivot – it is preserved exactly for all φ by the motor algebra. Only the velocity direction (eqn. (315)) must be enforced at each integration step.

1.7.4 Structure of inertia and inertia maps

The inertia map $I[\Omega]$ derived in the preceding subsections is a linear map that takes the velocity twist Ω of a rigid body and returns its momentum wrench $P = I[\Omega]$. Its matrix representation – shown in eqns. (305) and (307) – encodes mass, scalar moment of inertia, and the full classical inertia tensor in a single algebraic object. However, the classical quantities do not sit where standard intuition might suggest: mass does not appear on the diagonal, and moments of inertia do not appear in the block connecting forces to accelerations of the same type.

The placement of each entry is determined by two ingredients: the choice of basis for the input and output of the inertia map, and the complement structure of the respective algebra. The right complement $\overline{(\cdot)}$ maps each basis element to its complement in the algebra. In PGA the translational and rotational generators live in dual subspaces – ideal lines are dual to real lines – so the mass contribution, which couples velocity to momentum across dual subspaces, always lands in an off-diagonal block. The following paragraphs explain this structure first for $2D$, then for $3D$, and concludes with a unified comparison.

PGA2DP — **structure of the 2D inertia map:** In $2D$ PGA the velocity twist Ω is a vector and the momentum wrench P is a bivector, so the inertia map crosses two different grades:

$$I_{2D}[\Omega] : \text{Vec2dp} \to \text{BiVec2dp} \tag{317}$$

The basis elements and their physical roles are listed in Table 32.

Index	Vec2dp input (velocity twist Ω)	BiVec2dp output (momentum wrench P)
0	$\boldsymbol{e}_1$ – x-velocity	$\boldsymbol{e}_{31}$ – x-momentum
1	$\boldsymbol{e}_2$ – y-velocity	$\boldsymbol{e}_{32}$ – y-momentum
2	$\boldsymbol{e}_3$ – angular velocity	$\boldsymbol{e}_{12}$ – angular momentum

Table 32: Basis correspondence for the $2D$ PGA inertia map.

For a unitized point (X_x, X_y, X_z) with projective coordinate $X_z = 1$ and mass m, the inertia matrix from eqn. (305) simplifies to

$$I_{2D} = m \begin{bmatrix} 0 & 1 & -X_y \\ -1 & 0 & X_x \\ -X_x & -X_y & X_x^2 + X_y^2 \end{bmatrix} \tag{318}$$

where rows correspond to output components ($\boldsymbol{e}_{31}$, $\boldsymbol{e}_{32}$, $\boldsymbol{e}_{12}$) and columns to input components (Ω_x, Ω_y, Ω_z).

Why mass is off-diagonal: The mass contribution to the inertia map takes the form $I_{\text{mass}}(\Omega) = -m \cdot \overline{\Omega}$, where $\overline{(\cdot)}$ denotes the complement in $PGA2DP$. The complement of the translational basis vectors crosses indices:

$$\overline{\boldsymbol{e}_1} = \boldsymbol{e}_{32}, \qquad \overline{\boldsymbol{e}_2} = -\boldsymbol{e}_{31} \tag{319}$$

Applying $I_{\text{mass}} = -m \cdot \overline{(\cdot)}$ gives the following contributions:

- Ω_x (component along $\boldsymbol{e}_1$, column 0) maps to $-m \cdot \boldsymbol{e}_{32}$, landing at output row $1 \to I[1,0] = -m$.
- Ω_y (component along $\boldsymbol{e}_2$, column 1) maps to $-m \cdot (-\boldsymbol{e}_{31}) = +m \cdot \boldsymbol{e}_{31}$, landing at output row $0 \to I[0,1] = +m$.

The diagonal entries $I[0,0]$ and $I[1,1]$ are zero because the complement of $\boldsymbol{e}_1$ is $\boldsymbol{e}_{32}$ (not $\boldsymbol{e}_{31}$) and the complement of $\boldsymbol{e}_2$ is $-\boldsymbol{e}_{31}$ (not $\boldsymbol{e}_{32}$): there is no self-coupling. This is a direct consequence of the degenerate metric $\boldsymbol{e}_3^2 = 0$.

The resulting mass block is antisymmetric with $I[0,1] = +m$ and $I[1,0] = -m$. The sign structure is consistent with the kinematic encoding: the motor-generator Ω for a pure translation with classical velocity (v_x, v_y) is encoded in $2D$ PGA as $\Omega = (-v_y, v_x, 0)$. The two sign flips cancel and the classical linear momentum $p_x = mv_x$, $p_y = mv_y$ is correctly recovered. In any other context – position, force, acceleration – $\boldsymbol{e}_1$ always encodes the x-component and $\boldsymbol{e}_2$ the y-component.

Moment of inertia and cross-coupling terms: The entry $I[2,2] = m(X_x^2 + X_y^2) = mr^2$ maps angular velocity ($\boldsymbol{e}_3$, index 2) to angular momentum ($\boldsymbol{e}_{12}$, index 2), recovering the classical scalar moment of inertia $J = mr^2$ for a point mass at distance r from the rotation center. The complement of $\boldsymbol{e}_3$ is $\overline{\boldsymbol{e}_3} = -\boldsymbol{e}_{12}$, which connects the angular velocity generator directly to the angular momentum carrier; the sign is absorbed into the definition so that $J > 0$. The entries $I[0,2] = -mX_y$ and $I[1,2] = +mX_x$ couple angular velocity to linear momentum, encoding the classical relation $\vec{L}_{\text{linear}} = m(\vec{\omega} \times \vec{r})$ for a body rotating at offset (X_x, X_y) from the rotation center. These entries, together with their transposes $I[2,0]$ and $I[2,1]$, vanish when the point is at the origin.

Body-frame inertia matrix: Integrating the single-point formula over a rigid body with center of mass at the origin (body frame) gives:

$$I_{\text{body}} = \begin{bmatrix} 0 & m & 0 \\ -m & 0 & 0 \\ 0 & 0 & J \end{bmatrix} \tag{320}$$

where $J = \int_m (x^2 + y^2)\, dm$ is the classical scalar moment of inertia about the center of mass. The antisymmetric mass block (upper-left 2×2) and the moment of inertia entry (lower-right 1×1) are cleanly separated: translational and rotational degrees of freedom decouple at the center of mass, exactly as in classical mechanics.

General frame with center of mass at (c_x, c_y): When the chosen reference point does not coincide with the center of mass, the integrated inertia matrix is:

$$I = \begin{bmatrix} 0 & m & -mc_y \\ -m & 0 & mc_x \\ -mc_x & -mc_y & J_{\text{cm}} + m(c_x^2 + c_y^2) \end{bmatrix} \tag{321}$$

where J_{cm} is the scalar moment of inertia about the center of mass and $r^2 = c_x^2 + c_y^2$ is the squared distance from the center of mass to the reference point. The matrix decomposes into the following sub-blocks:

- *Upper-left* 2×2 $[0, m;\ -m, 0]$: the antisymmetric mass block. It is *frame-invariant* – its values are unchanged regardless of where the center of mass is located.
- *Upper-right* 2×1 $[-mc_y;\ mc_x]$: angular velocity → linear momentum coupling. Encodes $\vec{L}_{\text{linear}} = m(\vec{\omega} \times \vec{r}_{\text{cm}})$. Zero only when the center of mass coincides with the reference point.
- *Lower-left* 1×2 $[-mc_x,\ -mc_y]$: translational velocity → angular momentum coupling. Transpose of the upper-right block (up to sign convention).
- *Lower-right* 1×1: $J_{\text{cm}} + m(c_x^2 + c_y^2)$ – the classical parallel axis theorem applied to the scalar moment of inertia.

Uniform rectangular plate: As a verification, integrating over a uniform plate of width W (along $\boldsymbol{e}_1$), height H (along $\boldsymbol{e}_2$), total mass M, centered at

the origin gives:

$$I_{\text{plate}} = M \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & \dfrac{W^2 + H^2}{12} \end{bmatrix} \tag{322}$$

The entry $I[2,2] = M(W^2 + H^2)/12$ is the standard formula for the moment of inertia of a uniform rectangle about its centroid axis. For a discrete endpoint-inclusive grid of $n \times n$ equally spaced point masses, the relative error in $I[2,2]$ compared to the continuous value is $2/(n-1)$, independent of plate dimensions, converging from above as $n \to \infty$.

Table 33 summarizes the classical inertial quantities and their locations in the $2D$ PGA inertia matrix.

Classical quantity	$2D$ PGA entry	Condition
Total mass m	$I[0,1] = +m$, $I[1,0] = -m$	Always
Scalar moment J about CoM	$I[2,2] = J$	CoM at origin
CoM offset (c_x, c_y)	$I[0,2] = -mc_y$, $I[1,2] = +mc_x$ (and transposed)	General frame
Parallel axis correction	$I[2,2] = J_{\text{cm}} + m(c_x^2 + c_y^2)$	General frame

Table 33: Classical inertial quantities and their location in the $2D$ PGA 3×3 inertia matrix.

$3D$ PGA — **structure of the 3D inertia map:** In $3D$ PGA both the velocity twist Ω and the momentum wrench P are bivectors, so the inertia map acts within the same grade:

$$I_{3D}[\Omega] : \text{BiVec3dp} \to \text{BiVec3dp} \tag{323}$$

The six basis bivectors split into two groups: *ideal lines* ($\boldsymbol{e}_{41}$, $\boldsymbol{e}_{42}$, $\boldsymbol{e}_{43}$, containing the projective direction $\boldsymbol{e}_4$ which squares to zero) carrying translational information, and *real lines* ($\boldsymbol{e}_{23}$, $\boldsymbol{e}_{31}$, $\boldsymbol{e}_{12}$) of the Euclidean subspace carrying rotational information. Their physical roles are listed in Table 34.

Idx	BiVec3dp input (velocity twist Ω)	BiVec3dp output (momentum wrench P)
0	$\boldsymbol{e}_{41}$ – ideal line, x-velocity	$\boldsymbol{e}_{41}$ – linear momentum x
1	$\boldsymbol{e}_{42}$ – ideal line, y-velocity	$\boldsymbol{e}_{42}$ – linear momentum y
2	$\boldsymbol{e}_{43}$ – ideal line, z-velocity	$\boldsymbol{e}_{43}$ – linear momentum z
3	$\boldsymbol{e}_{23}$ – real line, angular velocity about $\boldsymbol{e}_1$	$\boldsymbol{e}_{23}$ – angular momentum about $\boldsymbol{e}_1$
4	$\boldsymbol{e}_{31}$ – real line, angular velocity about $\boldsymbol{e}_2$	$\boldsymbol{e}_{31}$ – angular momentum about $\boldsymbol{e}_2$
5	$\boldsymbol{e}_{12}$ – real line, angular velocity about $\boldsymbol{e}_3$	$\boldsymbol{e}_{12}$ – angular momentum about $\boldsymbol{e}_3$

Table 34: Basis correspondence for the $3D$ PGA inertia map.

The 6×6 inertia matrix from eqn. (307) for a unitized point with $X_w = 1$ is:

$$I_{3D} = m \begin{bmatrix} 0 & X_z & -X_y & 1 & 0 & 0 \\ -X_z & 0 & X_x & 0 & 1 & 0 \\ X_y & -X_x & 0 & 0 & 0 & 1 \\ X_y^2 + X_z^2 & -X_x X_y & -X_x X_z & 0 & -X_z & X_y \\ -X_x X_y & X_x^2 + X_z^2 & -X_y X_z & X_z & 0 & -X_x \\ -X_x X_z & -X_y X_z & X_x^2 + X_y^2 & -X_y & X_x & 0 \end{bmatrix} \tag{324}$$

where columns correspond to $(\Omega_{vx}, \Omega_{vy}, \Omega_{vz}, \Omega_{mx}, \Omega_{my}, \Omega_{mz})$ and rows to the same ordering for P.

Block structure and the complement in 3D: The 6×6 matrix decomposes into four 3×3 blocks with distinct physical roles:

- *Upper-left [0:3, 0:3]*: translational velocity → linear momentum. For a point at the origin this block is zero. For a point displaced to (X_x, X_y, X_z) it becomes the antisymmetric cross-product matrix $[X\times]$, encoding the linear momentum contribution due to the off-axis position.

- *Upper-right [0:3, 3:6]*: angular velocity → linear momentum. For a unitized point ($X_w = 1$) this block equals $m\mathbf{I}_3$ (mass times the 3×3 identity). This is the $3D$ analogue of the off-diagonal mass entries in the $2D$ case.

- *Lower-left [3:6, 0:3]*: translational velocity → angular momentum. This block carries the classical moments and products of inertia.

- *Lower-right [3:6, 3:6]*: angular velocity → angular momentum. Zero for a point at the origin; becomes antisymmetric for a displaced center of mass.

The placement of mass in the upper-right block follows from the complement in $3D$ *PGA*, which maps real lines to ideal lines:

$$\overline{\boldsymbol{e}_{23}} = \boldsymbol{e}_{41}, \quad \overline{\boldsymbol{e}_{31}} = \boldsymbol{e}_{42}, \quad \overline{\boldsymbol{e}_{12}} = \boldsymbol{e}_{43} \tag{325}$$

Applying $I_{\text{mass}} = -m \cdot \overline{(\cdot)}$ maps each angular velocity component to the corresponding linear momentum component, placing m on the diagonal of the upper-right block: $I[0,3] = I[1,4] = I[2,5] = m$. The diagonal entries of the upper-left and lower-right blocks are zero by the same degenerate-metric argument as in $2D$.

Body-frame inertia matrix: Integrating the single-point formula over a rigid body with center of mass at the origin gives:

$$I_{\text{body}} = \begin{bmatrix} 0 & 0 & 0 & m & 0 & 0 \\ 0 & 0 & 0 & 0 & m & 0 \\ 0 & 0 & 0 & 0 & 0 & m \\ I_{xx} & -I_{xy} & -I_{xz} & 0 & 0 & 0 \\ -I_{xy} & I_{yy} & -I_{yz} & 0 & 0 & 0 \\ -I_{xz} & -I_{yz} & I_{zz} & 0 & 0 & 0 \end{bmatrix} \tag{326}$$

written in block form as $I_{\text{body}} = [\mathbf{0},\ m\mathbf{I}_3;\ \mathbf{J}_{cm},\ \mathbf{0}]$, where $\mathbf{J}_{cm}$ is the classical 3×3 inertia tensor about the center of mass:

$$\mathbf{J}_{cm} = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{bmatrix} \tag{327}$$

with the classical integrals for the diagonal moments $I_{xx} = \int_m (y^2 + z^2)\, dm$, $I_{yy} = \int_m (x^2 + z^2)\, dm$, $I_{zz} = \int_m (x^2 + y^2)\, dm$ and the off-diagonal products of inertia $I_{xy} = \int_m xy\, dm$, $I_{xz} = \int_m xz\, dm$, $I_{yz} = \int_m yz\, dm$. Translational and rotational degrees of freedom decouple at the center of mass: the diagonal blocks are zero and all physical content sits in the off-diagonal blocks.

General frame with center of mass at $\vec{r}_{cm} = (c_x, c_y, c_z)$: In a general frame all four 3×3 blocks are non-zero:

$$I = \begin{bmatrix} 0 & mc_z & -mc_y & m & 0 & 0 \\ -mc_z & 0 & mc_x & 0 & m & 0 \\ mc_y & -mc_x & 0 & 0 & 0 & m \\ I_{xx} & -I_{xy} & -I_{xz} & 0 & -mc_z & mc_y \\ -I_{xy} & I_{yy} & -I_{yz} & mc_z & 0 & -mc_x \\ -I_{xz} & -I_{yz} & I_{zz} & -mc_y & mc_x & 0 \end{bmatrix} \tag{328}$$

where the lower-left inertia tensor entries include the classical parallel axis corrections:

$$\begin{aligned} I_{xx} &= I_{xx,cm} + m(c_y^2 + c_z^2), & I_{xy} &= I_{xy,cm} + mc_x c_y \\ I_{yy} &= I_{yy,cm} + m(c_x^2 + c_z^2), & I_{xz} &= I_{xz,cm} + mc_x c_z \\ I_{zz} &= I_{zz,cm} + m(c_x^2 + c_y^2), & I_{yz} &= I_{yz,cm} + mc_y c_z \end{aligned} \tag{329}$$

Two structural features carry over from the $2D$ case:

- *The upper-right block $m\,\mathbf{I}_3$ is frame-invariant.* It remains equal to $m\,\mathbf{I}_3$ regardless of the center-of-mass position. Mass is a dual coupling between ideal and real lines that does not depend on position. This is the $3D$ counterpart of the invariant antisymmetric mass block in $2D$.
- *The upper-left and lower-right blocks are conjugate antisymmetric matrices.* Upper-left $= m[\vec{r}_{cm}\times]^\top$ and lower-right $= m[\vec{r}_{cm}\times]$, where $[\vec{r}_{cm}\times]$ is the 3×3 antisymmetric cross-product matrix of the center-of-mass position vector. Both blocks are driven by the same offset $\vec{r}_{cm}$, reflecting that the translational-to-translational and angular-to-angular couplings induced by the offset are conjugate couplings.

This 6×6 structure is the *PGA* form of the classical spatial inertia matrix from rigid body dynamics [Dorst and De Keninck, 2023], expressed in the (translational, angular) velocity ordering of $3D$ *PGA*.

Uniform rectangular cuboid: For a uniform cuboid of width w (along $\boldsymbol{e}_1$), height h (along $\boldsymbol{e}_2$), depth d (along $\boldsymbol{e}_3$), total mass m, centered at the

origin, all cross-inertia and center-of-mass coupling terms vanish:

$$I_{\text{cuboid}} = m \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ \frac{h^2+d^2}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{w^2+d^2}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{w^2+h^2}{12} & 0 & 0 & 0 \end{bmatrix} \tag{330}$$

The three diagonal entries of the lower-left block are the classical rectangular moments of inertia: $I[3,0] = m(h^2+d^2)/12$ about the x-axis, $I[4,1] = m(w^2+d^2)/12$ about the y-axis, and $I[5,2] = m(w^2+h^2)/12$ about the z-axis – each determined by the two extents of the body perpendicular to the respective rotation axis, as in the classical formula.

Table 35 gives the correspondence between classical inertial quantities and their location in the $3D$ PGA inertia matrix.

Classical quantity	$3D$ PGA block	Entries (body frame, CoM at origin)
Total mass m	Upper-right diagonal ($m\mathbf{I}_3$)	$I[0,3] = I[1,4] = I[2,5] = m$
Diagonal moments I_{xx}, I_{yy}, I_{zz}	Lower-left diagonal	$I[3,0]$, $I[4,1]$, $I[5,2]$
Products of inertia I_{xy}, I_{xz}, I_{yz}	Lower-left off-diagonal	$I[3,1] = I[4,0] = -I_{xy}$, etc.
CoM offset (general frame)	All four 3×3 blocks	via parallel axis theorem

Table 35: Classical inertial quantities and their location in the $3D$ PGA 6×6 inertia matrix (body frame, center of mass at origin).

Unified view — 2D and 3D: In both $2D$ PGA and $3D$ PGA the mass contribution to the inertia map takes the same algebraic form:

$$I_{\text{mass}}(\Omega) = -m \cdot \overline{\Omega} \tag{331}$$

The complement maps each velocity generator to its momentum carrier in the dual subspace, placing mass in the off-diagonal block that connects dual

velocity and momentum spaces. The location of that block in the matrix depends on the basis ordering.

In $2D$ *PGA* the translational generators ($\boldsymbol{e}_1$, $\boldsymbol{e}_2$, indices 0–1) sit at lower indices and the rotational generator ($\boldsymbol{e}_3$, index 2) sits at the highest index. The complement maps translational inputs to translational outputs while crossing x and y, placing mass in the *upper-left* 2×2 sub-block (translational velocity → linear momentum).

In $3D$ *PGA* the translational generators ($\boldsymbol{e}_{41}$, $\boldsymbol{e}_{42}$, $\boldsymbol{e}_{43}$, indices 0–2) are ideal lines at lower indices, while the rotational generators ($\boldsymbol{e}_{23}$, $\boldsymbol{e}_{31}$, $\boldsymbol{e}_{12}$, indices 3–5) are real lines at higher indices. The complement maps real-line inputs (angular velocity) to ideal-line outputs (linear momentum), placing mass in the *upper-right* 3×3 sub-block (angular velocity → linear momentum).

Table 36 gives the side-by-side comparison for the body frame (center of mass at origin).

Classical quantity	$2D$ *PGA* (3×3)	$3D$ *PGA* (6×6)
Total mass m	Upper-left 2×2: $I[0,1] = +m$, $I[1,0] = -m$	Upper-right 3×3 diag.: $I[0,3] = I[1,4] = I[2,5] = m$
Scalar/diagonal moments	Lower-right 1×1: $I[2,2] = J$	Lower-left 3×3 diag.: $I[3,0], I[4,1], I[5,2]$
Products of inertia	n/a (scalar J in $2D$)	Lower-left 3×3 off-diagonal
CoM coupling	$I[0,2], I[1,2]$ and transposed	All four 3×3 blocks non-zero
Parallel axis	$I[2,2] = J_{\text{cm}} + mr^2$	Lower-left 3×3 (3D formula)

Table 36: Unified mapping of classical inertial quantities to the *PGA* inertia matrix (body frame, center of mass at origin).

The frame-invariance of the mass block and the decoupling at the center of mass are common to both dimensions. In the body frame with the origin at the center of mass – the natural choice for integrating eqn. (303) – the coupling blocks vanish, the mass block takes its simplest form, and the moment block reduces to J (in $2D$) or $\mathbf{J}_{cm}$ (in $3D$) without any parallel axis corrections. This matches the classical result that Newton's and Euler's equations decouple at the center of mass, and it is the reason the body frame with the center of mass at the origin is used as the preferred reference frame throughout this section. Note, however, that while the inertia map I_{body} is block-diagonal at the center of mass – translational and rotational blocks decoupling in the inertia matrix itself – the equations of motion re-

tain dynamical coupling through the $\mathrm{rcmt}(\Omega_b, I_b[\Omega_b])$ term. This is the *PGA* counterpart of the classical coupling terms $\vec{\omega} \times \vec{v}$ and $\vec{\omega} \times \mathbf{I}_b\vec{\omega}$, and it arises from the rotation of the body frame, not from the inertia structure.

Pivot-corrected inertia — using the pivot as reference point: The general-frame formulae derived above – eqn. (321) in $2D$ and eqn. (328) in $3D$ – apply directly when the reference point is chosen to coincide with the pivot rather than the center of mass. In that case the center-of-mass offset vector (c_x, c_y) resp. $\vec{r}_{cm}$ points from the pivot to the center of mass, and the inertia map automatically incorporates the full Steiner correction: $I[2,2] = J_{cm} + mr^2$ in $2D$ (eqn. (314)) and the complete tensor update of eqn. (329) in $3D$.

1.7.5 Moving coordinate systems

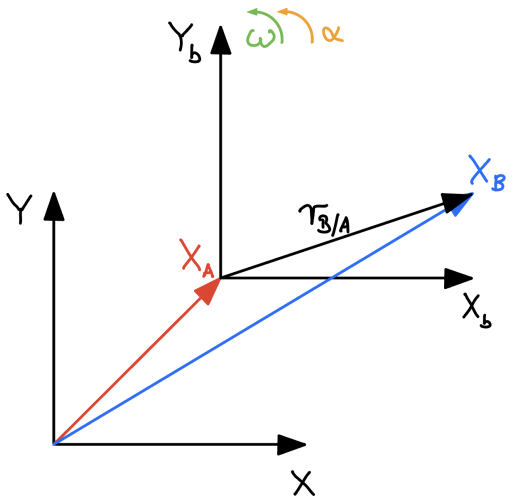


Figure 17: Moving frame: World or reference frame $\{X, Y\}$ (only $2D$ shown as examples, extends naturally to $3D$). Body frame $\{X_b, Y_b\}$. Point A might move with v_A and accelerate with a_A (not shown). Body frame turns with rotational speed ω and rotational acceleration α relative to the world frame.

When modelling motion in moving coordinate systems we look into relative motion. Typically, we use a world frame as reference frame and a body frame as frame moving relative to the world frame, as shown in fig. 17. x_A and x_B are absolute positions in the world frame.

Most of the time x_A is a reference position (e.g. the center of mass of a rigid body)[14], and we are interested in the movement of x_B relative to x_A. The relative position of B with respect to A is shown as $r_{B/A}$. If A is the center of mass of the rigid body then r are local (relative) coordinates of the points of the body in the body frame. So we define

$$r = x_B - x_A \quad \textit{(relative position)} \tag{332a}$$
$$\dot{r} = v_B - v_A \quad \textit{(relative velocity)} \tag{332b}$$
$$\ddot{r} = a_B - a_A \quad \textit{(relative acceleration)} \tag{332c}$$

as relative position, speed and acceleration.

[14]The reference position x_A can be any arbitrary point of the rigid body. However, it is often convenient to choose x_A to be located at the center of mass.

The relative position in a $3D$ frame is defined with (x, y, z) as local or relative coordinates

$$r = x\boldsymbol{e}_1 + y\boldsymbol{e}_2 + z\boldsymbol{e}_3 \tag{333}$$

The total change of the relative position has the following components

$$\dot{r} = \frac{dr}{dt} = \underbrace{\dot{x}\boldsymbol{e}_1 + \dot{y}\boldsymbol{e}_2 + \dot{z}\boldsymbol{e}_3}_{\text{changing speed: } \frac{dr}{dt} = v} + \underbrace{x\frac{d\boldsymbol{e}_1}{dt} + y\frac{d\boldsymbol{e}_2}{dt} + z\frac{d\boldsymbol{e}_3}{dt}}_{\text{changing frame: } \frac{dr}{dt} = \omega \times r} \tag{334}$$

leading to the relative speed $\dot{r}$

$$\dot{r} = \omega \times r_{B/A} + v_{B/A} \tag{335}$$

which can be transformed using eqn. (332b) resulting to the total speed at position B as

$$v_B = v_A + \omega \times r_{B/A} + v_{B/A} \tag{336}$$

We can get the relative acceleration from deriving eqn. (335) again with respect to t

$$\ddot{r} = \dot{\omega} \times r_{B/A} + \omega \times \dot{r}_{B/A} + \dot{v}_{B/A} \tag{337}$$

Using eqn. (335) directly for $\dot{r}_{B/A}$ and for $\dot{v}_{B/A} = \omega \times v_{B/A} + a_{B/A}$ this yields

$$\ddot{r} = \dot{\omega} \times r_{B/A} + \omega \times \omega \times r_{B/A} + \omega \times v_{B/A} + \omega \times v_{B/A} + a_{B/A} \tag{338}$$

when reordered for terms depending on position, relative speed and relative acceleration, we get

$$\ddot{r} = \underbrace{\omega \times \omega \times r_{B/A}}_{\text{centripetal accel. } = -|\omega|^2\, r_{B/A}} + \underbrace{2\,\omega \times v_{B/A}}_{\text{Coriolis acceleration}} + \underbrace{\dot{\omega} \times r_{B/A}}_{\text{frame acceleration}} + \underbrace{a_{B/A}}_{\text{particle accel.}} \tag{339}$$

Using eqn. (332c) this results in the total acceleration at position B as

$$a_B = a_A + \omega \times \omega \times r_{B/A} + 2\,\omega \times v_{B/A} + \dot{\omega} \times r_{B/A} + a_{B/A} \tag{340}$$

Now we can translate the various velocity and acceleration terms that describe physics using expressions in the geometric algebra language, separately for EGA and PGA. To get there, we mainly replace the cross-product expressions by bivectors in EGA and trivectors in PGA. They are duals of the momentary rotation axis. This directly applies to $2D$ in that form.[15]

[15] The $2D$ application is done in classical vector algebra by using the z-component of the $3D$ cross-product as replacement, even if the cross-product is not defined for $2D$. In the classical approach we temporarily have to include the third dimension and use the relevant z-component exclusively. However, this temporary inclusion of an additional dimension is not required in geometric algebra, since a rotation only requires two dimensions for its representation in geometric algebra.

Term		**Description**
Definition of vectors v and points P	VA	$v = (v_x, v_y, v_z)$, $P = (p_x, p_y, p_z)$
	EGA	$v = (v_x, v_y, v_z)$, $P = (p_x, p_y, p_z)$
	PGA	$v = (v_x, v_y, v_z, 0)$, $P = (p_x, p_y, p_z, 1)$
Definition of rotation	VA	rotation axis vector $\omega = (\omega_x, \omega_y, \omega_z)$ (a direction, axis always contains origin)
	EGA	rotation bivector $\Omega_E = \omega^\star = (\Omega_x, \Omega_y, \Omega_z)$ (bivector dual to direction vector ω – defines the plane of rotation, which always contains the origin)
	PGA	rotation direction $\omega = (\omega_x, \omega_y, \omega_z, 0)$ reference point on rotation axis $Q = (q_x, q_y, q_z, 1)$ rotation bivector $\Omega_P = Q \wedge \omega$ - a line, where $\Omega_P = (\Omega_{vx}, \Omega_{vy}, \Omega_{vz}, \Omega_{mx}, \Omega_{my}, \Omega_{mz})$ (a rotation axis modelled by a bivector, with arbitrary orientation *and* location; can also be created from any unitized line that is scaled with $\lvert\omega\rvert$)
Speed of rotation at r or R	VA	evaluation of velocity at $r = (r_x, r_y, r_z)$ (r is vector perpendicular to axis; points to evaluation point) $v = \omega \times r$
	EGA	evaluation of velocity at $r = (r_x, r_y, r_z)$ (see VA) $v = \mathrm{cmt}(r, \Omega_E) = \Omega_E \gg r$
	PGA	evaluation of velocity at $R = (r_x, r_y, r_z, 1)$ $v = \mathrm{rcmt}(\Omega_P, R)$
Centripetal acceleration	VA	$a_{cp} = \omega \times \omega \times r_{B/A}$
	EGA	$a_{cp} = \mathrm{cmt}(\mathrm{cmt}(r_{B/A}, \Omega_E), \Omega_E) = \Omega_E \gg (\Omega_E \gg r_{B/A})$
	PGA	$a_{cp} = \mathrm{rcmt}(\Omega_P, \mathrm{rcmt}(\Omega_P, r_{B/A}))$
Coriolis acceleration	VA	$a_{co} = 2\ \omega \times v_{B/A}$
	EGA	$a_{co} = 2\ \mathrm{cmt}(v_{B/A}, \Omega_E) = 2(\Omega_E \gg v_{B/A})$
	PGA	$a_{co} = 2\ \mathrm{rcmt}(\Omega_P, v_{B/A})$
Frame rotational acceleration	VA	$a_{frame\ rotation} = \dot{\omega} \times r_{B/A}$
	EGA	$a_{frame\ rotation} = \mathrm{cmt}(r_{B/A}, \dot{\Omega}_E) = (\dot{\Omega}_E \gg r_{B/A})$
	PGA	$a_{frame\ rotation} = \mathrm{rcmt}(\dot{\Omega}_P, r_{B/A})$

Table 37: Description of rotation and rotation speed evaluation for **VA** = Vector Algebra (classical approach using vectors); **EGA** = Euclidean Geometric Algebra; **PGA** = Projective (Euclidean) Geometric Algebra.

1.7.6 Frame trees: static, kinematic and dynamic systems

The preceding subsections developed the mechanics of a *single* rigid body: its pose evolves by a motor $M(t)$, its momentum wrench is the inertia map $P = I[\Omega]$ (subsubsection 1.7.4), and its motion obeys the body-frame Euler equation (303). Modelling assemblies – a vehicle with wheels, a robot arm, a pendulum chain – requires composing many such bodies, each described *relative* to another. The library provides this as a strict three-layer stack of classes, all sharing one geometry, a *tree of coordinate frames*. Each layer inherits the one below and adds exactly one new kind of state:

Layer	Adds per frame	Answers question
`static_system2dp`	a pose (relative to parent)	*Where* is each frame?
`kinematic_system2dp`	a velocity + acceleration twist	*How* is it moving? (no forces)
`dynamic_system2dp`	inertia + joints + gravity	How do *forces* move it?

Table 38: The three-layer system stack. Each layer inherits the previous one and adds a single kind of per-frame state.

The layering keeps each capability additive and independently testable: the transformation algebra at the static layer, velocity and acceleration transport at the kinematic layer, and energy conservation at the dynamic layer. A user who only needs coordinate transforms never pays for the dynamics. The same three classes exist for $3D$ (`static_system3dp`, `kinematic_system3dp`, `dynamic_system3dp`); their class structure, method names and algorithms are identical – only the representations change, as summarized at the end of this subsection. The tree describes *open* chains; closed kinematic loops (parallel mechanisms) are added by a separate layer on top of it, developed in subsection 1.7.10.

1.7.7 Pose, frame and the frame tree

A *pose* is the rigid placement of one frame relative to another: an origin and an orientation (in $2D$ a position vector and a scalar angle φ; in $3D$ a position and an axis-angle vector). It is the human-readable form of a *motor* M, and the two interconvert: the motor is built from the pose as in eqn. (175), and the pose is read back by decoding the origin from $M \vee O \vee \underset{\sim}{M}$ and the orientation from the rotated $\boldsymbol{e}_1$-axis (a constrained motor logarithm, exact on round-trip).

A `static_system2dp` holds a *tree* of frames, each recording its parent, with the invariant that a frame's pose is always stored relative to its parent, never

in world coordinates. The world (root) frame is its own parent. Adding a frame without naming a parent links it to the previously added one, so a plain sequence builds a linear chain $root \to f_1 \to f_2 \to \ldots$, while naming an earlier frame branches the tree. This provides three modelling capabilities: *branching* (a body frame with two arms, each arm chain parented off the body, the body off the possibly-moving world – any sub-assembly is a subtree); *name addressing* (frames carry unique names and every operation has a by-name overload); and *reposing* (moving a frame relative to its parent drags the whole subtree along automatically, because all descendant poses are relative).[16]

Tree walk: The transform from any frame to any other is composed along the tree path. Walking *up* from the source to the lowest common ancestor (LCA) composes child-to-parent steps (the regressive reverse of each frame's pose motor); walking *down* from the LCA to the target composes parent-to-child steps; the result is the regressive product of the two segments. Special cases fall out of the same procedure: a frame to itself yields the identity (the unit pseudoscalar, neutral for $\vee$); a pure ancestor/descendant relation makes one segment empty (a forward or backward transform); a linear chain is the degenerate tree where the LCA is simply the shallower frame. Because composition is motor multiplication with $\vee$, the entire tree machinery is dimension-neutral and lifts to $3D$ unchanged.

One sandwich for all objects: The transform between two frames is a single motor M, and *any* geometric object is moved by the *same* sandwich $X' = M \vee X \vee \underset{\sim}{M}$. Classical mechanics, by contrast, provides several different rules and the modeller must determine which one applies:

- a *point* (a location) transforms *affinely* – rotate *and* translate;
- a *free vector* (a direction, a velocity) transforms *linearly* – rotate only, no translation;
- *basis vectors* transform covariantly while a vector's *coordinates* transform contravariantly (the inverse-transpose), the source of much index bookkeeping.

[16] The frame tree is exercised end-to-end in the test cases of `ga_test/src/ga_appl2dp_mechanics_test.hpp` and its $3D$ analogue `ga_appl3dp_mechanics_test.hpp`: a frame transformation drives a `static_system2dp` directly, and a merry-go-round carries the tree one layer up as a rotating `kinematic_system2dp`.

In *PGA* there is a single operation, and the object itself carries the information that selects the right behaviour through its *homogeneous weight* (the projective component $\boldsymbol{e}_3$ in $2D$, $\boldsymbol{e}_4$ in $3D$): a point has weight 1 and is rotated *and* translated; a direction has weight 0 (a point at infinity) and is rotated *only*, since translating a point at infinity does nothing. No contravariant/covariant distinction remains: the motor sandwich is a single conjugation that transports every grade (points, lines, directions, and – see below – twists and wrenches) correctly. The choice of transformation law is thus handled by the algebra itself.

1.7.8 Transport of motion along a kinematic chain

The preceding subsections transport the twist of a *single* body between its body and world frames (eqn. (180a), eqn. (181)). The `kinematic_system2dp` generalizes this to a *chain* of frames: each frame carries a relative velocity twist and a relative acceleration twist with respect to its parent, and the layer answers velocity and acceleration queries for any point of any frame.

Adjoint: A twist B (a vector in $2D$, a bivector in $3D$) is the instantaneous motion (Lie-algebra) with the motor M defined as $M = \exp_{\vee}(\frac{1}{2}B)$ (twists *add*, motors *multiply*; see subsection 1.4.3). Re-expressing a twist B in another frame is done by the *adjoint* of the twist, and the geometric-algebra fact that makes the implementation compact is that the adjoint *is* the very sandwich used to move a point (or any other object):

$$\mathrm{Ad}_M(B) = M \vee B \vee \underset{\sim}{M} \tag{341}$$

This is eqn. (180a) read as a twist transport rather than an object transport. So the same sandwich serves double duty – point action and twist adjoint – because both are the conjugation $M \vee (\cdot) \vee \underset{\sim}{M}$. A separate 6×6 adjoint matrix, as the classical formulation requires, is unnecessary.

Velocity composition: This step goes beyond the single-body case. In a chain, each frame k contributes a relative twist ξ_k expressed in *its own* frame. Transporting each to the world frame by the adjoint and superposing – valid because twists live in a vector space – gives the world velocity twist of frame i:

$$V_i = \sum_{k \in \mathit{path}(\mathit{root} \to i)} \mathrm{Ad}_{M_k}(\xi_k) = \sum_k M_k \vee \xi_k \vee \underset{\sim}{M_k} \tag{342}$$

They add rather than compose precisely because they are infinitesimal: finite motions do not commute and must be multiplied, but to first order simultaneous infinitesimal motions sum. Equation (342) is the geometric-algebra

form of the robotics *spatial Jacobian* – a quantity a single moving body does not exhibit.

Velocity and acceleration fields: Given a frame's world velocity twist V (and acceleration twist A), the velocity of any point X of that frame is the regressive commutator $\dot{X} = \mathrm{rcmt}(V, X)$, and its acceleration is $\mathrm{rcmt}(A, X) + \mathrm{rcmt}(V, \mathrm{rcmt}(V, X))$ – exactly the frame (Euler) and centripetal terms of subsection 1.7.5, now evaluated per chain frame. Differentiating the adjoint along the moving chain yields the recursive Newton-Euler propagation

$$V_i = V_{parent} + \mathrm{Ad}_{M_i}(\xi_i) \tag{343a}$$
$$A_i = A_{parent} + \mathrm{Ad}_{M_i}(\dot{\xi}_i) + \left[\, V_i,\ \mathrm{Ad}_{M_i}(\xi_i) \,\right] \tag{343b}$$

where the second acceleration term – the Lie bracket of twists – is the Coriolis/centrifugal coupling, measuring how far two twists fail to commute. In this primal PGA that bracket is again a native product, $[A, B] = \mathrm{rcmt}(A, B)$ (the $\mathfrak{se}(2)/\mathfrak{se}(3)$ twist bracket). The single-frame transport theorem is the content of subsection 1.7.5; the recursion over a chain is what the kinematic layer adds.
The velocity/finite distinction underlying this – twists add in the Lie algebra while motors multiply non-commutatively, equal only when the generators commute (Chasles' theorem, the screw) – is developed in subsection 1.4.4. It is also the reason the parent-to-child pose of a frame is built by *composing* a rotation and a translation motor rather than exponentiating their summed generators.

1.7.9 Articulated coupled dynamics

The earlier subsections solve a *single* free rigid body (eqn. (303)) and a *single* revolute pivot (eqn. (316), eqn. (315)). The `dynamic_system2dp` adds the coupled dynamics: *several bodies coupled by joints*, integrated together in *reduced (joint) coordinates*.[17]

Spatial pairing and energy: Each body carries the inertia map I of subsection 1.7.4 (velocity twist $\to$ momentum wrench $P = I[\Omega]$). A twist

[17] The canonical example is the double pendulum – two revolute-jointed plates swinging chaotically under gravity while conserving energy, a `dynamic_system2dp` advanced via `step(dt)`. It is pinned by assertion-checked tests: the planar case in `ga_test/src/ga_appl2dp_mechanics_test.hpp` and the non-parallel-axes $3D$ case in `ga_appl3dp_mechanics_test.hpp`. Both check energy conservation to RK4 tolerance and the mass-matrix-versus-kinetic-energy identity.

is paired with a wrench through the dimension-agnostic *reciprocal (spatial) product*

$$\langle \xi, P \rangle = -\operatorname{rwdg}(\xi, P), \qquad \langle \xi, I[\xi] \rangle = 2T \tag{344}$$

i.e. the inertia-map quadratic form equals twice the kinetic energy T, carrying the mass and the angular terms uniformly with no $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ split. This one pairing drives the kinetic energy, the joint-space mass matrix and the forward dynamics, and lifts to $3D$ by replacing the 3×3 map with the 6×6 inertia map and rwdg with the bivector reciprocal product.

Joints: A joint is the reduced coordinate linking a body to its parent. There are two families. A *coordinate joint* carries one scalar q: *revolute* (a hinge), *prismatic* (a slider) or *helical* (a screw of pitch h). All three run identical code; the body-to-parent motor, the relative twist and the Jacobian column are always

$$M(q) = M_{rest} \vee \exp_{\vee}\!\left(\tfrac{1}{2}\, q\, S_b\right), \qquad \xi = \dot{q}\, S_b, \qquad v(\cdot) = \operatorname{rcmt}(S_b, \cdot) \tag{345}$$

and only the *screw generator* S_b differs: a *finite point* Q_b for a revolute joint (so $\exp_{\vee}$ is a rotation about Q_b) versus an *ideal point*/direction for a prismatic joint (so $\exp_{\vee}$ is a translation); in $3D$ the line plus h times its direction for a helical joint. The earlier single-pivot result (eqn. (316)) is the special case of one revolute joint; the generic screw machinery, the prismatic case, and the parallel-axis (Steiner) inertia emerging automatically from the spatial Jacobian are what the dynamic layer adds.

A *motor joint* carries k screws $S_1, \ldots, S_k$ in the body frame and k rates: *cylindrical* ($k = 2$), *spherical* ($k = 3$), *planar* ($k = 3$) and *free* ($k = 6$ in $3D$, $k = 3$ in the plane). The screws span a Lie subalgebra of the twists but do not commute, so no global scalar coordinates exist without singularities; the configuration is the relative motor M_J itself, and its twist and Jacobian columns are $\xi = \sum_k \dot{q}_k S_k$ and $\operatorname{rcmt}(S_k, \cdot)$ – one column per screw. A universal joint is not a type: its two axes do not span a subalgebra (the second turns with the first), so it is composed from two revolute joints with a massless carrier between them. A body added with mass and no joint is a free joint – the *floating base* – and takes part in the coupled solve like any other, so a chain hung on it loads it (tested: a hub carrying two swinging arms keeps the total momentum of the tree to $|\Delta P|/|P| < 10^{-12}$ over 3000 steps, in $2D$ and $3D$).

Forward dynamics: In reduced coordinates the joint itself *is* the constraint, so there are no explicit constraint forces, no drift, and clean energy conservation – at the cost of a small linear solve. By virtual work over the

bodies one assembles the joint-space equation of motion $M(q)\,\ddot{q} = \tau(q,\dot{q})$ with

$$M_{jk} = \sum_i \left\langle S_j^{(i)},\ I_i\big[\,S_k^{(i)}\,\big]\right\rangle, \qquad S_j^{(i)} = \underset{\sim}{M_i} \vee S_j \vee M_i \tag{346a}$$

$$\tau_j = \sum_i m_i\Big(\,v_{cm,i}(S_j)\cdot g \;-\; v_{cm,i}(S_j)\cdot b_{cm,i}\,\Big) \tag{346b}$$

where the sums run over bodies i that have joint j (respectively k) as an ancestor; S_j is the world joint screw; $S_j^{(i)}$ transports it into body i's frame, where its inertia map lives; $v_{cm,i}(S_j) = \mathrm{rcmt}(S_j, C_i)$ is the centre-of-mass velocity per unit joint rate (a column of the spatial Jacobian); $b_{cm,i}$ is the velocity-product (Coriolis/centripetal) cm-acceleration evaluated at $\ddot{q} = 0$; and g is gravity. The system is solved for $\ddot{q}$ by a small LU factorization. Each body still individually obeys the Euler equation (303); the mass matrix is its projection onto the joint coordinates, and the identity $\frac{1}{2}\,\dot{q}^{\top} M(q)\,\dot{q} = T$ ties the two together (a built-in consistency check). In $3D$ the right-hand side carries the full spatial bias wrench $I[A_{bias}] + \mathrm{rcmt}(V, I[V])$ – whose second term is the gyroscopic torque – which vanishes in the planar case to recover eqn. (346b).

The system is advanced in time by a Runge-Kutta integrator (RK4) over all its coordinates at once. A coordinate joint integrates $(q,\dot{q})$. A motor joint (the floating base included) is integrated on the group: its state is the pair (B,Ω) of a generator in the span of the joint's screws, restarted at zero every step, and the joint rates, with $M_J(t) = M_J \vee \exp_{\vee}(\frac{1}{2}B)$ and $\dot{B} = \Omega + \frac{1}{2}[B,\Omega] + \frac{1}{12}[B,[B,\Omega]]$ (the inverse differential of the exponential, truncated after the double bracket; see the twist entry of the glossary). The bracket is closed in the span because the span is a subalgebra. The two bracket terms are what make the scheme fourth order on the pose; with $\dot{B} = \Omega$ alone it is second order. Energy conservation to integrator tolerance over the motion – validated for the free body, the compound pendulum, the prismatic slider, the chaotic double pendulum and the free-floating chain, in both $2D$ and $3D$ – and the measured convergence order (16 per halving of the step) are the correctness metrics used throughout. The motor representation also carries a numerical advantage: a ball-jointed pendulum and the same pendulum built from three revolute joints with massless carriers reach the same trajectory (their difference falls by 16 per halving of the step), but at equal step the three-angle chain's own truncation error is about 10^5 times that of the spherical joint.

1.7.10 Closed-loop and parallel mechanisms

The frame tree of subsection 1.7.6 reaches every frame by a *unique* path from the root; that uniqueness is what makes the reduced (joint) coordinates well-defined. A closed kinematic chain breaks the premise: a four-bar linkage, a parallelogram, a Stewart-Gough platform each contain a frame reached through several branches at once, so no single parent assignment describes it. Such loops are admitted without giving up the energy-clean reduced coordinates by keeping the tree as a *spanning tree* and re-closing each loop with an explicit constraint between two of its frames – *coordinate partitioning* [Wehage and Haug, 1982], not the maximal coordinates weighed in subsection 1.7.12. The open-chain assembly is reused unchanged; the loop closure is an additive layer on top of it [Featherstone, 2008].[18]

Loop-closure constraints: A loop is closed by requiring two anchors – P_a carried by one branch, P_b by another – to coincide as points. In PGA coincidence is the vanishing of the join $P_a \vee P_b$, the line through the two points; realized in coordinates as the unitized point difference

$$g(q) = \hat{P}_a(q) - \hat{P}_b(q) = 0, \tag{347}$$

m scalar equations per loop ($m = 2$ in $2D$, $m = 3$ in $3D$). This is the spatial spherical (ball) joint that pins a Stewart-Gough leg to its platform, or the pin that closes a planar four-bar.

Constraint Jacobian: Differentiating eqn. (347) along the motion gives $\dot{g} = G(q)\,\dot{q}$ with $G = \partial g/\partial q$. Each column of G is the relative partial velocity of the two anchors,

$$G_{cj} = \frac{\partial(P_a - P_b)}{\partial q_j} = \mathrm{rcmt}(S_j, P_a) - \mathrm{rcmt}(S_j, P_b), \tag{348}$$

each term present only when joint j lies on the path to that anchor. But $v(\cdot) = \mathrm{rcmt}(S_j, \cdot)$ is the spatial-Jacobian column already built for the mass matrix and the bias force (eqn. (346a), eqn. (343)). The loop closure introduces no new geometry: the same joint screws S_j that assemble M and τ assemble G.

[18] The distinction in one mechanism: the same three-link arm under one shoulder drive, swinging freely as an open chain, and – with its hand pinned by a single coincidence constraint – re-solving its remaining joints to keep the hand planted.

Constrained equations of motion: The tree dynamics given by the equation $M(q)\,\ddot{q} = \tau$ (eqn. (346a)) gain the m constraints $g(q) = 0$. Differentiated twice they read $G\,\ddot{q} = -\dot{G}\,\dot{q}$; adjoining the constraint forces $G^{\mathsf{T}}\lambda$ gives the bordered (KKT) system solved each step,

$$\begin{bmatrix} M & G^{\mathsf{T}} \\ G & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} \tau \\ -\dot{G}\,\dot{q} \end{bmatrix}. \tag{349}$$

The Lagrange multipliers λ are the constraint wrenches – the leg forces. When rows of G are dependent (a locked knee, a body resting on two pins) the bordered matrix is singular; $\ddot{q}$ is still unique when the independent rows constrain the coordinates consistently, so the reduced system of an independent row subset is solved and λ is taken as the minimum-norm solution of $G^{\mathsf{T}}\lambda = \tau - M\ddot{q}$ over all rows – the constraint force is determined, its split over dependent rows is not. The subset is the best-conditioned one, the leading pivots of a column-pivoted QR factorization of G^{T}. Taking the rows in their given order instead, each one kept when it raises the rank, can keep a row whose independence is of the order of a small tilt: a box landing flat on its four corners – twelve rows on six coordinates – kept one independent only to $7 \cdot 10^{-11}$ while a subset independent to 0.25 existed, and the accelerations of the following step reached 10^{11}. M and τ are reused unchanged from the open-chain assembly (eqn. (346a)–(346b)); only G and the velocity-product term $\dot{G}\,\dot{q}$ – the relative anchor acceleration evaluated at $\ddot{q} = 0$, read from the same bias pass that yields b_{cm} – are new. The one constraint, differentiated zero, once and twice, supplies the three tiers of solve: the position assembly $g(q) = 0$ by Newton (the dependent joint angles from a driven one), the velocity distribution $G\,\dot{q} = 0$, and the acceleration distribution $G\,\ddot{q} = -\dot{G}\,\dot{q}$.
A loop closure is a joint between two tree frames [Featherstone, 2008], and its kind is the subspace of relative motion it removes – the rows it contributes to g and G. Three kinds are provided: the *coincidence* of two points (2 rows in the plane, 3 in space – a pin, a ball joint), the *distance* between two points (1 row – a rigid rod, a leg of fixed length, whose multiplier is the rod force), and the *coincidence of two frames* (3 and 6 rows) – a weld: the anchor points coincide and the two frames share their orientation, so the bodies are rigidly attached and no relative motion remains (a tool held in a hand, a foot flat on the ground, a rigid coupler); its multipliers are the force at the anchor and the couple transmitted across the attachment. For the frame kind the orientation residual is the weight of the generator of the world motor $M_a \vee \underset{\sim}{M_b}$ carrying frame b's orientation to frame a's (twice the rotational part of its logarithm), and its rows of G are the difference of the two world angular velocities. A constraint can also be switched off and on while the system

runs – a foot that lifts, a hand that lets go – in which case it contributes no rows and the system behaves as if it had not been added, while keeping its index so that a repeating contact sequence can toggle the same constraints. The multipliers of a weld are not the resultant of the distal weights: a chain pinned at its base and welded at its end is fixed by the moment balance about every joint, and the base carries what the weld does not.

Stabilisation and energy: Discretely integrated, the position and velocity constraints drift. They are restored after each step by projection onto the constraint manifold – a few Gauss-Newton corrections $q \leftarrow q - G^{+}g$ and $\dot{q} \leftarrow \dot{q} - G^{+}(G\,\dot{q})$ (the Gear-Gupta-Leimkuhler scheme [Gear et al., 1985]). Because the constraint forces $G^{\mathsf{T}}\lambda$ are orthogonal to the feasible velocities ($G\,\dot{q} = 0$) they do no work, and the projection corrects drift without injecting energy. The energy-conservation property of the reduced coordinates (subsection 1.7.12) therefore survives the loop closure – the objection that ruled out maximal coordinates, whose continuous stabilization trades energy for feasibility, does not bind on the few constraints a spanning-tree closure adds. Baumgarte stabilization [Baumgarte, 1972] (a term $2\alpha\,G\dot{q} + \beta^2 g$ on the right-hand side of eqn. (349)) is the simpler alternative, kept as a fallback.

Impacts: A constraint switched on while the velocities disagree with it – a foot touching down with a closing speed – is not drift, and the projection above is the wrong tool for it: G^{+} corrects every coordinate alike, whereas the physical jump is weighted by the inertia. Over the instant of the impact the configuration is frozen and momentum balance reads [Featherstone, 2008]

$$M\,(\dot{q}^{+} - \dot{q}^{-}) = -G^{\mathsf{T}}\Lambda, \qquad G\,\dot{q}^{+} = -e\,G\,\dot{q}^{-}, \tag{350}$$

with the impulses Λ in place of the forces λ and the coefficient of restitution e ($e = 0$ plastic: the rows stick; $e = 1$ elastic). Written for the jump $\Delta\dot{q}$ this is the bordered system (349) with a zero upper right-hand side and $-(1+e)\,G\,\dot{q}^{-}$ below, so the same rank-aware solve applies and a consistent state returns $\Delta\dot{q} = 0$ without residue. The kinetic energy changes by $-\frac{1}{2}\,\Delta\dot{q}^{\mathsf{T}} M\,\Delta\dot{q} \le 0$. A falling rod whose end meets a pin is the check: angular momentum about the pin is conserved through the impact, so a thin rod of length L arriving at speed v leaves rotating at $\omega = 3v/2L$ with three quarters of its kinetic energy, and the impulse equals its change of momentum (both $2D$ and $3D$, to 10^{-12}); a heavy body carrying a light link whose tip is stopped keeps its velocity to $4 \cdot 10^{-4}$ while the link swings – the case the unweighted projection gets wrong.

Unilateral contact: A foot on the ground is an inequality, not a joint: the point may not go below the ground and the ground can only push. Both halves are handled by events on the equality machinery above. The ground is a line in the plane (a plane in space), L; the signed height of a point is the incidence $P \wedge L$, zero on the ground, positive above, and a slope costs nothing more than a floor. A contact point crossing the ground downward is located by bisection on the last step: for each trial the system is reset to the pre-event state and stepped by the trial length, and once the crossing is found it is reset and stepped to the crossing time. The constraint is anchored at the projection of the crossing point onto the ground and switched on with the impact map (350), and the rest of the step continues from there, through as many events as the step contains. An active contact whose normal reaction turns to a pull is released where it is – Selig's lift-off criterion $\lambda_y \to 0$ – provided it would separate: the system with the constraint removed is stepped briefly and restored, and a contact that pulls but would sink straight back is a resting contact and stays (without this test a landed foot at zero load was released and landed again every millisecond). Two kinds: a *point* contact is a coincidence, the foot may rotate about it; a *flat* contact is a frame constraint to a ground frame posed under the foot, whose moment row is the ankle moment and whose centre of pressure is that moment over the normal force – a real foot tips when it leaves the sole. Every point of a mechanism that can reach the ground is registered as a contact (feet, knees, the corners of a torso), so a falling mechanism lands on whatever touches first and lies on the ground instead of sinking through it. Two rods and a biped test the layer: a rod falling onto the floor touches down at the free-fall time to 10^{-9}, its impulse is its change of momentum, its far end lands too and it lies flat carrying half the weight on each end; a flat landing is a cantilever with the moment $mgL/2$; a rod lands on a slope to 10^{-16}.

The touchdown search and the separation test step the system itself rather than a copy of it, so that a force evaluated from the state sees the trial's state (eqn. (365)). On a copy, a force that reads the state through a reference to the original is evaluated from the original, which at that moment already stands at the end of the step: the trial integrates a frozen force. A disc pulled onto the ground by a stiff spring ($\omega = 100\,\mathrm{s}^{-1}$, $\Delta t = 1\,\mathrm{ms}$) touched down $3.0 \cdot 10^{-6}\,\mathrm{s}$ late when the trials ran on copies; stepping the system itself, the error is the integrator's own, $6.8 \cdot 10^{-9}\,\mathrm{s}$, against the closed form. A point that reaches the ground almost parallel to it is a different matter: its height has a small slope in time, and its crossing time moves by orders of magnitude more than any perturbation of the motion (two runs of a falling mechanism that agreed to $10^{-10}\,\mathrm{s}$ in every other event landed a flat sole 22 ms apart).

Such a touchdown time is not a measurement.

The four-bar linkage: The planar four-bar is the spanning tree crank → coupler, with the rocker a second branch from the ground, closed by one coincidence of the coupler tip with the rocker tip: three revolute joints, two equations, one degree of freedom.[19] Driving the crank, the assembly solves the coupler and rocker angles from eqn. (347); the velocity and acceleration ratios follow from $G\,\dot{q} = 0$ and $G\,\ddot{q} = -\dot{G}\,\dot{q}$ and reproduce the classical four-bar relations. Released under gravity the single degree of freedom swings, with total energy conserved to integrator tolerance while $\|g\|$ stays bounded – the same metric the open-chain dynamics use.

The lift to $3D$: As for the open-chain tier, the step to $3D$ is a change of representation, not of method. The coincidence residual carries three components instead of two, the joint screws become lines (`twist3dp`, a bivector) and the anchor points carry the weight $\boldsymbol{e}_4$; eqn. (347)–(349) and the projection are unchanged, and the bordered solve is dimension-agnostic. A spatial loop – two arms meeting at a shared spherical joint, the kinematic core of a Stewart-Gough or delta platform[20] – then assembles, distributes its velocities, and conserves energy under the same machinery (`closed_loop_system3dp`).

1.7.11 The transition from $2D$ to $3D$

Everything above is presented for $2D$ *PGA* (`pga2dp`); the $3D$ stack (`pga3dp`) is the same classes, methods and algorithms, with only the representations changed:

[19] A motor-driven `closed_loop_system2dp` whose coupler point traces the classic coupler curve; pinned by assertion-checked tests: the planar four-bar in `ga_test/src/ga_appl2dp_mechanics_test.hpp` and a spatial two-arm loop ($3D$) in `ga_appl3dp_mechanics_test.hpp` – assembly, velocity and acceleration ratios, energy conservation across a long run, and a bounded closure error $\|g\|$.

[20] The planar analogue is a 5-bar (2-RRR) parallel manipulator – two two-link arms rising from two driven shoulders, their tips pinned to a shared end-effector by one coincidence (four revolute joints, two equations, two degrees of freedom). The two shoulders are driven by phase-shifted sinusoids; each frame re-solves the elbows from the loop closure (forward kinematics), so the effector traces a Lissajous curve. The same well-conditioning is pinned in `ga_test/src/ga_appl2dp_mechanics_test.hpp`; a true $3D$ delta is not built.

Concept	`pga2dp`	`pga3dp`
motor (pose)	even/odd versor `mvec2dp_u`	versor `mvec3dp_e`
pose orientation	scalar angle φ	axis-angle vector
velocity twist	a vector `Vec2dp` (grade 1)	a bivector `BiVec3dp` (grade 2)
point	vector, weight $\boldsymbol{e}_3 = 1$	vector, weight $\boldsymbol{e}_4 = 1$
joint screw	a finite/ideal point	a line / screw axis (a bivector)
inertia map	3×3	6×6
twist bracket	rcmt(vec, vec)	rcmt($bivec, bivec$)
spatial pairing	−rwdg($vec, bivec$)	−rwdg($bivec, bivec$)

Table 39: Representation changes in the $2D \to 3D$ lift of the system layer. The motor sandwich, the adjoint, the tree walk, the joint exponential, the inertia-map pairing and the Euler equation are *identical formulae* in both dimensions.

The motor sandwich, the adjoint, the tree walk, the joint exponential, the inertia-map pairing and the Euler equation $\dot{\Omega} = I^{-1}[\, W - \mathrm{rcmt}(\Omega, I[\Omega])\,]$ are identical formulae in $2D$ and $3D$, differing only in the grade the twist and wrench occupy. The $2D \to 3D$ step is thus a change of representation, not a rewrite; this follows from developing the $2D$ case fully first. The same holds for the closed-loop layer of subsection 1.7.10: the loop-closure residual gains a component and the joint screws become lines, while the constraint Jacobian and the bordered solve are unchanged.

1.7.12 Addendum: what geometric algebra contributes to rigid-body dynamics

This addendum weighs the approach against classical methods. It is not part of the modelling exposition; readers interested only in *how* the model works may skip it. It states where the PGA formulation is advantageous and where a classical method remains preferable. This library realizes the *primal* PGA ($\mathbb{G}(2,0,1)$, $\mathbb{G}(3,0,1)$, see subsection 1.7); the merits below are those of the geometric formulation as such and hold for the dual plane-based variant too. The fair baseline is not "geometric algebra versus nothing" but geometric algebra versus the standard tooling: 4×4 homogeneous $SE(3)$ matrices, or Featherstone's spatial vector algebra (six-component twists and wrenches with a hand-defined spatial cross product) [Featherstone, 2008]. Both are mature and fast; the question is whether the reorganization is worthwhile.

Properties of the geometric formulation:

1. *Pose is a motor, not a* 4×4 *matrix.* There is no orthonormality constraint to maintain and no renormalization drift; one exponential/logarithm covers rotation and translation together as a screw, and the pose evolves on the manifold exactly via $M \vee \exp_{\vee}(\frac{1}{2}\,\Omega\,dt)$.

2. *Twists, momenta and wrenches are bivectors, and the "spatial cross product" is the algebra's commutator.* A twist is a screw, the momentum wrench is the inertia-map image $P = I[\Omega]$, the Newton-Euler bias wrench is a bivector, and the spatial cross product is rcmt – a native product rather than a bespoke operator with conventional signs.

3. *The adjoint equals the point sandwich.* Transporting a twist between frames is the same operation as moving a point (eqn. (341)); the classical formulation needs a separate 6×6 adjoint matrix.

4. *A force is its line of action – a wrench.* The combined force-and-moment bivector $F = Q \wedge f$ (subsection 1.7.1) is the object the mechanics literature calls a *wrench*; moments are not a separate pseudovector built with an $r \times F$ cross product.

5. *The Euler equation is one dimension-agnostic line* (eqn. (303)), identical in $2D$ and $3D$ up to grade; the $2D \to 3D$ lift is the same formula, not a rewrite.

Coriolis coupling: The most involved part of a classical double-pendulum Lagrangian is the Coriolis/centrifugal coupling – the Christoffel symbols built from derivatives of the mass matrix. In the geometric formulation it collapses to one term, the Lie bracket of twists in the acceleration recursion (343): the Coriolis force is the geometric fact that twists do not commute. These objects are also drawable in a way six-component vectors are not: each body's instantaneous centre of rotation is a point read off its world twist, the joint screw is the joint point, and the momentum wrench is a bivector. The dynamical state is itself geometry, so the visualization is not a separate construction.

Limitations: Inertia remains a linear operator (3×3 in $2D$, 6×6 in $3D$); for an anisotropic body it does not collapse to a single multivector. The joint-space mass matrix $M(q)$ is still a small numerical solve – reducing to generalized coordinates is a standard robotics step, and the final $\ddot{q} = M^{-1}\tau$ is ordinary linear algebra. Performance is not improved either: mature spatial-vector and matrix codes are highly optimized, so the geometric formulation changes the representation, not the run time.

Reduced vs. maximal coordinates: A more GA-native formulation was deliberately not used: maximal coordinates, in which every body stays a full motor-and-twist and joints become constraint wrenches. That formulation is more uniform, but its constraint stabilization injects and removes energy, which conflicts with an energy-conservation test. The reduced (joint) coordinates chosen here make the joint itself the constraint, so there are no explicit constraint forces, no drift, and clean energy – at the cost of the small $M(q)$ solve. This retains most of the modelling value while conserving energy. The closed-loop extension (subsection 1.7.10) keeps the same choice: a reduced-coordinate spanning tree plus a few explicit loop constraints, rather than maximal coordinates throughout, so its constraint forces do no work and the energy behaviour is preserved.
The recursive Newton-Euler *algorithm* is, of course, classical and not original here. What geometric algebra contributes is the *representation and the concepts* – screws, motors, bivectors and the commutator in place of matrices, six-vectors and hand-defined operators – together with three properties the standard tooling does not provide directly: the same primitives scale unchanged from one body to a coupled tree; the state is visualizable geometry; and the $2D \to 3D$ transition is a change of representation rather than a rewrite.
Because those primitives carry the semantics, the system classes are thin orchestration over them – a frame tree and a Newton-Euler recursion – and reconstruct faithfully in any language that exposes the primitives. The `ga_py.systems` bindings rebuild the `static_/kinematic_/dynamic_system` classes from the bound motor algebra, twist velocity field, inertia map and solver, and reproduce the same energies, momenta and forward dynamics; the reconstruction does no geometry of its own, so it inherits the results of the compiled operations.

1.7.13 Addendum: further reading

The concepts underlying this layer – motors, the sandwich, the adjoint, twists as a Lie algebra, the screw unification of rotation and translation, the inertia map, the Coriolis commutator – are each standard results with a primary source. The references below progress from the algebra itself, through the plane-based specialization of PGA, to rigid-body mechanics, and finally to the robotics / Lie-theory and classical viewpoints that the geometric formulation translates.
A standing caveat applies to every PGA source: this library uses the *primal* $\mathbb{G}(2,0,1)/\mathbb{G}(3,0,1)$ (Lengyel's convention – points and lines as primary objects, motions carried by the regressive product), whereas Gunn and Dorst &

De Keninck use the *dual* plane-based $\mathbb{G}^*$ (planes primary, motions carried by the geometric product). The two are dual-equivalent (subsection 1.7); their *concepts* carry over directly, but their *formulae* are the dual of those used here and must be translated through the complement relation before being matched against the code.

- **Geometric algebra foundations.**
  - [Macdonald, 2010] – a gentle, rigorous first contact with the geometric product, blades and rotors.
  - [Hestenes, 1986] – the origin of doing mechanics in geometric algebra; its chapter on rigid-body mechanics already points toward *PGA*.
  - [Browne, 2012] – the exterior-algebra background for the wedge and regressive products, complements, and the force and momentum lines.
  - [Perwass, 2009] – a broad engineering-oriented treatment of geometric algebra, covering conformal geometric algebra, numerical and uncertainty aspects, and the interactive visualization tool CLUViz.
- **Plane-based *PGA* and the primal/dual choice.**
  - [Lengyel, 2024b] – the convention this library follows (rationale in [Lengyel, 2024a] and [Lengyel, 2025]).
  - [Gunn, 2020] and [Dorst and De Keninck, 2022] – the foundational and the most accessible plane-based (dual) treatments; read for structure, dualize the formulae.
- **Rigid-body mechanics in *PGA*.**
  - [Dorst and De Keninck, 2023] – the direct (dual) analogue of the dynamics developed here: momentum bivector, the "forque", the motor equation of motion.
  - [Gunn, 2011] – the most complete treatment of rigid-body kinematics and dynamics with motors and bivector twists in (dual) projective geometric algebra.
- **The Lie-theory / robotics translation** (the same groups $SE(2)/SE(3)$ as matrices).

  - [Solà et al., 2018] – the most compact modern account of the group/algebra correspondence, the adjoint and the bracket.
  - [Lynch and Park, 2017] – the clearest intuition for twists, the adjoint and the velocity/acceleration recursion.
  - [Murray et al., 1994] – the canonical thorough treatment.
  - [Selig, 2005] – bridges classical screw theory and the Lie-group view.

- **The classical baseline.**
  - [Featherstone, 2008] – spatial vector algebra: six-component twists and wrenches and the spatial cross product, the Plücker-coordinate cousin of the bivector twists and rcmt used here, and the reference for the forward-dynamics algorithm.

- **Constrained multibody dynamics** (the closed-loop layer of subsection 1.7.10).
  - [Wehage and Haug, 1982] – coordinate partitioning: the split into a reduced (spanning-tree) set and the dependent coordinates fixed by the loop constraints.
  - [Gear et al., 1985] – the projection (GGL) scheme that restores the position and velocity constraints after each step without injecting energy.
  - [Baumgarte, 1972] – the simpler constraint stabilization kept as a fallback.

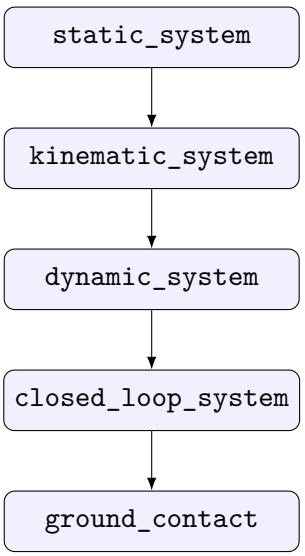
1.8 Modelling robots in *PGA*

A robot is a mechanism with actuators and a task. The mechanism is what subsection 1.7 describes: bodies, joints, loop closures and contacts, advanced in time in joint coordinates. A robotics application asks further questions of the same model: how joint rates move a frame or a point (differential kinematics), which joint values place a frame at a desired pose (inverse kinematics), where the centre of mass is and how the momentum of the whole mechanism changes, what the equation of motion looks like to a controller, and whether a stance on the ground can hold. This subsection collects the concepts, the central equations and the library facilities for these questions. The robotics literature develops each topic in depth [Lynch and Park, 2017, Murray et al., 1994, Featherstone, 2008, Tedrake, 2024]; the emphasis here is on the *PGA* form of the concepts and on how to use them with the library. Control laws built on these quantities belong to the applications and are not part of the library.

1.8.1 From mechanism to robot

The mechanics layer is a stack of five classes, each building on the one before it and adding one kind of state (fig. 18). The robotics quantities attach to the layer that owns the state they are computed from: the Jacobian and the whole-body quantities need joints and masses and are members of `dynamic_system`, the constraint Jacobian and the mobility of a closed chain belong to `closed_loop_system`, the ground reaction and the zero-moment point to `ground_contact`. Every class exists for $2D$ (suffix `2dp`) and $3D$ (`3dp`). The formulae of this subsection are written once and hold in both, with the twist a vector in $2D$ and a bivector in $3D$ (subsubsection 1.7.11).

All five classes are value types. A copy of a model is a complete and independent model, so a forecast, a trial step or a finite-difference probe runs on a copy and leaves the original untouched. A copy carries the force functions registered on the original along with it, which decides how a force that depends on the state is written (paragraph *Force functions and force laws* in subsubsection 1.8.5).



frame tree of poses (motors): where each frame is

twists along the chain: velocity and acceleration of any point of any frame

bodies, joints, gravity, force elements, actuators: $M\ddot{q} = \text{RHS} + \tau$; the Jacobian, the centre of mass, the momentum of the whole mechanism

loop closures, impacts: the constraint Jacobian G, its rank and the mobility, the multipliers λ

unilateral ground contacts as events: the ground reaction, the zero-moment point, the centre of pressure

Figure 18: The mechanics layer as a stack. Each class builds on the one above it and adds one kind of state; the right column lists what a robotics application reads from each layer.

1.8.2 Building a robot model

A robot model is a frame tree whose frames carry bodies and joints, closed by constraints where the mechanism forms loops and where it touches the ground (fig. 19). It is assembled in five steps.

1. *Root.* `add_body` with a mass adds a body on a free joint, the floating base of a mobile or legged robot. A fixed-base manipulator hangs its first joint from the world frame instead.

2. *Chains.* Each link is added with its joint relative to its parent, one builder per joint type (`add_revolute_body`, `add_prismatic_body` and likewise for the helical, cylindrical, spherical and planar joint). Every joint moves its body by a motor, for a one-parameter joint the motor of eqn. (345). A universal joint is two revolute joints with a massless carrier between them.

3. *Bodies.* `make_plate_body` and `make_cuboid_body` for shapes, `make_point_body2dp` and `make_point_body3dp` for a point mass, `make_body_from_inertia` for a mass and an inertia tensor given as data, which is the form a robot description file supplies.

4. *Force elements.* Springs and dampers on the joints, springs from body points to the ground, wrenches given as functions of time and actuator torques at the joints (`set_joint_spring_damper`, `add_grounded_spring`, `set_applied_wrench`, `set_joint_torque`). Wrenches and torques can also be given as laws of the state, eqn. (365).

5. *Closures and contacts.* A `closed_loop_system` adds loop constraints between two frames of the tree: coincidence of two points, a fixed distance, or coincidence of two frames. A `ground_contact` registers every point that can touch the ground as a unilateral point contact or a flat contact.

A configuration is set from outside the integrator with `set_joint` and `set_joint_rate`, which is where the result of an inverse-kinematics step is written, and `assemble` brings a closed chain back onto its constraints. Frames are addressed by index or by name.

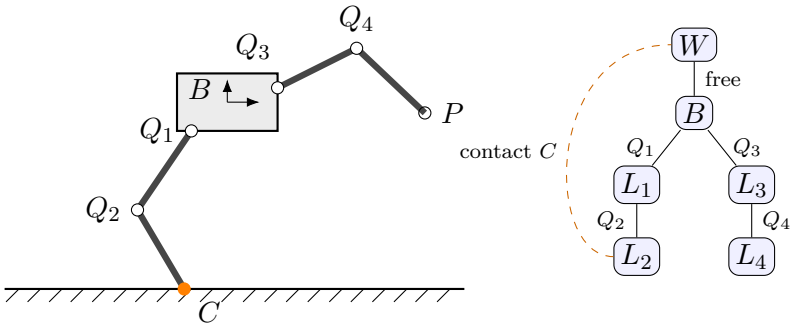


Figure 19: A robot model and its frame tree. A floating base B on a free joint carries two limbs of two links each, jointed at $Q_1, \dots, Q_4$; the lower limb touches the ground at C, the upper one carries a tool point P. The tree (right) reaches every frame from the world W on a unique path; the contact is a constraint between the last link and the world (dashed), which closes a loop while it is active.

1.8.3 Differential kinematics: the Jacobian

Columns are joint screws: Let frame f sit at the end of a chain of joints $j = 1, \dots, n$ with rates $\dot{q}_j$. Each joint contributes its screw, expressed in the world and scaled by its rate, and the contributions add (eqn. (342)):

$$V_f = \sum_{j=1}^{n} \dot{q}_j \, S_j = J(q)\,\dot{q}, \qquad J(q) = \begin{bmatrix} S_1 & S_2 & \cdots & S_n \end{bmatrix}, \qquad S_j = M_j \vee S_{b,j} \vee \underset{\sim}{M_j} \tag{351}$$

$S_{b,j}$ is the screw of joint j in its own frame and M_j the motor from that frame to the world at the current configuration. The columns of the *Jacobian* are the joint axes in their current position. In $3D$ a revolute joint contributes its axis line, a prismatic joint an ideal line (its direction), a helical joint of pitch h its axis line plus h times the direction. In $2D$ a twist is a vector: a revolute column is the pivot point Q_j, a prismatic column an ideal point. A joint that is not on the path from the root to f contributes a zero column, and a joint with several screws (spherical, planar, free) contributes one column per screw.

Velocity of a point: A point P fixed to frame f moves with the velocity $\dot{P} = \operatorname{rcmt}(V_f, P)$ of eqn. (181). Inserting eqn. (351) splits it into one contribution per joint,

$$\dot{P} = \sum_{j=1}^{n} \dot{q}_j \operatorname{rcmt}(S_j, P) \tag{352}$$

each the velocity P would have if only joint j moved. For a revolute joint that velocity is perpendicular to the lever from the axis to P and proportional to its length (fig. 20). The translational Jacobian of a point, a separate $3 \times n$ matrix in classical robotics, is this sum written in coordinates: the columns act on any point through the same product.

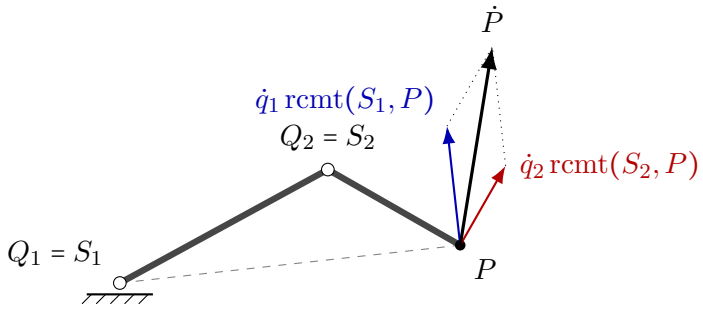


Figure 20: The Jacobian of a planar two-link arm acting on its tip P. In $2D$ the column of a revolute joint is its pivot point, $S_j = Q_j$; each column contributes the velocity $\dot{q}_j \operatorname{rcmt}(S_j, P)$, perpendicular to the lever from Q_j to P (dashed), and the tip velocity is their sum, eqn. (352).

Space and body form: Eqn. (351) expresses the columns in the world frame, the *space Jacobian*. The *body Jacobian* expresses them in frame f. Since the adjoint is the motor sandwich (eqn. (341)), the change is one sandwich per column,

$$J_b = \underset{\sim}{M_f} \veebar J \veebar M_f, \qquad V_{b,f} = J_b \dot{q} \tag{353}$$

with M_f the motor of frame f and $V_{b,f}$ its twist seen from itself. The classical formulation converts with a 6×6 adjoint matrix; here the product that moves a point moves the columns.

Library: `jacobian_columns(f,body_form)` returns the columns as twists for geometric use, and `jacobian(f,body_form)` returns the same columns as a flat row-major matrix for the solvers: $3\times n$ in $2D$ and $6\times n$ in $3D$, with the rows in the component order of the twist (in $3D$ the three angular components first, then the three linear ones). `jacobian(f,joints,body_form)` returns the columns of a chosen set of joints. The library's tests check the analytic columns against finite differences of the motor logarithm.

Statics: A wrench W acting on frame f does work at the rate $\langle V_f, W\rangle$, with the pairing $\langle \xi, W\rangle = -\operatorname{rwdg}(\xi, W)$ of eqn. (344). With $V_f = \sum_j \dot{q}_j S_j$ it exerts the generalised force $\langle S_j, W\rangle$ on joint j, and the joint torque that balances it is

$$\tau_j = -\langle S_j, W\rangle = \operatorname{rwdg}(S_j, W) \tag{354}$$

the classical $\tau = J^\top F$ for the wrench $F = -W$ that the frame exerts on its surroundings, with each column paired with the wrench directly. In $3D$, for a force line W and a revolute axis S_j, $\operatorname{rwdg}(S_j, W)$ is the mutual moment of the two lines. It vanishes when the lines intersect or are parallel: a force whose line of action meets a joint axis, or runs parallel to it, loads that joint with no torque.

Singularities: A configuration is singular where the columns become linearly dependent, $\operatorname{rank} J < 6$ in $3D$ (< 3 in $2D$) for a frame that should reach every twist. The rank is linear algebra (`matrix_rank` on `jacobian()`); the geometry says which screws became dependent: two revolute axes that are collinear, three parallel revolute axes in one plane, four revolute axes through one point [Lynch and Park, 2017]. The minimum-norm and damped solves of subsubsection 1.8.4 stay defined there.

Screw axis of a motion: By Chasles' theorem a rigid motion is a rotation about a line combined with a translation along it. With the full generator $B = 2\log_{\mathrm{v}}(M)$ of a motor M, read on the shortest of $\pm M$, and the unit line l,

$$B = \theta\, l + d\, \operatorname{att}(l), \qquad \text{pitch} = d/\theta \tag{355}$$

`screw_axis(M)` returns l, the angle $\theta \le \pi$, the distance d and the pitch, the same for M and $-M$, which are one motion; applied to a twist it returns the

instantaneous axis with the angular and linear rates. A pose error between two frames is thus one screw motion with a definite axis.

Reading the shortest of $\pm M$ bounds the angle and lets the direction of l carry the sense of the turn. A motion by more than half a turn is therefore reported as the complementary angle about the reversed axis, with the distance measured along that reversed axis: a rotation by 270° about l with $d = 0.35$ reads as 90° about $-l$ with $d = -0.35$, the same motion. A motion that turns steadily past π sees its reported axis flip there. Where the application fixes the axis direction u — a joint axis, the vertical of a turn counter — the signed angle about it is θ times the sign of the scalar product of u with the direction of l (its weight), in $[-\pi, \pi]$, and a count past a half turn is continued with `rlog_continued` (test case "pga2dp / pga3dp: screw_axis reads the motion, not the motor's sheet" in `ga_test/src/ga_pga_contract_test.hpp`).

Mobility: The screws of a set of joints span a subspace Δ_1 of the twists; its dimension is the instantaneous mobility. Closing Δ_1 under the bracket rcmt gives Δ_∞. Where $\Delta_\infty = \Delta_1$ the screws form a subalgebra and the joints keep their mobility through finite motion; a closure that grows marks a mobility that exists only instantaneously [Selig, 2005]. `screw_system(screws)` returns both dimensions. For a closed chain the mobility at the current configuration is $n - \operatorname{rank} G$, with the constraint Jacobian G of eqn. (348) (`constraint_rank()`). The rank changes as the mechanism moves, which is why the constrained solve determines it at every step (subsubsection 1.7.10).

Notation in the robotics literature: The standard texts write the same objects as six-component vectors and matrices, with differing orderings and symbols. Table 40 relates them to the library, which makes the formulae of each source translate term by term. Selig [Selig, 2005] works in the plane-based algebra with the opposite signature, where points are trivectors; there the outer product is the meet of this library and the formulae translate through the complement.

	Lynch & Park	Feather-stone	Murray, Li, Sastry	library
twist	$\mathcal{V} = (\omega, v)$	$\hat{v}$, angular first	$\xi = (v, \omega)$	bivector ($3D$), vector ($2D$)
twist transport	$[\mathrm{Ad}_T]\,\mathcal{V}$	$X\,\hat{v}$	$\mathrm{Ad}_g\,\xi$	$M \vee V \vee \underset{\sim}{M}$
bracket	$[\mathrm{ad}_{\mathcal{V}_1}]\,\mathcal{V}_2$	$\hat{v}_1 \times \hat{v}_2$	$\mathrm{ad}_{\xi_1}\,\xi_2$	$\mathrm{rcmt}(V_1, V_2)$
wrench	$\mathcal{F} = (m, f)$	$\hat{f}$, moment first	$F = (f, \tau)$	$\sum_i Q_i \wedge f_i$ plus couples
wrench transport	$[\mathrm{Ad}_T]^\top \mathcal{F}$	$X^* \hat{f}$	$\mathrm{Ad}^\top_{g^{-1}} F$	$M \vee W \vee \underset{\sim}{M}$
power	$\mathcal{V}^\top \mathcal{F}$	$\hat{v} \cdot \hat{f}$	$F^\top \xi$	$-\,\mathrm{rwdg}(V, W)$
exponential	$e^{[\mathcal{S}]\theta}$	—	$e^{\hat{\xi}\theta}$	$\exp_\vee(\frac{1}{2}\,\theta\,S)$

Table 40: The same quantities in three robotics texts and in the library. The four transport maps of the classical notation are one sandwich here, the two brackets one product, and the exponential takes the half generator.

1.8.4 Inverse kinematics

Inverse kinematics asks for joint values that place a frame at a desired motor M_d. Two routes are in use: an iteration on the Jacobian, which works for any chain, and closed-form solutions for arms whose geometry decouples.

Numerical: With the current motor M of the frame the motion still missing is $\underset{\sim}{M} \vee M_d$, and its generator

$$E_b = 2\log_\vee\big(\underset{\sim}{M} \vee M_d\big) \tag{356}$$

is the body twist that carries the frame to M_d in unit time. A Newton step solves $J_b\,\Delta q = E_b$ in the least-squares sense,

$$\Delta q = J_b^+ E_b + N z, \qquad q \leftarrow q + \Delta q \tag{357}$$

and is repeated until E_b vanishes. J_b^+ is the Moore–Penrose pseudo-inverse (`minnorm_solve`, defined at every shape and rank); near a singularity a damped solve (`lstsq_solve` with a damping factor) bounds the step. For a redundant chain the columns of N span the null space of J_b (`nullspace_basis`), and z moves the joints without moving the frame, towards a preferred posture or away from a joint limit. In $3D$ the error is one bivector, so position and orientation are components of the same screw rather than two residuals weighted against each other. The library supplies the Jacobian, the logarithm and the solves; the iteration itself is written in the application.

Closed form, in *PGA* and in *CGA*: For an arm whose wrist axes meet in one point, position and orientation decouple, and the position problem reduces to a triangle [Lynch and Park, 2017]. A link of length L_1 from a base point b and a link of length L_2 to a tip point t form a triangle with the side $d = \mathrm{dist}(b,t)$, and the joint j is its third corner (fig. 21). Both algebras solve it in a few lines.

In *PGA* the joint is the point at distance L_1 from b on the line towards t, turned about b by the angle φ that the law of cosines gives at b:

$$u = \frac{t-b}{d}, \qquad \cos\varphi = \frac{L_1^2 + d^2 - L_2^2}{2\,L_1\,d}$$
$$j_\pm = M_\pm \vee (b + L_1 u) \vee \underset{\sim}{M_\pm}, \qquad M_\pm = \exp_\vee\!\big(\pm\tfrac{1}{2}\,\varphi\,b\big) \tag{358}$$

with b and t unitized, so that $t-b$ is a direction and b itself is the generator of a rotation about b (eqn. (345)). The tip is reachable if $|\cos\varphi| \le 1$. In $3D$ the rotation about the point b becomes a rotation about the line $l = b \wedge n$ through b along the normal direction n of the arm plane π, $M_\pm = \exp_\vee(\pm\frac{1}{2}\,\varphi\,l)$, and the rest is unchanged.

In *CGA* the joint lies on the circles $c(b, L_1)$ and $c(t, L_2)$, so it is one of the two points of the dipole

$$d = c(b,L_1) \vee c(t,L_2) \quad (2D), \qquad d = s(b,L_1) \vee s(t,L_2) \vee \pi \quad (3D) \tag{359}$$
$$j_\pm = \mathrm{cen}(d) \pm \sqrt{r^2}\,\mathrm{att}(d)$$

with the spheres s, the unitized dipole d and its squared radius r^2 (eqn. (241)). The tip is reachable if $r^2 \ge 0$. The library provides this route as `two_link_ik2dc` and `two_link_ik3dc` (subsubsection 1.5.4).

Set side by side, the two routes differ less than their notation:

- Both return the two configurations as a $\pm$ pair and both carry one reachability condition. The conditions are the same: the radius of the dipole is half the distance between j_+ and j_-, so $r^2 = L_1^2 \sin^2\varphi = L_1^2(1-\cos^2\varphi)$.
- The difference is where the triangle relation lives. The *PGA* route states the law of cosines; the *CGA* route forms the intersection and leaves that relation inside the meet. This saves one derivation, not a case distinction. In $3D$ each route grows by one object: the plane in the meet, the axis line in the motor.

- The *CGA* route is slower. Measured on a three-joint positioner, both routes returning both elbow positions to 10^{-14}, the meet chain took 20.7 ns per solve against 5.2 ns for the triangle written in coordinates (68.7 ns in the textbook form with its trigonometric round trips). It also brings a second algebra beside the *PGA* model.

- Neither route returns joint angles; they follow from atan2 or from the angle between two lines. Neither removes the shoulder singularity, a target on the base axis where the arm plane is undefined. Both cover only the decoupled case: a general six-joint arm has up to sixteen solutions and no closed form.

For inverse kinematics *CGA* therefore offers a different notation for the same triangle rather than a simpler solution. The *PGA* route is as short, uses the algebra the robot model is written in, and numerical inverse kinematics in motor space needs no *CGA* either. The Paden–Kahan subproblems [Murray et al., 1994] relate the two views: the rotation of a point about an axis onto another point is the motor of eqn. (358), and the subproblems of two successive rotations and of a rotation to a given distance are the circle meets of eqn. (359).

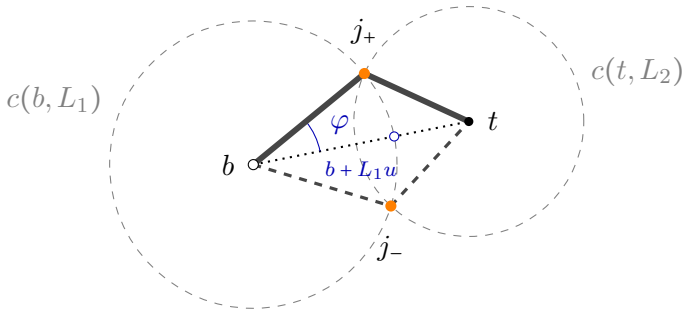


Figure 21: Two-link inverse kinematics by both routes. *PGA*: the point $b + L_1 u$ on the line from b to t (dotted), turned about b by $\pm\varphi$ (eqn. (358)). *CGA*: the joint lies on both circles, so the dipole $c(b, L_1) \vee c(t, L_2)$ carries both configurations $j_\pm$ (solid and dashed). A tip out of reach gives $|\cos\varphi| > 1$ in the one route and an imaginary radius of the dipole in the other.

1.8.5 Whole-body quantities and actuation

Centre of mass: The bodies' centres of mass C_i, unitized and weighted by their masses, add to one projective point,

$$\sum_i m_i C_i = M C, \qquad M = \sum_i m_i \tag{360}$$

whose weight is the total mass; unitizing the sum divides it out and yields the centre of mass C (`total_mass`, `centre_of_mass`). Its velocity and acceleration are the same weighting of the bodies' own (`centre_of_mass_velocity`, `centre_of_mass_acceleration`).

Gravity as one wrench: The weights are force lines through the bodies' centres of mass. By the linearity of the wedge and eqn. (360) their sum is the weight line of the whole mechanism,

$$W_g = \sum_i C_i \wedge m_i\, g = C \wedge M g \tag{361}$$

(`gravity_wrench`). Its moment about C vanishes, and its moment about any other point is one `moment_about` (eqn. (280)).

Momentum of the mechanism: The momentum of the whole mechanism is the sum of the bodies' momentum bivectors (subsubsection 1.7.2),

$$P = \sum_i P_i, \qquad \mathrm{att}(P) = M\,\dot{C} \tag{362}$$

(`total_momentum`). Its attitude is the linear momentum of the total mass moving with the centre of mass, and its moment about a point R, `moment_about(R,P)`, is the angular momentum about R. Newton–Euler for the mechanism is one bivector equation,

$$\dot{P} = W_g + W_R + W_a \tag{363}$$

with the ground reaction W_R (subsubsection 1.8.6) and the applied wrenches W_a. Joint torques and springs act equally and oppositely on the two bodies they connect and cancel in the sum. Taken about C the gravity term drops out, so the rate of the angular momentum about the centre of mass is the moment of the contact and applied wrenches about it; in flight it is zero, and no motion of the joints changes the angular momentum of an airborne mechanism about its centre of mass. A sum over only some of the bodies obeys no such law, since the joints exchange momentum with the rest.

Actuators and passive elements: An actuator at a joint applies equal and opposite torques to the parent and the child body, which in joint coordinates is the generalised force τ_j on the coordinate q_j itself (`set_joint_torque`; a joint with several screws takes one value per screw). It changes the total energy by $\int \tau_j\, \dot{q}_j\, dt$ and leaves the momentum of the mechanism unchanged. A passive stiffness and damping enter the same way, $\tau_j = -k\,(q_j -$

$q_{0,j}) - c\,\dot{q}_j$ (`set_joint_spring_damper`). A spherical or cylindrical joint has no coordinate q_j; its spring acts on the coordinates θ_k of its motion from the motor M_0 it holds when the spring is attached,

$$2\,\mathrm{rlog}\big(\underset{\sim}{M_0} \vee M\big) = \textstyle\sum_k \theta_k\, S_k, \qquad \tau_k = -k\,\theta_k - c\,\dot{q}_k, \tag{364}$$

with the joint's screws S_k and the potential $\frac{1}{2}k\,|\theta|^2$. The force is that potential's: the generator is unchanged by the exponential's Jacobian, and the cylindrical joint's two screws commute. On a planar or free joint, whose rotation and translation do not commute, the same law is not, and the energy drifts by an amount that does not shrink with the step; it is not provided there. Past a full turn the two joints differ. A cylindrical joint can wind any number of turns, and its spring is a torsion spring: θ is continued from the last step (`rlog_continued`), since the motor holds the pose and not the turns. A spherical joint's spring is a potential on the motor itself, maximal at a full turn about any axis, where the motor is -1; wound past it the joint unwinds towards two full turns.

Force functions and force laws: A torque or a wrench is registered in one of two forms. A *function of time* gives the force from the clock alone, a *law* from the clock and the state s of the system it acts on:

$$\tau_j = f(t), \quad W = f(t) \qquad \text{or} \qquad \tau_j = \ell(s,t), \quad W = \ell(s,t) \tag{365}$$

A feedback is written as a law: a hold $\tau_j = -k\,(q_j - q_{0,j}) - c\,\dot{q}_j$ reads q_j and $\dot{q}_j$ from s, and a push along a force line through a body's current position reads that position from s. The integrator evaluates the force at every Runge-Kutta stage, and s is that stage's state, the motor of a motor joint included (within a step a motor joint's own motor stays at the step's start, with the stage's motion accumulated on top of it, so the stage's motor is reported in its place). The contact layer's event search steps the system itself, reset to the pre-event state for each trial, and s is the trial's state (subsubsection 1.7.10). A copy of the system carries its laws, and in the copy s is the copy.

A function of time can read the state as well, through a reference to the system captured when it was registered. It then reads that system wherever it is evaluated. On the system it was registered on this agrees with the law; in a copy it does not, since the copy evaluates the force from the original's state, which does not change while the copy is advanced, and integrates a frozen force. A pendulum under such a hold, copied for a forecast of 0.3 s, shows the difference: with the hold written as a law the forecast equals the subsequent motion of the original bit for bit, with a captured reference it is

1.25 rad off (1.59 rad in $3D$). The law is therefore the form for every force that depends on the state, and the function of time the form for a force of the clock alone. Both are evaluated at every stage and on every trial, so both must be functions of the state and the clock: a force that accumulates a quantity internally counts the trials. The state is passed read-only (the read-only accessors of the system are `const`), so a law cannot change the system it acts on.

What a joint may do: A joint's restriction and its drive are part of its own specification, shaped like its degrees of freedom. A revolute or prismatic joint carries an interval $[q_{\min}, q_{\max}]$ on its coordinate, a joint with several screws one bound per screw (`set_joint_range`); its drive carries the largest generalised force $\tau_{\max}$, rate and rate change it delivers, and a rotor inertia added to the diagonal of the mass matrix (`set_joint_drive`). Both have a neutral value – the unbounded interval, the infinite drive – so a model that states nothing behaves as before, and restricting a joint is a change of specification, not of kind. A range is enforced at the model, by a spring stop of a firmness the step can integrate; a drive is enforced by the controller, which may not ask for more than it has. A joint without an actuator is marked passive in its drive; the model refuses an actuator torque on it, and a controller leaves it out of the torques it computes.

The equation of motion as input to a controller: A model-based controller uses the equation of motion at the current state rather than its solution:

$$M(q)\,\ddot{q} = \mathrm{RHS}(q, \dot{q}) + \tau - G^{\mathsf{T}}\lambda, \qquad G\,\ddot{q} = -\dot{G}\,\dot{q} \tag{366}$$

`mass_bias()` returns M and RHS, the latter collecting gravity, the velocity-product terms, springs, dampers and applied wrenches. The registered actuator torques are excluded, so a law that computes τ from these quantities does not see its own output. `constraint_jacobian()` returns the active G of a closed chain and `constraint_row_offset(c)` the rows of one constraint; the multipliers come from `joint_accelerations(&lambda)`. The force a constraint exerts on its first anchor is $-\lambda$, so a foot pressing on the ground has a negative vertical multiplier. After a step of a closed chain `sync_accelerations()` writes the constrained accelerations into the frames, so that the point and centre-of-mass accelerations report the constrained motion.

The simplest use is gravity compensation: with $\dot{q} = 0$ the torque $\tau = -\mathrm{RHS}$ holds an open chain at rest. Computed torque, task-space control and prior-

itized whole-body control [Lynch and Park, 2017, Tedrake, 2024] are formulated on the same quantities. They are linear algebra on M, RHS, G and task Jacobians, and are left to the application.

The stack under the drives' box: With drives stated, each level of the stack is a least-squares problem in the torques under the box $-\tau_{\max} \le u \le \tau_{\max}$, and where the box binds, the two ways of imposing it differ. Solving the level unconstrained and clipping u to the box leaves a saturated joint's acceleration to the dynamics: the joint is passive, and a task solved exactly through a passive light link grows without bound – measured on a three-link arm, the two other torques doubled every half millisecond, at a rate that converged as the step shrank. Solving in the joints' accelerations instead, each confined to the interval its own drive delivers whatever the other joints do, prescribes a saturated joint's acceleration; the torques follow by inverse dynamics through Ψ, and no joint is ever passive. Both are special cases of one program,

$$\min_{\ddot{q},\,\nu} \tfrac{1}{2}\,\|A\,\ddot{q} - b\|^2 \quad \text{level by level,} \quad -\tau_{\max} \le \Psi^{+}(\ddot{q} - \ddot{q}_0) + N\nu \le \tau_{\max} \tag{367}$$

with the torque box exact, every further bound – a viability bound on a range, a load kept on a contact – a linear inequality in the same variables, and N the null space of Ψ, the load split between closed contacts, as variables of its own (`qp_ls_solve` solves each level). The three agree wherever no bound binds. Where one does they answer different questions: the clipped solve drops the saturated joint's share of the demand, the program delivers the demand through whatever remains. The choice belongs to the machine's specification – an arm whose stated drives are small is not to be handed the share a saturated knee cannot carry – and every number tuned on one of them is a number about that one.

A demand the drives cannot deliver: The reachable part of a task row is the support function of the torque box,

$$h(n) = \sum_{c} \tau_{\max,c}\,|n^{\top} A\,e_c|, \tag{368}$$

exact for a box; a demand b is deliverable along its own direction when $\|b\| \le h(b/\|b\|)$. A controller that reads this verdict answers a shortfall first with time. The reference $x_{\mathrm{ref}}(s(\tau))$ runs on its own clock, $\dot{\tau} = r$ with $r \in [r_{\min}, 1]$ driven toward r times the deliverable fraction while the demand is short, so that

$$\dot{x}_{\mathrm{ref}} = x'\,s'\,r, \qquad \ddot{x}_{\mathrm{ref}} = x''(s'\,r)^2 + x'\left(s''\,r^2 + s'\,\dot{r}\right) \tag{369}$$

are fed forward exactly. Only a shortfall that persists with $r = r_{\min}$ refuses the reach, with the demand, the bound and their ratio, instead of approximating silently. Refusing at once stopped 126 of 360 reaches on a grid at 5 and 6 kg, and 93 of them arrived once slowed: most shortfalls are a path taken faster than the drives allow, not a target beyond them. The price is 12 of the 360 reaches, whose free joints wind up while the clock slows; the divergence guard ends those, where the immediate refusal had stopped them before.

The solver's guarantees: `qp_ls_solve` is a primal active-set method with a null-space step, from a feasible start. A lexicographic level has fewer independent rows than free variables as a rule, so its objective is singular and the usual inference – a negative multiplier means the step leaves that row – does not hold. Two rules keep the iteration finite: the step's rank is judged at the level's own scale rather than the projected matrix's, and a row released at a stationary point that blocks the next step at zero length is pinned until the point moves. Exhausting the iteration cap throws. The tests check optimality by an independent oracle, the KKT system over the whole tight set solved as a non-negative least squares (`bvls_solve`): a zero residual is sufficient for a convex program whatever the degeneracy of the tight set. One known instance stops off its optimum and is gated as such.

1.8.6 Contact and balance

The contact layer of subsubsection 1.7.10 treats ground contact as events on the loop constraints. For a robot it is set up once. The ground is a line in $2D$ or a plane in $3D$ (`ground_line`, `ground_plane`; an inclined ground is no different from a level one). Every point that can touch is registered with `add` as a point contact (a pin) or as a flat contact (a weld under a foot of given extent), and `step` advances the model with each touchdown located by bisection and applied with the impact map. A contact can change its kind while active (`set_kind`, a foot settling onto its sole) and can be made bilateral for a controlled transfer (`set_unilateral`).

Segment contact: A sole, or any edge of a body, is registered in $2D$ as a *segment* between two points (`spec.segment` with `end_b`). It switches its own kind: the first end to touch is a pin, the other end's touchdown welds the segment inside the same impact map, and an end whose reaction turns to a pull rolls the weld back onto a pin at the other end. The segment never holds more than three rows, where two independent pins on one rigid foot would hold four on three degrees of freedom and the heel strike would be overdetermined. Its weld reaction is reported split into the two end forces,

and `pinned_end` names the end it stands on. The polygon sole of $3D$ is the same idea and is not built.

Ground reaction: The forces of the active contacts are force lines through the contact points P_i, and a flat contact adds a couple T_k at its ankle. Their sum is the ground reaction as one wrench,

$$W_R = \sum_i P_i \wedge f_i + \sum_k T_k \tag{370}$$

(`reaction_wrench`), the W_R of eqn. (363). The force is its attitude and its moment about any point one `moment_about`.

Zero-moment point: The *zero-moment point* (ZMP) is the point of the ground about which the ground reaction has no moment tangential to the ground. With the ground L (a line in $2D$, a plane in $3D$) it is the meet

$$Z = W_R \vee L \tag{371}$$

(`zmp`, fig. 22). In $2D$ the reaction wrench is the line of action of the resultant force, and Z is the point where it crosses the ground. In $3D$ the reaction of feet that push and slide is a general screw with no line of action; the definition then asks only for the two tangential moment components to vanish. The meet delivers that, because a couple about the ground normal satisfies $T_n \vee L = 0$ while a tangential couple does not. The same expression therefore holds in both dimensions and on inclined ground. On level ground without friction it reduces to the centre-of-pressure formula $Z = \sum_i p_i N_i \,/\, \sum_i N_i$ with the normal forces N_i [Tedrake, 2024]. A resultant parallel to the ground meets it at infinity. A single flat contact has its own centre of pressure (`cop`), and `tipping` reports when it leaves the foot.

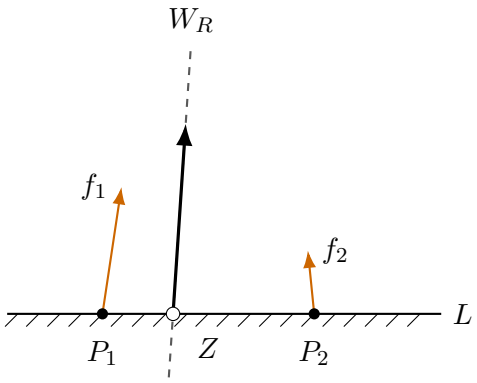


Figure 22: The zero-moment point in $2D$. The contact forces f_1, f_2 at P_1, P_2 sum to the reaction wrench W_R, which is the line of action of their resultant (dashed, resultant force drawn bold); $Z = W_R \vee L$ is where that line meets the ground.

Balance: A stance on flat contacts can hold only while Z lies inside the support polygon of the contacts (in $2D$ the support segment). Eqn. (363) ties Z to the motion: the acceleration of the centre of mass and the rate of the angular momentum fix $W_R = \dot{P} - W_g$, and with it Z. Commanding the centre of mass is therefore also commanding the zero-moment point, which is how the balance condition enters a controller.

1.8.7 The algebra's role in robotics

Table 41 summarizes where the geometric formulation carries the robotics topics of this subsection and where the work is linear algebra or numerics written as such.

topic	*GA* form	contribution
joints, floating base, integration	$M = \exp_\vee(\frac{1}{2}\sum_k q_k S_k)$; motor joints integrated on the group	one description for all joint types; no angle singularities; fourth order with two bracket terms
transport of twists and wrenches	one sandwich; bracket rcmt; pairing −rwdg	replaces the transport matrices and cross-product operators of the classical notation
Jacobian, statics	columns are joint screws; body form by one sandwich; $\tau_j = \mathrm{rwdg}(S_j, W)$	the columns are axes and lines of action; joint loads are incidences
centre of mass, gravity, momentum	weighted point sum; sum of force lines; sum of momentum bivectors	whole-body statements as single objects; Newton–Euler as one equation
zero-moment point	meet of the reaction wrench with the ground	one expression in $2D$, in $3D$ and on inclined ground
closed-form inverse kinematics	*CGA* meet of circles, spheres and a plane	both branches and reachability from one object; slower; decoupled arms only
rank, null space, least squares, *KKT*	—	linear algebra
controllers, time scaling, stability analysis	—	linear algebra, optimisation and numerics
recursive $O(n)$ dynamics	—	not implemented; the assembly is cubic in the number of joints

Table 41: The role of the geometric formulation in the robotics topics of this subsection.

1.9 Modelling magnetostatics

The static field of currents and of magnetised bodies, and the force and torque it exerts on them. The two halves fall into the two algebras of subsection 1.10.7: a field value belongs to the point where it is evaluated and is an EGA quantity; a force belongs to its line of action and is a PGA wrench. The primitives of this subsection are those of `ga_usr_magnetostatics.hpp`, in two and three dimensions, and table 44 at its end lists them. The bivector reading follows [Jancewicz, 1988], [Doran and Lasenby, 2003] and [Arthur, 2013], and the classical forms the primitives are checked against are those of [Griffiths, 2017].

1.9.1 The grades of the field

In $3D$ the flux density is carried here as the vector b; the current element is the vector $I\,dl$, a magnetisation and a dipole moment are vectors m and μ. The other common convention carries the induction as a bivector, the dual of the field vector (subsection 1.2.4, `dual(b)`),

$$B = b^\star, \tag{372}$$

and the dipole moment accordingly as $m = \mu^\star$. Relating a direction to the plane element perpendicular to it uses the metric, so the dual is the operation here; in EGA it evaluates as the complement, because the extended metric is the identity. A formula written in one convention translates into the other by taking duals. A formula linear in the field and linear in the moment keeps its form; a formula in which the field meets itself changes sign, since for vectors b and n of $3D$ EGA

$$b^\star \wedge n \wedge b^\star = -\,b \wedge n \wedge b \tag{373}$$

(an identity of the product and dual tables, subsection 1.2.9; the sign is that of the Euclidean metric). The dipole field below is of the first kind, the stress of the second.

The distinction between dual and complement becomes visible in STA, where the extended metric is not the identity and the two differ in sign on the bivectors $\boldsymbol{g}_{k4}$. These relative vectors square to +1 and span a copy of $EGA3D$ ($e_k \to \boldsymbol{g}_{k4}$). The dual of $\boldsymbol{g}_{14}$ is the image of ${e_1}^\star = e_{23}$, the complement its negative. With $F = \sum_k b_k\,{\boldsymbol{g}_{k4}}^\star$ the magnetic part of the field bivector carries the sign the Lorentz force fixes, and $\nabla_\wedge F = \mu_0 J$ (eqn. (116)) holds; built with the complement, the same field returns the current reversed.

A $2D$ field is the field of a current distribution that does not depend on z, read in the plane $z = \text{const}$. The flux density lies in the plane and stays a vector b. What points along z is a multiple of e_3, which is not in the plane's algebra; it enters through its $3D$ dual, $e_3{}^\star = e_{12}$: the current density as $J^\star = J_z\, e_{12}$, a line current as $I^\star = I_z\, e_{12}$ and the vector potential as $A^\star = A_z\, e_{12}$, the pseudoscalars the $2D$ functions take. A magnetisation in the plane is a vector m; the surface current it implies at an edge points along z as well and enters as $K^\star = m \wedge n$.

1.9.2 The field equation

With the vector derivative of subsection 1.3, the two magnetostatic laws $\nabla \cdot b = 0$ and $\nabla \times b = \mu_0 J$ are one equation,

$$\nabla b = \mu_0\, J^\star, \tag{374}$$

in $2D$ and $3D$ alike, its scalar part the vanishing divergence and its bivector (in $2D$ pseudoscalar) part the curl, $\nabla \wedge b = (\nabla \times b)^\star$. The field is the vector derivative of the dual potential taken backwards,

$$b = -\nabla A^\star, \qquad \nabla^2 A^\star = -\mu_0\, J^\star, \tag{375}$$

the classical $b = \operatorname{curl} A$, in $2D$ $(\partial_y A_z, -\partial_x A_z)$. In $3D$ it holds in the Coulomb gauge $\nabla \cdot A = 0$, which the potential of a closed current satisfies; otherwise $\nabla A^\star$ carries the divergence as a trivector part. In $2D$ the gauge holds identically. The flux per unit length through a path from P to Q is a difference of the potential, $\Phi = A_z(Q) - A_z(P)$, through the normal $\nu = t_\star$ (the left dual turns the tangent by $-90°$, table 1); field lines are the level sets of A_z.

1.9.3 Sources

The line current: The field of a line current $I^\star = I_z\, e_{12}$ through the point p is the dual of the vector inverse,

$$b(r) = \frac{\mu_0 I_z}{2\pi}\left((r-p)^{-1}\right)^\star, \qquad A^\star(r) = -\frac{\mu_0}{2\pi}\ln|r-p|\; I^\star, \tag{376}$$

with $(r-p)^{-1} = (r-p)/|r-p|^2$: the $2D$ Green's function of the vector derivative, $\nabla\big(r/(2\pi|r|^2)\big) = \delta$. The right dual of a $2D$ vector turns it by $+90°$ (table 1), which is the classical $\hat{\varphi}/\rho$ direction; any current distribution in the plane is a sum of these kernels, the Cauchy kernel of complex analysis read in GA.

The current segment: In $3D$ a straight segment from a to b with unit direction u carries I. With ρ the rejection of $r-a$ from the segment's line,

$$b(r) = \frac{\mu_0 I}{4\pi}\,(c_a - c_b)\,\big(u \wedge \rho^{-1}\big)^\star, \qquad c_a = \frac{u\cdot(r-a)}{|r-a|}, \quad c_b = \frac{u\cdot(r-b)}{|r-b|}, \tag{377}$$

the Biot–Savart law integrated in closed form, $(u\wedge v)^\star = u\times v$. For a segment of infinite length $c_a - c_b \to 2$ and the field is the $2D$ kernel of eqn. (376): the line current is this segment seen in the plane. Its vector potential points along the segment, $A(r) = \frac{\mu_0 I}{4\pi}\,\ln\frac{|r-b|-u\cdot(r-b)}{|r-a|-u\cdot(r-a)}\,u$. A closed loop is a polygon of segments; the field of a circle of N chords converges to the circle's as N^{-2}.

The dipole: Far from a small loop, the field is that of its moment $\mu = I\,(\text{area})\,n$. The classical form $b(r) = \frac{\mu_0}{4\pi r^3}\big(3\,\hat r(\mu\cdot\hat r) - \mu\big)$ is a single reflection sandwich,

$$b(r) = \frac{\mu_0}{8\pi\,r^3}\,\big(\mu + 3\,\hat r \wedge \mu \wedge \hat r\big), \qquad B(r) = \frac{\mu_0}{8\pi\,r^3}\,\big(m + 3\,\hat r \wedge m \wedge \hat r\big), \tag{378}$$

with $\hat r\wedge\mu\wedge\hat r$ the reflection of μ in the separation direction.[21] The second form is the first in the bivector convention of eqn. (372): the dual of a reflection sandwich is the sandwich of the dual, $(\hat r \wedge \mu \wedge \hat r)^\star = \hat r \wedge \mu^\star \wedge \hat r$, so it is the same expression with no sign change. The two terms carry the field's anisotropy between them: on the equator of the moment ($\hat r \perp \mu$) the reflection reproduces $-\mu$ and the bracket is -2μ, on the axis it reproduces $+\mu$ and the bracket is 4μ — the axial-to-equatorial ratio of -2, read off one reflection. Eqn. (378) is the case $d = 3$ of the dipole field in d dimensions,

$$b(r) = \frac{\mu_0}{2S_d\,r^d}\,\big((d-2)\,\mu + d\,\hat r \wedge \mu \wedge \hat r\big), \tag{379}$$

with S_d the area of the unit sphere, $S_2 = 2\pi$ and $S_3 = 4\pi$. In $2D$ the identity term drops out with $d-2$: the line dipole (moment μ per unit length) is one reflection, $b(r) = \frac{\mu_0}{2\pi r^2}\,\hat r \wedge \mu \wedge \hat r$.

Magnetised bodies: A magnetisation m is equivalent to the surface currents

$$K^\star = m \wedge n \tag{380}$$

[21] The rewrite uses the axis reflection $\hat r \wedge v \wedge \hat r = 2(\hat r\cdot v)\,\hat r - v$ (the reflection of a vector v in the axis $\hat r$; the mirror across the plane $\perp \hat r$ is its negative, which is why no overall minus sign appears). The two-term bracket regroups as $3(\hat r\cdot\mu)\,\hat r - \mu = \frac{1}{2}\big(\mu + 3\,\hat r \wedge \mu \wedge \hat r\big)$, the factor $\frac{1}{2}$ turning the 4π into 8π.

on its boundary with outward normal n (and the volume current $J^\star = \nabla \wedge m$, zero for a uniform one), in $2D$ and $3D$ alike: in $3D$ the dual applied twice is the identity, so $K = (m \wedge n)^\star = m \times n$. Two bodies have closed forms that link the sources above: a uniformly magnetised disc of radius R has the field $\frac{1}{2}\mu_0 m$ inside and outside the line dipole of moment $\pi R^2 m$; a uniformly magnetised sphere has $\frac{2}{3}\mu_0 m$ inside and outside the dipole of moment $\frac{4}{3}\pi R^3 m$, exactly. For any other bounded magnetisation the dipole is the leading term of the multipole expansion, the corrections falling off by further powers of size over distance.

1.9.4 Forces and couples

The force on a current element is one contraction in $2D$ and $3D$ alike,

$$dF = (I\,dl)^\star \lfloor b, \tag{381}$$

in $3D$ equal to $(I\,dl \wedge b)^\star = I\,dl \times b$. In $2D$ the force per unit length on a line current $I^\star = I_z\, e_{12}$ is $F = I^\star \lfloor b = I_z\, b^\star$, the field turned by $+90°$, the classical $I\, e_z \times b$. For a dipole the coupling energy, the torque and the force all read off the one field object:

$$U = -\,m \cdot B, \qquad T = \mathrm{cmt}(B, m), \qquad F = \nabla_r\,(m \cdot B), \tag{382}$$

the energy from the scalar product of the library (positive definite on bivectors in EGA; the scalar part of the geometric product of two bivectors is its negative, $\langle m \wedge B \rangle_0 = -\,m \cdot B$), the torque as the commutator $\frac{1}{2}(B \wedge m - m \wedge B)$ — a bivector, the grade GA assigns to rotational quantities — and the force as the gradient of the coupling. Two identities connect these to the classical forms: the scalar product carries over the dual, $a^\star \cdot b^\star = a \cdot b$, so U equals $-\mu \cdot b$; and the commutator of two duals is the dual of the cross product, $\mathrm{cmt}(b^\star, a^\star) = (a \times b)^\star$, so $T = (\mu \times b)^\star$, component for component. The closed-form dipole–dipole force and torque of [Yung et al., 1998] and [Landecker et al., 1999] are the references. Table 42 sets the two formulations side by side.

Quantity	Classical (vector calculus)	GA formulation
dipole moment	axial vector $\vec{\mu}$	bivector $m = \vec{\mu}^{\star}$ $(EGA3D)$
field	$\vec{B} = \frac{\mu_0}{4\pi r^3}\big(3\hat{r}(\vec{\mu}\cdot\hat{r}) - \vec{\mu}\big)$	$B = \frac{\mu_0}{8\pi r^3}\big(m + 3\hat{r} \wedge m \wedge \hat{r}\big)$
coupling energy	$-\vec{\mu}\cdot\vec{B}$	$-\,m\cdot B$ (same value)
torque	$\vec{\mu}\times\vec{B}$ (axial vector)	$\mathrm{cmt}(B, m)$ (bivector)
force	closed form [Yung et al., 1998]	$\nabla_r(m\cdot B)$
force + torque bookkeeping	three objects: $\vec{F}$, $\vec{T}$, lever $\vec{r}\times\vec{F}$ about a chosen reference	one wrench $W = \mathrm{wdg}(P, F) + \mathrm{couple}(T)$
action = reaction	$\vec{r}\times\vec{F} + \vec{T}_{ij} + \vec{T}_{ji} = 0$	$W_{ij} + W_{ji} = 0$
change of reference point	re-sum all lever terms	one motor sandwich $\mathrm{move3dp}(W, \underset{\sim}{M})$

Table 42: Dipole–dipole interaction: classical vector calculus versus geometric algebra.

1.9.5 The Maxwell stress as one sandwich

The force the field exerts on everything inside a closed surface is the integral of the traction over that surface. The traction on a surface element with unit normal n is one sandwich,

$$t(n) = \frac{1}{2\mu_0}\, b \wedge n \wedge b = \frac{1}{\mu_0}\Big((b\cdot n)\, b - \tfrac{1}{2} b^2 n\Big), \tag{383}$$

Maxwell's tensor applied to n, because $b \wedge n \wedge b = 2(b\cdot n)\, b - b^2 n$ for vectors. The identity holds in any dimension, so eqn. (383) is the same expression in $2D$ and $3D$ (in $3D$ its trivector part $b \wedge n \wedge b$ vanishes). In the bivector convention the traction is $-\frac{1}{2\mu_0}\, B \wedge n \wedge B$: the field meets itself, and eqn. (373) changes the sign.

1.9.6 The wrench of a field on a body

The tractions act at points on the surface, so their sum is a sum of force lines, one PGA bivector (subsection 1.7.1):

$$W = \oint_{\partial V} \mathrm{wdg}\big(P,\, t\big)\, dS, \tag{384}$$

with P the point lifted to PGA, the traction as an ideal direction, and dS the boundary's measure, the arc length of a closed contour in $2D$ and the area element of a closed surface in $3D$. In $2D$ the sum is a line of action: the

force, its direction and the line it acts on, which carries the torque about any point. In $3D$ force lines through different points sum to a general *screw*, a force plus a couple. The couple needs no term of its own: for a dipole in the field of a second one, the stress integral over a sphere around it is the force line through the dipole plus the couple of eqn. (382),

$$W = \mathrm{wdg}(P, F) + \mathrm{couple}(T), \tag{385}$$

to $3 \cdot 10^{-14}$ in the library's tests, the couple coming out of the sum of the force lines. The lever torque $\vec{r} \times \vec{F}$ of the classical bookkeeping is what the wrench reads as at a different reference point, and a change of reference is the motor sandwich move3dp$(W, \underset{\sim}{M})$. Action and reaction between two bodies are one bivector identity, $W_{ij} + W_{ji} = 0$ — the forces between two dipoles are not central and the couples differ, and the identity still holds as one statement. Paired with the twist S of a degree of freedom, the reciprocal product $\langle S, W \rangle$ of subsection 1.7.9 is the generalised force the field exerts on it, the mechanical power per unit rate.

Packing a force (in N) and the torque it generates (in Nm) into one bivector does not sum unlike quantities: the weight of W carries the force, $\|W\|_\circ^2 = \|f\|^2$ in N^2, and the bulk the torque about the reference point, $\|W\|_\bullet^2$ in $(\mathrm{Nm})^2$. No norm of the library adds the two — the bulk norm of a bivector is a scalar of the algebra, the weight norm a pseudoscalar — and the motor sandwich that transports a wrench moves force and torque together without combining them. The guarantee is geometric, not a unit type: the components are plain numbers, and "never sum a force component against a torque component" remains a convention, the same one the classical lever bookkeeping requires.

1.9.7 Where the algebra helps

The gains are in the formulation. The two magnetostatic laws are one equation, eqn. (374); the stress is one sandwich, the same in $2D$ and $3D$; the force and torque of a field on a body are one wrench with its line of action, where the classical form computes torque, force and line of action as separate integrals; the torque on a dipole has the grade of the field. The potential and the flux function carry the same content as their classical counterparts — GA names the 90° turn as the dual, nothing more.

The arithmetic is not reduced. The graded types store the classical component counts — a field or a moment is three numbers, a wrench six — and the products compile to the same scalar expressions, so the classical and the GA forms agree to the last bit and cost the same (table 43). The same holds for

the numerical methods built on the field: the element matrices of a first-order finite-element model, written with the gradient of a barycentric coordinate as the dual of an edge, equal the classical Jacobian form and cost the same; the mesh predicates are the classical determinants. Where the field has to be computed rather than written down, GA explains the construction and leaves the numerics as they are.

Operation	Classical	GA
dipole field $B(r)$	3.60	3.78
torque $\mathrm{cmt}(B, m)$	4.46	4.74
wrench assembly	1.25	1.17

Table 43: Per-operation cost in nanoseconds (two million evaluations, one machine, an optimized build). The wrench-assembly row compares the classical lever cross product with the GA wedge-plus-couple, from identical force and torque inputs.

Quantity	Function (`ga_usr_magnetostatics.hpp`)
vacuum permeability	`mu0`
line current: field, potential, a set	`b_line_current`, `A_line_current`, `b_line_currents`
current segment: field, potential, a set	`b_segment`, `A_segment`, `b_segments`; a loop as chords `circle_loop`
dipole field (vector / bivector), coupling	`b_dipole`, `dipole_coupling`
Lorentz force ($2D$ / $3D$)	`lorentz_force`
surface current of a magnetisation	`surface_current`
magnetised disc, sphere as currents	`magnetised_disc_currents`, `magnetised_sphere_currents`
Maxwell stress	`traction`
closed contour, closed surface	`circle_contour`, `sphere_surface`
wrench of a field on a body	`stress_wrench`

Table 44: The magnetostatic primitives and their functions (namespace `hd::ga`; bound to Python and Lua).

1.10 Approaches with increasing expressiveness

1.10.1 Classic approach vector algebra (VA)

- Use of scalars and vectors, exclusively
- Use of the cross-product between vectors to create directions perpendicular to both vectors (a concept that can only be defined in three-dimensional space)
- Rotations are defined by a rotation axis (a concept that is only defined in three-dimensional space)[22]

1.10.2 General observations for Geometric Algebra (GA)

- The GA approach is coordinate-free!
- W.r.t. a vector, the dot product (or interior product) takes into account the parts parallel to the vector, exclusively. Thus, it is not invertible itself.
- W.r.t. a vector, the wedge product (or outer product) takes into account the parts perpendicular to the vector, exclusively. Thus, it is not invertible itself.
- Only the geometric product taking both parts into account is invertible.

1.10.3 Euclidean geometric algebra (EGA)

- Use of scalars, vectors, bivectors, ..., pseudoscalars (higher grade objects up to order n depend on the dimension n of the modelled space).
- Use of the outer product (wedge product) between vectors, which defines the space spanned by both vectors (can be defined in any dimension, applied as join operation).
- Use of the regressive outer product (regressive wedge product) which defines the space shared by both objects (represents the intersection of the spaces used by the objects, applied as meet operation).
- Modelled objects cover scalars, directions, planes (in $3D$) and the whole space directly.

[22] The last two concepts cannot be extended to higher-dimensional spaces.

- All objects modelled in the algebra contain the (non-distinguishable) origin $O = (0, 0, ..., 0)$. Objects outside the origin must be modeled by explicitly adding translations (leading to objects that cannot directly be manipulated as easily as objects that directly represent subspaces within the algebra).
- Orthogonal transformations in the algebra can directly model reflections and rotations. Rotations are defined in a plane spanned by two vectors defining the rotation.

1.10.4 Projective (Euclidean) geometric algebra (PGA)

- Like EGA, but in addition having a distinguishable origin encoded in an additional dimension $O = (0, 0, ..., 0, 1)$ (in a projective space with an additional dimension projected onto a fixed value, typically 1).
- Use of the outer and regressive outer products as join and meet operations.
- Directly models points, directions and planes (in $3D$), as well as the whole space, including objects located in the horizon, i.e. at infinity (points at infinity are modeled as directions towards those points).
- Can directly model higher grade objects by products between objects in the algebra that do not contain the origin, e.g. the wedge product between two arbitrary points defines a line.
- Orthogonal transformations in the algebra can directly model reflections, rotations and translations.

1.10.5 Conformal geometric algebra (CGA)

- Extends the projective approach by a second additional dimension, the round dimension, so that $2D$ Euclidean space is modelled in four and $3D$ Euclidean space in five dimensions.
- Euclidean points sit on a null cone: an embedded point squares to zero, and the dot product of two of them returns their squared distance directly.
- Directly models the round objects — point pairs, circles and spheres — as blades of the algebra, alongside the flat objects already available in PGA, which reappear as the rounds that contain the point at infinity.

- The metric is non-degenerate. Every dual is therefore invertible, the bulk/weight split of the duals disappears, and the split of the objects becomes a four-part one instead, by origin and infinity.
- Orthogonal transformations extend from the rigid motions to the whole conformal group: dilations, inversions in a sphere and transversions join rotation and translation, all as the same sandwich product.
- Intersections need no case distinction: the meet of two rounds always exists, and the sign of its squared radius reports whether real points do.

1.10.6 Space-time algebra (STA)

- Extends the Euclidean approach to space-time: $\mathbb{G}(1,3,0)$ with three space-like and one time-like basis vector (Minkowski signature).
- Bivectors carry both kinds of transformation generators: space-like planes generate rotations, time-like planes generate Lorentz boosts. The even-grade rotors implement both through the same sandwich product as in EGA.
- Quantities that are separate objects in $3D$ combine into single space-time objects (events and four-velocities as vectors; the electric and magnetic fields as one field bivector), and frame changes act on all of them by rotor conjugation.
- Suitable when time is part of the modelled physics itself: fields of moving sources, induction, waves, invariance arguments. For static and quasi-static problems EGA with time as an external parameter remains sufficient.

1.10.7 Choosing the algebra

The algebras are layers with distinct responsibilities rather than competing formalisms. EGA describes the *value* of a physical quantity at a point: its metric is invertible, which provides norms, angles and gradients, and with them field calculus. Its objects carry no position – the evaluation point is a parameter, not part of the object. PGA describes *located* objects – points, lines of action, planes, poses – and rigid motion; its degenerate metric intentionally gives up field calculus to gain translations as sandwich products. CGA keeps everything PGA models and adds the objects with a radius and

the transformations that change size; it pays for that with two extra dimensions, 32 components in $3D$, and with the round objects mixing position and radius in one component, which makes moving them more expensive than moving a flat one. STA describes quantities whose physics couples space and time.

A quantity therefore belongs in EGA when its position is merely where it is evaluated (a field value, an energy density), and in PGA when its position is part of what it is (a force on its line of action, the pose of a body). The two connect through the ideal sector: an EGA direction embeds as an ideal point, an EGA bivector as an ideal line (a couple), so a local result computed in EGA is bound to its location by a single wedge product – e.g. $P \wedge f$ turns a force vector evaluated at a point into the wrench acting on that point's line of action. STA enters when time can no longer stay an external parameter.

Both directions are one call. The embedding is a construction, and the weight decides the kind: a PGA vector built with unit weight is the point at v, one built with vanishing weight is the direction v, and a line is assembled from a direction and a moment – `Vec3dp(v,1.0)`, `Vec3dp(v,0.0)` and `BiVec3dp(dir,mom)`. The way back reads the Euclidean part off the object: `to_vec3d`, `to_bivec3d` and `to_scalar3d` in $3D$, `to_vec2d` and `to_pscalar2d` in $2D$. The requested type selects which part is meant, since each grade holds at most one part of each Euclidean type – a line has one direction and one moment, a plane one normal n and one offset d, the latter two satisfying $n \cdot p + d = 0$ for every point p of the plane. A vector is the one case where the weight has to be read: it is a point when it carries weight and a direction when it is ideal, so the conversion divides the weight out in the first case and returns the bulk unchanged in the second. The test is the one `unitize` and `try_unitize` apply, so all three agree on which objects carry weight.

A quantity belongs in CGA when circles, spheres or angle-preserving maps are part of the problem rather than a construction on top of it: intersections of curved objects, inversive geometry, or a chain of transformations that includes scaling. Where the objects stay flat and the motions rigid, PGA answers the same questions with half the components.

In short: EGA answers *what* – the value of a quantity at a point. PGA answers *where* – location and rigid motion. CGA answers *how big and how curved* – rounds and the conformal maps. STA answers *when* – coupling across space and time.

1.11 Glossary

Pose

- A pose is the rigid placement of one frame relative to another: an origin and an orientation (in $2D$ a position and a scalar angle φ; in $3D$ a position and an axis-angle vector).
- It is a point of the rigid-motion group, carried by a unit motor M; the origin/orientation form and the motor interconvert exactly (see 1.7.7).
- A pose is a finite placement (where a frame is), as opposed to a twist (how it is moving).

Motor and rotor

- In PGA a motor M performs a rigid motion by the sandwich product $X' = M \vee X \vee \underset{\sim}{M}$, applied identically to every object: points, lines, twists, wrenches.
- It is the exponential of a twist, $M = \exp_{\vee}(\frac{1}{2}B)$ (see eqn. (175)); motors multiply (compose) where their generators add.
- A unit motor in $3D$ PGA encodes a screw motion (rotation about and translation along an axis), so one motor covers rotation, translation and their combination.
- The EGA counterpart, built from the geometric product $\wedge$ instead of the regressive $\vee$ and acting only through the origin, is the rotor (rotations only).
- In CGA the same sandwich carries the whole conformal group, so a motor there may also dilate, invert or apply a transversion. Its generators follow the parity of the algebra: bivectors in the four-dimensional $CGA2DC$, trivectors in the five-dimensional $CGA3DC$, where the motors are odd-grade.

Screw (Chasles' theorem)

- By Chasles' theorem every rigid motion is a screw: a rotation about an axis combined with a translation along it.
- In PGA the screw axis is a line (a bivector in $3D$); the principal quantities are all screws on such an axis: the twist (a velocity screw), the wrench (a force screw), and the motor (the finite screw motion $\exp_{\vee}$ of a twist).

- A pure rotation and a pure translation are the degenerate screws of zero and infinite pitch.

Twist

- A twist B is the instantaneous motion of a frame: the Lie-algebra generator that builds its motor through $M = \exp_{\vee}(\frac{1}{2}B)$ (see eqn. (175)). Its grade is dimension-dependent: a vector in $2D$ PGA, a bivector in $3D$ PGA.

- The rate Ω is recovered from a moving motor by $\Omega = 2\,\dot{M} \vee \underset{\sim}{M}$ (see eqn. (177)). It equals the time derivative of the generator, $\Omega = \dot{B}$, only where B and $\dot{B}$ commute — for a constant rate the generator grows linearly, $B(t) = \Omega t$, and the motor is $M(t) = \exp_{\vee}(\frac{1}{2}\Omega t)$. In general the two are related through the differential of the exponential, $\Omega = \mathrm{dexp}_B \dot{B}$, whose inverse is the commutator series $\dot{B} = \Omega + \frac{1}{2}[B,\Omega] + \frac{1}{12}[B,[B,\Omega]] + \ldots$ (the bracket is the regressive commutator, `rcmt`). A time integrator that advances the generator must use this series: with $\dot{B} = \Omega$ alone a fourth-order scheme on the pair (B,Ω) degrades to second order (measured in the free-body integrator, error ratio 4 instead of 16 per halving of the step).

- Twists add (they live in a vector space) where motors multiply.

- A twist is a velocity screw inducing the point velocity $\dot{X} = \mathrm{rcmt}(\Omega, X)$.

Pose vs. twist

- Pose is the group element (where a frame is, the motor M); twist is the tangent/rate (how it moves, the generator B with rate $\Omega = \dot{B}$).

- They interconvert by integration and differentiation: $M(t) = M_0 \vee \exp_{\vee}(\frac{1}{2}\int \Omega\, dt)$ and conversely $\Omega = 2\,\dot{M} \vee \underset{\sim}{M}$.

- Pose lives in the Lie group, twist in the Lie algebra (see 1.4.4); cf. the entry on infinitesimal vs. finite movements.

Lie group vs. Lie algebra

- The motors form a Lie group: the manifold of finite rigid motions, composed by $\vee$.

- The twists form its Lie algebra: the tangent space at the identity, a vector space closed under the bracket $[A,B] = \mathrm{rcmt}(A,B)$.

- The two are linked by $\exp_{\vee}$ (algebra to group) and its logarithm (group to algebra); twists add and commute to first order, motors multiply and do not (see Sec. 1.4.4).

Infinitesimal vs. finite movements

- An infinitesimal movement is a twist (a Lie-algebra generator); a finite movement is the motor it integrates to (a Lie-group element).
- Simultaneous infinitesimal motions superpose (twists add), but the finite motors they generate do not commute and must be multiplied in order.
- This is why a parent-to-child pose is built by composing a rotation and a translation motor, not by exponentiating their summed generators (see 1.4.4).

Adjoint

- The adjoint Ad_M re-expresses a twist in another frame.
- In *PGA* it is the same sandwich that moves any object, $\mathrm{Ad}_M(B) = M \vee B \vee \underline{M}$ (see eqn. (341)).
- Point action and twist transport are thus one operation; no separate 6×6 adjoint matrix is needed (see 1.7.8).

Commutator (Lie bracket)

- The Lie bracket (commutator) of two twists measures how far they fail to commute.
- In *PGA* it is a native product, $[A, B] = \mathrm{rcmt}(A, B)$: the $\mathfrak{se}(2)/\mathfrak{se}(3)$ twist bracket.
- It supplies the Coriolis/centrifugal coupling in the Newton-Euler recursion (see eqn. (343)) and the bias term $\mathrm{rcmt}(\Omega, I[\Omega])$ in the body-frame Euler equation.

Central element

- An element z is *central* if it commutes with every element of the algebra, $z \wedge X = X \wedge z$ for all X; equivalently its commutator vanishes identically, $\mathrm{cmt}(z, X) = 0$.

- For a grade-k element A_k in n dimensions

$$\mathbb{1} \wedge A_k = (-1)^{k(n-1)}\, A_k \wedge \mathbb{1}\,, \tag{386}$$

so $\mathbb{1}$ is central exactly when n is odd. For even n it commutes with the even grades and anticommutes with the odd ones.

- Centrality follows from the dimension alone, while the metric fixes $\mathbb{1}^2$ independently; the two combine per algebra as

algebra	n	$\mathbb{1}^2$	$\mathbb{1}$ central	centre = span$\{1, \mathbb{1}\}$
$EGA2D$	2	-1	no	—
$EGA3D$	3	-1	yes	$\cong \mathbb{C}$
$PGA2DP$	3	0	yes	$\cong$ dual numbers
$PGA3DP$	4	0	no	—
$STA4DS$	4	-1	no	—
$CGA2DC$	4	-1	no	—
$CGA3DC$	5	-1	yes	$\cong \mathbb{C}$

In odd dimensions the centre is spanned by the scalar 1 and the pseudoscalar $\mathbb{1}$ alone; in even dimensions it reduces to the scalars. Its *shape* is always a scalar plus a pseudoscalar — a dual number in the sense of 1.2.3 — but the algebra that shape forms is fixed by $\mathbb{1}^2$: for $\mathbb{1}^2 = -1$ it is $\mathbb{C}$, with $\mathbb{1}$ in the role of i and every non-zero element invertible; for $\mathbb{1}^2 = 0$ it is the dual numbers proper, where $\mathbb{1}$ is nilpotent and only elements with a non-zero scalar part are invertible. The same split governs the exponential: $\cos\alpha + \mathbb{1}\sin\alpha$ in the first case, $1 + \alpha\mathbb{1}$ in the second.

- A unit central versor generates no motion: $E \wedge X \wedge \widetilde{E} = X \wedge E \wedge \widetilde{E} = X$. Such an E acts by multiplication rather than by conjugation, which is what makes $\exp(\alpha\mathbb{1}) = \cos\alpha + \mathbb{1}\sin\alpha$ a duality rotation in $EGA3D$ and a rotor in $EGA2D$.

- Electrodynamics is where the $EGA3D$ case is put to work: with $\mathbb{1}$ central the electromagnetic multivector $f = e + \mathbb{1} \wedge b$ behaves as a complex vector, a duality rotation carries solutions of the field equation to solutions, and the energy density and flux carried by $f \wedge \widetilde{f}$ are invariant under it. In the code the same property surfaces as $\mathrm{cmt}(X, \mathbb{1}) = 0$: the commutator cannot express a product against a central pseudoscalar.

Duality rotation

- The one-parameter family generated by the pseudoscalar,

$$X \mapsto \exp(\alpha \mathbb{1}) \wedge X = (\cos\alpha + \mathbb{1}\sin\alpha) \wedge X\,, \tag{387}$$

  defined where $\mathbb{1}^2 = -1$. It is a rotation in the two-dimensional space spanned by X and $\mathbb{1} \wedge X$, not a rotation of space: at $\alpha = 0$ it is the identity and at $\alpha = \pi/2$ it is multiplication by $\mathbb{1}$, which is the left dual of the reverse, $\mathbb{1} \wedge X = (\widetilde{X})_\star$ (see 1.2.4); in EGA the dual and the complement coincide.

- It acts by multiplication, not by the sandwich, so it is not a motion — the contrast to a rotor or motor. On a pair $X + \mathbb{1} \wedge Y$ of an object and a dual partner it turns the two into each other by the angle α, leaving $|X|^2 + |Y|^2$ fixed.

- Where it is available follows the pseudoscalar's centrality. In $EGA3D$ $\mathbb{1}$ is central, so the rotation acts on every grade. In $STA4DS$ it commutes with the bivectors only, so the rotation is defined there on bivector quantities and not on vector ones. Where $\mathbb{1}^2 = 0$, as in PGA, the exponential degenerates to $1 + \alpha\mathbb{1}$ and the family shears instead of rotating.

Inner and outer derivative

- The vector derivative ∇ acts on a field through the geometric product, and its two grade parts are named separately (subsection 1.3 develops this at length):

$$\nabla_\wedge X = \nabla \cdot X + \nabla \wedge X\,, \tag{388}$$

  the *inner* derivative lowering the grade of a k-vector field to $k-1$ and the *outer* derivative raising it to $k+1$.

- The classical operators are single grades of that pair rather than operators in their own right: grad is the outer derivative of a *scalar* field, div the inner derivative of a *vector* field, and curl the *dual* of the outer derivative of a vector field. In $3D$:

field	$\nabla \cdot X$	$\nabla \wedge X$
scalar	0	vector = grad
vector	scalar = div	bivector = $(\text{curl})_\star$
bivector	vector	trivector
trivector	bivector	0

- So $\nabla \wedge A$ is not the curl but its dual: the same coordinates carrying a different object, a plane rather than an axis. The classical vector is recovered as $(\nabla \wedge A)_\star$ — by the dual, which carries the metric, and not by the complement, which does not. *EGA* cannot separate the two: under an identity metric $\underline{X} = X_\star$ at every grade, so either name reads correctly there while only one of them survives a change of algebra. The two part on precisely those blades whose extended-metric entry is -1 — in *STA4DS* exactly half of the sixteen. Multiplication by $\mathbb{1}$ is a third map again, agreeing with the dual only up to grade one: $\mathbb{1} \wedge X = (\widetilde{X})_\star$ carries a reversion (see **Duality rotation**). The curl is also the only member of the trio tied to three dimensions — the outer derivative of a vector field is a pseudoscalar in $2D$ and a trivector in *STA4DS*, whereas $\nabla\cdot$ and $\nabla\wedge$ mean the same in every algebra.

- "Gradient equals divergence plus curl" is a misreading of eqn. (388). That identity holds for a *vector* field, its left-hand side is the vector derivative rather than the gradient, and its second term is the dual of the curl. For a scalar field the inner part vanishes identically, so there $\nabla\varphi$ *is* the gradient and nothing is added to it.

Moment and torque

- Torque is the rotational counterpart of linear force: a force applied at a distance from a reference point generates a moment (torque) about that point equal to the force times the perpendicular distance.

- The terms are often used interchangeably; in mechanics, *torque* usually refers to the turning effect in a dynamic system, *moment* in a static one. Both are calculated as force times perpendicular distance from the pivot.

- When modelling mechanics in *PGA*, force and torque do not need to be tracked separately. The line of action $F = P \wedge f$ (eqn. (272)) carries both: the force f itself and the torque it generates about any reference point. The torque at the origin is $bulk(F)$; the free moment at any other point follows from the same object F.

Wrench

- A wrench is a force screw — the bivector encoding a resultant force and the torque it induces, bound to a line of action (the "forque" of 1.7.1). It is the force-space counterpart of the velocity twist: the two pair through the spatial product $\langle \xi, P \rangle = -\,\mathrm{rwdg}(\xi, P)$ to give power.

- The momentum line of Sec. 1.7.2 is likewise the wrench dual to the velocity twist.
- A twist ξ pairs with a wrench $P = I[\xi]$ through the spatial product to give twice the kinetic energy (see eqn. (344)).

Momentum

- Momentum[23] characterizes the movement of an object: it has a direction (that of the velocity) and a magnitude (mass times speed). A heavier object moving at the same speed has more momentum; momentum changes when either speed or direction changes.
- In PGA the momentum of a body is a bivector (a momentum wrench): the image $P = I[\Omega]$ of the velocity twist Ω under the inertia map.
- It combines linear and angular momentum in one object, just as a wrench combines force and torque; the two are recovered as bulk and weight parts of P (see 1.7.2).
- Newton's second law takes the form $\dot{P} = W$ (net wrench equals rate of change of momentum wrench), from which the Euler equation follows on inverting the inertia map.

Inertia map

- The inertia map $I[\cdot]$ is the linear operator mapping a body's velocity twist Ω to its momentum wrench, $P = I[\Omega]$ (see 1.7.4).
- It is a 3×3 map (vector to bivector) in $2D$ and a 6×6 map (bivector to bivector) in $3D$, encoding mass, scalar moment of inertia and the full inertia tensor in one object; the mass contribution is $-m\,\overline{\Omega}$, the complement placing it in the dual block.
- It is additive over the parts of a body and is inverted to set up the Euler equation $\dot{\Omega} = I^{-1}[\, W - \mathrm{rcmt}(\Omega, I[\Omega])\,]$.

Jacobian

- The Jacobian of a frame maps the joint rates to the frame's twist, $V = J\,\dot{q}$ (see eqn. (351)).

[23] German: *Impuls*, *Bewegungsgröße* — not to be confused with *Moment* (torque), the entry above.

- Its columns are the joint screws in their current world position: axis lines of revolute joints, directions of prismatic joints in $3D$, pivot points in $2D$. The body form follows by one motor sandwich per column.
- Paired with a wrench W acting on the frame it gives the joint torques that balance it, the classical $\tau = J^\top F$ with $F = -W$ (see 1.8.3):

$$\tau_j = \mathrm{rwdg}(S_j, W)$$

Mobility

- The number of independent motions a mechanism can make at its current configuration.
- For a set of joints it is the dimension of the span of their screws; the mobility persists through finite motion when that span is closed under the bracket rcmt.
- For a closed chain it is $n - \operatorname{rank} G$, with n joint coordinates and the constraint Jacobian G.

Zero-moment point and centre of pressure

- The zero-moment point is the point of the ground about which the ground reaction has no tangential moment; in PGA the meet $Z = W_R \vee L$ of the reaction wrench with the ground (see eqn. (371)).
- The centre of pressure is the point where the pressure distribution of a contact area acts; on level ground without friction the zero-moment point is the centre of pressure of all contacts.
- A stance on flat contacts holds only while the zero-moment point lies inside the support polygon.

Round and flat objects

- The two families of geometric objects in CGA. A *round* object has a radius — round point, dipole, circle, sphere — and a *flat* object is a round that contains the point at infinity: flat point, line, plane.
- The *codimension-one* round is the largest round of the algebra — the circle in $CGA2DC$, the sphere in $CGA3DC$ — one dimension short of the modelled Euclidean space, and it plays among the rounds the role the hyperplane plays among the flats. The name is dimension-independent, which is what lets one statement cover both algebras; a circle in $CGA3DC$ is a round of codimension two. The library's comments write it `codim-1`.

- The grade decides which kind: in $CGA3DC$ a vector is a round point, a bivector a dipole or flat point, a trivector a circle or line, a quadvector a sphere or plane.
- Both are built by the same join: a round from the points on it, the corresponding flat from the same points and e_∞.

Dipole

- The point pair of CGA: the round object of grade two, the join of two round points, and the meet of a sphere with a line or of two circles.
- Its two points are recovered as $p_\pm = \mathrm{cen}(d) \pm \sqrt{r^2}\,\mathrm{att}(d)$ for a unitized d.
- A dipole with $r^2 < 0$ has an imaginary radius and no real points — the result of meeting objects that do not intersect.

Carrier, cocarrier and container

- The *carrier* $\mathrm{car}(u) = u \wedge e_\infty$ is the lowest-dimensional flat object containing u: the plane of a circle, the line of a dipole. It vanishes for an object that is already flat.
- The *cocarrier* $\mathrm{ccr}(u)$ is the carrier of the antidual and enters the definition of the center.
- The *container* $\mathrm{con}(u)$ is the smallest codimension-one round containing u — the sphere around a circle. Carrier and container factor the object between them, $u = \mathrm{car}(u) \vee \mathrm{con}(u)$.

Center, partner and attitude

- The *center* $\mathrm{cen}(u)$ is returned as a round point and carries the radius of u as well as its position, so a round point is its own center.
- The *partner* $\mathrm{par}(u)$ shares center and carrier with u and has r^2 negated, exchanging real and imaginary objects; it satisfies $u \cdot \mathrm{par}(u) = 0$.
- The *attitude* $\mathrm{att}(u)$ removes the origin content and returns the direction data one grade lower — the axis of a dipole, the normal of a circle. It is the PGA attitude taken with the complement of the algebra's projective basis vector.

Four-part split (round and flat bulk and weight)

- The *CGA* counterpart of the bulk/weight split of *PGA*. With two distinguished basis vectors instead of one, a component is *flat* when its blade contains e_∞ and *weight* when it contains e_o, which gives the four parts $u_\bullet + u_\circ + u_\blacksquare + u_\square$.
- Each part has its norm; unitization scales the *round weight* norm to one, and the center norm is built from the round bulk and the flat weight as the two legs of a right triangle.

Antidot square (radius norm)

- The regressive dot product of a round object with itself, $u \circ u$, which for a unitized object is the squared radius. It is signed, and its sign classifies the object as real, degenerate or imaginary.
- Its counterpart under the ordinary dot product carries the opposite sign, $u \cdot u = -r^2$, so which product is used decides the sign, not the object.

Flector

- An orientation-reversing operator of *CGA*, applied by the sandwich $-s \vee u \vee \underset{\sim}{s}$ with a sphere or circle s as the operator itself: the inversion in that sphere.
- Two inversions compose into a motor, which is how the dilation arises — from two inversions in concentric spheres.
- An inversion maps spheres to spheres, except that a sphere through the center of s becomes a plane: its round weight vanishes.

Dilation, transversion and loxodromic motion

- The conformal transformations that are not rigid motions. A *dilation* scales about a center by σ, entering the generator as the rapidity $\delta = -\frac{1}{2}\ln\sigma$; a *transversion* is a translation conjugated by an inversion.
- A *loxodromic* motion fixes the two points of a dipole and spirals between them, combining a rotation about the carrier line with a hyperbolic flow along it. Its generator is not simple, so its motor is the product of two simple exponentials.
- The ratio of the two parameters is the loxodromic parameter: it degenerates to a pure rotation at one end and to a pure hyperbolic motion at the other.

Total thickness variation (TTV)

- The total thickness variation of a wafer is the spread of its thickness over the surface, TTV = $\max t - \min t$: the difference between its thickest and thinnest points, and the principal flatness metric of a thinned wafer.
- In the surface model the thickness is the ground upper surface minus the chuck profile, a difference of two height fields; TTV is the peak-to-peak of that difference.

1.12 Links to get started

Overview to easily get started (high-level motivation and initial introductions on youtube)[24]:

- *https://www.thestrangeloop.com* (Jack Rusher, Funny intro to GA with historical perspective)
- *https://www.youtube.com/@sudgylacmoe* (sudgylacmoe, A swift intro to GA)
- *https://www.youtube.com/@Wildberger* (Norman Wildberger - Projective geometry)
- *https://www.youtube.com/@sudgylacmoe-GAoperations* (sudgylacmoe, Overview of GA ops.)
- *https://www.youtube.com/@AllAngles/GA* (All Angles - GA Intro)
- *https://www.youtube.com/@bivector/GAME23* (De Keninck and Roelfs, GAME 2023 - GA Intro)

Sources for GA and PGA as a mathematical topic:

- *https://projectivegeometricalgebra.org* (Eric Lengyel)
- *https://terathon.com/blog/poor-foundations-ga.html* (Eric Lengyel)
- *https://bivector.net* (bivector home page, Leo Dorst et.al.)
- *https://www.youtube.com/@bivector* (Youtube channel bivector)
- *https://en.wikipedia.org/wiki/Bivector* (Wikipedia Bivector)
- *https://www.youtube.com/@sudgylacmoe* (Youtube channel sudgylacmoe)
- *https://www.youtube.com/@hestenes* (Hestenes, Tutorial Geometric Calculus (relevant formulas))
- *https://en.wikipedia.org/PGA* (Plane-based GA in $R^*(3,0,1)$ instead of $R(3,0,1)$ as used here)
- *https://terathon.com/blog/dual-pga.html* (Lengyel's argumentation for using $R(3,0,1)$)

Sources for GA, PGA and STA in more specific applications:

- *https://www.youtube.com/DynamicsPGA* (Leo Dorst, Dynamics in PGA)
- *https://en.wikipedia.org/wiki/Newton-Euler* (Newton-Euler equations)
- *https://en.wikipedia.org/wiki/Rigidbody* (Rigid body dynamics)
- *https://en.wikipedia.org/wiki/Inertia* (Moment of inertia)
- *https://en.wikipedia.org/wiki/ScrewTheory* (Screw theory, twist and wrench)
- https://en.wikipedia.org/wiki/Chasles (Chasles' theorem)
- http://web.cs.iastate.edu/Pluecker (Plücker coordinates in projective spaces)
- *https://www.youtube.com/@sudgylacmoe-STA* (sudgylacmoe, A swift intro to spacetime algebra)

[24] URLs partly edited for improved readability. Don't use the printed URL directly, but rather click the link to go to the linked target.

1.13 Literature

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